DIVISORIAL EIGENVALUATIONS OF SUPERATTRACTING GERMS IN HIGHER DIMENSIONS

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Abstract. We answer the three parts of Question 6.1 of Luo and Meng on eigenvaluations of superattracting germs in dimension at least 3. We prove that any finite superattracting holomorphic germ in the first two parts, and not only an iterate of it, has a primitive divisorial eigenvaluation. For the third part, on totally invariant subvarieties of the Jacobian exceptional set of a polarized endomorphism of projective space, we construct a counterexample in each dimension at least 3.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

For a normal projective variety X, a surjective endomorphism $f : X \to X$ is *q-polarized* if $f^*H \sim qH$ for an ample Cartier divisor H on X and an integer $q > 1$. When $X = \mathbb{P}^n$, the condition reads

$$f^*\mathcal{O}_{\mathbb{P}^n}(1) \cong \mathcal{O}_{\mathbb{P}^n}(q).$$

A conjecture of Gongyo [Men23, Conjecture 1.3] predicts that, for any q-polarized endomorphism f of a smooth projective variety X, there exists an integer $m \geq 1$ such that $(X, R(f^m)/(q^m - 1))$ is lc, where $R(f^m)$ is the ramification divisor of f^m. Luo and Meng proved Gongyo's conjecture for $X = \mathbb{P}^2$ [LM26a, Theorem 1.3], and Yoshikawa proved that, for any normal $\mathbb{Q}$-Gorenstein projective variety X with a q-polarized endomorphism f, there exists an integer $r_0 \geq 1$ such that $(X, R(f^{mr_0})/(q^{mr_0} - 1))$ is lc for any integer $m \geq 1$ [Yos26, Theorem 1.4]. In particular, Gongyo's conjecture holds. For any normal projective variety X, not necessarily $\mathbb{Q}$-Gorenstein,

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and any q-polarized endomorphism f of X, there exists an integer $s \geq 1$ such that for every integer $m \geq 1$ the pair $(X, R(f^{ms})/(q^{ms}-1))$ is lc and $K_X + R(f^{ms})/(q^{ms}-1) \sim_{\mathbb{Q}} 0$ [Liu26b, Theorem 1.5]. Any normal projective variety X with a surjective endomorphism h such that $h^*H - H$ is ample for an ample Cartier divisor H on X is of Calabi–Yau type, that is, there exists a $\mathbb{Q}$-divisor $B \geq 0$ on X such that (X, B) is lc and $K_X + B \sim_{\mathbb{Q}} 0$ [Liu26b, Theorem 1.3]. In particular, any normal projective variety admitting a q-polarized endomorphism for some integer $q > 1$ is of Calabi–Yau type, as conjectured by Broustet and Gongyo [Liu26b, Corollary 1.4]. For $X = \mathbb{P}^2$, the pair $(\mathbb{P}^2, R(f^N)/(q^N-1))$ is lc for $N := 17280 \cdot 2^{200}$, any integer $q \geq 2$, and any q-polarized endomorphism f [Liu26a, Theorem 1.5].

Let $n \geq 1$ be an integer, let $R := \mathbb{C}\{z_1, \ldots, z_n\}$, and let $\mathfrak{m} := (z_1, \ldots, z_n)$ be the maximal ideal of R. For a finite holomorphic germ $F : (\mathbb{C}^n, 0) \to (\mathbb{C}^n, 0)$ with pullback $F^* : R \to R$, the *local degree* of F and the *attraction rates* of the iterates F^j are

$$d(F) := \ell_R(R/F^*(\mathfrak{m})R), \qquad c(F^j) := \max\{i \in \mathbb{N} \mid (F^j)^*(\mathfrak{m})R \subset \mathfrak{m}^i\} \quad (j \in \mathbb{N}),$$

and the *asymptotic attraction rate* of F is

$$c_\infty(F) := \lim_{j \to +\infty} c(F^j)^{1/j},$$

which exists since $c(F^{j+k}) \geq c(F^j)c(F^k)$ for all $j, k \in \mathbb{N}$. The germ F is *superattracting* if the differential dF_0 is nilpotent, equivalently, $c(F^j) \to +\infty$ as $j \to +\infty$. For a real valuation v of $\mathrm{Frac}(R)$, we let $F_*v := v \circ F^*$, where F^* is extended to $\mathrm{Frac}(R)$. Similarly, $f_*v := v \circ f^*$ for a finite surjective endomorphism f of a normal projective variety X and a real valuation v of the function field of X. Luo and Meng define $d(F)$, $c(F^j)$, $c_\infty(F)$, superattracting germs, and F_*v when $n = 2$ [LM26a, Section 2.3]. The definitions above extend theirs to any $n \geq 1$, and we read Question 1.1 below with them.

Let $F : (\mathbb{C}^2, 0) \to (\mathbb{C}^2, 0)$ be a finite superattracting germ, so that $n = 2$ and $R = \mathbb{C}\{z_1, z_2\}$. Favre and Jonsson proved that there exists an element $\nu_\star$ of the valuative tree of R, that is, a semivaluation $\nu_\star$ on R with $\nu_\star(\mathfrak{m}) = 1$, such that $\nu_\star \circ F^* = c_\infty(F)\nu_\star$ [FJ07, Theorem 4.2] (see [LM26a, Proposition 2.10(i)]). The semivaluation $\nu_\star$ is quasi-monomial (see Definition 2.3 for quasi-monomial and primitive divisorial valuations), or an analytic curve valuation, which takes the value $+\infty$ on an equation of a curve, or an infinitely singular valuation. Favre and Jonsson also proved that there exists a real number b with $0 < b \leq 1$ and

$$bc_\infty(F)^j \leq c(F^j) \leq c_\infty(F)^j \quad \text{for any integer } j \geq 1$$

[FJ07, Theorem A]. The lower bound in [FJ07, Theorem A] is the 2-dimensional form of the bound $c(F^j) \geq bq^j$ in Theorem 1.4(a). By a theorem of Gignac and Ruggiero [GR14, Theorem 3.1] (see [LM26a, Proposition 2.11]), either F has a unique eigenvaluation in the valuative tree of R in the sense of [LM26a, Section 2.3], or there exists a non-degenerate segment in the valuative tree of R consisting of quasi-monomial valuations ν with $(F^2)_*\nu = c_\infty(F)^2\nu$. For a finite non-invertible germ h at a normal surface singularity x, with $\mathfrak{m}_x$ the maximal ideal of the local ring at x, Gignac and Ruggiero extended the classification of [GR14, Theorem 3.1]: if the ramification divisor of h is non-zero, then x is a smooth point or a quotient singularity, and a point or a segment of semivaluations ν with $\nu(\mathfrak{m}_x) = 1$, fixed pointwise by the square of the map $\nu \to \nu \circ h^*/\nu(h^*\mathfrak{m}_x)$, attracts every quasi-monomial valuation ν with $\nu(\mathfrak{m}_x) = 1$ under the iterates of that square [GR21, Theorem B and Theorem 4.41]. Using [GR14, Theorem 3.1], Luo and Meng proved that, if $d(F) = c_\infty(F)^2 = q^2$ for an integer $q > 1$, then there exists a primitive divisorial valuation v centered at the origin such that $(F^2)_*v = q^2v$ [LM26a, Theorem 4.2]. In dimension 2, for a q-polarized endomorphism f of a smooth projective surface X, a point $p \in X$ with $f^{-1}(p) = \{p\}$, and a primitive divisorial valuation v centered at p with $f_*v = qv$, Luo and Meng construct a projective birational morphism $\pi : Y \to X$ from a normal projective surface Y whose exceptional locus is a prime divisor E with $\mathrm{ord}_E = v$, and a finite q-polarized endomorphism $g : Y \to Y$ with $\pi \circ g = f \circ \pi$ and $g^*E = qE$ [LM26a, Corollary 4.5], and prove that $(X, R(f^m)/(q^m-1))$ is lc near p for some integer $m \geq 1$ [LM26a, Proposition 4.7].

Let $f : \mathbb{P}^n \to \mathbb{P}^n$ be a q-polarized endomorphism. A closed subvariety $Z \subset \mathbb{P}^n$ is *totally invariant* if $f^{-1}(Z) = Z$. Every totally invariant irreducible closed subvariety of $\mathbb{P}^n$ is a linear subspace [LM26b, Linearity Theorem]. The *Jacobian exceptional set* of f is

$$\mathcal{E}_{\mathrm{Jac}}(f) := \bigcup_{\delta>0} \bigcap_{j\geq 1} \{x \in \mathbb{P}^n \mid \mathrm{mult}_x R(f^j) \geq \delta q^j\},$$

introduced by Luo and Meng [LM26a, Section 1] following [Par11, Theorem 4.3]. In [LM26a, Section 6.1], Luo and Meng explain that their proof in dimension 2 uses the divisorial eigenvaluation of [LM26a, Theorem 4.2] and the equivariant extraction of [LM26a, Corollary 4.5], and that "Neither step has a direct higher-dimensional analogue." By [LM26a, Theorem 1.4], for a q-polarized endomorphism $f : \mathbb{P}^n \to \mathbb{P}^n$ and any integer $m \gg 0$, the pair $(\mathbb{P}^n, R(f^m)/(q^m - 1))$ is klt on $\mathbb{P}^n \backslash \mathcal{E}_{\mathrm{Jac}}(f)$. On the other hand, if $m \geq 1$ is an integer and E is a prime divisor over $\mathbb{P}^n$ with $(f^m)_* \operatorname{ord}_E = q^m \operatorname{ord}_E$, then Lemma 3.2 with $B = 0$ gives

$$q^m a(\operatorname{ord}_E, \mathbb{P}^n) = a((f^m)_* \operatorname{ord}_E, \mathbb{P}^n) = a(\operatorname{ord}_E, \mathbb{P}^n) + \mathrm{mult}_E R(f^m),$$

and hence $a(\operatorname{ord}_E, \mathbb{P}^n, R(f^m)/(q^m - 1)) = 0$ (see Definition 2.3(5) for log discrepancies). Luo and Meng asked the following question.

Question 1.1 ([LM26a, Question 6.1])**.** Let $n \geq 3$ and let $F : (\mathbb{C}^n, 0) \to (\mathbb{C}^n, 0)$ be a finite superattracting germ satisfying $d(F) = c_\infty(F)^n = q^n$, where $q > 1$ is an integer. Does there exist an integer $m \geq 1$ and a quasi-monomial valuation v centered at the origin such that $(F^m)_* v = q^m v$? Can v always be taken to be primitive divisorial? More generally, if $f : \mathbb{P}^n \to \mathbb{P}^n$ is q-polarized and $Z \subset \mathcal{E}_{\mathrm{Jac}}(f)$ is a totally invariant irreducible subvariety of codimension at least 2, can one find an integer $s \geq 1$ and a primitive divisorial valuation v centered at Z such that $(f^s)_* v = q^s v$?

For $n = 2$, a theorem of Luo and Meng [LM26a, Theorem 4.2] answers the analogues of the first two parts of Question 1.1 affirmatively, with $m = 2$.

Yoshikawa answered the third part of Question 1.1 affirmatively under an additional hypothesis [Yos26, Remark 5.11]. Let $f : \mathbb{P}^n \to \mathbb{P}^n$ be a q-polarized endomorphism, so that $\deg f = q^n$, and let

$$T_{f,0} := \sum_{j\geq 1} [q^{-nj} (f^j)_* R(f)]$$

[Yos26, Definition 3.1(4) and Theorem 4.5], where $(f^j)_*$ is the pushforward of divisors and $[\cdot]$ is the current of integration. The current $T_{f,0}$ is a closed positive $(1,1)$-current on $\mathbb{P}^n$. Yoshikawa defines the non-klt locus $\mathrm{Nklt}(\mathbb{P}^n, 0; T_{f,0})$ through multiplier ideals of currents [Yos26, Definition 4.2], which are not the multiplier ideals of ideals used from Section 4 on. If $Z \subset \mathbb{P}^n$ is a totally invariant subvariety and an irreducible component of $\mathrm{Nklt}(\mathbb{P}^n, 0; T_{f,0})$, then there exist an integer $m \geq 1$ and a prime divisor E over $\mathbb{P}^n$ with $\mathrm{center}_{\mathbb{P}^n} E = Z$ and $(f^m)_* \operatorname{ord}_E = q^m \operatorname{ord}_E$ [Yos26, Remark 5.11].

Our proof of Theorem 1.4(a) and (b) adapts the proof of the first of the following two results and uses the second.

Theorem 1.2 ([Yos26, Proposition 5.9])**.** *Let $k_0 \supset \mathbb{C}$ be an algebraically closed field, let (Y, B_Y) be a klt pair of finite type over k_0 with $B_Y \geq 0$ a $\mathbb{Q}$-divisor, let $y \in Y$ be a closed point, and let $S := \mathcal{O}_{Y,y}$ and $\mathfrak{n} := \mathfrak{m}_y S$. Let $\mathfrak{b}_\bullet$ be a linearly bounded $\mathfrak{n}$-filtration of S with*

$$\mathrm{lct}(Y \ni y, B_Y; \mathfrak{b}_\bullet) = 1,$$

and let $\mathcal{W}$ be the set of real valuations u of the function field of Y centered at y with $a(u, Y, B_Y) = u(\mathfrak{b}_\bullet) = 1$. Then $\mathcal{W} \neq \emptyset$, and vol_S has a unique minimizer on $\mathcal{W}$. The minimizer is quasi-monomial.

Theorem 1.3 ([BLQ24, Proposition 3.12])**.** *Let $(A, \mathfrak{m}_A)$ be an analytically irreducible Noetherian local domain, and let $\mathfrak{a}_\bullet \subset \mathfrak{b}_\bullet$ be $\mathfrak{m}_A$-filtrations with $e(\mathfrak{a}_\bullet) = e(\mathfrak{b}_\bullet)$. Then $v(\mathfrak{a}_\bullet) = v(\mathfrak{b}_\bullet)$ for any real valuation v of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$ with $\mathrm{vol}_A(v) > 0$.*

We do not know of an earlier answer to the first two parts of Question 1.1 or of an earlier counterexample to the third part. The goal of this paper is to answer the three parts of Question 1.1.

Main results.

Theorem 1.4. *Let $n \geq 3$ and $q \geq 2$ be integers.*

(a) Let $R := \mathbb{C}\{z_1, \dots, z_n\}$ with maximal ideal $\mathfrak{m}$, and let $F : (\mathbb{C}^n, 0) \to (\mathbb{C}^n, 0)$ be a finite holomorphic germ with $d(F) = c_\infty(F)^n = q^n$. Then there exists a real number $b > 0$ with $c(F^j) \geq bq^j$ for every integer $j \geq 0$, and there exists a quasi-monomial valuation v of $\mathrm{Frac}(R)$ centered at $\mathfrak{m}$ with $F_*v = qv$.

(b) Let R, $\mathfrak{m}$, and F be as in (a). Then there exists a primitive divisorial valuation w of $\mathrm{Frac}(R)$ centered at $\mathfrak{m}$ with $F_*w = qw$. Moreover, w is the order of vanishing along a prime divisor of a proper bimeromorphic modification of a neighborhood of 0 in $\mathbb{C}^n$.

(c) Let r be an integer with $3 \leq r \leq n$, let $f : \mathbb{P}^n \to \mathbb{P}^n$ be given by

$$f([U_0 : \cdots : U_n]) = [U_0^q : U_1U_0^{q-1} - U_2^q : U_1^q : U_3^q : \cdots : U_n^q],$$

and let $Z_r := \{U_1 = \cdots = U_r = 0\}$. Then f is a finite q-polarized endomorphism, Z_r is a totally invariant irreducible subvariety with $\dim Z_r = n - r$ and $Z_r \subset \mathcal{E}_{\mathrm{Jac}}(f)$, and for every integer $k \geq 1$ no real valuation v of the function field of $\mathbb{P}^n$ centered at the generic point of Z_r satisfies $(f^k)_*v = q^kv$.

Parts (a) and (b) answer the first two parts of Question 1.1 affirmatively, with $m = 1$. The germ F in (a) is superattracting, since $c_\infty(F) = q \geq 2$ implies $c(F^j) \to +\infty$ as $j \to +\infty$. Part (c) answers the third part negatively for any $n \geq 3$: since

$$\operatorname{codim} Z_r = r \geq 3,$$

the subvariety Z_r is in the range of the third part. For $r = n$, the subvariety $Z_n = \{[1 : 0 : \cdots : 0]\}$ is a closed point, so the negative answer given by (c) does not depend on how "centered at Z" is read. Moreover, the germ F of f at $[1 : 0 : \cdots : 0]$ satisfies $c_\infty(F) = 1 < q$, since $(f^j)^*(U_1/U_0) - U_1/U_0 \in \mathfrak{m}^2$ for any integer $j \geq 1$, where $\mathfrak{m}$ is the maximal ideal of the local ring of $\mathbb{P}^n$ at $[1 : 0 : \cdots : 0]$. Thus F is not in the range of (a). For $n > r$, we have $\dim Z_r = n - r > 0$, and we read "centered at Z" in Question 1.1 as centered at the generic point of Z_r. Here the reading cannot be relaxed to "centered at a point of Z_r": in the coordinates $U_0/U_n, \dots, U_r/U_n$, the monomial valuation v with weights $2, 1, \dots, 1$ is centered at the generic point of $Z_r \cap \{U_0 = 0\}$, is primitive divisorial, and satisfies $f_*v = qv$, since f^* maps U_0/U_n, U_1/U_n, U_2/U_n, U_i/U_n $(3 \leq i \leq n-1)$ to $(U_0/U_n)^q$, $(U_1/U_n)(U_0/U_n)^{q-1} - (U_2/U_n)^q$, $(U_1/U_n)^q$, $(U_i/U_n)^q$, whose v-initial forms $(U_0/U_n)^q$, $-(U_2/U_n)^q$, $(U_1/U_n)^q$, $(U_i/U_n)^q$ (as $2q - 1 > q$) are monomials in distinct variables of q times the weights. Real valuations and their centers, including centers at the generic point of a subvariety, are as in Definition 2.3(1). Since $H_3 := \{U_3 = 0\} \subset \mathrm{Nklt}(\mathbb{P}^n, 0; T_{f,0})$ (Lemma 13.1) and $Z_r \subsetneq H_3$, the subvariety Z_r is not an irreducible component of $\mathrm{Nklt}(\mathbb{P}^n, 0; T_{f,0})$, so Theorem 1.4(c) is outside the hypothesis of [Yos26, Remark 5.11].

Sketch of the proof. Parts (a) and (b) of Theorem 1.4 depend on Theorem 6.5 and the bound $c(F^j) \geq bq^j$ (Section 13.1). Theorem 6.5 considers a local ring $(B, \mathfrak{n})$ as in Set-up 4.8, so that K_B is $\mathbb{Q}$-Cartier and $\operatorname{Spec} B$ is klt, a finite injective local $\mathbb{C}$-algebra homomorphism $h : B \to B$, an integer $\Lambda \geq 2$, and a real number $b > 0$ with

$$\begin{aligned} &[\mathrm{Frac}(B) : h(\mathrm{Frac}(B))] = \Lambda^{\dim B}[B/\mathfrak{n} : h_{B/\mathfrak{n}}(B/\mathfrak{n})] \\ &\text{and} \quad c(h^j) \geq b\Lambda^j \text{ for any integer } j \geq 0 \end{aligned} \tag{1.1}$$

(Definition 2.10), and yields a primitive divisorial valuation w of $\mathrm{Frac}(B)$ centered at $\mathfrak{n}$ with $h_*w = \Lambda w$. For $(B, h, \Lambda) = (R, F^*, q)$, the equality in (1.1) reads $d(F) = q^n$ (Lemma 2.12), so if there exists a real number $b > 0$ with $c(F^j) \geq bq^j$ for any integer $j \geq 0$, then Theorem 6.5 for (R, F^*, q) yields (a) with $v := w$ and (b). The bimeromorphic modification in (b) is the

analytification of the extraction in Construction-Proposition 9.4. If $\inf_{j\geq 0} c(F^j)/q^j = 0$, then there exist an integer $a \geq 1$ and prime ideals $0 \neq C \subsetneq P$ of R, totally invariant under $(F^a)^*$, such that

- $A := (R/C)_{P/C}$ is of klt type and $\dim A \geq 2$,
- the endomorphism h of A induced by $(F^a)^*$ satisfies (1.1) with (A, h, q^a) in place of (B, h, Λ), and
- for any integer $i \geq 1$ and any $\mathfrak{m}_A$-primary ideal I, $h^i(I)A \not\subset \overline{I^{q^{ai}}}$, where $\overline{J}$ denotes the integral closure of an ideal J

(Construction-Theorem 10.6). There exist integers $m, N \geq 1$, a finite lift $g : Y \to Y$ of $\mathrm{Spec}(h^m)$ to a small $\mathbb{Q}$-factorial model Y of $\mathrm{Spec}\, A$, and an irreducible closed subset $V \subset Y$ over $\mathfrak{m}_A$ with generic point η_V and $(g^N)^{-1}(V) = V$ such that $(\mathcal{O}_{Y,\eta_V}, (g^N)^*, q^{amN})$ satisfies Set-up 4.8 (Construction-Theorems 11.1 and 12.2). Theorem 6.5 for $\mathcal{O}_{Y,\eta_V}$ then yields an integer $i \geq 1$ and an $\mathfrak{m}_A$-primary ideal I with $h^i(I)A \subset I^{q^{ai}}$ (Proposition 9.3), a contradiction.

Theorem 6.5 follows [Yos26, Section 5], where $B = \mathcal{O}_{X,\eta_Z}$ and $h = f^*$ for an int-amplified endomorphism f of a projective variety X [Yos26, Definition 3.1(2)] and a subvariety $Z \subset X$ with $f^{-1}(Z) = Z$ which is an irreducible component of the non-klt locus of [Yos26, Definition 4.2] of the current of [Yos26, Theorem 4.5] not contained in the non-klt locus of the boundary [Yos26, Notation 5.2]. In our notation, the proof runs as follows.

(1) As in [Yos26, Proposition 5.8], an invariant probability measure ρ of $v \to \Lambda^{-1}h_*v$ on the closure, for pointwise convergence, of the orbit of a divisorial valuation, with $\int a(v, \mathrm{Spec}\, B)d\rho(v) = 1$, defines $\theta := \int v d\rho(v)$ on $B\backslash\{0\}$, and the intersections over $0 < t' < t$ of the ideals generated by $\theta^{-1}([t', +\infty))$, for $t > 0$, form an $\mathfrak{n}$-filtration $\mathfrak{b}_\bullet$ with $\mathrm{lct}(B; \mathfrak{b}_\bullet) = 1$. Since the $\mathfrak{n}$-filtrations $(h(\mathfrak{b}_t)B)_t \subset (\mathfrak{b}_{\Lambda t})_t$ have multiplicity $\Lambda^{\dim B} e(\mathfrak{b}_\bullet)$, $(h_*v)(\mathfrak{b}_\bullet) = \Lambda v(\mathfrak{b}_\bullet)$ for any real valuation v centered at $\mathfrak{n}$ with $a(v, \mathrm{Spec}\, B) < +\infty$ (Theorem 1.3). The set $\mathcal{W}$ of real valuations v centered at $\mathfrak{n}$ with $a(v, \mathrm{Spec}\, B) = v(\mathfrak{b}_\bullet) = 1$ is compact, and $\Lambda u \in \mathcal{W}$ whenever $h_*u \in \mathcal{W}$ (Construction-Proposition 4.10).
(2) vol_B has a unique minimizer v_0 on $\mathcal{W}$, and v_0 is quasi-monomial (Section 6.3).
(3) Since h_* is surjective (**Step 4** of the proof of Lemma 3.2), there exists a real valuation u with $h_*u = v_0$. Then $\Lambda u \in \mathcal{W}$ and

$$\mathrm{vol}_B(v_0) = \mathrm{vol}_B(\Lambda^{-1}h_*(\Lambda u)) \geq \mathrm{vol}_B(\Lambda u)$$

(Lemma 3.3), so $\Lambda u = v_0$ by (2), and $h_*v_0 = \Lambda v_0$.
(4) Rational weights in the monomial coordinates of v_0 yield w (Lemma 6.4), as in [Yos26, proof of Theorem 5.1].

The differences from [Yos26, Section 5] are in (1) and (2).

(i) In [Yos26, Section 5], the orbit is bounded, $c(h^j) \geq b\Lambda^j$ for a real number $b > 0$ [Yos26, Proposition 5.6], and $\mathrm{lct}(B; \mathfrak{b}_\bullet) \geq 1$, all by [Yos26, Theorem 3.6]. For Theorem 6.5, the bound $c(h^j) \geq b\Lambda^j$ is a hypothesis, and the inequality $v(R(h)) \geq (\Lambda - 1)v(\mathfrak{b}_\bullet)$ for any real valuation v centered at $\mathfrak{n}$ with $a(v, \mathrm{Spec}\, B) < +\infty$ (Construction-Proposition 4.10(2)) replaces [Yos26, Theorem 3.6].
(ii) [Yos26, Proposition 5.9] (Theorem 1.2) obtains the minimizer through [Xu20, Theorem 1.1] and [Blu18, Proposition 3.7 and Section 5.1], proved for varieties over a field, and B need not be essentially of finite type over a field (for example $B = \mathbb{C}\{z_1, \ldots, z_n\}$ with $h(z_i) = z_i^2$ and $\Lambda = 2$). Section 6.3 replaces them by the convexity of multiplicities [BLQ24, Section 4] and of log canonical thresholds (Lemma 4.7) along geodesics (Definition 2.8(4)), a convergence of multiplicities uniform over intersections of the filtrations $\mathfrak{a}_\bullet(v)$ of valuation ideals (Definitions 2.3(2) and 2.8(2)) for $v \in \mathcal{W}$ (Lemma 6.2), and Theorem 5.4, which carries the argument of [Xu20, Theorem 3.3 and Lemma 3.7] over to B through a strict henselization of B and complements of bounded index [Bir19, Theorem 1.7].

For A and φ as in Set-up 7.1 such that $Q := c_\infty(\varphi) \geq 2$ is an integer, (A, φ, Q) satisfies the equality in (1.1), and $\inf_j c(\varphi^j)/Q^j = 0$, there exist an integer $a \geq 1$ and a prime ideal $0 \neq P \neq \mathfrak{m}_A$, totally invariant under φ^a, such that $(A/P, (\varphi^a)_{A/P}, Q^a)$ satisfies the equality and $(A_P, (\varphi^a)_{A_P}, Q^a)$ the bound in (1.1) (Theorem 8.2). The canonical log discrepancy $a_{\mathrm{can}}(\cdot, A)$ of Definition-Theorem 7.3, built from nef envelopes [BdFF12], satisfies

$$a_{\mathrm{can}}(\varphi_* v, A) = a_{\mathrm{can}}(v, A) + v(\mathbf{R}_\varphi) \quad \text{with} \quad v(\mathbf{R}_\varphi) \geq 0,$$

where $\mathbf{R}_\varphi$ is the ramification $\boldsymbol{b}$-divisor of Definition-Theorem 7.3. For an integer $a_0 \geq 1$ with $c(\varphi^{a_0}) \geq 2$, we normalize the $(\varphi^{a_0})_*$-iterates of a divisorial valuation centered at $\mathfrak{m}_A$ by a_{can}, take an invariant probability measure of their limits, which has all its mass over periodic prime ideals other than $\mathfrak{m}_A$ (Lemmas 7.4 and 8.1, based on [Gig14, Theorem 1.2 and Proposition 6.4]), and, by the multiplicity formula over the periodic prime ideals, compare the integral of $\log(1 + u(\mathbf{R}_{\varphi^{a_0}}))$ over the valuations u centered at the periodic prime ideals with $a_0 \log Q$ times the total mass (Theorem 8.2).

Construction-Theorem 10.6 minimizes first $\operatorname{ht} P$ and then $\dim(S/\mathfrak{c})$. We take an integer $e \geq 1$ and a prime ideal P of R with $\operatorname{ht} P$ minimal such that Set-up 9.7 holds with $((F^e)^*, q^e)$ in place of (φ, Q), set $S := R_P$, and, after replacing e with a multiple, take a primitive divisorial valuation w centered at a prime ideal of R properly contained in P with $(F^e)_* w = q^e w$ (Theorems 8.2 and 6.5 and Construction-Proposition 9.4) such that the minimal lc center $V(\mathfrak{c})$ of the lc pair on $\operatorname{Spec} S$ defined by w in Construction-Proposition 9.8 has $\dim(S/\mathfrak{c})$ minimal. Then, by subadjunction (cf. [FG12, Theorem 7.2]), $S/\mathfrak{c}$ is of klt type (see Lemma 9.6), the third item holds (see Lemma 10.1 and Set-up 9.7), and the bound in the second item follows from the minimality of $\dim(S/\mathfrak{c})$ and the minimality of $\operatorname{ht} P$ (see Theorem 8.2 and Lemma 10.3), so Construction-Theorem 10.6 holds with $a := e$, $C := \mathfrak{c} \cap R$, and $A := S/\mathfrak{c} = (R/C)_{P/C}$.

We pass to Y since Theorem 6.5 uses that K_B is $\mathbb{Q}$-Cartier (for $R(h)$ in (i)), whereas K_A need not be $\mathbb{Q}$-Cartier. Up to isomorphism, the small $\mathbb{Q}$-factorial models of $\operatorname{Spec} A$ are finitely many (cf. [LM22, Theorem 17.3]), and the normalized base change by $\operatorname{Spec} h$ takes small $\mathbb{Q}$-factorial models of $\operatorname{Spec} A$ to small $\mathbb{Q}$-factorial models of $\operatorname{Spec} A$, so some $\operatorname{Spec}(h^m)$ lifts to g (Construction-Theorem 11.1). For the probability measure μ of Construction-Proposition 11.2, invariant under $v \to q^{-am} g_* v$, there exists an irreducible closed subset V of Y over $\mathfrak{m}_A$ with generic point η_V such that the set of valuations whose center on Y is η_V has positive μ-measure and $g^N(\eta_V) = \eta_V$ for an integer $N \geq 1$ (Lemma 8.1), and V satisfies $(g^N)^{-1}(V) = V$ and (1.1) for $(\mathcal{O}_{Y,\eta_V}, (g^N)^*, q^{amN})$ (Construction-Theorem 12.2).

For Part (c), let $G(x, y) := (x - y^q, x^q)$. On $\{U_0 \neq 0\}$, f^k acts by G^k on $(x, y) := (U_1/U_0, U_2/U_0)$ and by $U_i/U_0 \to (U_i/U_0)^{q^k}$ for $3 \leq i \leq n$. Thus H_3 has coefficient $q^k - 1$ in $R(f^k)$, so $Z_r \subset H_3 \subset \mathcal{E}_{\mathrm{Jac}}(f)$, and $x_i := (f^i)^* x$ satisfies $x_{i+1} = x_i - x_{i-1}^{q^2}$ for any integer $i \geq 1$. If a real valuation v centered at the generic point of Z_r satisfies $(f^k)_* v = q^k v$, then $v(x_{i+k}) = q^k v(x_i)$ for any $i \geq 0$, whereas the ultrametric inequality forces $v(x_i)$ to be constant for $i \gg 0$ or $v(x_i) = q^2 v(x_{i-1})$ for any $i \geq 1$. Neither alternative is compatible with $v(x_{i+k}) = q^k v(x_i)$, since $v(x_i) > 0$ and $1 \neq q^k \neq q^{2k}$ (Section 13.2).

Structure of the paper. Section 2 collects preliminaries on valuations, filtrations, volumes, local endomorphisms, and invariant measures. Section 3 studies log discrepancies and volumes of real valuations under finite morphisms. Section 4 constructs an invariant filtration with log canonical threshold 1 on a klt local ring with a finite endomorphism satisfying analogues of $d(F) = q^n$ and $c(F^j) \geq bq^j$. Section 5 finds quasi-monomial valuations computing log canonical thresholds of graded sequences of ideals on klt local rings. Section 6 proves that the endomorphism in Section 4 has a primitive divisorial eigenvaluation. Section 7 defines a canonical log discrepancy on local rings of klt type. Section 8 finds a totally invariant prime ideal when the analogue of $c(F^j) \geq bq^j$ fails for every $b > 0$. Section 9 extracts the prime divisor of a primitive divisorial eigenvaluation equivariantly and studies the minimal lc center of the associated ideal pair. Section 10 reduces the failure of $c(F^j) \geq bq^j$ for every $b > 0$ to a local ring of klt type.

Section 11 constructs an equivariant small model and an invariant measure for the local ring of Section 10. Section 12 finds a totally invariant periodic center on the small model. Section 13 proves the main theorem.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B).

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

The definitions of pair, sub-pair, log discrepancy, (sub-)lc, (sub-)klt, lc place and lc center in the first two paragraphs are used also when X is a normal integral excellent scheme of characteristic zero admitting a dualizing complex, where K_X is given by the dualizing complex and log resolutions exist by [Tem18, Theorem 1.1.6].

Let A be a normal excellent local domain of characteristic zero admitting a dualizing complex. If there exists a $\mathbb{Q}$-divisor $\Delta \geq 0$ on $\operatorname{Spec} A$ such that $K_{\operatorname{Spec} A} + \Delta$ is $\mathbb{Q}$-Cartier and $(\operatorname{Spec} A, \Delta)$ is klt, then we say that A is *of klt type*.

Lemma 2.2. *Let $n \geq 1$ be an integer and A a normal local domain essentially of finite type over $\mathbb{C}\{z_1, \ldots, z_n\}$. Then A is excellent, admits a dualizing complex, and is analytically irreducible.*

Proof. Since $\mathbb{C}\{z_1, \ldots, z_n\}$ is excellent [Mat80, §40, Theorem 102 and the Remark after it] and rings of fractions and algebras of finite type over an excellent ring are excellent [Gro65, Scholie 7.8.3 (ii)], A is excellent. Since $\mathbb{C}\{z_1, \ldots, z_n\}$ is a regular local ring, hence Gorenstein, $\mathbb{C}\{z_1, \ldots, z_n\}[0]$ is a dualizing complex [Sta, Tag 0AWX], and any algebra essentially of finite type over a Noetherian ring with a dualizing complex admits a dualizing complex [Sta, Tag 0A7K]. Thus A admits a dualizing complex. The completion $\widehat{A}$ is normal [Gro65, Scholie 7.8.3 (v)], hence a domain, that is, A is analytically irreducible. □

2.2. Valuations.

Definition 2.3. Let $(A, \mathfrak{m}_A)$ be a Noetherian local domain.

(1) A *real valuation* of $\operatorname{Frac}(A)$ is a map $v : \operatorname{Frac}(A)^* \to \mathbb{R}$ with $v(fg) = v(f) + v(g)$ for all $f, g \in \operatorname{Frac}(A)^*$, $v(f + g) \geq \min\{v(f), v(g)\}$ for all $f, g \in \operatorname{Frac}(A)^*$ with $f + g \neq 0$, and $v(\operatorname{Frac}(A)^*) \neq \{0\}$, such that $v(\mathbb{C}^*) = \{0\}$ when $\mathbb{C} \subset A$. For a prime ideal P of A, the valuation v is *centered at* P if $v(f) \geq 0$ for any $f \in A \backslash \{0\}$ and

$$P \backslash \{0\} = \{f \in A \backslash \{0\} \mid v(f) > 0\}.$$

For an integral scheme X and a real valuation v of the function field of X, a point $x \in X$ is a *center* of v on X, and v is *centered at* x, if v is centered at the maximal ideal of

$\mathcal{O}_{X,x}$. If X is separated, then v has at most one center on X, denoted by $\operatorname{center}_X v$ when it exists, and $\operatorname{center}_X v$ exists when there exists a proper birational morphism $X \to \operatorname{Spec} A$ and $v \geq 0$ on $A\backslash\{0\}$. For a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L on X and v centered at $x \in X$, $v(L) := v(g)/N$, where $N \geq 1$ is an integer with NL Cartier and g is a local equation of NL at x.

(2) For a real valuation v of $\operatorname{Frac}(A)$ with $v \geq 0$ on $A\backslash\{0\}$, a real number t, and a non-zero ideal I of A,

$$\mathfrak{a}_t(v) := \{0\} \cup \{f \in A\backslash\{0\} \mid v(f) \geq t\}, \qquad v(I) := \min\{v(f) \mid f \in I\backslash\{0\}\}.$$

For a finite injective homomorphism $\varphi : A \to A$, which is a $\mathbb{C}$-algebra homomorphism when $\mathbb{C} \subset A$, and a real valuation v of $\operatorname{Frac}(A)$, $\varphi_* v$ is the real valuation $f \to v(\varphi(f))$ of $\operatorname{Frac}(A)$, where φ is extended to $\operatorname{Frac}(A)$, and v is an *eigenvaluation* of φ if $\varphi_* v = \lambda v$ for a real number $\lambda > 0$. Similarly, for a finite surjective morphism $h : X' \to X$ of integral schemes over $\mathbb{C}$ and a real valuation v of the function field of X', $h_* v := v \circ h^*$, where h^* is the induced homomorphism of function fields.

(3) For a prime divisor E on a normal integral scheme Y with a proper birational morphism $Y \to \operatorname{Spec} A$, the order of vanishing along E is a real valuation of $\operatorname{Frac}(A)$. A real valuation v of $\operatorname{Frac}(A)$ is *divisorial* if $v = c \operatorname{ord}_E$ for some real number $c > 0$ and some such E, and *primitive divisorial* if moreover $v(\operatorname{Frac}(A)^*) = \mathbb{Z}$.

(4) A real valuation v of $\operatorname{Frac}(A)$ is *quasi-monomial* if there exist a regular integral scheme Y with a proper birational morphism $Y \to \operatorname{Spec} A$ and a simple normal crossing (snc) divisor on Y such that v is monomial with positive real weights in local equations of components of that divisor at the generic point of a stratum [JM12, Section 3.1].

(5) Let X be a normal integral excellent scheme of characteristic zero admitting a dualizing complex, B a $\mathbb{Q}$-divisor on X with $K_X + B$ $\mathbb{Q}$-Cartier, and v a real valuation of the function field of X with a center on X. Then $a(v, X, B) := a(v, Y) - v(B_Y)$, where $Y \to X$ is a log resolution of (X, B), $K_Y + B_Y$ is the pullback of $K_X + B$, and $a(v, Y)$ is the log discrepancy $A_Y(v)$ of [JM12, Section 5]. We write $a(v, X) := a(v, X, 0)$.

By [JM12, Proposition 5.1(i) and Remark 5.6] and [Tem18, Theorem 1.1.6], $a(v, X, B)$ in Definition 2.3(5) does not depend on Y, and $a(v, X, B) = ca(E, X, B)$ for any real number $c > 0$, any prime divisor E over X, and v equal to c times the order of vanishing along E.

Lemma 2.4. *Let X be a separated normal integral excellent scheme of characteristic zero admitting a dualizing complex, and let B be a $\mathbb{Q}$-divisor on X with $K_X + B$ $\mathbb{Q}$-Cartier. Then $v \to a(v, X, B)$ is lower semicontinuous on the set of real valuations of the function field of X with a center on X, for the topology of pointwise convergence on the non-zero elements of the function field.*

Proof. Let $\pi : Y \to X$ be a log resolution of (X, B) [Tem18, Theorem 1.1.6], and let $K_Y + B_Y := \pi^*(K_X + B)$. Since π is proper, any real valuation v of the function field of X with a center on X has a center on Y, and

$$a(v, X, B) = a(v, Y) - v(B_Y)$$

by Definition 2.3(5). The topology of pointwise convergence is the topology of [JM12, Section 4.1] [JM12, Lemma 4.1], for which $v \to a(v, Y)$ is lower semicontinuous [JM12, Lemma 5.7] and $v \to v(\mathfrak{b})$ is continuous for any non-zero ideal sheaf $\mathfrak{b}$ on Y. Since Y is regular, there exist an integer $N \geq 1$ and Cartier divisors $G^+ \geq 0$, $G^- \geq 0$ on Y with $NB_Y = G^+ - G^-$, so

$$v(B_Y) = N^{-1}\big(v(\mathcal{O}_Y(-G^+)) - v(\mathcal{O}_Y(-G^-))\big)$$

is continuous in v. The lemma follows. □

Lemma 2.5. *(1)* [JM12, Proposition 5.10] *Let X be a separated regular integral excellent scheme over $\mathbb{Q}$ and v a real valuation of the function field of X with $\xi := \operatorname{center}_X v$ and*

$a(v,X) < +\infty$, where $a(v,X)$ is the log discrepancy $A_X(v)$ of [JM12, Section 5]*, and let $\mathfrak{m}_\xi$ be the maximal ideal of $\mathcal{O}_{X,\xi}$. Then*

$$\mathfrak{m}_\xi^{\lceil t/v(\mathfrak{m}_\xi)\rceil} \subset \mathfrak{a}_t(v) \subset \mathfrak{m}_\xi^{\lceil t/a(v,X)\rceil}$$

for any real number $t > 0$, where $\mathfrak{a}_t(v)$ is taken in $\mathcal{O}_{X,\xi}$.

(2) [RS14, Theorem 1.3 and Remark 1.6] *Let $(A, \mathfrak{m}_A)$ be an analytically irreducible Noetherian local domain and v a divisorial valuation of* $\mathrm{Frac}(A)$ *centered at $\mathfrak{m}_A$. Then there exists a real number $\beta > 0$ with*

$$\mathfrak{a}_t(v) \subset \mathfrak{m}_A^{\lceil t/\beta\rceil}$$

for any real number $t > 0$.

For a Noetherian local ring $(S, \mathfrak{m}_S)$, $f \in S\backslash\{0\}$ and a non-zero ideal I of S, let

$$\mathrm{ord}_{\mathfrak{m}_S}(f) := \max\{j \in \mathbb{N} \mid f \in \mathfrak{m}_S^j\}, \qquad \mathrm{ord}_{\mathfrak{m}_S}(I) := \max\{j \in \mathbb{N} \mid I \subset \mathfrak{m}_S^j\}.$$

For a point ξ of a locally Noetherian scheme X, with $\mathfrak{m}_\xi$ the maximal ideal of $\mathcal{O}_{X,\xi}$, we write $\mathrm{ord}_\xi := \mathrm{ord}_{\mathfrak{m}_\xi}$ on $\mathcal{O}_{X,\xi}$. If S is regular, $\mathrm{ord}_{\mathfrak{m}_S}$ is a valuation. The inclusions in Lemma 2.5(1) are equivalent to

$$v(\mathfrak{m}_\xi)\,\mathrm{ord}_\xi \le v \le a(v,X)\,\mathrm{ord}_\xi \quad \text{on } \mathcal{O}_{X,\xi}\backslash\{0\}.$$

Lemma 2.5(2) is the Izumi–Rees theorem [RS14, Theorem 1.3 and Remark 1.6]. Here $c^{-1}v$ is a divisorial valuation in the sense of [RS14, Definition 1.2] for $v = c\,\mathrm{ord}_E$, by the dimension formula [Sta, Tag 02IJ], since A is universally catenary [Sta, Tag 0AW6].

Lemma 2.6 (cf. [RS14, Lemma 2.8])**.** *Let $T \subset S$ be Noetherian local domains with S finite over T, and let* Nm *be the norm of the field extension* $\mathrm{Frac}(S)/\mathrm{Frac}(T)$.

(1) If T is normal, then $\mathrm{Nm}(x) \in T \cap xS$ for any $x \in S\backslash\{0\}$.

(2) If T is regular and S is analytically irreducible, then there exists an integer $N \ge 1$ such that $x \in \mathfrak{m}_S^{\lceil j/N\rceil}$ for any integer $j \ge 0$ and any $x \in S\backslash\{0\}$ with $\mathrm{Nm}(x) \in \mathfrak{m}_T^j$.

Proof. (1) Let $m := [\mathrm{Frac}(S) : \mathrm{Frac}(T)]$, and let $y^m + c_{m-1}y^{m-1} + \cdots + c_0$ be the characteristic polynomial of multiplication by x on $\mathrm{Frac}(S)$ over $\mathrm{Frac}(T)$, so $c_0 = (-1)^m\,\mathrm{Nm}(x)$. Since x is integral over the normal ring T, $c_0, \ldots, c_{m-1} \in T$. In particular $\mathrm{Nm}(x) \in T$. Substituting x for y and writing $c_m := 1$, we get

$$\mathrm{Nm}(x) = (-1)^{m+1}x\sum_{i=1}^{m} c_i x^{i-1} \in xT[x] \subset xS.$$

(2) Since T is regular, T is normal and analytically irreducible, and $\mathrm{ord}_{\mathfrak{m}_T}$ is a valuation. By (1), $\mathrm{Nm}(S\backslash\{0\}) \subset T$. Thus conditions (i) and (ii) of [RS14, Lemma 2.8] both hold, and [RS14, Lemma 2.8] implies that there exists an integer $N \ge 1$ with

$$\mathrm{ord}_{\mathfrak{m}_T}(\mathrm{Nm}(x)) \le N\,\mathrm{ord}_{\mathfrak{m}_S}(x) \quad \text{for any } x \in S\backslash\{0\}.$$

If $\mathrm{Nm}(x) \in \mathfrak{m}_T^j$, then $j \le N\,\mathrm{ord}_{\mathfrak{m}_S}(x)$, so $x \in \mathfrak{m}_S^{\lceil j/N\rceil}$. □

For a real valuation v of a field $\mathbb{F}$, let

$$\mathrm{ratrk}(v) := \dim_{\mathbb{Q}}(v(\mathbb{F}^*) \otimes_{\mathbb{Z}} \mathbb{Q}),$$

and let $\kappa(v)$ be the residue field of the valuation ring of v. Let X be a separated regular integral excellent scheme over $\mathbb{Q}$, and v a real valuation of the function field of X with $\xi := \mathrm{center}_X\, v$. Then v is quasi-monomial in the sense of [JM12, Section 3.1] (for $X = \mathrm{Spec}\,A$, in the sense of Definition 2.3(4)) if and only if

$$\mathrm{ratrk}(v) + \mathrm{trdeg}(\kappa(v)/\kappa(\xi)) = \dim \mathcal{O}_{X,\xi}$$

[JM12, Proposition 3.7].

Definition-Lemma 2.7. Let Y be a separated regular integral excellent scheme over $\mathbb{Q}$, and let $D = \sum_{k=1}^m D_k$ be a reduced snc divisor on Y. Let $\mathrm{QM}(Y, D)$ be the set of real valuations of the function field of Y that are monomial in (Y, D) [JM12, Section 3.1], and let $r_{Y,D}$ be the retraction onto $\mathrm{QM}(Y, D)$ [JM12, Section 4.3]. Then any real valuation u of the function field of Y with a center on Y satisfies $a(u, Y) \geq \sum_{k=1}^m u(D_k)$, and equality holds if and only if $u \in \mathrm{QM}(Y, D)$.

Proof. If $\mathrm{center}_Y u \notin \operatorname{Supp} D$, then $\sum_{k=1}^m u(D_k) = 0 < a(u, Y)$ [JM12, Section 5.2] and $u \notin \mathrm{QM}(Y, D)$. Assume that $\mathrm{center}_Y u \in \operatorname{Supp} D$. Then $r_{Y,D}(u)(D_k) = u(D_k)$ for $k = 1, \ldots, m$ [JM12, Section 4.3], so

$$a(r_{Y,D}(u), Y) = \sum_{k=1}^m u(D_k)$$

by [JM12, (5.1)]. By [JM12, Corollary 5.4], $a(r_{Y,D}(u), Y) \leq a(u, Y)$, and equality holds if and only if $u = r_{Y,D}(u)$, that is, $u \in \mathrm{QM}(Y, D)$. □

2.3. **Filtrations, volumes and thresholds.**

Definition 2.8. Let $(A, \mathfrak{m}_A)$ be a Noetherian local domain with $d := \dim A \geq 1$.

(1) An $\mathfrak{m}_A$-*filtration* is a family $\mathfrak{a}_\bullet = (\mathfrak{a}_t)_{t\in\mathbb{R}}$ of ideals of A satisfying the following [BLQ24, Definition 2.1].
   (i) $\mathfrak{a}_t = A$ for any $t \leq 0$, and $\mathfrak{a}_t$ is $\mathfrak{m}_A$-primary for any $t > 0$.
   (ii) $\mathfrak{a}_{t'} \subset \mathfrak{a}_t$ for any real numbers $t \leq t'$.
   (iii) $\mathfrak{a}_t\mathfrak{a}_{t'} \subset \mathfrak{a}_{t+t'}$ for any real numbers t, t'.
   (iv) $\mathfrak{a}_t = \bigcap_{t'<t} \mathfrak{a}_{t'}$ for any $t > 0$.
   The $\mathfrak{m}_A$-filtration $\mathfrak{a}_\bullet$ is *linearly bounded* if there exists a real number $\alpha > 0$ such that $\mathfrak{a}_t \subset \mathfrak{m}_A^{\lceil \alpha t \rceil}$ for any $t > 0$ [BLQ24, Definition 2.2]. An $\mathfrak{m}_A$-primary ideal $\mathfrak{a}$ is regarded as the $\mathfrak{m}_A$-filtration with $\mathfrak{a}_t = \mathfrak{a}^{\lceil t \rceil}$ for any $t > 0$. A *graded sequence* is a family $\mathfrak{a}_\bullet = (\mathfrak{a}_j)_{j\geq 1}$ of $\mathfrak{m}_A$-primary ideals such that $\mathfrak{a}_j\mathfrak{a}_{j'} \subset \mathfrak{a}_{j+j'}$ for any integers $j, j' \geq 1$ [JM12, Section 2.1].

(2) Let v be a real valuation of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$. For an $\mathfrak{m}_A$-filtration $\mathfrak{a}_\bullet$ and for a graded sequence $\mathfrak{a}_\bullet$ respectively, we let

$$v(\mathfrak{a}_\bullet) := \inf_{t>0} \frac{v(\mathfrak{a}_t)}{t} \quad \text{and} \quad v(\mathfrak{a}_\bullet) := \inf_{j\geq 1} \frac{v(\mathfrak{a}_j)}{j}.$$

   For an $\mathfrak{m}_A$-filtration $\mathfrak{a}_\bullet$, we let

$$e(\mathfrak{a}_\bullet) := \lim_{t\to+\infty} \frac{d!\ell_A(A/\mathfrak{a}_t)}{t^d}$$

   whenever the limit exists. If A is analytically irreducible, then by [Cut14, Theorem 6.5] and [BLQ24, (2.1)], the limit exists and

$$e(\mathfrak{a}_\bullet) = \lim_{t\to+\infty} \frac{e(\mathfrak{a}_t)}{t^d}.$$

   Since $v(\mathfrak{m}_A) > 0$, $\mathfrak{a}_\bullet(v) := (\mathfrak{a}_t(v))_{t\in\mathbb{R}}$ is an $\mathfrak{m}_A$-filtration, and $v(\mathfrak{a}_\bullet(v)) = 1$ since $v(\mathfrak{a}_t(v)) \geq t$ for any $t > 0$, with equality for $t = v(x)$ and $x \in \mathfrak{m}_A\backslash\{0\}$. We call $\mathrm{vol}_A(v) := e(\mathfrak{a}_\bullet(v))$ the *volume* of v.

(3) Let X be a normal integral excellent scheme of characteristic zero admitting a dualizing complex, $x \in X$ a point with $\mathcal{O}_{X,x} = A$, and B a $\mathbb{Q}$-divisor on X such that (X, B) is a klt pair. For an $\mathfrak{m}_A$-filtration or a graded sequence $\mathfrak{a}_\bullet$, we let

$$\mathrm{lct}(X \ni x, B; \mathfrak{a}_\bullet) := \inf_v \frac{a(v, X, B)}{v(\mathfrak{a}_\bullet)},$$

   where v runs over the real valuations of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$, and the quotient is read as $+\infty$ when $v(\mathfrak{a}_\bullet) = 0$. When $X = \operatorname{Spec} A$, $x = \mathfrak{m}_A$ and $B = 0$, we write

$$\mathrm{lct}(A; \mathfrak{a}_\bullet) := \mathrm{lct}(\operatorname{Spec} A \ni \mathfrak{m}_A, 0; \mathfrak{a}_\bullet).$$

(4) Let $\mathfrak{a}_\bullet$ and $\mathfrak{c}_\bullet$ be $\mathfrak{m}_A$-filtrations and $0 < \sigma < 1$ a real number. The *geodesic* between $\mathfrak{a}_\bullet$ and $\mathfrak{c}_\bullet$ at σ is the $\mathfrak{m}_A$-filtration $\mathfrak{g}_\bullet$ with

$$\mathfrak{g}_t = \sum_{(1-\sigma)t_0+\sigma t_1=t} (\mathfrak{a}_{t_0} \cap \mathfrak{c}_{t_1}) \quad \text{for any } t \in \mathbb{R},$$

where the sum runs over the real numbers t_0, t_1 with $(1-\sigma)t_0 + \sigma t_1 = t$ [BLQ24, Definition 4.1].

Lemma 2.9. *Let $(A, \mathfrak{m}_A)$ be as in Definition 2.8 and $\mathfrak{a}_\bullet$ an $\mathfrak{m}_A$-filtration. Then $(\mathfrak{a}_j)_{j\geq 1}$ is a graded sequence, and $v(\mathfrak{a}_\bullet) = v((\mathfrak{a}_j)_{j\geq 1})$ for any real valuation v of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$. In particular, in Definition 2.8(3),*

$$\mathrm{lct}(X \ni x, B; \mathfrak{a}_\bullet) = \mathrm{lct}(X \ni x, B; (\mathfrak{a}_j)_{j\geq 1}).$$

Proof. The ideals $\mathfrak{a}_j$ with $j \geq 1$ are $\mathfrak{m}_A$-primary and $\mathfrak{a}_j\mathfrak{a}_{j'} \subset \mathfrak{a}_{j+j'}$, so $(\mathfrak{a}_j)_{j\geq 1}$ is a graded sequence. The inequality $\leq$ holds since

$$v(\mathfrak{a}_\bullet) = \inf_{t>0} \frac{v(\mathfrak{a}_t)}{t} \leq \inf_{j\geq 1} \frac{v(\mathfrak{a}_j)}{j} = v((\mathfrak{a}_j)_{j\geq 1}).$$

Since $\mathfrak{a}_{i/j}^j \subset \mathfrak{a}_i$ for any integers $i, j \geq 1$ and $\mathfrak{a}_\bullet$ is decreasing,

$$v(\mathfrak{a}_t) \geq v(\mathfrak{a}_{i/j}) \geq \frac{v(\mathfrak{a}_i)}{j} \geq \frac{i}{j} v((\mathfrak{a}_k)_{k\geq 1}) \quad \text{for any } t > 0 \text{ and any rational number } 0 < \frac{i}{j} \leq t,$$

so letting i/j tend to t implies the inequality $\geq$. The equality of log canonical thresholds follows from Definition 2.8(3). □

2.4. **Local endomorphisms.**

Definition 2.10. Let $(A, \mathfrak{m}_A)$ be a Noetherian local domain, $k := A/\mathfrak{m}_A$, and $\varphi : A \to A$ an injective local homomorphism.

(1) If $\dim A \geq 1$, then we let

$$c(\varphi) := \max\{m \in \mathbb{N} \mid \varphi(\mathfrak{m}_A)A \subset \mathfrak{m}_A^m\} \quad \text{and} \quad c_\infty(\varphi) := \lim_{j\to+\infty} c(\varphi^j)^{1/j} \in [1, +\infty].$$

(2) A prime ideal P of A is *totally invariant* under φ if $\varphi^{-1}(P) = P$ and $\sqrt{\varphi(P)A} = P$. If moreover φ is finite, then $\varphi_{A_P} : A_P \to A_P$ and $\varphi_{A/P} : A/P \to A/P$ denote the induced local homomorphisms, and $\varphi_k := \varphi_{A/\mathfrak{m}_A} : k \to k$.

The limit defining $c_\infty(\varphi)$ in Definition 2.10(1) exists by Lemma 2.13.

Lemma 2.11. *Let $(A, \mathfrak{m}_A)$ be a Noetherian local domain, let $\varphi : A \to A$ be a finite injective local homomorphism, and let P be a prime ideal of A totally invariant under φ. Then P is the only prime ideal P' of A with $\varphi^{-1}(P') = P$, and φ_{A_P} and $\varphi_{A/P}$ are finite injective local homomorphisms.*

Proof. Let P' be a prime ideal of A with $\varphi^{-1}(P') = P$. Then $P' \supset \varphi(P)A$, so $P' \supset \sqrt{\varphi(P)A} = P$. Since A is integral over $\varphi(A)$ and $P \cap \varphi(A) = P' \cap \varphi(A) = \varphi(P)$, incomparability implies $P' = P$. Thus the localization of A at $\varphi(A\backslash P)$ is finite over the local ring $\varphi(A)_{\varphi(P)}$ and has P as the only prime ideal lying over $\varphi(P)$, so it is local and equal to A_P, and φ_{A_P} is finite. Moreover, A/P is finite over $\varphi(A)/\varphi(P)$, so $\varphi_{A/P}$ is finite. Finally, φ_{A_P} and $\varphi_{A/P}$ are injective and local since A is a domain, $\varphi^{-1}(P) = P$ and $\varphi^{-1}(\mathfrak{m}_A) = \mathfrak{m}_A$. □

Lemma 2.12. *Let $(S, \mathfrak{n})$ be a Noetherian local domain with $d := \dim S \geq 1$, let $k := S/\mathfrak{n}$, and let $\varphi : S \to S$ be a finite injective local homomorphism with $D := [\mathrm{Frac}(S) : \varphi(\mathrm{Frac}(S))]$ and $\delta := [k : \varphi_k(k)]$. Then*

$$e(\varphi(I)S) = \frac{D}{\delta} e(I)$$

for any $\mathfrak{n}$-primary ideal I, and

$$c(\varphi^j) \le \left(\frac{D}{\delta}\right)^{j/d}$$

for any integer $j \ge 0$. If S is regular, then

$$\ell_S(S/\varphi(I)S) = \frac{D}{\delta}\ell_S(S/I)$$

for any $\mathfrak{n}$-primary ideal I.

Proof. Let M be the S-module whose underlying group is S and on which $s \in S$ acts by multiplication by $\varphi(s)$. Since φ is finite, M is a finite S-module, and $\operatorname{rank}_S M = D$ since $M \otimes_S \operatorname{Frac}(S) = \operatorname{Frac}(S)$ with $\operatorname{Frac}(S)$ acting through φ. Let I be an $\mathfrak{n}$-primary ideal. Since S is integral over $\varphi(S)$ and $\mathfrak{n}$ is the only maximal ideal of S, $\varphi(I)S$ is $\mathfrak{n}$-primary. For any integer $m \ge 1$, $I^m M = (\varphi(I)S)^m$, and any composition factor of the S-module $S/(\varphi(I)S)^m$ is k, which has length δ as an S-module through φ. Thus

$$\ell_S(M/I^m M) = \delta\ell_S\big(S/(\varphi(I)S)^m\big). \tag{2.1}$$

Let

$$e(I;M) := \lim_{m\to+\infty} \frac{d!\ell_S(M/I^m M)}{m^d}.$$

Since S is a domain, [Mat89, Theorem 14.8] implies $e(I;M) = \operatorname{rank}_S M \cdot e(I) = De(I)$, and (2.1) implies $e(I;M) = \delta e(\varphi(I)S)$. Thus $e(\varphi(I)S) = (D/\delta)e(I)$.

Let $j \ge 0$ be an integer. Then φ^j is a finite injective local homomorphism with

$$[\operatorname{Frac}(S) : \varphi^j(\operatorname{Frac}(S))] = D^j \quad \text{and} \quad [k : \varphi_k^j(k)] = \delta^j,$$

since φ maps the extension $\operatorname{Frac}(S)/\varphi(\operatorname{Frac}(S))$ isomorphically onto the extension

$$\varphi(\operatorname{Frac}(S))/\varphi^2(\operatorname{Frac}(S)),$$

and φ_k maps $k/\varphi_k(k)$ isomorphically onto $\varphi_k(k)/\varphi_k^2(k)$. Since $\varphi^j(\mathfrak{n})S \subset \mathfrak{n}^{c(\varphi^j)}$, the equality $e(\varphi(I)S) = (D/\delta)e(I)$ applied to φ^j and $I = \mathfrak{n}$ implies

$$c(\varphi^j)^d e(\mathfrak{n}) = e(\mathfrak{n}^{c(\varphi^j)}) \le e(\varphi^j(\mathfrak{n})S) = \left(\frac{D}{\delta}\right)^j e(\mathfrak{n}).$$

Since $e(\mathfrak{n}) > 0$, $c(\varphi^j) \le (D/\delta)^{j/d}$.

Assume that S is regular. Since S is Cohen–Macaulay and $\dim S/\varphi(\mathfrak{n})S = 0$, φ is flat [Mat89, Theorem 23.1], so M is a free S-module of rank D. Thus $\ell_S(M/IM) = D\ell_S(S/I)$, and (2.1) with $m = 1$ implies $\ell_S(S/\varphi(I)S) = (D/\delta)\ell_S(S/I)$. □

Lemma 2.13. *Let $(A, \mathfrak{m}_A)$ and φ be as in Definition 2.10(1). Then $c(\varphi^j)$ is a positive integer for any integer $j \ge 0$, $c(\varphi^{i+j}) \ge c(\varphi^i)c(\varphi^j)$ for any integers $i, j \ge 0$, and the limit defining $c_\infty(\varphi)$ exists in $[1, +\infty]$. If φ is finite, then $c_\infty(\varphi) < +\infty$.*

Proof. Let $j \ge 0$ be an integer. Then $c(\varphi^j) \ge 1$ since $\varphi^j(\mathfrak{m}_A) \subset \mathfrak{m}_A$, and $c(\varphi^j) < +\infty$ since

$$\varphi^j(\mathfrak{m}_A) \ne 0 = \bigcap_{m\ge1} \mathfrak{m}_A^m.$$

For any integers $i, j \ge 0$,

$$\varphi^{i+j}(\mathfrak{m}_A)A \subset \varphi^i\big(\mathfrak{m}_A^{c(\varphi^j)}\big)A = \big(\varphi^i(\mathfrak{m}_A)A\big)^{c(\varphi^j)} \subset \mathfrak{m}_A^{c(\varphi^i)c(\varphi^j)},$$

so $c(\varphi^{i+j}) \ge c(\varphi^i)c(\varphi^j)$, and the limit defining $c_\infty(\varphi)$ exists in $[1, +\infty]$ by Fekete's lemma for the superadditive sequence $(\log c(\varphi^j))_{j\ge0}$. If φ is finite, then $c_\infty(\varphi) < +\infty$ by Lemma 2.12. □

Lemma 2.14. *Let $(S, \mathfrak{n})$ be a Noetherian local domain with $\dim S \ge 1$, let $\varphi : S \to S$ be a finite injective local homomorphism, and let $\mathfrak{a}_\bullet$ be an $\mathfrak{n}$-filtration. Then $(\varphi(\mathfrak{a}_t)S)_{t\in\mathbb{R}}$ is an $\mathfrak{n}$-filtration.*

Proof. Let $\mathfrak{b}_t := \varphi(\mathfrak{a}_t)S$ for any $t \in \mathbb{R}$. Then $\mathfrak{b}_t = S$ for any $t \le 0$, $\mathfrak{b}_{t'} \subset \mathfrak{b}_t$ for any real numbers $t \le t'$, and $\mathfrak{b}_t\mathfrak{b}_{t'} \subset \mathfrak{b}_{t+t'}$ for any real numbers t, t', by Definition 2.8(1)(i)–(iii) for $\mathfrak{a}_\bullet$. Since $S/\varphi(\mathfrak{n})S$ is a finite-dimensional vector space over the field $\varphi(S)/\varphi(\mathfrak{n})$, $\varphi(\mathfrak{n})S$ is $\mathfrak{n}$-primary. For $t > 0$ and an integer $k \ge 1$ with $\mathfrak{n}^k \subset \mathfrak{a}_t$, we have

$$(\varphi(\mathfrak{n})S)^k \subset \mathfrak{b}_t \subset \mathfrak{n},$$

so $\mathfrak{b}_t$ is $\mathfrak{n}$-primary. Let $t > 0$. Since $S/\mathfrak{a}_t$ is Artinian, the set of the ideals $\mathfrak{a}_{t'}$ with $0 < t' < t$, each containing $\mathfrak{a}_t$, has a minimal element $\mathfrak{a}_{t_0}$. Then $\mathfrak{a}_{t'} = \mathfrak{a}_{t_0}$ for any $t' \in [t_0, t)$, so $\mathfrak{a}_{t_0} = \mathfrak{a}_t$ by Definition 2.8(1)(iv), and hence

$$\bigcap_{t'<t} \mathfrak{b}_{t'} = \mathfrak{b}_{t_0} = \mathfrak{b}_t.$$

□

2.5. **Invariant measures.**

Lemma 2.15. *Let $\mathcal{K}$ be a compact Hausdorff space, $f : \mathcal{K} \to \mathcal{K}$ a continuous map, and $x \in \mathcal{K}$. For any integer $N \ge 1$, let μ_N be the Radon probability measure on $\mathcal{K}$ with*

$$\int g d\mu_N = \frac{1}{N} \sum_{j=0}^{N-1} g(f^j(x))$$

for any continuous function $g : \mathcal{K} \to \mathbb{R}$. Then $(\mu_N)_{N\ge1}$ has a subnet converging in the weak-$*$ topology, and the limit μ of any such subnet is a Radon probability measure on $\mathcal{K}$ with $\mu(f^{-1}(E)) = \mu(E)$ for any Borel set $E \subset \mathcal{K}$.

Proof. This is the Krylov–Bogolyubov theorem [Wal82, Theorem 6.9]. The proof is the standard one via the Banach–Alaoglu theorem. □

3. Log discrepancies and volumes under finite morphisms

In this section we compare a real valuation with the pushforward under a finite morphism. For a finite surjective morphism $f : X' \to X$ between normal integral excellent schemes of characteristic zero admitting dualizing complexes, the *ramification divisor* of f is $R(f) := \sum_P (m_P - 1)P$, where P runs over the prime divisors on X' and m_P is the ramification index of f along P. We choose $K_{X'}$ and K_X with $K_{X'} = f^*K_X + R(f)$.

Lemma 3.1. *Let $f : X' \to X$ be a finite surjective morphism of normal integral excellent schemes, let $\mu : X_1 \to X$ be a proper birational morphism from a normal integral scheme X_1, and let E be a prime divisor on X_1. Let u be a real valuation of the function field of X' with a center on X' such that the restriction of u to the function field of X is c times the order of vanishing along E for a real number $c > 0$, and let $g : X_1' \to X_1$ be the normalization of X_1 in the function field of X'. Then X_1' is proper and birational over X', and there exist a prime divisor E' on X_1' with $g(E') = E$ and a real number $c' > 0$ such that u is c' times the order of vanishing along E' and $c = c'm_{E'}$, where $m_{E'}$ is the ramification index of g along E'.*

Proof. Since X_1 is excellent, g is finite. Since X' is normal and finite over X, it is the normalization of X in the function field of X', so there exists an induced morphism $\mu' : X_1' \to X'$ with $f \circ \mu' = \mu \circ g$, and μ' is proper and birational since $\mu \circ g$ is proper. Since g is finite, u has a center ξ' on X_1' over the generic point of E. Since g is finite and X_1, X_1' are normal, ξ' is the generic point of a prime divisor E', and $g(E') = E$. The local ring $\mathcal{O}_{X_1',E'}$ of X_1' at the generic point of E' is a DVR dominated by the valuation ring V_u of u, so $\mathcal{O}_{X_1',E'} = V_u$ and $u = c' \operatorname{ord}_{E'}$ for some real number $c' > 0$. Since $m_{E'} = \operatorname{mult}_{E'} g^*E$, the restriction of u to the function field of X_1 is $c'm_{E'} \operatorname{ord}_E$, so $c = c'm_{E'}$. The lemma follows. □

Lemma 3.2 (cf. [Yos26, Lemma 3.2]). *Let $f : X' \to X$ be a finite surjective morphism between normal integral excellent schemes of characteristic zero admitting dualizing complexes, let B be a $\mathbb{Q}$-divisor on X with K_X+B $\mathbb{Q}$-Cartier, and let $B' := f^*B - R(f)$, so that $K_{X'}+B' = f^*(K_X+B)$. Let u be a real valuation of the function field of X' with a center on X', and let v be the restriction of u to the function field of X. Then*

$$a(v, X, B) = a(u, X', B'),$$

where both sides may be $+\infty$.

Proof. For an integral scheme Y, we let Val_Y be the set of real valuations of the function field of Y with a center on Y. An element $w \in \mathrm{Val}_Y$ is *divisorial* if $w = c\,\mathrm{ord}_E$ for some real number $c > 0$ and some prime divisor E on a normal integral scheme with a proper birational morphism to Y. The valuation v is centered at $f(\mathrm{center}_{X'} u)$. Possibly shrinking X to an affine open neighborhood of $\mathrm{center}_X v$ and replacing X' with $f^{-1}(X)$, we may assume that $X = \operatorname{Spec} A$ and $X' = \operatorname{Spec} A'$. Then Val_X is the set of real valuations of $\mathrm{Frac}(A)$ that are non-negative on $A\backslash\{0\}$, and $v = f_*u$, where $f_* : \mathrm{Val}_{X'} \to \mathrm{Val}_X$ is the restriction to $\mathrm{Frac}(A)$. We equip Val_X with the weakest topology such that $w \to w(\mathfrak{a})$ is continuous for any non-zero ideal $\mathfrak{a}$ of A [JM12, Section 4.1, Lemma 4.1]. Since any ideal of A is finitely generated, it is the topology of pointwise convergence on $\mathrm{Frac}(A)^*$. If Y is a regular scheme with a proper birational morphism $Y \to X$, then $\mathrm{Val}_Y = \mathrm{Val}_X$, and by [JM12, Lemma 4.1] the topology on Val_X is also the weakest one such that $w \to w(\mathfrak{b})$ is continuous for any non-zero ideal sheaf $\mathfrak{b}$ on Y. For any normal integral affine scheme X'' finite over X, we equip $\mathrm{Val}_{X''}$ with the weakest topology such that $w \to w(\mathfrak{a})$ is continuous for any non-zero ideal $\mathfrak{a}$ of the coordinate ring of X''.

Step 1. We show that for any $w \in \mathrm{Val}_X$,

$$a(w, X, B) = \sup_U \inf\{a(w_1, X, B) \mid w_1 \in U \text{ is divisorial}\}, \tag{3.1}$$

where U runs over the open neighborhoods of w in Val_X, and that (3.1) holds for (X', B') in place of (X, B). Let $\pi : Y \to X$ be a log resolution of (X, B) [Tem18, Theorem 1.1.6] and let $K_Y + B_Y := \pi^*(K_X + B)$, so that $a(w_1, X, B) = a(w_1, Y) - w_1(B_Y)$ for any $w_1 \in \mathrm{Val}_X$. By Lemma 2.4, $a(\cdot, X, B)$ is lower semicontinuous: for any real number $t < a(w, X, B)$, the set $\{w_1 \mid a(w_1, X, B) > t\}$ is an open neighborhood of w, so the right side of (3.1) is at least $a(w, X, B)$.

Conversely, assume that $a(w, X, B) < +\infty$, and let U be an open neighborhood of w and $\epsilon > 0$. Let (Y_1, D) run over the log smooth pairs over Y [JM12, Definition 3.2] such that D contains the support of the pullback of B_Y, and let $r_{Y_1,D}$ be the retraction of Definition-Lemma 2.7. Then $r_{Y_1,D}(w) \to w$ [JM12, Theorem 4.9] and

$$a(r_{Y_1,D}(w), Y) \le a(w, Y)$$

[JM12, Corollary 5.4], so the lower semicontinuity of $a(\cdot, Y)$ [JM12, Lemma 5.7] implies $a(r_{Y_1,D}(w), Y) \to a(w, Y)$. Since B_Y is a $\mathbb{Q}$-linear combination of Cartier divisors on the regular scheme Y, $w_1 \to w_1(B_Y)$ is continuous, so there exists (Y_1, D) with

$$w_0 := r_{Y_1,D}(w) \in U \quad \text{and} \quad a(w_0, X, B) < a(w, X, B) + \epsilon.$$

The valuation w_0 is monomial in local equations of the components $D_1, \dots, D_m$ of D through the generic point of a stratum of D, with weights $\alpha := (w_0(D_i))_{i=1}^m \in \mathbb{R}^m_{>0}$. For $\alpha' \in \mathbb{R}^m_{>0}$, let $w_{\alpha'}$ be the valuation monomial in the local equations of $D_1, \dots, D_m$ with weights α'. For any $x \in A\backslash\{0\}$, $\alpha' \to w_{\alpha'}(x)$ is the minimum of finitely many linear functions of α', so $\alpha' \to w_{\alpha'}$ is continuous. By [JM12, Proposition 5.1, (5.1)], and since the pullback of B_Y is supported on D,

$$a(w_{\alpha'}, X, B) = \sum_{i=1}^m \alpha'_i\big(a(\mathrm{ord}_{D_i}, Y) - \mathrm{ord}_{D_i}(B_Y)\big).$$

For $\alpha' \in \mathbb{Q}^m_{>0}$ close to α, $a(w_{\alpha'}, X, B) < a(w, X, B) + 2\epsilon$, $w_{\alpha'} \in U$, and $w_{\alpha'}$ is divisorial [JM12, Remark 3.9]. This proves (3.1). Replacing (X, B) with (X', B'), the argument above proves (3.1) for (X', B').

Step 2. Let $v_1 = c \operatorname{ord}_E \in \operatorname{Val}_X$, where $c > 0$ and E is a prime divisor on a normal integral scheme X_1 with a proper birational morphism $\mu : X_1 \to X$. We show that any $u_1 \in \operatorname{Val}_{X'}$ with $f_* u_1 = v_1$ is divisorial and

$$a(u_1, X', B') = a(v_1, X, B).$$

By Lemma 3.1, $u_1 = c' \operatorname{ord}_{E'}$ for a prime divisor E' on the normalization X'_1 of X_1 in $\operatorname{Frac}(A')$ and a real number $c' > 0$ with $c = c' m_{E'}$, where $m_{E'}$ is the ramification index of $g : X'_1 \to X_1$ along E'. Since X'_1 is proper and birational over X', u_1 is divisorial. Let $\mu' : X'_1 \to X'$ be the induced morphism, let $K_{X_1} + B_{X_1} := \mu^*(K_X + B)$, and let $K_{X'_1} + B_{X'_1} := \mu'^*(K_{X'} + B')$. Since $K_{X'_1} + B_{X'_1} = g^*(K_{X_1} + B_{X_1})$ and $\operatorname{mult}_{E'} R(g) = m_{E'} - 1$ by definition,

$$a(u_1, X', B') = c'(1 - \operatorname{mult}_{E'} B_{X'_1}) = c' m_{E'}(1 - \operatorname{mult}_E B_{X_1}) = a(v_1, X, B).$$

Step 3. Let $u_1 \in \operatorname{Val}_{X'}$ be divisorial. We show that $f_* u_1$ is divisorial. Let L be a Galois closure of $\operatorname{Frac}(A')$ over $\operatorname{Frac}(A)$, let $G := \operatorname{Gal}(L/\operatorname{Frac}(A))$, and let $X'' = \operatorname{Spec} A''$ be the normalization of X in L. Since X is excellent, X'' is finite over X and over X'. The group G acts on X'' over X, and $X''/G = X$ since X is normal. Let u'' be a valuation of L extending u_1. The valuation u'' is real since $u''(L^*)/u_1(\operatorname{Frac}(A')^*)$ is finite [Sta, Tag 0ASH]. By **Step 2** applied to $X'' \to X'$, $u'' = c'' \operatorname{ord}_{E''_0}$ for some $c'' > 0$ and a prime divisor E''_0 on a normal integral scheme X_2 with a proper birational morphism $X_2 \to X''$. The local ring of X_2 at the generic point of E''_0 is the valuation ring $V_{u''}$ of u'', and it is a localization of $A''[x_1/x_0, \dots, x_m/x_0]$ for some $x_0, \dots, x_m \in A'' \backslash \{0\}$. Let $I := (x_0, \dots, x_m)A''$, let $I^G := \prod_{\sigma \in G} \sigma(I)$, and let X''_1 be the normalization of the blow-up of I^G. Since $I^G \mathcal{O}_{X''_1}$ is invertible, $I\mathcal{O}_{X''_1}$ is invertible, so X''_1 admits a morphism over X'' to the blow-up of I. On the chart $\operatorname{Spec} A''[x_1/x_0, \dots, x_m/x_0]$ of the blow-up of I, the center of u'' has local ring $V_{u''}$. Hence the local ring of X''_1 at the center of u'' dominates $V_{u''}$ and is dominated by $V_{u''}$, so it equals $V_{u''}$ and the center of u'' on X''_1 is the generic point of a prime divisor F''.

Since I^G is G-invariant, G acts on X''_1 over X. As X''_1 is projective over the affine scheme X, any G-orbit in X''_1 is contained in an affine open subset, so the quotient X''_1/G exists [GR71, Exp. V, Proposition 1.8]. The scheme X''_1/G is normal, and the induced morphism $X''_1/G \to X''/G = X$ is proper birational. The image F of F'' is a prime divisor on X''_1/G, so the local ring $\mathcal{O}_{X''_1/G,F}$ of X''_1/G at the generic point of F is a DVR. It is dominated by the valuation ring $V_{f_* u_1}$ of the restriction $f_* u_1$ of u'' to $\operatorname{Frac}(A)$, so $\mathcal{O}_{X''_1/G,F} = V_{f_* u_1}$ and $f_* u_1$ is a positive multiple of ord_F.

Step 4. Let L, G, and $X'' = \operatorname{Spec} A''$ be as in **Step 3**, let $f' : X'' \to X'$ be the induced morphism, and let $f'' := f \circ f'$. We show that f_* is continuous, surjective and open. For $x \in A \backslash \{0\}$ and $u_1 \in \operatorname{Val}_{X'}$, $(f_* u_1)(x) = u_1(x)$, so f_* is continuous. Similarly, f'_* and f''_* are continuous. Any $w \in \operatorname{Val}_X$ extends to a valuation of L, which is real by [Sta, Tag 0ASH] and non-negative on A'' since A'' is integral over A. Thus f''_* is surjective, and similarly f'_* and f_* are surjective.

We show that the map f''_* is closed. Let C be a closed subset of $\operatorname{Val}_{X''}$ and let $(w_j)_j$ be a net in C with $f''_* w_j \to v_0 \in \operatorname{Val}_X$. Let $y \in L^*$, let $y^m + c_1 y^{m-1} + \cdots + c_m = 0$ be the minimal equation of y over $\operatorname{Frac}(A)$, and let $c_0 := 1$. For any j, the minimum of $w_j(c_i y^{m-i})$ over the indices i with $c_i \neq 0$ is attained at two indices $i_1 < i_2$, so

$$w_j(y) = \frac{(f''_* w_j)(c_{i_2}) - (f''_* w_j)(c_{i_1})}{i_2 - i_1}.$$

Thus $(w_j(y))_j$ is eventually bounded. A subnet of $(w_j)_j$ converges in $[-\infty, +\infty]^{L^*}$ to a function w, which is finite-valued. The conditions $w(yy') = w(y) + w(y')$, $w(y + y') \geq \min\{w(y), w(y')\}$ for $y, y', y + y' \in L^*$, and $w \geq 0$ on $A'' \backslash \{0\}$ are closed, and w is non-trivial since $w = v_0$ on $\operatorname{Frac}(A)^*$, so $w \in \operatorname{Val}_{X''}$, hence $w \in C$ and $v_0 = f''_* w \in f''_*(C)$.

Any $\sigma \in G$ acts on $\mathrm{Val}_{X''}$ by the homeomorphism $w \to w \circ \sigma$. If $w_1, w_2 \in \mathrm{Val}_{X''}$ and $f''_* w_1 = f''_* w_2$, then by [EP05, Conjugation Theorem 3.2.15] there exists $\sigma \in G$ such that the valuation rings of w_2 and $w_1 \circ \sigma$ are equal. Then $w_2 = \iota \circ w_1 \circ \sigma$ for an order-preserving isomorphism ι between subgroups of $\mathbb{R}$, and ι is the identity on $(f''_* w_1)(\mathrm{Frac}(A)^*)$, which has finite index in $w_1(L^*)$ [Sta, Tag 0ASH]. Hence ι is the identity and $w_2 = w_1 \circ \sigma$. Therefore the fibers of f''_* are G-orbits. Since f''_* is a continuous closed surjection, for any open subset U'' of $\mathrm{Val}_{X''}$ the set $f''_*(U'')$ is open, because

$$(f''_*)^{-1}(f''_*(U'')) = \bigcup_{\sigma \in G} \{w \circ \sigma \mid w \in U''\}$$

is open. Finally, for any open subset U' of $\mathrm{Val}_{X'}$, $f_*(U') = f''_*((f'_*)^{-1}(U'))$ is open since f'_* is surjective.

Step 5. For a subset U of Val_X and a subset U' of $\mathrm{Val}_{X'}$, let

$$\Phi(U) := \inf\{a(w_1, X, B) \mid w_1 \in U \text{ is divisorial}\},$$
$$\Phi'(U') := \inf\{a(u_1, X', B') \mid u_1 \in U' \text{ is divisorial}\}.$$

By **Steps 2** and 3, f_* maps the divisorial valuations in U' onto the divisorial valuations in $f_*(U')$ and $a(u_1, X', B') = a(f_* u_1, X, B)$ for any divisorial u_1, so $\Phi'(U') = \Phi(f_*(U'))$. If U' is an open neighborhood of u, then $f_*(U')$ is an open neighborhood of v by **Step 4**. If U is an open neighborhood of v, then $(f_*)^{-1}(U)$ is an open neighborhood of u and $f_*((f_*)^{-1}(U)) = U$. By (3.1) for (X', B') and u, and (3.1) for (X, B) and v,

$$a(u, X', B') = \sup_{U'} \Phi(f_*(U')) \le a(v, X, B) = \sup_{U} \Phi'((f_*)^{-1}(U)) \le a(u, X', B'),$$

where U' and U run over the open neighborhoods of u and v respectively. The lemma follows. □

Lemma 3.3 (cf. [LM26a, Lemma 4.1] and [Yos26, Lemma 5.7]). *Let $(S, \mathfrak{n})$ be a normal excellent local domain containing $\mathbb{C}$ with $d := \dim S \ge 1$, let $k := S/\mathfrak{n}$, let $\varphi : S \to S$ be a finite injective local $\mathbb{C}$-algebra homomorphism, and let $\lambda > 0$ be a real number with*

$$[\mathrm{Frac}(S) : \varphi(\mathrm{Frac}(S))] = \lambda^d [k : \varphi_k(k)].$$

Let u be a real valuation of $\mathrm{Frac}(S)$ centered at $\mathfrak{n}$. Then the following hold.

(1) $\mathrm{vol}_S(\lambda^{-1}\varphi_* u) \ge \mathrm{vol}_S(u)$.

(2) If $\mathrm{vol}_S(\lambda^{-1}\varphi_* u) = \mathrm{vol}_S(u)$ *and there exists a real number $\beta > 0$ with*

$$\mathfrak{a}_t(\lambda^{-1}\varphi_* u) \subset \mathfrak{n}^{\lceil t/\beta \rceil} \quad \text{for any real number } t > 0,$$

then u is the only real valuation z of $\mathrm{Frac}(S)$ with $\varphi_* z = \varphi_* u$.

Proof. Since S is normal and excellent, $\widehat{S}$ is a normal domain, so S is analytically irreducible, and by Definition 2.8(2), for any $\mathfrak{n}$-filtration $\mathfrak{a}_\bullet$,

$$e(\mathfrak{a}_\bullet) = \lim_{t \to +\infty} \frac{e(\mathfrak{a}_t)}{t^d}.$$

Let $w := \lambda^{-1}\varphi_* u$. Since φ is an injective local $\mathbb{C}$-algebra homomorphism, w is a real valuation of $\mathrm{Frac}(S)$ centered at $\mathfrak{n}$. For any real number t, let

$$\mathfrak{c}_t := \varphi(\mathfrak{a}_t(w))S \quad \text{and} \quad \mathfrak{d}_t := \mathfrak{a}_{\lambda t}(u).$$

If $x \in \mathfrak{a}_t(w)$, then $u(\varphi(x)) = \lambda w(x) \ge \lambda t$, so $\mathfrak{c}_t \subset \mathfrak{d}_t$. By Definition 2.8(2), $\mathfrak{a}_\bullet(w)$ and $\mathfrak{a}_\bullet(u)$ are $\mathfrak{n}$-filtrations, so $\mathfrak{d}_\bullet$ is one, and so is $\mathfrak{c}_\bullet$ by Lemma 2.14. By Lemma 2.12, $e(\varphi(I)S) = \lambda^d e(I)$ for any $\mathfrak{n}$-primary ideal I, so

$$e(\mathfrak{c}_\bullet) = \lim_{t \to +\infty} \frac{\lambda^d e(\mathfrak{a}_t(w))}{t^d} = \lambda^d \mathrm{vol}_S(w) \quad \text{and} \quad e(\mathfrak{d}_\bullet) = \lambda^d \mathrm{vol}_S(u),$$

the second equality by the definition of vol_S.

(1) Since $\mathfrak{c}_t \subset \mathfrak{d}_t$, $\ell_S(S/\mathfrak{c}_t) \geq \ell_S(S/\mathfrak{d}_t)$ for any t, so $e(\mathfrak{c}_\bullet) \geq e(\mathfrak{d}_\bullet)$ and $\mathrm{vol}_S(w) \geq \mathrm{vol}_S(u)$.

(2) Let z be a real valuation of $\mathrm{Frac}(S)$ with $\varphi_* z = \varphi_* u$, that is, $z = u$ on $\varphi(\mathrm{Frac}(S))$. We show that $z = u$.

Since S is integral over $\varphi(S)$ and $z = u$ on $\varphi(S)$, z is centered at $\mathfrak{n}$.

Let Nm be the norm from $\mathrm{Frac}(S)$ to $\varphi(\mathrm{Frac}(S))$. Since $\varphi(S) \cong S$ is normal and S is finite over $\varphi(S)$, Lemma 2.6(1) implies that for any $x \in S\backslash\{0\}$,

$$\mathrm{Nm}(x) \in \varphi(S) \quad \text{and} \quad \mathrm{Nm}(x)/x \in S. \tag{3.2}$$

Next, $\mathrm{vol}_S(z) > 0$. Let $x_1, \dots, x_d$ be a system of parameters of S and let $c \geq 1$ be an integer with $\mathfrak{n}^c \subset (x_1, \dots, x_d)$. Let k' be a coefficient field of $\widehat{\varphi(S)}$ [Sta, Tag 032A], let

$$T := k'[[\varphi(x_1), \dots, \varphi(x_d)]] \subset \widehat{\varphi(S)},$$

and let $\mathfrak{m}_T$ be the maximal ideal of T. Since $\varphi(x_1), \dots, \varphi(x_d)$ is a system of parameters of $\varphi(S)$, T is a regular complete local ring of dimension d and $\widehat{\varphi(S)}$ is finite over T by [Mat89, Theorem 8.4]. Since S is finite over $\varphi(S)$ and $\varphi(\mathfrak{n})S$ is $\mathfrak{n}$-primary, $\widehat{S} = S \otimes_{\varphi(S)} \widehat{\varphi(S)}$ is finite over $\widehat{\varphi(S)}$. Thus $T \subset \widehat{\varphi(S)} \subset \widehat{S}$ are finite extensions of complete local domains, and

$$\mathrm{Frac}(\widehat{S}) = \mathrm{Frac}(S) \otimes_{\varphi(\mathrm{Frac}(S))} \mathrm{Frac}(\widehat{\varphi(S)}).$$

Let $\mathrm{Nm}_{\widehat{S}/T}$ and $\mathrm{Nm}_{\widehat{\varphi(S)}/T}$ be the norms of the extensions of fraction fields. Then for $x \in S\backslash\{0\}$,

$$\mathrm{Nm}_{\widehat{S}/T}(x) = \mathrm{Nm}_{\widehat{\varphi(S)}/T}(\mathrm{Nm}(x)).$$

Let $t > 0$, let $M := \lceil t/(\lambda\beta) \rceil$, and let $x \in \mathfrak{a}_t(z)\backslash\{0\}$. By (3.2) and since $z \geq 0$ on S,

$$z(\mathrm{Nm}(x)) = z(x) + z(\mathrm{Nm}(x)/x) \geq t.$$

Write $\mathrm{Nm}(x) = \varphi(y)$ with $y \in S$. Then

$$w(y) = \lambda^{-1} u(\varphi(y)) = \lambda^{-1} z(\mathrm{Nm}(x)) \geq t/\lambda,$$

so

$$y \in \mathfrak{a}_{t/\lambda}(w) \subset \mathfrak{n}^M \subset (x_1, \dots, x_d)^{\lfloor M/c \rfloor},$$

so

$$\mathrm{Nm}(x) = \varphi(y) \in (\varphi(x_1), \dots, \varphi(x_d))^{\lfloor M/c \rfloor} \widehat{\varphi(S)} = \mathfrak{m}_T^{\lfloor M/c \rfloor} \widehat{\varphi(S)}.$$

Since T is regular, $\mathrm{ord}_{\mathfrak{m}_T}$ is a valuation of $\mathrm{Frac}(T)$. Extend it to a normal closure of $\mathrm{Frac}(\widehat{\varphi(S)})$ over $\mathrm{Frac}(T)$. The extended valuation is non-negative on elements integral over T, so any conjugate of $\mathrm{Nm}(x)$ over $\mathrm{Frac}(T)$ has value at least $\lfloor M/c \rfloor$, and

$$\mathrm{ord}_{\mathfrak{m}_T} \mathrm{Nm}_{\widehat{S}/T}(x) = \mathrm{ord}_{\mathfrak{m}_T} \mathrm{Nm}_{\widehat{\varphi(S)}/T}(\mathrm{Nm}(x)) \geq \lfloor M/c \rfloor.$$

Since T is regular and $\widehat{S}$ is a complete local domain, Lemma 2.6(2) for $T \subset \widehat{S}$ and $\mathfrak{n}^j \widehat{S} \cap S = \mathfrak{n}^j$ imply that there exists a positive integer N such that for any $x' \in S\backslash\{0\}$,

$$x' \in \mathfrak{n}^{\lceil \mathrm{ord}_{\mathfrak{m}_T} \mathrm{Nm}_{\widehat{S}/T}(x')/N \rceil}.$$

With $\gamma := Nc\lambda\beta$ we obtain

$$x \in \mathfrak{n}^{\lceil \lfloor M/c \rfloor / N \rceil} \subset \mathfrak{n}^{\lceil t/\gamma \rceil - 1} \quad \text{since} \quad \frac{\lfloor M/c \rfloor}{N} \geq \frac{t}{\gamma} - 1.$$

As $t > 0$ and $x \in \mathfrak{a}_t(z)\backslash\{0\}$ are arbitrary, $\mathrm{vol}_S(z) \geq e(\mathfrak{n})/\gamma^d > 0$.

Next, $z \geq u$ on S. By the assumption $\mathrm{vol}_S(w) = \mathrm{vol}_S(u)$,

$$e(\mathfrak{c}_\bullet) = \lambda^d \mathrm{vol}_S(w) = \lambda^d \mathrm{vol}_S(u) = e(\mathfrak{d}_\bullet),$$

so Theorem 1.3 implies $z(\mathfrak{c}_\bullet) = z(\mathfrak{d}_\bullet)$. For $t > 0$, $\mathfrak{c}_t$ is generated by the elements $\varphi(y)$ with $y \in \mathfrak{a}_t(w)$, and $z(\varphi(y)) = u(\varphi(y)) = \lambda w(y)$, so $z(\mathfrak{c}_t) = \lambda w(\mathfrak{a}_t(w))$. By Definition 2.8(2), $w(\mathfrak{a}_\bullet(w)) = 1$. Therefore

$$\lambda = z(\mathfrak{c}_\bullet) = z(\mathfrak{d}_\bullet) = \lambda z(\mathfrak{a}_\bullet(u)),$$

so $z(\mathfrak{a}_t(u)) \geq t$ for any $t > 0$. For $x \in S\backslash\{0\}$ with $u(x) > 0$, $x \in \mathfrak{a}_{u(x)}(u)$, so $z(x) \geq u(x)$. If $u(x) = 0$, then $z(x) \geq 0 = u(x)$.

Finally, let $x \in S\backslash\{0\}$. Since $z = u$ on $\varphi(S)$, (3.2) implies

$$z(x) + z(\mathrm{Nm}(x)/x) = z(\mathrm{Nm}(x)) = u(\mathrm{Nm}(x)) = u(x) + u(\mathrm{Nm}(x)/x).$$

As $z(x) \geq u(x)$ and $z(\mathrm{Nm}(x)/x) \geq u(\mathrm{Nm}(x)/x)$, we get $z(x) = u(x)$. Thus $z = u$ on S, hence on $\mathrm{Frac}(S)$. The lemma follows. □

4. An invariant filtration

In this section, for the data of Set-up 4.8, we construct the filtration $\mathfrak{b}_\bullet$ of Construction-Proposition 4.10 as in [Yos26, Proposition 5.8].

4.1. Log canonical thresholds of filtrations.

Definition 4.1. Let $(A, \mathfrak{m}_A)$ be a normal excellent local domain of characteristic zero admitting a dualizing complex such that K_A is $\mathbb{Q}$-Cartier, and let $I_1, \dots, I_m$ be non-zero ideals of A.

(1) A *log resolution* of $I_1, \dots, I_m$ is a morphism $\pi : Z \to \mathrm{Spec}\, A$ satisfying the following.
   (i) π is a projective birational morphism from a regular integral scheme Z.
   (ii) $I_i\mathcal{O}_Z = \mathcal{O}_Z(-F_i)$ for a divisor $F_i \geq 0$ on Z for any i.
   (iii) There exists an snc divisor $\sum_k D_k$ on Z whose support contains $\mathrm{Supp}\, F_i$ for any i and any π-exceptional prime divisor.

   Log resolutions exist by [Tem18, Theorem 1.1.6].

(2) For a log resolution $\pi : Z \to \mathrm{Spec}\, A$ of $I_1, \dots, I_m$, a canonical divisor K_Z with $\pi_* K_Z = K_A$, and real numbers $t_1, \dots, t_m \geq 0$, let $L := \lceil K_Z - \pi^* K_A - \sum_i t_i F_i \rceil$. The *multiplier ideal* (cf. [Laz04, Definition 9.2.3 and Generalization 9.2.8]) is

$$\mathcal{J}(A, \prod_i I_i^{t_i}) := \pi_* \mathcal{O}_Z(L).$$

   It does not depend on π by Lemma 4.2(1).

(3) For a graded sequence $\mathfrak{a}_\bullet$ and a real number $t \geq 0$, the *asymptotic multiplier ideal* $\mathcal{J}(A, t \cdot \mathfrak{a}_\bullet)$ (cf. [Laz04, Definition 11.1.15]) is the largest of the ideals $\mathcal{J}(A, \mathfrak{a}_j^{t/j})$ with $j \geq 1$. It exists since A is Noetherian and

$$\mathcal{J}(A, \mathfrak{a}_j^{t/j}) \subset \mathcal{J}(A, \mathfrak{a}_{jj'}^{t/(jj')}) \quad \text{for any integers } j, j' \geq 1$$

   by $\mathfrak{a}_j^{j'} \subset \mathfrak{a}_{jj'}$ and Lemma 4.2(2).

Lemma 4.2. *Let $(A, \mathfrak{m}_A)$ and $I_1, \dots, I_m$ be as in Definition 4.1, let $\pi : Z \to \mathrm{Spec}\, A$ be a log resolution of $I_1, \dots, I_m$, and let $t_1, \dots, t_m \geq 0$ be real numbers.*

(1) For any $f \in A\backslash\{0\}$,

$$f \in \mathcal{J}(A, \prod_i I_i^{t_i}) \iff v(f) > \sum_i t_i v(I_i) - a(v, \mathrm{Spec}\, A) \tag{4.1}$$

for any divisorial valuation v of $\mathrm{Frac}(A)$.

In particular, $\mathcal{J}(A, \prod_i I_i^{t_i})$ does not depend on π.

(2) Let I be a non-zero ideal of A and $t \geq 0$ a real number with $\sum_i t_i v(I_i) \geq t v(I)$ for any divisorial valuation v of $\mathrm{Frac}(A)$. *Then*

$$\mathcal{J}(A, \prod_i I_i^{t_i}) \subset \mathcal{J}(A, I^t). \tag{4.2}$$

Proof. (1) Let $I_i\mathcal{O}_Z = \mathcal{O}_Z(-F_i)$ and the snc divisor $\sum_k D_k$ be as in Definition 4.1(1), and let L be as in Definition 4.1(2). Any divisorial valuation of $\mathrm{Frac}(A)$ is $c\,\mathrm{ord}_E$ for a real number $c > 0$ and a prime divisor E over $\operatorname{Spec} A$, and $a(c\,\mathrm{ord}_E, \operatorname{Spec} A) = ca(E, \operatorname{Spec} A)$, so it suffices to consider $v = \mathrm{ord}_E$. If E is on Z, then $\mathrm{mult}_E(K_Z - \pi^*K_A) = a(E, \operatorname{Spec} A) - 1$, $v(I_i) = \mathrm{mult}_E F_i$, and $v(f) = \mathrm{mult}_E \operatorname{div}(f)$, so, as $v(f)$ is an integer, the inequality in (4.1) for v is equivalent to

$$\mathrm{mult}_E \operatorname{div}(f) + \mathrm{mult}_E L \geq 0.$$

This proves the implication $\Leftarrow$. Assume that $f \in \mathcal{J}(A, \prod_i I_i^{t_i})$ and let E be a prime divisor over $\operatorname{Spec} A$ which is not on Z. Write

$$L = K_Z - \pi^*K_A - \sum_i t_iF_i + \sum_k \delta_k D_k, \quad 0 \leq \delta_k < 1.$$

As $f \in \mathcal{J}(A, \prod_i I_i^{t_i})$,

$$\operatorname{div}(f) + L \geq 0.$$

Since in addition $a(v, \operatorname{Spec} A) = a(v, Z) + v(K_Z - \pi^*K_A)$ for $v = \mathrm{ord}_E$ by Definition 2.3(5), we have

$$v(f) - \sum_i t_i v(I_i) + a(v, \operatorname{Spec} A) \geq a(v, Z) - \sum_k \delta_k v(D_k) > 0,$$

where the last inequality follows from $a(v, Z) > 0$ [JM12, Proposition 5.1(i)] if $v(D_k) = 0$ for any k, and otherwise from Definition-Lemma 2.7 as $\delta_k < 1$ for any k. The right-hand side of (4.1) does not involve π, so $\mathcal{J}(A, \prod_i I_i^{t_i})$ does not depend on π.

(2) By (1) for $I_1, \dots, I_m$ and for I, (4.2) follows from $\sum_i t_i v(I_i) - a(v, \operatorname{Spec} A) \geq tv(I) - a(v, \operatorname{Spec} A)$ for any divisorial valuation v of $\mathrm{Frac}(A)$. □

Lemma 4.3 (cf. [JM08, Theorem 3.6])**.** *Let $(A, \mathfrak{m}_A)$ and $I_1, \dots, I_m$ be as in Definition 4.1, and let $t \geq 0$ be a real number. Then*

$$\mathcal{J}\big(A, (I_1 + \cdots + I_m)^t\big) = \sum_{t_1 + \cdots + t_m = t,\ t_i \geq 0} \mathcal{J}(A, \prod_i I_i^{t_i}). \tag{4.3}$$

Proof. First, let $\pi : Z \to \operatorname{Spec} A$ be a log resolution of $I_1, \dots, I_m$ with $I_i\mathcal{O}_Z = \mathcal{O}_Z(-F_i)$ and snc divisor $\sum_k D_k$ as in Definition 4.1(1), let $t_1, \dots, t_m \geq 0$ be real numbers, and let L be as in Definition 4.1(2). We show that

$$R^q\pi_*\mathcal{O}_Z(L) = 0 \quad \text{for any integer } q > 0. \tag{4.4}$$

Since $\lceil x' \rceil = \lceil x \rceil$ for any real numbers $x' \leq x$ close to x and the set of the D_k is finite, there exist rational numbers $t_i' \geq t_i$ with

$$\Big\lceil K_Z - \pi^*K_A - \sum_i t_i' F_i \Big\rceil = L.$$

Let $G := -\pi^*K_A - \sum_i t_i'F_i$ and $\Delta := \lceil K_Z + G \rceil - K_Z - G$. Since $K_Z - \pi^*K_A$ is π-exceptional, $\Delta \geq 0$ is a $\mathbb{Q}$-divisor with $\operatorname{Supp}\Delta \subset \operatorname{Supp}\sum_k D_k$ and $\lfloor \Delta \rfloor = 0$. Since NK_A is a principal divisor on the local scheme $\operatorname{Spec} A$ for some integer $N \geq 1$, $\pi^*K_A \sim_{\mathbb{Q}} 0$ over $\operatorname{Spec} A$. Since $\mathcal{O}_Z(-F_i) = I_i\mathcal{O}_Z$ is generated by global sections, $-F_i$ is π-nef. Thus G is π-nef, and G is π-big as π is birational. Since

$$L = K_Z + \Delta + G,$$

π is proper and birational, $\operatorname{Spec} A$ admits a dualizing complex, and Z is regular, [Mur25, Theorem A(i)] implies (4.4).

If $\sum_i t_i = t$, then for any divisorial valuation v of $\mathrm{Frac}(A)$,

$$\sum_i t_i v(I_i) \geq t \min_i v(I_i) = tv(I_1 + \cdots + I_m),$$

so the inclusion $\supset$ follows from (4.2). For $\subset$, let $\pi : Z \to \operatorname{Spec} A$ be a log resolution of the ideals I_i $(1 \leq i \leq m)$ and $I_i + I_{i'}$ $(1 \leq i < i' \leq m)$. Let $z \in Z$. Since $(I_i + I_{i'})\mathcal{O}_{Z,z}$ is principal for any $i < i'$,

the principal ideals $I_i\mathcal{O}_{Z,z}$ are totally ordered by inclusion, and $(I_1+\cdots+I_m)\mathcal{O}_Z=\mathcal{O}_Z(-F)$ for a divisor $F\geq 0$ which coincides with one of $F_1,\dots,F_m$ near any point of Z. In particular, π is a log resolution of $I_1+\cdots+I_m$. By [JM08, Proposition 2.2, Remark 3.3 and Theorem 3.6], the coefficients of $K_Z-\pi^*K_A$ and $F_1,\dots,F_m$ along the D_k determine a complex of sheaves on Z

$$0\to\mathcal{C}_{m-1}\to\cdots\to\mathcal{C}_0\to\mathcal{O}_Z\big(\lceil K_Z-\pi^*K_A-tF\rceil\big)\to 0, \tag{4.5}$$

where each $\mathcal{C}_j$ is a finite direct sum of sheaves $\mathcal{O}_Z(\lceil K_Z-\pi^*K_A-\sum_i t_iF_i\rceil)$ with $t_i\geq 0$ and $\sum_i t_i=t$, whose push-forwards are the ideals in [JM08, Remark 3.3]. Let $d:=\dim\mathcal{O}_{Z,z}$ and let $x_1,\dots,x_d$ be a regular system of parameters of $\mathcal{O}_{Z,z}$ containing local equations of the D_k through z. Then $\widehat{\mathcal{O}}_{Z,z}=\kappa(z)[[x_1,\dots,x_d]]$, and (4.5) tensored with $\widehat{\mathcal{O}}_{Z,z}$ is the base change along the flat homomorphism

$$\kappa(z)[x_1,\dots,x_d]\to\kappa(z)[[x_1,\dots,x_d]]$$

of the cellular free resolution of a Laurent monomial module, which is exact by [JM08, Proposition 2.2]. Since $\widehat{\mathcal{O}}_{Z,z}$ is faithfully flat over $\mathcal{O}_{Z,z}$, (4.5) is exact at z. By (4.4), all terms of (4.5) are π-acyclic, so π_* of (4.5) is exact. In particular, the map

$$\pi_*\mathcal{C}_0\to\mathcal{J}\big(A,(I_1+\cdots+I_m)^t\big)$$

is surjective, and $\pi_*\mathcal{C}_0$ is a finite direct sum of ideals $\mathcal{J}(A,\prod_i I_i^{t_i})$ with $t_i\geq 0$ and $\sum_i t_i=t$. This proves $\subset$. □

Definition-Lemma 4.4. Let $(A,\mathfrak{m}_A)$ be a Noetherian local domain with $\dim A\geq 1$, let $\alpha_0>0$ be a real number, and let $\omega: A\backslash\{0\}\to\mathbb{R}$ be a function. Let $\mathcal{V}(\alpha_0,\omega)$ be the set of real valuations v of $\mathrm{Frac}(A)$ with $0\leq v(f)\leq\omega(f)$ for any $f\in A\backslash\{0\}$ and $v(\mathfrak{m}_A)\geq\alpha_0$. Then any $v\in\mathcal{V}(\alpha_0,\omega)$ is centered at $\mathfrak{m}_A$, and $\mathcal{V}(\alpha_0,\omega)$ is compact for the topology of pointwise convergence on $\mathrm{Frac}(A)^*$.

Proof. Let $v\in\mathcal{V}(\alpha_0,\omega)$. For a unit f of A, $v(f)=-v(f^{-1})\leq 0$, so $v(f)=0$, and $v(\mathfrak{m}_A)\geq\alpha_0$ implies $v>0$ on $\mathfrak{m}_A\backslash\{0\}$. Thus v is centered at $\mathfrak{m}_A$. Let $f_1,\dots,f_N$ generate $\mathfrak{m}_A$, and let W be the set of $w\in\prod_{f\in A\backslash\{0\}}[0,\omega(f)]$ with

$$w(fg)=w(f)+w(g),\quad w(f+g)\geq\min\{w(f),w(g)\}\text{ if }f+g\neq 0,\quad\text{and}\quad\min_i w(f_i)\geq\alpha_0$$

for any $f,g\in A\backslash\{0\}$. Any $w\in W$ extends to a real valuation of $\mathrm{Frac}(A)$ by $w(f/g):=w(f)-w(g)$, which vanishes on $\mathbb{C}^*$ when $\mathbb{C}\subset A$ since $\mathbb{C}^*$ consists of units. Hence restriction to $A\backslash\{0\}$ is a homeomorphism from $\mathcal{V}(\alpha_0,\omega)$ onto W. Since W is closed in the product, which is compact by Tychonoff's theorem, $\mathcal{V}(\alpha_0,\omega)$ is compact. □

Lemma 4.5 (cf. [JM12, Corollary 6.9])**.** *Let $(A,\mathfrak{m}_A)$ be a normal local domain essentially of finite type over $\mathbb{C}\{z_1,\dots,z_n\}$ such that K_A is $\mathbb{Q}$-Cartier and $\mathrm{Spec}\,A$ is klt.*

(1) For any non-zero ideal $I\subset\mathfrak{m}_A$ there exists a real number $\gamma>0$ with $a(v,\mathrm{Spec}\,A)\geq\gamma v(I)$ for any real valuation v of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$. If moreover I is $\mathfrak{m}_A$-primary, then $\mathrm{lct}(A;I)=a(v,\mathrm{Spec}\,A)/v(I)$ for some divisorial valuation v of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$.

(2) Let $\beta\geq\alpha>0$ be real numbers and $\mathfrak{a}_\bullet$ a graded sequence with

$$\mathfrak{m}_A^{\lceil\beta j\rceil}\subset\mathfrak{a}_j\subset\mathfrak{m}_A^{\lceil\alpha j\rceil}\quad\text{for any integer } j\geq 1.$$

Then

$$0<\mathrm{lct}(A;\mathfrak{a}_\bullet)=\sup_{j\geq 1}j\,\mathrm{lct}(A;\mathfrak{a}_j)<+\infty,$$

and there exists a real valuation v of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$ with $a(v,\mathrm{Spec}\,A)=1$ and $v(\mathfrak{a}_\bullet)=1/\mathrm{lct}(A;\mathfrak{a}_\bullet)$.

Proof. Any real valuation of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$ has a center on any scheme proper and birational over $\mathrm{Spec}\,A$ (Definition 2.3(1)).

(1) Let $\pi: Z\to\mathrm{Spec}\,A$ be a log resolution of I with $I\mathcal{O}_Z=\mathcal{O}_Z(-F)$ and snc divisor $\sum_k D_k$ as in Definition 4.1(1) (A is excellent and admits a dualizing complex by Lemma 2.2). We let

$$a_k:=a(D_k,\mathrm{Spec}\,A)\quad\text{and}\quad f_k:=\mathrm{mult}_{D_k}F.$$

Let Δ_Z be the $\mathbb{Q}$-divisor on Z with $K_Z+\Delta_Z=\pi^*K_A$. Then $\Delta_Z=\sum_k(1-a_k)D_k$ by Definition 2.1, $a_k>0$ for any k since $\operatorname{Spec}A$ is klt, and $\max_k f_k>0$ since $I\subset\mathfrak{m}_A$. For any real valuation v of $\operatorname{Frac}(A)$ centered at $\mathfrak{m}_A$, Definition 2.3(5) and Definition-Lemma 2.7 imply

$$a(v,\operatorname{Spec}A)=a(v,Z)-v(\Delta_Z)\geq\sum_k a_kv(D_k)\quad\text{and}\quad v(I)=v(F)=\sum_k f_kv(D_k). \tag{4.6}$$

Thus $\gamma:=\min_k a_k/\max_k f_k$ satisfies $a(v,\operatorname{Spec}A)\geq\gamma v(I)$. Assume that I is $\mathfrak{m}_A$-primary. By Definition 2.8(1), I is regarded as the $\mathfrak{m}_A$-filtration $(I^{\lceil t\rceil})_{t\in\mathbb{R}}$, and

$$v\big((I^{\lceil t\rceil})_{t\in\mathbb{R}}\big)=\inf_{t>0}\frac{\lceil t\rceil v(I)}{t}=v(I),\quad\text{so}\quad\operatorname{lct}(A;I)=\inf_v\frac{a(v,\operatorname{Spec}A)}{v(I)}.$$

For any k with $f_k>0$, $\pi(D_k)=\{\mathfrak{m}_A\}$, so ord_{D_k} is centered at $\mathfrak{m}_A$. Since $v(I)>0$, (4.6) implies

$$\frac{a(v,\operatorname{Spec}A)}{v(I)}\geq\frac{\sum_k a_kv(D_k)}{\sum_k f_kv(D_k)}\geq\min_{k:f_k>0}\frac{a_k}{f_k}=\min_{k:f_k>0}\frac{a(\operatorname{ord}_{D_k},\operatorname{Spec}A)}{\operatorname{ord}_{D_k}(I)}.$$

Thus $\operatorname{lct}(A;I)$ is attained by ord_{D_k} for some k with $f_k>0$.

(2) For any real valuation v of $\operatorname{Frac}(A)$ centered at $\mathfrak{m}_A$ and any integer $j\geq1$, $\mathfrak{a}_j\subset\mathfrak{m}_A^{\lceil\alpha j\rceil}$ implies $v(\mathfrak{a}_\bullet)\geq\alpha v(\mathfrak{m}_A)>0$, and $v(\mathfrak{a}_\bullet)\leq v(\mathfrak{a}_j)/j$, so

$$\frac{a(v,\operatorname{Spec}A)}{v(\mathfrak{a}_\bullet)}\geq\frac{ja(v,\operatorname{Spec}A)}{v(\mathfrak{a}_j)}\geq j\operatorname{lct}(A;\mathfrak{a}_j). \tag{4.7}$$

Let $\tau:=\sup_{j\geq1}j\operatorname{lct}(A;\mathfrak{a}_j)$. By (4.7), $\operatorname{lct}(A;\mathfrak{a}_\bullet)\geq\tau$, and $\tau<+\infty$ by (4.7) for one divisorial valuation v centered at $\mathfrak{m}_A$, for which $a(v,\operatorname{Spec}A)<+\infty$. By (1), $\tau\geq\operatorname{lct}(A;\mathfrak{a}_1)>0$. For integers $j,j'\geq1$, $\mathfrak{a}_j^{j'}\subset\mathfrak{a}_{jj'}$ implies $jj'\operatorname{lct}(A;\mathfrak{a}_{jj'})\geq j\operatorname{lct}(A;\mathfrak{a}_j)$. Since any $k\geq1$ divides $j!$ for $j\geq k$, the sequence $(j!\operatorname{lct}(A;\mathfrak{a}_{j!}))_{j\geq1}$ is increasing with limit τ.

By (1), for any $j\geq1$ there exists a divisorial valuation v_j of $\operatorname{Frac}(A)$ centered at $\mathfrak{m}_A$ with

$$a(v_j,\operatorname{Spec}A)=1\quad\text{and}\quad v_j(\mathfrak{a}_{j!})=\frac{1}{\operatorname{lct}(A;\mathfrak{a}_{j!})};$$

here we rescale, using $0<a(v_j,\operatorname{Spec}A)<+\infty$. Since $\mathfrak{m}_A^{\lceil\beta j!\rceil}\subset\mathfrak{a}_{j!}$ and $\lceil\beta j!\rceil\leq(\beta+1)j!$,

$$v_j(\mathfrak{m}_A)\geq\frac{v_j(\mathfrak{a}_{j!})}{\lceil\beta j!\rceil}\geq\frac{1}{(\beta+1)j!\operatorname{lct}(A;\mathfrak{a}_{j!})}\geq\frac{1}{(\beta+1)\tau}.$$

For $f\in\mathfrak{m}_A\backslash\{0\}$, by (1) for the ideal fA, there exists a real number $\gamma_f>0$ with $v_j(f)\leq1/\gamma_f$ for any j, since $a(v_j,\operatorname{Spec}A)=1$. For $f\in A\backslash\mathfrak{m}_A$, $v_j(f)=0$. Let $\omega(f):=1/\gamma_f$ for $f\in\mathfrak{m}_A\backslash\{0\}$ and $\omega(f):=0$ for $f\in A\backslash\mathfrak{m}_A$. Then $v_j\in\mathcal{V}(1/((\beta+1)\tau),\omega)$ for any j, so by Definition-Lemma 4.4, a subnet of $(v_j)_{j\geq1}$ converges pointwise on $\operatorname{Frac}(A)^*$ to some $v\in\mathcal{V}(1/((\beta+1)\tau),\omega)$, which is a real valuation of $\operatorname{Frac}(A)$ centered at $\mathfrak{m}_A$. By Lemma 2.4, $a(v,\operatorname{Spec}A)\leq1$. Fix an integer $k\geq1$. For $j\geq k$, $\mathfrak{a}_k^{j!/k}\subset\mathfrak{a}_{j!}$ implies

$$v_j(\mathfrak{a}_k)\geq\frac{k}{j!}v_j(\mathfrak{a}_{j!})=\frac{k}{j!\operatorname{lct}(A;\mathfrak{a}_{j!})}\geq\frac{k}{\tau},$$

so $v(g)\geq k/\tau$ for any $g\in\mathfrak{a}_k\backslash\{0\}$, that is, $v(\mathfrak{a}_k)\geq k/\tau$. Hence $v(\mathfrak{a}_\bullet)\geq1/\tau$. Since (4.7) implies $a(v,\operatorname{Spec}A)\geq\tau v(\mathfrak{a}_\bullet)$,

$$1\geq a(v,\operatorname{Spec}A)\geq\tau v(\mathfrak{a}_\bullet)\geq1.$$

Thus $a(v,\operatorname{Spec}A)=1$ and $v(\mathfrak{a}_\bullet)=1/\tau$, so

$$\operatorname{lct}(A;\mathfrak{a}_\bullet)\leq\frac{a(v,\operatorname{Spec}A)}{v(\mathfrak{a}_\bullet)}=\tau.$$

Together with $\operatorname{lct}(A;\mathfrak{a}_\bullet)\geq\tau$, (2) follows. □

Lemma 4.6. *Let $(A, \mathfrak{m}_A)$ be a normal local domain essentially of finite type over $\mathbb{C}\{z_1, \dots, z_n\}$ such that K_A is $\mathbb{Q}$-Cartier, and let u be a real valuation of $\operatorname{Frac}(A)$ centered at $\mathfrak{m}_A$ with $a(u, \operatorname{Spec} A) < +\infty$. Then there exists a real number $\kappa_u > 0$ with*

$$\mathfrak{a}_t(u) \subset \mathfrak{m}_A^{\lceil t/\kappa_u \rceil} \quad \text{for any real number } t > 0.$$

In particular, $\operatorname{vol}_A(u) > 0$.

Proof. Let $\pi : Z \to \operatorname{Spec} A$ be a log resolution of $\mathfrak{m}_A$ [Tem18, Theorem 1.1.6] (A is excellent and admits a dualizing complex by Lemma 2.2), let $\xi := \operatorname{center}_Z u$, and let Δ_Z be the $\mathbb{Q}$-divisor on Z with $K_Z + \Delta_Z = \pi^* K_A$. By Definition 2.3(5), $a(u, Z) = a(u, \operatorname{Spec} A) + u(\Delta_Z) < +\infty$. By Lemma 2.5(1),

$$u(g) \le a(u, Z) \operatorname{ord}_\xi(g) \quad \text{for any } g \in \mathcal{O}_{Z,\xi} \backslash \{0\},$$

where ord_ξ is the order function of the regular local ring $\mathcal{O}_{Z,\xi}$. In particular, $a(u, Z) > 0$, since u is centered at ξ. The valuation ord_ξ is the order of vanishing along the exceptional prime divisor over ξ of the normalized blow-up of Z along the closure of ξ (along the closure of ξ itself if $\dim \mathcal{O}_{Z,\xi} = 1$), so ord_ξ is a divisorial valuation of $\operatorname{Frac}(A)$, and it is centered at $\pi(\xi) = \mathfrak{m}_A$. Since A is analytically irreducible by Lemma 2.2, Lemma 2.5(2) gives a real number $\kappa' > 0$ with

$$\mathfrak{a}_t(\operatorname{ord}_\xi) \subset \mathfrak{m}_A^{\lceil t/\kappa' \rceil} \quad \text{for any real number } t > 0.$$

Let $\kappa_u := a(u, Z)\kappa'$. Then for any real number $t > 0$,

$$\mathfrak{a}_t(u) \subset \mathfrak{a}_{t/a(u,Z)}(\operatorname{ord}_\xi) \subset \mathfrak{m}_A^{\lceil t/\kappa_u \rceil},$$

so, with $d := \dim A$,

$$\operatorname{vol}_A(u) \ge e(\mathfrak{m}_A)\kappa_u^{-d} > 0.$$

□

The following convexity yields the upper bound $\operatorname{lct}(B; \mathfrak{b}_\bullet) \le 1$ in Construction-Proposition 4.10 and the uniqueness of the volume minimizer in Section 6.

Lemma 4.7 (cf. [XZ21, Lemma 3.12] and [BLQ24, Proposition 5.1])**.** *Let $(A, \mathfrak{m}_A)$ be a normal local domain essentially of finite type over $\mathbb{C}\{z_1, \dots, z_n\}$ such that K_A is $\mathbb{Q}$-Cartier and* $\operatorname{Spec} A$ *is klt, let $\beta \ge \alpha > 0$ be real numbers, let $\mathfrak{a}_\bullet$ and $\mathfrak{c}_\bullet$ be $\mathfrak{m}_A$-filtrations with*

$$\mathfrak{m}_A^{\lceil \beta t \rceil} \subset \mathfrak{a}_t \subset \mathfrak{m}_A^{\lceil \alpha t \rceil} \quad \text{and} \quad \mathfrak{m}_A^{\lceil \beta t \rceil} \subset \mathfrak{c}_t \subset \mathfrak{m}_A^{\lceil \alpha t \rceil}$$

for any real number $t > 0$, and let $0 < \sigma < 1$. Then the geodesic $\mathfrak{g}_\bullet$ between $\mathfrak{a}_\bullet$ and $\mathfrak{c}_\bullet$ at σ satisfies

$$\operatorname{lct}(A; \mathfrak{g}_\bullet) \le (1 - \sigma) \operatorname{lct}(A; \mathfrak{a}_\bullet) + \sigma \operatorname{lct}(A; \mathfrak{c}_\bullet).$$

Proof. **Step 1.** We let

$$\mathfrak{a}'_j := \mathfrak{a}_{j/(1-\sigma)}, \quad \mathfrak{c}'_j := \mathfrak{c}_{j/\sigma} \quad \text{and} \quad \mathfrak{d}_j := \sum_{i=0}^{j} (\mathfrak{a}'_i \cap \mathfrak{c}'_{j-i})$$

for any integer $j \ge 0$, so that $\mathfrak{a}'_0 = \mathfrak{c}'_0 = \mathfrak{d}_0 = A$. Since $\mathfrak{a}_t \mathfrak{a}_{t'} \subset \mathfrak{a}_{t+t'}$ and $\mathfrak{c}_t \mathfrak{c}_{t'} \subset \mathfrak{c}_{t+t'}$, $\mathfrak{a}'_\bullet$, $\mathfrak{c}'_\bullet$, and $\mathfrak{d}_\bullet$ are graded sequences. Let $\alpha' := \alpha/2$ and $\beta' := \beta/(\sigma(1-\sigma))$. For any $j \ge 1$, $\mathfrak{d}_j \supset \mathfrak{a}'_j$ and

$$\mathfrak{a}'_i \cap \mathfrak{c}'_{j-i} \subset \mathfrak{m}_A^{\max\{\lceil \alpha i \rceil, \lceil \alpha(j-i) \rceil\}} \subset \mathfrak{m}_A^{\lceil \alpha' j \rceil},$$

so for $\mathfrak{e}_\bullet = \mathfrak{a}'_\bullet$, $\mathfrak{c}'_\bullet$, or $\mathfrak{d}_\bullet$,

$$\mathfrak{m}_A^{\lceil \beta' j \rceil} \subset \mathfrak{e}_j \subset \mathfrak{m}_A^{\lceil \alpha' j \rceil} \quad \text{for any integer } j \ge 1.$$

Let I be an $\mathfrak{m}_A$-primary ideal and $t \ge 0$ a real number. By (4.1) for $f = 1$ (A is excellent and admits a dualizing complex by Lemma 2.2), $\mathcal{J}(A, I^t) = A$ if and only if $a(v, \operatorname{Spec} A) > tv(I)$ for any divisorial valuation v of $\operatorname{Frac}(A)$. If $t < \operatorname{lct}(A; I)$, then this holds: for v centered at $\mathfrak{m}_A$ since $a(v, \operatorname{Spec} A) \ge \operatorname{lct}(A; I)v(I)$ and $v(I) > 0$, and for the other v since $v(I) = 0$ and $a(v, \operatorname{Spec} A) > 0$

as $\operatorname{Spec} A$ is klt. Conversely, by Lemma 4.5(1) there exists a divisorial valuation v of $\operatorname{Frac}(A)$ centered at $\mathfrak{m}_A$ with $\operatorname{lct}(A;I) = a(v,\operatorname{Spec} A)/v(I)$, so $\mathcal{J}(A,I^t) = A$ implies $t < \operatorname{lct}(A;I)$. Let $\mathfrak{e}_\bullet$ be one of $\mathfrak{a}'_\bullet$, $\mathfrak{c}'_\bullet$, and $\mathfrak{d}_\bullet$, and let $t \geq 0$ be a real number. As $\mathcal{J}(A, t\cdot\mathfrak{e}_\bullet)$ is the largest of the ideals $\mathcal{J}(A,\mathfrak{e}_j^{t/j})$ and each $\mathfrak{e}_j$ with $j \geq 1$ is $\mathfrak{m}_A$-primary, $\mathcal{J}(A, t\cdot\mathfrak{e}_\bullet) = A$ if and only if $t < j\operatorname{lct}(A;\mathfrak{e}_j)$ for some integer $j \geq 1$. Since $\operatorname{lct}(A;\mathfrak{e}_\bullet) = \sup_j j\operatorname{lct}(A;\mathfrak{e}_j)$ by Lemma 4.5(2), it follows that

$$\mathcal{J}(A, t\cdot\mathfrak{e}_\bullet) = A \iff t < \operatorname{lct}(A;\mathfrak{e}_\bullet). \tag{4.8}$$

Let $t \geq 0$ be a real number, let $j \geq 1$ be an integer with

$$\mathcal{J}(A, t\cdot\mathfrak{d}_\bullet) = \mathcal{J}(A, \mathfrak{d}_j^{t/j}),$$

and let $I_i := \mathfrak{a}'_i \cap \mathfrak{c}'_{j-i}$ for $0 \leq i \leq j$. By Lemma 4.3, $\mathcal{J}(A,\mathfrak{d}_j^{t/j})$ is the sum of the ideals $\mathcal{J}(A, \prod_{i=0}^j I_i^{t_i})$ with $t_i \geq 0$ and $\sum_i t_i = t/j$. Fix such $t_0,\dots,t_j$ and let

$$\lambda := \sum_{i=0}^j it_i \quad\text{and}\quad \lambda' := \sum_{i=0}^j (j-i)t_i, \quad\text{so that}\quad \lambda+\lambda' = t.$$

For any divisorial valuation v of $\operatorname{Frac}(A)$, since $\mathfrak{a}_i'^{j!/i} \subset \mathfrak{a}'_{j!}$ for $1 \leq i \leq j$,

$$\sum_{i=0}^j t_i v(I_i) \geq \sum_{i=1}^j t_i v(\mathfrak{a}'_i) \geq \sum_{i=1}^j t_i \frac{i}{j!} v(\mathfrak{a}'_{j!}) = \frac{\lambda}{j!} v(\mathfrak{a}'_{j!}),$$

so (4.2) implies

$$\mathcal{J}(A, \prod_i I_i^{t_i}) \subset \mathcal{J}(A, \mathfrak{a}_{j!}'^{\lambda/j!}) \subset \mathcal{J}(A, \lambda\cdot\mathfrak{a}'_\bullet).$$

Similarly, since $\mathfrak{c}_{j-i}'^{j!/(j-i)} \subset \mathfrak{c}'_{j!}$ for $0 \leq i \leq j-1$,

$$\sum_{i=0}^j t_i v(I_i) \geq \sum_{i=0}^{j-1} t_i v(\mathfrak{c}'_{j-i}) \geq \frac{\lambda'}{j!} v(\mathfrak{c}'_{j!}),$$

so

$$\mathcal{J}(A, \prod_i I_i^{t_i}) \subset \mathcal{J}(A, \lambda'\cdot\mathfrak{c}'_\bullet).$$

Therefore

$$\mathcal{J}(A, t\cdot\mathfrak{d}_\bullet) \subset \sum_{\lambda+\lambda'=t,\ \lambda,\lambda'\geq 0} \big(\mathcal{J}(A,\lambda\cdot\mathfrak{a}'_\bullet) \cap \mathcal{J}(A,\lambda'\cdot\mathfrak{c}'_\bullet)\big). \tag{4.9}$$

Now let

$$t := \operatorname{lct}(A;\mathfrak{a}'_\bullet) + \operatorname{lct}(A;\mathfrak{c}'_\bullet).$$

If $\lambda+\lambda' = t$, then $\lambda \geq \operatorname{lct}(A;\mathfrak{a}'_\bullet)$ or $\lambda' \geq \operatorname{lct}(A;\mathfrak{c}'_\bullet)$, so by (4.8),

$$\mathcal{J}(A,\lambda\cdot\mathfrak{a}'_\bullet) \cap \mathcal{J}(A,\lambda'\cdot\mathfrak{c}'_\bullet) \subset \mathfrak{m}_A.$$

By (4.9), $\mathcal{J}(A, t\cdot\mathfrak{d}_\bullet) \neq A$, and (4.8) implies

$$\operatorname{lct}(A;\mathfrak{d}_\bullet) \leq \operatorname{lct}(A;\mathfrak{a}'_\bullet) + \operatorname{lct}(A;\mathfrak{c}'_\bullet). \tag{4.10}$$

Step 2. Let v be a real valuation of $\operatorname{Frac}(A)$ centered at $\mathfrak{m}_A$. For any real number $\tau > 0$, we show that

$$v(\mathfrak{a}_\bullet) = \inf_{j\geq 1} \frac{v(\mathfrak{a}_{j\tau})}{j\tau}. \tag{4.11}$$

The inequality $\leq$ follows from the definition. Since $\mathfrak{a}_\bullet$ is decreasing and multiplicative, $t \to v(\mathfrak{a}_t)$ is increasing and

$$v(\mathfrak{a}_{t+t'}) \leq v(\mathfrak{a}_t) + v(\mathfrak{a}_{t'}).$$

Let $t > 0$ be a real number. For any integer $N \geq 1$, $j_N := \lceil Nt/\tau \rceil$ satisfies $Nt \leq j_N\tau < Nt + \tau$, so

$$\frac{v(\mathfrak{a}_{j_N\tau})}{j_N\tau} \leq \frac{v(\mathfrak{a}_{Nt+\tau})}{Nt} \leq \frac{Nv(\mathfrak{a}_t) + v(\mathfrak{a}_\tau)}{Nt},$$

which tends to $v(\mathfrak{a}_t)/t$ as $N \to +\infty$. This proves (4.11). By (4.11) with $\tau = 1/(1-\sigma)$, and by (4.11) for $\mathfrak{c}_\bullet$ with $\tau = 1/\sigma$,

$$v(\mathfrak{a}'_\bullet) = \frac{v(\mathfrak{a}_\bullet)}{1-\sigma} \quad \text{and} \quad v(\mathfrak{c}'_\bullet) = \frac{v(\mathfrak{c}_\bullet)}{\sigma},$$

so

$$\operatorname{lct}(A; \mathfrak{a}'_\bullet) = (1-\sigma)\operatorname{lct}(A; \mathfrak{a}_\bullet) \quad \text{and} \quad \operatorname{lct}(A; \mathfrak{c}'_\bullet) = \sigma \operatorname{lct}(A; \mathfrak{c}_\bullet).$$

Next we show that $v(\mathfrak{g}_\bullet) \geq v(\mathfrak{d}_\bullet)$. Let $t > 0$, let t_0, t_1 be real numbers with $(1-\sigma)t_0 + \sigma t_1 = t$, and let $I := \mathfrak{a}_{t_0} \cap \mathfrak{c}_{t_1}$. If $t_0 < 0$, then $t_1 > t/\sigma$ and

$$I = \mathfrak{c}_{t_1} \subset \mathfrak{a}_0 \cap \mathfrak{c}_{t/\sigma}.$$

If $t_1 < 0$, then $I \subset \mathfrak{a}_{t/(1-\sigma)} \cap \mathfrak{c}_0$. Since a lower bound for v on a larger ideal bounds $v(I)$, possibly replacing (t_0, t_1) with $(0, t/\sigma)$ or $(t/(1-\sigma), 0)$, we may assume that $t_0, t_1 \geq 0$. Let $N \geq 2/t$ be an integer. Since

$$\lfloor N(1-\sigma)t_0 \rfloor + \lfloor N\sigma t_1 \rfloor \geq \lfloor Nt \rfloor - 1 \geq 1,$$

there exist integers $0 \leq i \leq \lfloor N(1-\sigma)t_0 \rfloor$ and $0 \leq i' \leq \lfloor N\sigma t_1 \rfloor$ with $i + i' = \lfloor Nt \rfloor - 1$. Then $i/(1-\sigma) \leq Nt_0$ and $i'/\sigma \leq Nt_1$, so

$$I^N \subset \mathfrak{a}_{Nt_0} \cap \mathfrak{c}_{Nt_1} \subset \mathfrak{a}'_i \cap \mathfrak{c}'_{i'} \subset \mathfrak{d}_{\lfloor Nt \rfloor - 1}.$$

Hence

$$Nv(I) \geq v(\mathfrak{d}_{\lfloor Nt \rfloor - 1}) \geq (\lfloor Nt \rfloor - 1)v(\mathfrak{d}_\bullet),$$

and letting $N \to +\infty$ yields $v(I) \geq tv(\mathfrak{d}_\bullet)$. Since $\mathfrak{g}_t$ is the sum of such ideals I, $v(\mathfrak{g}_t) \geq tv(\mathfrak{d}_\bullet)$ for any $t > 0$, so $v(\mathfrak{g}_\bullet) \geq v(\mathfrak{d}_\bullet) > 0$.

Therefore, by (4.10),

$$\operatorname{lct}(A; \mathfrak{g}_\bullet) \leq \operatorname{lct}(A; \mathfrak{d}_\bullet) \leq \operatorname{lct}(A; \mathfrak{a}'_\bullet) + \operatorname{lct}(A; \mathfrak{c}'_\bullet) = (1-\sigma)\operatorname{lct}(A; \mathfrak{a}_\bullet) + \sigma \operatorname{lct}(A; \mathfrak{c}_\bullet).$$

The lemma follows. □

4.2. **Construction of the filtration.**

Set-up 4.8. Let A_0, $\mathfrak{q}$, Y, y, B, $\mathfrak{n}$, r, h, Λ, and $R(h)$ be as follows.

(1) A_0 is a normal complex analytic local domain, $\mathfrak{q}$ is a prime ideal of A_0, Y is a normal integral scheme with a birational morphism of finite type $Y \to \operatorname{Spec}(A_0)_\mathfrak{q}$, and $y \in Y$.
(2) K_B is $\mathbb{Q}$-Cartier and $\operatorname{Spec} B$ is klt, where $B := \mathcal{O}_{Y,y}$, $\mathfrak{n} := \mathfrak{m}_B$, and $r := \dim B \geq 1$.
(3) $h : B \to B$ is a finite injective local $\mathbb{C}$-algebra homomorphism, $\Lambda \geq 2$ is an integer with

$$[\operatorname{Frac}(B) : h(\operatorname{Frac}(B))] = \Lambda^r [B/\mathfrak{n} : h_{B/\mathfrak{n}}(B/\mathfrak{n})],$$

and there exists a real number $b > 0$ with $c(h^j) \geq b\Lambda^j$ for any integer $j \geq 0$.
(4) $R(h)$ is the ramification divisor of $\operatorname{Spec} h$.

Since A_0 is a quotient of $\mathbb{C}\{z_1, \ldots, z_n\}$ for some n and Y is of finite type over $\operatorname{Spec}(A_0)_\mathfrak{q}$, B is a normal local domain essentially of finite type over $\mathbb{C}\{z_1, \ldots, z_n\}$, so B is excellent, admits a dualizing complex, and is analytically irreducible (Lemma 2.2).

Lemma 4.9. *Let $(S, \mathfrak{n})$ be an analytically irreducible Noetherian local domain containing $\mathbb{C}$ with $r := \dim S \geq 1$, let $\varphi : S \to S$ be a finite injective local $\mathbb{C}$-algebra homomorphism, and let $Q \geq 2$ be an integer with*

$$[\operatorname{Frac}(S) : \varphi(\operatorname{Frac}(S))] = Q^r [S/\mathfrak{n} : \varphi_{S/\mathfrak{n}}(S/\mathfrak{n})].$$

Assume that there exists a real number $b > 0$ with $c(\varphi^j) \geq bQ^j$ for any integer $j \geq 0$. Then there exists a real number $C_1 > 0$ such that $f \in \mathfrak{n}^{\lceil k/(C_1 Q^j) \rceil}$ for any integers $j, k \geq 0$ and any $f \in S$ with $\varphi^j(f) \in \mathfrak{n}^k$.

Proof. For any $\mathfrak{n}$-primary ideal I and any integer $j \geq 1$, $\varphi^{j-1}(I)S$ is $\mathfrak{n}$-primary because S is finite over $\varphi^{j-1}(S)$, and

$$\varphi^j(I)S = \varphi(\varphi^{j-1}(I)S)S.$$

By Lemma 2.12, the degree identity in the statement, and induction on j,

$$e(\varphi^j(I)S) = Q^{rj}e(I) \quad \text{for any } \mathfrak{n}\text{-primary ideal } I \text{ of } S \text{ and any integer } j \geq 0. \tag{4.12}$$

We let L be a coefficient field of $\widehat{S}$ containing $\mathbb{C}$ [Mat89, Theorem 28.3], and let $x_1, \dots, x_r \in S$ be a system of parameters of S. Since $\widehat{S}/(x_1, \dots, x_r)\widehat{S}$ is a finite-dimensional L-vector space, the L-algebra homomorphism $L[[X_1, \dots, X_r]] \to \widehat{S}$, $X_i \to x_i$, makes $\widehat{S}$ a finite $L[[X_1, \dots, X_r]]$-module [Mat89, Theorem 8.4], and its kernel is 0 since $\dim \widehat{S} = r$. Thus $L[[x_1, \dots, x_r]] \subset \widehat{S}$ is a formal power series ring, and $\widehat{S}$ is a finite $L[[x_1, \dots, x_r]]$-module. Let $y_1, \dots, y_{N_0} \in \widehat{S}$ generate $\widehat{S}$ as an $L[[x_1, \dots, x_r]]$-module. For $f \in \widehat{S}\backslash\{0\}$, let $\mathrm{Nm}(f)$ be the norm of f for the finite field extension

$$\mathrm{Frac}(L[[x_1, \dots, x_r]]) \subset \mathrm{Frac}(\widehat{S}).$$

By Lemma 2.6 for $L[[x_1, \dots, x_r]] \subset \widehat{S}$ (the first ring is regular and $\widehat{S}$ is a complete local domain), $\mathrm{Nm}(f) \in L[[x_1, \dots, x_r]] \cap f\widehat{S}$, and there exists an integer $C_0 \geq 1$ with

$$\mathrm{ord}_{(x)}(\mathrm{Nm}(f)) \leq C_0 \,\mathrm{ord}_{\widehat{\mathfrak{n}}}(f) \quad \text{for any } f \in \widehat{S}\backslash\{0\},$$

where (x) is the maximal ideal of $L[[x_1, \dots, x_r]]$ and $\widehat{\mathfrak{n}} := \mathfrak{n}\widehat{S}$, by Lemma 2.6(2) with $\mathrm{ord}_{(x)}(\mathrm{Nm}(f))$ in place of j. Since $\mathfrak{n}^m\widehat{S} \cap S = \mathfrak{n}^m$ for any m, we have $\mathrm{ord}_{\widehat{\mathfrak{n}}}(f) = \mathrm{ord}_{\mathfrak{n}}(f)$ for $f \in S\backslash\{0\}$. Let

$$C_1 := N_0 C_0 b^{1-r}.$$

Fix $f \in S\backslash\{0\}$ and an integer $j \geq 0$. We show

$$\mathrm{ord}_{\mathfrak{n}}(\varphi^j(f)) \leq C_1 Q^j \,\mathrm{ord}_{\mathfrak{n}}(f).$$

If f is a unit, then so is $\varphi^j(f)$, and both sides are 0. Thus we may assume that $f \in \mathfrak{n}$. Let $\Phi_f \in L[x_1, \dots, x_r]$ be the homogeneous part of $\mathrm{Nm}(f)$ of degree $\mathrm{ord}_{(x)}(\mathrm{Nm}(f))$, so $\Phi_f \neq 0$. Writing the coefficients of Φ_f in a basis of L over $\mathbb{C}$, we find $\lambda \in \mathbb{C}^r$ with $\Phi_f(\lambda) \neq 0$, since $\mathbb{C}$ is infinite. Let $x'_1, \dots, x'_{r-1}$ be linearly independent $\mathbb{C}$-linear combinations of $x_1, \dots, x_r$ vanishing at λ. Then $x'_i \in S$, $L[[x_1, \dots, x_r]]/(x'_1, \dots, x'_{r-1})$ is a formal power series ring $L[[\zeta]]$ in which x_i maps to $\lambda_i\zeta$ for any i, and the image of $\mathrm{Nm}(f)$ in $L[[\zeta]]$ is $\Phi_f(\lambda)\zeta^{\mathrm{ord}_{(x)}(\mathrm{Nm}(f))}$ plus terms of higher degree. Let

$$I := (f, x'_1, \dots, x'_{r-1})S \quad \text{and} \quad J := (\mathrm{Nm}(f), x'_1, \dots, x'_{r-1})\widehat{S} \subset I\widehat{S}.$$

Since L is the residue field of both $\widehat{S}$ and $L[[x_1, \dots, x_r]]$, and $y_1, \dots, y_{N_0}$ generate $\widehat{S}$ over $L[[x_1, \dots, x_r]]$,

$$\ell_{\widehat{S}}(\widehat{S}/J) \leq N_0 \ell\big(L[[x_1, \dots, x_r]]/(\mathrm{Nm}(f), x'_1, \dots, x'_{r-1})\big) = N_0 \,\mathrm{ord}_{(x)}(\mathrm{Nm}(f)) \leq N_0 C_0 \,\mathrm{ord}_{\mathfrak{n}}(f).$$

In particular, J is generated by a system of parameters of $\widehat{S}$, and I is $\mathfrak{n}$-primary. Since $J \subset I\widehat{S}$, multiplicities do not change under completion, and the multiplicity of an ideal generated by a system of parameters is at most its colength [Mat89, Theorem 14.10],

$$e(I) = e(I\widehat{S}) \leq e(J) \leq \ell_{\widehat{S}}(\widehat{S}/J) \leq N_0 C_0 \,\mathrm{ord}_{\mathfrak{n}}(f).$$

On the other hand, let $q_j := \mathrm{ord}_{\mathfrak{n}}(\varphi^j(f))$. Since S is finite over $\varphi^j(S)$, the ideal $\varphi^j(I)S$ is $\mathfrak{n}$-primary, so $\varphi^j(f), \varphi^j(x'_1), \dots, \varphi^j(x'_{r-1})$ is a system of parameters of S. Since $\varphi^j(f) \in \mathfrak{n}^{q_j}$ and $\varphi^j(x'_i) \in \mathfrak{n}^{c(\varphi^j)}$ for any i,

$$\big(\varphi^j(f)^{c(\varphi^j)}, \varphi^j(x'_1)^{q_j}, \dots, \varphi^j(x'_{r-1})^{q_j}\big)S \subset \mathfrak{n}^{c(\varphi^j)q_j}.$$

The multiplicity of the ideal on the left is $c(\varphi^j)q_j^{r-1}e(\varphi^j(I)S)$ [Mat89, Theorem 14.12], and

$$e(\mathfrak{n}^{c(\varphi^j)q_j}) = (c(\varphi^j)q_j)^r e(\mathfrak{n}) \geq (c(\varphi^j)q_j)^r.$$

Thus $q_j c(\varphi^j)^{r-1} \le e(\varphi^j(I)S)$. By (4.12) and $c(\varphi^j) \ge bQ^j$,

$$q_j \le c(\varphi^j)^{1-r} Q^{rj} e(I) \le (bQ^j)^{1-r} Q^{rj} N_0 C_0 \operatorname{ord}_{\mathfrak{n}}(f) = C_1 Q^j \operatorname{ord}_{\mathfrak{n}}(f).$$

Since $\operatorname{ord}_{\mathfrak{n}}$ takes integer values, for any integer $k \ge 0$ with $\varphi^j(f) \in \mathfrak{n}^k$ we get $\operatorname{ord}_{\mathfrak{n}}(f) \ge k/(C_1 Q^j)$, so $f \in \mathfrak{n}^{\lceil k/(C_1 Q^j)\rceil}$. For $f = 0$ there is nothing to prove. □

Construction-Proposition 4.10. Notation and conditions as in Set-up 4.8. Then there exist an $\mathfrak{n}$-filtration $\mathfrak{b}_\bullet$ of B and real numbers $\beta \ge \alpha > 0$ with

$$\mathfrak{n}^{\lceil \beta t\rceil} \subset \mathfrak{b}_t \subset \mathfrak{n}^{\lceil \alpha t\rceil} \quad \text{for any real number } t > 0$$

satisfying the following.

(1) $\operatorname{lct}(B; \mathfrak{b}_\bullet) = 1$.
(2) $(h_* v)(\mathfrak{b}_\bullet) = \Lambda v(\mathfrak{b}_\bullet)$ and $v(R(h)) \ge (\Lambda - 1) v(\mathfrak{b}_\bullet)$ for any real valuation v of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$ with $a(v, \operatorname{Spec} B) < +\infty$.
(3) The set $\mathcal{W}$ of real valuations v of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$ with $a(v, \operatorname{Spec} B) = v(\mathfrak{b}_\bullet) = 1$ is non-empty and compact for the topology of pointwise convergence, and $\Lambda u \in \mathcal{W}$ for any real valuation u of $\operatorname{Frac}(B)$ with $h_* u \in \mathcal{W}$.

Proof. Since K_B is $\mathbb{Q}$-Cartier and $R(h) \sim_{\mathbb{Q}} K_B - (\operatorname{Spec} h)^* K_B$, $R(h)$ is $\mathbb{Q}$-Cartier. Since moreover $R(h) \ge 0$ and B is a normal local domain, there exist an integer $m_0 \ge 1$ and an element $g_0 \in B\backslash\{0\}$ with $m_0 R(h) = \operatorname{div}(g_0)$. Thus

$$u(R(h)) = u(g_0)/m_0 \in [0, +\infty)$$

for any real valuation u of $\operatorname{Frac}(B)$ with $u \ge 0$ on $B\backslash\{0\}$.

Step 1. We show that for any real valuation u of $\operatorname{Frac}(B)$ with $u \ge 0$ on $B\backslash\{0\}$,

$$a(h_* u, \operatorname{Spec} B) = a(u, \operatorname{Spec} B) + u(R(h)), \tag{4.13}$$

where both sides may be $+\infty$, and that for any real valuation u of $\operatorname{Frac}(B)$, $h_* u$ is centered at $\mathfrak{n}$ if and only if u is centered at $\mathfrak{n}$.

The valuation u has a center on $\operatorname{Spec} B$, and $\operatorname{Spec} h$ is finite and surjective since h is finite and injective. Lemma 3.2 for $f = \operatorname{Spec} h$ and the zero $\mathbb{Q}$-divisor on $\operatorname{Spec} B$, and Definition 2.3(5) since $R(h)$ is $\mathbb{Q}$-Cartier, imply

$$a(h_* u, \operatorname{Spec} B) = a(u, \operatorname{Spec} B, -R(h)) = a(u, \operatorname{Spec} B) + u(R(h)),$$

which proves (4.13).

Since h is local, $h^{-1}(\mathfrak{n}) = \mathfrak{n}$, so $h_* u$ is centered at $\mathfrak{n}$ if u is. Conversely, if $h_* u$ is centered at $\mathfrak{n}$, then u is centered at $\mathfrak{n}$ since B is integral over $h(B)$. In this case (4.13) and $u(R(h)) \ge 0$ imply

$$a(u, \operatorname{Spec} B) \le a(h_* u, \operatorname{Spec} B). \tag{4.14}$$

Step 2. Since B is analytically irreducible (Lemma 2.2), Lemma 4.9 with B, h, and Λ in place of S, φ, and Q (the degree identity is Set-up 4.8(3)) gives a real number $C_1 > 0$ with

$$b\Lambda^j \operatorname{ord}_{\mathfrak{n}}(f) \le \operatorname{ord}_{\mathfrak{n}}(h^j(f)) \le C_1 \Lambda^j \operatorname{ord}_{\mathfrak{n}}(f) \tag{4.15}$$

for any $f \in B\backslash\{0\}$ and any integer $j \ge 0$. Here the first inequality holds since $h^j(\mathfrak{n})B \subset \mathfrak{n}^{c(h^j)}$ and $c(h^j) \ge b\Lambda^j$, and the second follows from Lemma 4.9 with $k = \operatorname{ord}_{\mathfrak{n}}(h^j(f))$.

Step 3. We construct a compact set $\mathcal{K}$ of real valuations of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$, real numbers $\kappa_2 \ge \kappa_1 > 0$, and a Radon probability measure ρ on $\mathcal{K}$ satisfying the following.

(i) $\Lambda^{-1} h_* v \in \mathcal{K}$ for any $v \in \mathcal{K}$.
(ii) We have

$$\kappa_1 \operatorname{ord}_{\mathfrak{n}}(f) \le v(f) \le \kappa_2 \operatorname{ord}_{\mathfrak{n}}(f) \quad \text{for any } v \in \mathcal{K} \text{ and any } f \in B\backslash\{0\}. \tag{4.16}$$

(iii) $\rho(E) = \rho(\{v \in \mathcal{K} \mid \Lambda^{-1} h_* v \in E\})$ for any Borel set $E \subset \mathcal{K}$.
(iv) $a(\cdot, \operatorname{Spec} B)$ is bounded on $\mathcal{K}$.

(v) We have

$$\int_{\mathcal{K}} a(v,\operatorname{Spec} B)d\rho(v)=1 \quad\text{and}\quad \int_{\mathcal{K}} v(R(h))d\rho(v)=\Lambda-1. \tag{4.17}$$

For a real number $\alpha_0>0$ and a function $\omega: B\backslash\{0\}\to\mathbb{R}$, let $\mathcal{V}(\alpha_0,\omega)$ be the set of Definition-Lemma 4.4. It is compact, and its elements are real valuations of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$.

Let v_0 be a divisorial valuation of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$, for instance the order of vanishing along an irreducible component of the preimage of the closed point in the normalized blow-up of $\mathfrak{n}$. By Lemma 2.5(2), there exists a real number $\kappa_0>0$ with $v_0(f)\le\kappa_0\operatorname{ord}_{\mathfrak{n}}(f)$ for any $f\in B\backslash\{0\}$, and $v_0(f)\ge v_0(\mathfrak{n})\operatorname{ord}_{\mathfrak{n}}(f)$. For any integer $j\ge 0$, let

$$v_j:=\Lambda^{-j}(h^j)_*v_0,$$

so $v_{j+1}=\Lambda^{-1}h_*v_j$. By (4.15), for any $f\in B\backslash\{0\}$,

$$bv_0(\mathfrak{n})\operatorname{ord}_{\mathfrak{n}}(f)\le\Lambda^{-j}v_0(\mathfrak{n})\operatorname{ord}_{\mathfrak{n}}(h^j(f))\le v_j(f)\le\Lambda^{-j}\kappa_0\operatorname{ord}_{\mathfrak{n}}(h^j(f))\le\kappa_0C_1\operatorname{ord}_{\mathfrak{n}}(f),$$

so $v_j\ge 0$ on $B\backslash\{0\}$ and $v_j(\mathfrak{n})\ge bv_0(\mathfrak{n})$, and hence $v_j\in\mathcal{V}_0$, where

$$\mathcal{V}_0:=\mathcal{V}(bv_0(\mathfrak{n}),\kappa_0C_1\operatorname{ord}_{\mathfrak{n}}).$$

Let $\mathcal{K}_0$ be the closure of $\{v_j\mid j\ge 0\}$ in $\mathcal{V}_0$. Then $\mathcal{K}_0$ is compact. The map $v\to\Lambda^{-1}h_*v$ is continuous, since $(\Lambda^{-1}h_*v)(f)=\Lambda^{-1}v(h(f))$, and maps v_j to v_{j+1}. Thus it maps $\mathcal{K}_0$ into $\mathcal{K}_0$.

Let $C_2:=\kappa_0C_1\operatorname{ord}_{\mathfrak{n}}(g_0)/m_0$, so $v_j(R(h))=v_j(g_0)/m_0\le C_2$ for any j. Since v_j is centered at $\mathfrak{n}$, and $a(\mu v,\operatorname{Spec}B)=\mu a(v,\operatorname{Spec}B)$ for any real valuation v of $\operatorname{Frac}(B)$ with a center on $\operatorname{Spec}B$ and any real number $\mu>0$ [JM12, Section 4.3, (5.1), (5.2)], (4.13) for $u=v_j$ implies

$$\Lambda a(v_{j+1},\operatorname{Spec}B)=a(v_j,\operatorname{Spec}B)+v_j(R(h)).$$

Since v_0 is divisorial, $a(v_0,\operatorname{Spec}B)<+\infty$. Let $C_3:=a(v_0,\operatorname{Spec}B)+C_2/(\Lambda-1)$. If $a(v_j,\operatorname{Spec}B)\le C_3$, then $a(v_{j+1},\operatorname{Spec}B)\le(C_3+C_2)/\Lambda\le C_3$. Thus $a(v_j,\operatorname{Spec}B)\le C_3$ for any j, and $a(v,\operatorname{Spec}B)\le C_3$ for any $v\in\mathcal{K}_0$ by Lemma 2.4. By Lemma 4.5(1) for $I=\mathfrak{n}$, there exists a real number $\gamma_0>0$ with

$$a(v,\operatorname{Spec}B)\ge\gamma_0v(\mathfrak{n})\ge\gamma_0bv_0(\mathfrak{n})\quad\text{for any } v\in\mathcal{K}_0.$$

By Lemma 2.15 for the map $\mathcal{K}_0\to\mathcal{K}_0$, $v\to\Lambda^{-1}h_*v$, and the point v_0, there exists a Radon probability measure ρ_0 on $\mathcal{K}_0$ with

$$\rho_0(E)=\rho_0(\{v\in\mathcal{K}_0\mid\Lambda^{-1}h_*v\in E\})\quad\text{for any Borel set } E\subset\mathcal{K}_0.$$

Hence $\int\chi(\Lambda^{-1}h_*v)d\rho_0(v)=\int\chi d\rho_0$ for any bounded Borel function χ on $\mathcal{K}_0$. Since $a(\cdot,\operatorname{Spec}B)$ is lower semicontinuous on $\mathcal{K}_0$, it is Borel, and

$$\tau:=\int_{\mathcal{K}_0}a(v,\operatorname{Spec}B)d\rho_0(v)\in[\gamma_0bv_0(\mathfrak{n}),C_3].$$

Let $\mathcal{K}:=\{\tau^{-1}v\mid v\in\mathcal{K}_0\}$, let ρ be the image of ρ_0 under the homeomorphism $\mathcal{K}_0\to\mathcal{K}$, $v\to\tau^{-1}v$, and let

$$\kappa_1:=bv_0(\mathfrak{n})/\tau\quad\text{and}\quad\kappa_2:=\kappa_0C_1/\tau.$$

Then $\mathcal{K}$, κ_1, κ_2, and ρ satisfy (i)–(iii), $a(v,\operatorname{Spec}B)\le C_3/\tau$ for any $v\in\mathcal{K}$, and $\int_{\mathcal{K}}a(v,\operatorname{Spec}B)d\rho(v)=1$. Since $\mathcal{K}\ne\emptyset$, (4.16) for any $f\in\mathfrak{n}\backslash\{0\}$ gives $\kappa_2\ge\kappa_1$. The function $v\to v(R(h))=v(g_0)/m_0$ is continuous and bounded on $\mathcal{K}$. By (4.13),

$$a(\Lambda^{-1}h_*v,\operatorname{Spec}B)=\Lambda^{-1}\big(a(v,\operatorname{Spec}B)+v(R(h))\big)\quad\text{for any } v\in\mathcal{K}.$$

Integrating over $\mathcal{K}$ and using the invariance of ρ, we obtain

$$1=\Lambda^{-1}\Big(1+\int_{\mathcal{K}}v(R(h))d\rho(v)\Big),$$

which is (4.17).

Step 4. We construct $\mathfrak{b}_\bullet$, α, and β. For $f \in B\backslash\{0\}$, let

$$\theta(f) := \int_{\mathcal{K}} v(f) d\rho(v),$$

the integral of a continuous bounded function. By (4.16), the invariance of ρ and (4.17), for any $f, g \in B\backslash\{0\}$,

$$\begin{aligned} \kappa_1 \operatorname{ord}_{\mathfrak{n}}(f) \le \theta(f) \le \kappa_2 \operatorname{ord}_{\mathfrak{n}}(f), \quad \theta(fg) = \theta(f) + \theta(g), \\ \theta(h(f)) = \Lambda\theta(f), \quad \text{and} \quad \theta(g_0) = m_0(\Lambda - 1). \end{aligned} \tag{4.18}$$

For a real number $t > 0$, let $\mathfrak{b}_t^0$ be the ideal of B generated by $\{f \in B\backslash\{0\} \mid \theta(f) \ge t\}$. Let $\mathfrak{b}_t := B$ for $t \le 0$, and

$$\mathfrak{b}_t := \bigcap_{0<t'<t} \mathfrak{b}_{t'}^0 \quad \text{for } t > 0.$$

Let $\alpha := \kappa_2^{-1}$ and $\beta := \kappa_1^{-1}$, so $\beta \ge \alpha > 0$.

For $t > 0$, any non-zero $f \in \mathfrak{n}^{\lceil \beta t \rceil}$ satisfies $\theta(f) \ge \kappa_1 \lceil \beta t \rceil \ge t$, and any $f \in B\backslash\{0\}$ with $\theta(f) \ge t$ satisfies $\kappa_2 \operatorname{ord}_{\mathfrak{n}}(f) \ge t$. Thus

$$\mathfrak{n}^{\lceil \beta t \rceil} \subset \mathfrak{b}_t^0 \subset \mathfrak{n}^{\lceil \alpha t \rceil} \quad \text{for any real number } t > 0.$$

Since $\mathfrak{b}_t^0 \subset \mathfrak{b}_{t'}^0$ for $0 < t' \le t$, and $\lceil \alpha t' \rceil = \lceil \alpha t \rceil$ for any $t' < t$ close to t, we obtain

$$\mathfrak{n}^{\lceil \beta t \rceil} \subset \mathfrak{b}_t \subset \mathfrak{n}^{\lceil \alpha t \rceil} \quad \text{for any real number } t > 0.$$

In particular, $\mathfrak{b}_t$ is $\mathfrak{n}$-primary for $t > 0$. By definition, $\mathfrak{b}_{t'} \subset \mathfrak{b}_t$ for $t \le t'$, and $\mathfrak{b}_t = \bigcap_{t'<t} \mathfrak{b}_{t'}$ for $t > 0$. Since $\theta(fg) = \theta(f) + \theta(g)$,

$$\mathfrak{b}_t^0 \mathfrak{b}_{t'}^0 \subset \mathfrak{b}_{t+t'}^0 \quad \text{for any real numbers } t, t' > 0.$$

For $t, t' > 0$ and $0 < t_0 < t + t'$, write $t_0 = t_1 + t_2$ with $0 < t_1 < t$ and $0 < t_2 < t'$. Then

$$\mathfrak{b}_t \mathfrak{b}_{t'} \subset \mathfrak{b}_{t_1}^0 \mathfrak{b}_{t_2}^0 \subset \mathfrak{b}_{t_0}^0,$$

so $\mathfrak{b}_t \mathfrak{b}_{t'} \subset \mathfrak{b}_{t+t'}$. If $t \le 0$ or $t' \le 0$, then $\mathfrak{b}_t \mathfrak{b}_{t'} \subset \mathfrak{b}_{t+t'}$ since $\mathfrak{b}_{t''} \subset \mathfrak{b}_{t+t'}$ for $t'' \ge t + t'$. Thus $\mathfrak{b}_\bullet$ is an $\mathfrak{n}$-filtration of B with the bounds in the statement.

We show that for any real valuation u of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$,

$$u(\mathfrak{b}_\bullet) = \inf_{f \in \mathfrak{n}\backslash\{0\}} \frac{u(f)}{\theta(f)}. \tag{4.19}$$

Let η be the right-hand side. If $t > 0$ and $\theta(f) \ge t$, then $f \in \mathfrak{n}$ because $\theta = 0$ on the units of B, so $u(f) \ge \eta\theta(f) \ge \eta t$. Thus $u(\mathfrak{b}_t^0) \ge \eta t$, and hence $u(\mathfrak{b}_t) \ge \eta t$ for $t > 0$. Hence $u(\mathfrak{b}_\bullet) \ge \eta$. Conversely, for $f \in \mathfrak{n}\backslash\{0\}$ and $t := \theta(f) > 0$, we have $f \in \mathfrak{b}_t^0 \subset \mathfrak{b}_t$, so

$$u(\mathfrak{b}_\bullet) \le u(\mathfrak{b}_t)/t \le u(f)/\theta(f).$$

Finally, if $t > 0$ and $\theta(f) \ge t$, then $\theta(h(f)) = \Lambda\theta(f) \ge \Lambda t$. Thus $h(\mathfrak{b}_t^0)B \subset \mathfrak{b}_{\Lambda t}^0$, and

$$h(\mathfrak{b}_t)B \subset \bigcap_{0<t'<t} \mathfrak{b}_{\Lambda t'}^0 = \mathfrak{b}_{\Lambda t} \quad \text{for any real number } t > 0. \tag{4.20}$$

Step 5. We prove (2). For any real valuation u of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$, since $\theta(g_0) = m_0(\Lambda - 1) > 0$, we have $g_0 \in \mathfrak{n}$, and (4.19) implies

$$u(\mathfrak{b}_\bullet) \le u(g_0)/\theta(g_0) = u(R(h))/(\Lambda - 1).$$

Thus

$$u(R(h)) \ge (\Lambda - 1) u(\mathfrak{b}_\bullet) \quad \text{for any real valuation } u \text{ of } \operatorname{Frac}(B) \text{ centered at } \mathfrak{n}. \tag{4.21}$$

Let u be a real valuation of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$ with $a(u, \operatorname{Spec} B) < +\infty$. By Lemma 4.6, $\operatorname{vol}_B(u) > 0$.

Let $\mathfrak{c}_t := h(\mathfrak{b}_t)B$ and $\mathfrak{d}_t := \mathfrak{b}_{\Lambda t}$ for $t \in \mathbb{R}$. Then $\mathfrak{d}_\bullet$ is an $\mathfrak{n}$-filtration, and $\mathfrak{c}_t \subset \mathfrak{d}_t$ by (4.20). By Lemma 2.14, $\mathfrak{c}_\bullet$ is an $\mathfrak{n}$-filtration. Since B is analytically irreducible,

$$e(\mathfrak{e}_\bullet) = \lim_{m\to+\infty} e(\mathfrak{e}_m)/m^r \quad \text{for any } \mathfrak{n}\text{-filtration } \mathfrak{e}_\bullet \text{ of } B$$

(Definition 2.8(2)). By Lemma 2.12 and Set-up 4.8(3), $e(h(I)B) = \Lambda^r e(I)$ for any $\mathfrak{n}$-primary ideal I, so by the definition of $e(\cdot)$,

$$e(\mathfrak{c}_\bullet) = \lim_{m\to+\infty} \frac{\Lambda^r e(\mathfrak{b}_m)}{m^r} = \Lambda^r e(\mathfrak{b}_\bullet) = e(\mathfrak{d}_\bullet).$$

By Theorem 1.3, $u(\mathfrak{c}_\bullet) = u(\mathfrak{d}_\bullet)$. Since h_*u is centered at $\mathfrak{n}$ by **Step 1** and $u(h(\mathfrak{b}_t)B) = (h_*u)(\mathfrak{b}_t)$, we have $u(\mathfrak{c}_\bullet) = (h_*u)(\mathfrak{b}_\bullet)$. Also $u(\mathfrak{d}_\bullet) = \Lambda u(\mathfrak{b}_\bullet)$. Thus

$$(h_*u)(\mathfrak{b}_\bullet) = \Lambda u(\mathfrak{b}_\bullet). \tag{4.22}$$

This proves (2).

Step 6. We show that $a(u, \operatorname{Spec} B) \geq u(\mathfrak{b}_\bullet)$ for any real valuation u of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$. In particular, $\operatorname{lct}(B; \mathfrak{b}_\bullet) \geq 1$.

Let u_0 be a real valuation of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$. Since $u_0(\mathfrak{b}_\bullet) \leq u_0(\mathfrak{b}_1) < +\infty$, we may assume that $a(u_0, \operatorname{Spec} B) < +\infty$. For $j \geq 1$, we choose inductively a real valuation u_j of $\operatorname{Frac}(B)$ with $h_*u_j = u_{j-1}$: since $\operatorname{Frac}(B)$ is a finite extension of $h(\operatorname{Frac}(B))$, the valuation $h(f) \to u_{j-1}(f)$ of $h(\operatorname{Frac}(B))$ extends to $\operatorname{Frac}(B)$ [EP05, Theorem 3.1.2], the value group of the extension contains that of u_{j-1} with finite index [Sta, Tag 0ASH], so the extension is a real valuation u_j, and $u_j = 0$ on $\mathbb{C}^*$ because h is a $\mathbb{C}$-algebra homomorphism. By **Step 1**, u_j is centered at $\mathfrak{n}$ and $a(u_j, \operatorname{Spec} B) \leq a(u_{j-1}, \operatorname{Spec} B) < +\infty$. By (4.22), $\Lambda^j u_j(\mathfrak{b}_\bullet) = u_0(\mathfrak{b}_\bullet)$. Since $a(\cdot, \operatorname{Spec} B) \geq 0$ by Lemma 4.5(1), (4.13) and (4.21) imply that for any integer $m \geq 1$,

$$a(u_0, \operatorname{Spec} B) = a(u_m, \operatorname{Spec} B) + \sum_{j=1}^{m} u_j(R(h)) \geq (\Lambda - 1) \sum_{j=1}^{m} \Lambda^{-j} u_0(\mathfrak{b}_\bullet) = (1 - \Lambda^{-m}) u_0(\mathfrak{b}_\bullet).$$

Letting $m \to +\infty$, we obtain $a(u_0, \operatorname{Spec} B) \geq u_0(\mathfrak{b}_\bullet)$. By Definition 2.8(3), $\operatorname{lct}(B; \mathfrak{b}_\bullet) \geq 1$.

Step 7. We show that $\operatorname{lct}(B; \mathfrak{b}_\bullet) \leq 1$, which together with **Step 6** proves (1).

By Lemma 2.9 and the bounds in the statement with $t = m$, $(\mathfrak{b}_m)_{m\geq 1}$ is a graded sequence with

$$\operatorname{lct}(B; (\mathfrak{b}_m)_{m\geq 1}) = \operatorname{lct}(B; \mathfrak{b}_\bullet) \quad \text{and} \quad \mathfrak{n}^{\lceil \beta m \rceil} \subset \mathfrak{b}_m \subset \mathfrak{n}^{\lceil \alpha m \rceil} \quad \text{for any integer } m \geq 1,$$

so Lemma 4.5(2) implies

$$\operatorname{lct}(B; \mathfrak{b}_\bullet) = \sup_{m\geq 1} m \operatorname{lct}(B; \mathfrak{b}_m). \tag{4.23}$$

Fix an integer $m \geq 1$ and a real number $0 < \epsilon < 1$, and let $t := (1-\epsilon)m$. Then $\mathfrak{b}_m \subset \mathfrak{b}_t^0$, and there exist $f_1, \dots, f_n \in B \setminus \{0\}$ generating $\mathfrak{b}_t^0$ with $\theta(f_j) \geq t$ for any j. Since ρ is a probability measure on $\mathcal{K}$, the point

$$\big(\theta(f_1), \dots, \theta(f_n), 1\big) = \int_{\mathcal{K}} \big(v(f_1), \dots, v(f_n), a(v, \operatorname{Spec} B)\big) d\rho(v)$$

belongs to the closed convex hull of the bounded set $\{(v(f_1), \dots, v(f_n), a(v, \operatorname{Spec} B)) \mid v \in \mathcal{K}\}$. Thus for any real number $0 < \delta < t$, there exist $w_1, \dots, w_N \in \mathcal{K}$ and real numbers $p_1, \dots, p_N > 0$ with $\sum_{i=1}^N p_i = 1$,

$$\sum_{i=1}^{N} p_i w_i(f_j) \geq t - \delta \quad \text{for } 1 \leq j \leq n, \quad \text{and} \quad \sum_{i=1}^{N} p_i a(w_i, \operatorname{Spec} B) \leq 1 + \delta.$$

By (4.16),

$$\mathfrak{n}^{\lceil \beta t' \rceil} \subset \mathfrak{a}_{t'}(w_i) \subset \mathfrak{n}^{\lceil \alpha t' \rceil} \quad \text{for any } i \text{ and any real number } t' > 0,$$

and, by Definition 2.8(2), $\mathfrak{a}_\bullet(w_i)$ is an $\mathfrak{n}$-filtration with $w_i(\mathfrak{a}_\bullet(w_i)) = 1$. Hence $\operatorname{lct}(B; \mathfrak{a}_\bullet(w_i)) \le a(w_i, \operatorname{Spec} B)$. Let $\mathfrak{g}_\bullet^{(1)} := \mathfrak{a}_\bullet(w_1)$, and for $2 \le i \le N$ let $\mathfrak{g}_\bullet^{(i)}$ be the geodesic between $\mathfrak{g}_\bullet^{(i-1)}$ and $\mathfrak{a}_\bullet(w_i)$ at $\sigma_i := p_i/(p_1 + \cdots + p_i)$. For $i = 1$, (a) and (c) hold by (4.16) and the definition of $\mathfrak{a}_\bullet(w_1)$, and (b) is $\operatorname{lct}(B; \mathfrak{a}_\bullet(w_1)) \le a(w_1, \operatorname{Spec} B)$. For $2 \le i \le N$, assuming (a)–(c) for $i-1$:

(a) For any real number $t' > 0$,

$$\mathfrak{n}^{\lceil \beta t' \rceil} \subset \mathfrak{g}_{t'}^{(i)} \subset \mathfrak{n}^{\lceil \alpha t' \rceil},$$

since the term $\mathfrak{g}_{t'}^{(i-1)} \cap \mathfrak{a}_{t'}(w_i)$ contains $\mathfrak{n}^{\lceil \beta t' \rceil}$, and for any term $\mathfrak{g}_{t_0}^{(i-1)} \cap \mathfrak{a}_{t_1}(w_i)$ with $(1-\sigma_i)t_0 + \sigma_i t_1 = t'$,

$$\mathfrak{g}_{t_0}^{(i-1)} \cap \mathfrak{a}_{t_1}(w_i) \subset \mathfrak{n}^{\lceil \alpha \max\{t_0, t_1\} \rceil} \subset \mathfrak{n}^{\lceil \alpha t' \rceil}.$$

(b) By (a) for $i-1$ and Lemma 4.7,

$$\operatorname{lct}(B; \mathfrak{g}_\bullet^{(i)}) \le \frac{\sum_{i'=1}^{i} p_{i'} a(w_{i'}, \operatorname{Spec} B)}{p_1 + \cdots + p_i}.$$

(c) For any $f \in B \backslash \{0\}$, let

$$t'_f := \frac{\sum_{i'=1}^{i} p_{i'} w_{i'}(f)}{p_1 + \cdots + p_i} \quad \text{and} \quad t_f := \frac{\sum_{i'=1}^{i-1} p_{i'} w_{i'}(f)}{p_1 + \cdots + p_{i-1}}.$$

Then $f \in \mathfrak{g}_{t'_f}^{(i)}$, since $f \in \mathfrak{g}_{t_f}^{(i-1)} \cap \mathfrak{a}_{w_i(f)}(w_i)$ and $(1-\sigma_i)t_f + \sigma_i w_i(f) = t'_f$.

Let $\mathfrak{g}_\bullet := \mathfrak{g}_\bullet^{(N)}$. By (c), $f_j \in \mathfrak{g}_{t-\delta}$ for any j, so $\mathfrak{b}_t^0 \subset \mathfrak{g}_{t-\delta}$, and hence $\mathfrak{b}_m \subset \mathfrak{g}_{t-\delta}$. By (b), $\operatorname{lct}(B; \mathfrak{g}_\bullet) \le 1 + \delta$. For any real valuation v of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$,

$$(t-\delta)v(\mathfrak{g}_\bullet) \le v(\mathfrak{g}_{t-\delta}) \le v(\mathfrak{b}_m).$$

Thus

$$m \operatorname{lct}(B; \mathfrak{b}_m) \le \frac{m \operatorname{lct}(B; \mathfrak{g}_\bullet)}{t-\delta} \le \frac{m(1+\delta)}{(1-\epsilon)m - \delta}.$$

Letting $\delta \to 0$ and then $\epsilon \to 0$, we obtain $m \operatorname{lct}(B; \mathfrak{b}_m) \le 1$ for any m, so $\operatorname{lct}(B; \mathfrak{b}_\bullet) \le 1$ by (4.23).

Step 8. We prove (3). By (1), Lemma 4.5(2) for $(\mathfrak{b}_m)_{m \ge 1}$, and Lemma 2.9, there exists a real valuation v of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$ with $a(v, \operatorname{Spec} B) = 1$ and $v(\mathfrak{b}_\bullet) = 1$, so $\mathcal{W} \ne \emptyset$.

For $f \in \mathfrak{n} \backslash \{0\}$, Lemma 4.5(1) for $I = fB$ implies that there exists a real number $\gamma_f > 0$ with $a(v, \operatorname{Spec} B) \ge \gamma_f v(f)$ for any real valuation v of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$. Let $\omega(f) := 1/\gamma_f$ for $f \in \mathfrak{n} \backslash \{0\}$ and $\omega(f) := 0$ for the units f of B. Let $v \in \mathcal{W}$. Since $v(\mathfrak{b}_t) \ge tv(\mathfrak{b}_\bullet) = t$ and $\mathfrak{n}^{\lceil \beta t \rceil} \subset \mathfrak{b}_t$, we have $\lceil \beta t \rceil v(\mathfrak{n}) \ge t$ for any $t > 0$. Taking $t \in \beta^{-1}\mathbb{N}^+$, we obtain $v(\mathfrak{n}) \ge \beta^{-1}$, and $v \ge 0$ on $B \backslash \{0\}$ since v is centered at $\mathfrak{n}$. Also $v(f) \le a(v, \operatorname{Spec} B)/\gamma_f = \omega(f)$ for $f \in \mathfrak{n} \backslash \{0\}$. Thus $\mathcal{W} \subset \mathcal{V}(\beta^{-1}, \omega)$, which is compact by Definition-Lemma 4.4. By **Step 6**,

$$\mathcal{W} = \big\{ v \in \mathcal{V}(\beta^{-1}, \omega) \;\big|\; a(v, \operatorname{Spec} B) \le 1 \text{ and } v(\mathfrak{b}_\bullet) \ge 1 \big\}.$$

The function $a(\cdot, \operatorname{Spec} B)$ is lower semicontinuous by Lemma 2.4. For any $t > 0$, $v \to v(\mathfrak{b}_t)$ is the minimum of v over finitely many generators of $\mathfrak{b}_t$, hence continuous. Since $v(\mathfrak{b}_\bullet) = \inf_{t>0} v(\mathfrak{b}_t)/t$, the function $v \to v(\mathfrak{b}_\bullet)$ is upper semicontinuous. Thus $\mathcal{W}$ is closed in $\mathcal{V}(\beta^{-1}, \omega)$, hence compact.

Let u be a real valuation of $\operatorname{Frac}(B)$ with $h_* u \in \mathcal{W}$. By **Step 1**, u is centered at $\mathfrak{n}$, and $a(u, \operatorname{Spec} B) < +\infty$ by (4.14). By (4.22), $u(\mathfrak{b}_\bullet) = \Lambda^{-1}$. By (4.13) and (4.21),

$$1 = a(h_* u, \operatorname{Spec} B) = a(u, \operatorname{Spec} B) + u(R(h)) \ge a(u, \operatorname{Spec} B) + (\Lambda - 1)\Lambda^{-1},$$

so

$$a(\Lambda u, \operatorname{Spec} B) = \Lambda a(u, \operatorname{Spec} B) \le 1 = (\Lambda u)(\mathfrak{b}_\bullet).$$

By **Step 6**, $a(\Lambda u, \operatorname{Spec} B) \ge (\Lambda u)(\mathfrak{b}_\bullet)$. Thus $a(\Lambda u, \operatorname{Spec} B) = (\Lambda u)(\mathfrak{b}_\bullet) = 1$, and $\Lambda u \in \mathcal{W}$ since Λu is centered at $\mathfrak{n}$. This proves (3). □

5. Quasi-monomial valuations computing thresholds

In this section we prove Theorem 5.4.

5.1. Complements over the strict henselization.

Lemma 5.1. *Let $(B, \mathfrak{n})$ be a normal local domain essentially of finite type over $\mathbb{C}\{z_1, \dots, z_n\}$ such that K_B is $\mathbb{Q}$-Cartier and $\operatorname{Spec} B$ is klt, let B' be a strict henselization of B, and let $\mathfrak{n}' := \mathfrak{n}B'$. Then the following hold.*

(1) B' is a strictly henselian normal excellent local domain of characteristic zero with maximal ideal $\mathfrak{n}'$, and $\dim B' = \dim B$. Moreover, B' admits a dualizing complex, $K_{B'}$ is $\mathbb{Q}$-Cartier, and $\operatorname{Spec} B'$ is klt. Finally, B' contains a coefficient field, which is algebraically closed.

(2) For any $\mathfrak{n}$-primary ideal $\mathfrak{a}$ of B,

$$\operatorname{lct}(B; \mathfrak{a}) = \operatorname{lct}(B'; \mathfrak{a}B').$$

(3) Let w be a quasi-monomial valuation of $\operatorname{Frac}(B')$ centered at $\mathfrak{n}'$, and let v be the restriction of w to $\operatorname{Frac}(B)$. Then v is a quasi-monomial valuation of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$, $v(\mathfrak{a}) = w(\mathfrak{a}B')$ for any non-zero ideal $\mathfrak{a}$ of B, and

$$a(v, \operatorname{Spec} B) \leq a(w, \operatorname{Spec} B').$$

Proof. Since B is a localization of a finitely generated $\mathbb{C}\{z_1, \dots, z_n\}$-algebra, there exist a non-negative integer m, a prime ideal $\mathfrak{p}$ of $\mathbb{C}\{z_1, \dots, z_n\}[x_1, \dots, x_m]$, and a prime ideal $\mathfrak{q}_B$ of

$$S := (\mathbb{C}\{z_1, \dots, z_n\}[x_1, \dots, x_m])_{\mathfrak{p}}$$

such that $B = S/\mathfrak{q}_B$. The ring S is an excellent regular local ring with residue field $k := B/\mathfrak{n}$. Let k' be the residue field of B', which is a separable closure of k, and let S^{sh} be the strict henselization of S with respect to k'. Since B is a finite S-algebra and $S^{\mathrm{sh}}/\mathfrak{q}_B S^{\mathrm{sh}}$ is a local ring,

$$B' = S^{\mathrm{sh}}/\mathfrak{q}_B S^{\mathrm{sh}}$$

by [Gro67, Proposition 18.8.10]. By [Gro67, Proposition 18.8.8 (iii), (iv)], the homomorphisms $S \to S^{\mathrm{sh}}$ and $B \to B'$ are faithfully flat, $\mathfrak{n}'$ is the maximal ideal of B', and S^{sh} and B' are Noetherian. Since $\operatorname{char} k = 0$, k' is an algebraic closure of k. Since $B \to B'$ is faithfully flat and $\mathfrak{n}B'$ is the maximal ideal of B', $\dim B' = \dim B$. By [Gro67, Proposition 18.8.12 (i)], B' is normal, hence a domain, and by [Gro67, Corollaire 18.8.13], S^{sh} is regular. Since B is excellent, B' is a G-ring [Sta, Tag 07QR] and universally catenary [Gro67, Proposition 18.8.17], hence excellent [Gro65, Scholie 7.8.3 (i)].

Dualizing complex and canonical divisor. Since S^{sh} is a regular local ring, S^{sh} is a dualizing complex of itself, so

$$\mathbf{R}\operatorname{Hom}_{S^{\mathrm{sh}}}(B', S^{\mathrm{sh}})$$

is a dualizing complex of B'. Let $c := \dim S - \dim B = \dim S^{\mathrm{sh}} - \dim B'$. The canonical modules of B and B' are

$$\omega_B = \operatorname{Ext}^c_S(B, S) \quad \text{and} \quad \omega_{B'} = \operatorname{Ext}^c_{S^{\mathrm{sh}}}(B', S^{\mathrm{sh}}).$$

Since B has a finite free resolution over the regular local ring S and $S \to S^{\mathrm{sh}}$ is flat, $\omega_{B'} \cong \omega_B \otimes_B B'$. Let N be a positive integer such that NK_B is Cartier, that is, $(\omega_B^{\otimes N})^{**} \cong B$. Since tensor products and duals of finitely presented modules commute with flat base change, $(\omega_{B'}^{\otimes N})^{**} \cong B'$, so $NK_{B'}$ is Cartier.

Base change of log resolutions. Let $\pi : Y \to \operatorname{Spec} B$ be a log resolution and Γ a reduced snc divisor on Y such that π is an isomorphism over $\operatorname{Spec} B \setminus \pi(\operatorname{Supp} \Gamma)$ and every π-exceptional prime divisor is a component of Γ (such π and Γ exist by [Tem18, Theorem 1.1.6]), and write $K_Y + \Delta_Y = \pi^* K_B$ with $\pi_* K_Y = K_B$. Let

$$\widetilde{Y} := Y \times_{\operatorname{Spec} B} \operatorname{Spec} B',$$

let $\widetilde{\pi} : \widetilde{Y} \to \operatorname{Spec} B'$ be the projection, and let $\widetilde{\Gamma}$ and $\Delta_{\widetilde{Y}}$ be the pullbacks of Γ and Δ_Y to $\widetilde{Y}$. Since B' is a filtered colimit of localizations of étale B-algebras, $\widetilde{Y} \to Y$ is a filtered limit of étale morphisms. Thus $\widetilde{Y}$ is regular, and for any point $\widetilde{y} \in \widetilde{Y}$ with image $y \in Y$, local equations at y of the components of Γ through y form part of a regular system of parameters of $\mathcal{O}_{\widetilde{Y},\widetilde{y}}$, and each of them defines exactly one component of $\widetilde{\Gamma}$ through $\widetilde{y}$. In particular, $\widetilde{\Gamma}$ is a reduced snc divisor and $\widetilde{\pi}$ is an isomorphism over $\operatorname{Spec} B' \backslash \widetilde{\pi}(\operatorname{Supp} \widetilde{\Gamma})$. The morphism $\widetilde{Y} \to Y$ is flat, and the fiber over the generic point of Y is the spectrum of a localization of B'. Hence $\widetilde{Y}$ is integral and $\widetilde{\pi}$ is birational. Since the dualizing complexes of $\operatorname{Spec} B$ and Y pull back to those of $\operatorname{Spec} B'$ and $\widetilde{Y}$ (for Y see [Sta, Tag 0AA8]), we have $K_{\widetilde{Y}} + \Delta_{\widetilde{Y}} = \widetilde{\pi}^* K_{B'}$. Therefore, if $\widetilde{\Gamma}_j$ is a component of $\widetilde{\Gamma}$ and Γ_i is the component of Γ whose generic point is the image of the generic point of $\widetilde{\Gamma}_j$, then

$$\operatorname{mult}_{\widetilde{\Gamma}_j} \Delta_{\widetilde{Y}} = \operatorname{mult}_{\Gamma_i} \Delta_Y \quad \text{and} \quad a(\widetilde{\Gamma}_j, \operatorname{Spec} B') = a(\Gamma_i, \operatorname{Spec} B). \tag{5.1}$$

Since $\operatorname{Spec} B$ is klt, the coefficients of $\Delta_{\widetilde{Y}}$ are less than 1, so $\operatorname{Spec} B'$ is klt by [JM12, Proposition 5.1, (5.1) and Corollary 5.4].

Coefficient field. Since B' is henselian, contains $\mathbb{Q}$, and has the algebraically closed residue field k', Hensel's lemma extends a lift of a transcendence basis of k' over $\mathbb{C}$ to a coefficient field of B', which is isomorphic to k' and hence algebraically closed. This proves (1).

Proof of (2). Let $\mathfrak{a}$ be an $\mathfrak{n}$-primary ideal of B. Let $(\pi : Y \to \operatorname{Spec} B, \Gamma)$ be as in the paragraph **Base change of log resolutions** with moreover $\mathfrak{a}\mathcal{O}_Y = \mathcal{O}_Y(-F)$ for a Cartier divisor F with $\operatorname{Supp} F \subset \operatorname{Supp} \Gamma$ [Tem18, Theorem 1.1.6]. Let u be a real valuation of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$, and let $u' := r_{Y,\Gamma}(u)$ (Definition-Lemma 2.7). Then $u'(\mathfrak{a}) = u(\mathfrak{a})$ [JM12, Corollary 4.8], and by [JM12, (5.1) and Corollary 5.4],

$$a(u, \operatorname{Spec} B) \geq a(u', \operatorname{Spec} B) = \sum_i u(\Gamma_i) a(\Gamma_i, \operatorname{Spec} B), \qquad u(\mathfrak{a}) = \sum_i u(\Gamma_i) \operatorname{mult}_{\Gamma_i} F,$$

where Γ_i runs over the components of Γ. Since $a(\Gamma_i, \operatorname{Spec} B) > 0$ for any i,

$$\operatorname{lct}(B; \mathfrak{a}) = \min \left\{ \frac{a(\Gamma_i, \operatorname{Spec} B)}{\operatorname{mult}_{\Gamma_i} F} \;\middle|\; \operatorname{mult}_{\Gamma_i} F > 0 \right\}. \tag{5.2}$$

Let $\widetilde{Y}$, $\widetilde{\pi}$, and $\widetilde{\Gamma}$ be the base change of (Y, Γ) as in the paragraph **Base change of log resolutions**. Let $\widetilde{F}$ be the pullback of F to $\widetilde{Y}$. Then

$$\mathfrak{a}B'\mathcal{O}_{\widetilde{Y}} = \mathcal{O}_{\widetilde{Y}}(-\widetilde{F}),$$

$\operatorname{Supp} \widetilde{F} \subset \operatorname{Supp} \widetilde{\Gamma}$, and $a(\widetilde{\Gamma}_j, \operatorname{Spec} B') > 0$ for any j by (5.1), so, possibly replacing (Y, Γ, F) and B with $(\widetilde{Y}, \widetilde{\Gamma}, \widetilde{F})$ and B', the argument giving (5.2) [JM12, Lemma 4.7, (5.1) and Corollary 5.4] gives

$$\operatorname{lct}(B'; \mathfrak{a}B') = \min \left\{ \frac{a(\widetilde{\Gamma}_j, \operatorname{Spec} B')}{\operatorname{mult}_{\widetilde{\Gamma}_j} \widetilde{F}} \;\middle|\; \operatorname{mult}_{\widetilde{\Gamma}_j} \widetilde{F} > 0 \right\}.$$

If $\widetilde{\Gamma}_j$ and Γ_i are as in (5.1), then $\operatorname{mult}_{\widetilde{\Gamma}_j} \widetilde{F} = \operatorname{mult}_{\Gamma_i} F$, and since $\widetilde{Y} \to Y$ is surjective, any Γ_i arises from some $\widetilde{\Gamma}_j$. By (5.1),

$$\left\{ \frac{a(\widetilde{\Gamma}_j, \operatorname{Spec} B')}{\operatorname{mult}_{\widetilde{\Gamma}_j} \widetilde{F}} \;\middle|\; \operatorname{mult}_{\widetilde{\Gamma}_j} \widetilde{F} > 0 \right\} = \left\{ \frac{a(\Gamma_i, \operatorname{Spec} B)}{\operatorname{mult}_{\Gamma_i} F} \;\middle|\; \operatorname{mult}_{\Gamma_i} F > 0 \right\},$$

so $\operatorname{lct}(B'; \mathfrak{a}B') = \operatorname{lct}(B; \mathfrak{a})$ by (5.2). This proves (2).

Proof of (3). Since $B \to B'$ is local and faithfully flat, $\mathfrak{n}' \cap B = \mathfrak{n}$. Since $w \geq 0$ on $B'\backslash\{0\}$ and $w > 0$ exactly on $\mathfrak{n}'\backslash\{0\}$, v is centered at $\mathfrak{n}$. If $g_1, \dots, g_\ell$ generate a non-zero ideal $\mathfrak{a}$ of B, then they generate $\mathfrak{a}B'$, so

$$w(\mathfrak{a}B') = \min_i w(g_i) = \min_i v(g_i) = v(\mathfrak{a}).$$

We show that v is quasi-monomial. For a Noetherian local domain $(A, \mathfrak{m}_A)$ which is excellent of characteristic zero and a real valuation u of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$, let $Y \to \operatorname{Spec} A$ be a proper birational morphism with Y regular [Tem18, Theorem 1.1.6] and ξ the center of u on Y. By the criterion for quasi-monomial valuations in Subsection 2.2 and the dimension formula

$$\dim \mathcal{O}_{Y,\xi} + \mathrm{trdeg}(\kappa(\xi)/\kappa(\mathfrak{m}_A)) = \dim A,$$

which holds since A is excellent, hence universally catenary [Mat89, Theorem 15.6],

$$u \text{ is quasi-monomial} \iff \mathrm{ratrk}(u) + \mathrm{trdeg}(\kappa(u)/\kappa(\mathfrak{m}_A)) = \dim A. \tag{5.3}$$

Since B' is a filtered colimit of localizations of étale B-algebras, $\mathrm{Frac}(B')$ is algebraic over $\mathrm{Frac}(B)$, so [Sta, Tag 0ASH] for the finite extensions $\mathrm{Frac}(B)(y)$, $y \in \mathrm{Frac}(B')$, implies that $w(\mathrm{Frac}(B')^*)/v(\mathrm{Frac}(B)^*)$ is a torsion group and $\kappa(w)$ is algebraic over $\kappa(v)$. Since k' is algebraic over k and $\kappa(v) \subset \kappa(w)$,

$$\mathrm{ratrk}(v) = \mathrm{ratrk}(w) \quad \text{and} \quad \mathrm{trdeg}(\kappa(v)/k) = \mathrm{trdeg}(\kappa(w)/k) = \mathrm{trdeg}(\kappa(w)/k').$$

Since w is quasi-monomial and $\dim B = \dim B'$, (5.3) implies that v is quasi-monomial.

It remains to prove the inequality. For a real valuation u of $\mathrm{Frac}(B)$ centered at $\mathfrak{n}$, by [JM12, (5.1), (5.2), Lemma 4.7 and Remark 5.6],

$$a(u, \operatorname{Spec} B) = \sup_{(Y,\Gamma)} \sum_i u(\Gamma_i) a(\Gamma_i, \operatorname{Spec} B),$$

where (Y, Γ) runs over the pairs as in the paragraph **Base change of log resolutions** and Γ_i runs over the components of Γ. Let (Y, Γ) be such a pair, and let $(\widetilde{Y}, \widetilde{\Gamma})$ be the base change of (Y, Γ). The center $\widetilde{y}$ of w on $\widetilde{Y}$ maps to the center y of v on Y, and the components $\widetilde{\Gamma}_j$ of $\widetilde{\Gamma}$ through $\widetilde{y}$ correspond bijectively to the components Γ_i of Γ through y, and the pullback of a local equation of Γ_i at y is a local equation at $\widetilde{y}$ of the corresponding $\widetilde{\Gamma}_j$. Thus $w(\widetilde{\Gamma}_j) = v(\Gamma_i)$ for corresponding i and j, and by (5.1) and [JM12, Corollary 5.4],

$$\sum_i v(\Gamma_i) a(\Gamma_i, \operatorname{Spec} B) = \sum_j w(\widetilde{\Gamma}_j) a(\widetilde{\Gamma}_j, \operatorname{Spec} B') \leq a(w, \operatorname{Spec} B').$$

Taking the supremum over (Y, Γ), we obtain $a(v, \operatorname{Spec} B) \leq a(w, \operatorname{Spec} B')$. The lemma follows. □

Lemma 5.2. *Let $(B', \mathfrak{n}')$ be a strictly henselian normal excellent local domain of characteristic zero admitting a dualizing complex, and let $k' \subset B'$ be a coefficient field. Assume that $\dim B' \geq 2$, $K_{B'}$ is $\mathbb{Q}$-Cartier, and $\operatorname{Spec} B'$ is klt. Then there exists a positive integer N with the following property: for any real number $\epsilon > 0$, there exists a k'-vector subspace $V_\epsilon \subset B'$ with $\dim_{k'} V_\epsilon < +\infty$ such that for any $\mathfrak{n}'$-primary ideal $\mathfrak{a}$ of B', there exists a divisorial valuation v of $\mathrm{Frac}(B')$ centered at $\mathfrak{n}'$ satisfying (1) and (2) below.*

(1) $a(v, \operatorname{Spec} B') = 1$ and $v(\mathfrak{a}) = 1/\operatorname{lct}(B'; \mathfrak{a})$.

(2) If $v(\mathfrak{n}') \geq \epsilon$, then there exists a non-zero element $f \in V_\epsilon$ such that $(\operatorname{Spec} B', N^{-1} \operatorname{div}(f))$ is lc and v is a positive multiple of the order of vanishing along an lc place of $(\operatorname{Spec} B', N^{-1} \operatorname{div}(f))$.

Proof. The proof follows [Xu20, Proof of Proposition 3.5]. Let $X := \operatorname{Spec} B'$ and $d := \dim B' \geq 2$. Let r be a positive integer such that rK_X is Cartier. Since B' is local, $rK_X = \operatorname{div}(\kappa)$ for some $\kappa \in \mathrm{Frac}(B')^*$. Since B' is strictly henselian of characteristic zero, the residue field $B'/\mathfrak{n}'$, which is isomorphic to k' via the coefficient field, is algebraically closed. Let N be the positive integer of [Bir19, Theorem 1.7] for dimension $d-1$ and the finite set $\{1\}$. Possibly replacing N with rN, we may assume that $r \mid N$. Then N depends only on d and r, hence only on B'. For a real

number $\epsilon > 0$, let J be a positive integer with $J\epsilon > N$, and let $V_\epsilon \subset B'$ be a k'-vector subspace mapping isomorphically onto $B'/\mathfrak{n}'^J$. Since k' is a coefficient field and $B'/\mathfrak{n}'^J$ has finite length, $\dim_{k'} V_\epsilon < +\infty$.

We use the following consequence of Definition-Lemma 2.7. For any separated regular integral excellent scheme W of characteristic zero admitting a dualizing complex and any $\mathbb{Q}$-divisor $\Delta = \sum_i b_i \Delta_i$ on W, where the Δ_i are distinct prime divisors, $\operatorname{Supp}\Delta$ is snc, and $b_i \le 1$ for any i, the following hold. For any prime divisor Q over W, let $\operatorname{mult}_Q \Delta_i$ be the coefficient of Q in the pullback of the Cartier divisor Δ_i to a birational model of W on which Q is a divisor. Then

$$a(Q, W, \Delta) \ge \sum_i \operatorname{mult}_Q \Delta_i (1 - b_i) \ge 0, \tag{5.4}$$

since $a(Q, W, \Delta) = a(Q, W) - \sum_i b_i \operatorname{mult}_Q \Delta_i$, equality holds in the first inequality if and only if ord_Q is monomial in local equations of the components of $\operatorname{Supp}\Delta$ through the center of Q, and $a(Q, W, \Delta) = a(Q, W) > 0$ if $\operatorname{mult}_Q \Delta_i = 0$ for any i. Hence:

(S1) If $b_i < 1$ for any i, then $a(Q, W, \Delta) > 0$ for any Q.
(S2) If $b_{i_0} = 1$ for exactly one i_0, then Δ_{i_0} is the only prime divisor Q over W with $a(Q, W, \Delta) = 0$.

Step 1. In this step we construct v and a plt extraction $g : Z \to X$ of the corresponding prime divisor. Here a pair (T, Δ_T) as in Definition 2.1 is *plt* if $a(Q, T, \Delta_T) > 0$ for any prime divisor Q exceptional over T. Let $\mathfrak{a}$ be an $\mathfrak{n}'$-primary ideal of B' and $t := \operatorname{lct}(B'; \mathfrak{a})$. Since X is klt and $\mathfrak{a} \ne B'$, $0 < t < +\infty$.

Let $Y_1 \to X$ be the blow-up of $\mathfrak{a}\mathfrak{n}'$. By [Tem18, Theorem 1.1.6], there exists a sequence of blow-ups $Y = Y_m \to Y_{m-1} \to \cdots \to Y_1$ with regular centers such that Y is regular, the inverse image in Y of the Cartier divisor defined by $\mathfrak{a}\mathfrak{n}'\mathcal{O}_{Y_1}$ together with the exceptional loci is an snc divisor, and any center is disjoint from the inverse image of the locus where $(Y_1, V(\mathfrak{a}\mathfrak{n}'\mathcal{O}_{Y_1}))$ is semi-regular [Tem18, §2.1.12], which contains the regular locus of $Y_1 \backslash V(\mathfrak{a}\mathfrak{n}'\mathcal{O}_{Y_1})$. Let $\pi : Y \to X$ be the composite, let $Y_0 := X$, let $C_1 := V(\mathfrak{a}\mathfrak{n}') \subset X$, and for $2 \le k \le m$ let $C_k \subset Y_{k-1}$ be the center of $Y_k \to Y_{k-1}$. Thus $C_1 = \{\mathfrak{n}'\}$ set-theoretically. Let

$$\Sigma := \{\mathfrak{n}'\} \cup \operatorname{Sing} X,$$

a closed subset with $\operatorname{codim}_X \Sigma \ge 2$, since X is normal and $d \ge 2$. Since $Y_1 \to X$ is an isomorphism over $X \backslash \{\mathfrak{n}'\}$ and $V(\mathfrak{a}\mathfrak{n}'\mathcal{O}_{Y_1})$ maps to $\{\mathfrak{n}'\}$, the image of any C_k in X is contained in Σ. Since Y is regular and $Y \to X$ is an isomorphism near any point of Y whose image in Y_{k-1} is not in C_k for any k, any point of Y over $\operatorname{Sing} X$ maps into some C_k. Thus π is an isomorphism over $X \backslash \Sigma$, and $\Gamma := \pi^{-1}(\Sigma)$ with the reduced structure is an snc divisor whose components E_i are exactly the exceptional prime divisors of π. Moreover, $\mathfrak{a}\mathcal{O}_Y = \mathcal{O}_Y(-F)$ and $\mathfrak{n}'\mathcal{O}_Y = \mathcal{O}_Y(-F_0)$ for Cartier divisors F and F_0 with

$$\operatorname{Supp} F \subset \operatorname{Supp} F_0 = \pi^{-1}(\mathfrak{n}') \subset \operatorname{Supp} \Gamma.$$

For $1 \le k \le m$, let D_k be the pullback to Y of the Cartier divisor $C_k \times_{Y_{k-1}} Y_k$. Then $-C_k \times_{Y_{k-1}} Y_k$ is ample over Y_{k-1}, and $\operatorname{Supp} \sum_k D_k = \operatorname{Supp} \Gamma$. Hence for rational numbers $c_1 \gg c_2 \gg \cdots \gg c_m > 0$, the $\mathbb{Q}$-divisor $P := -\sum_k c_k D_k$ is π-ample, and

$$-P = \sum_i \alpha_i E_i \quad \text{with } \alpha_i \in \mathbb{Q}_{>0} \text{ for any } i.$$

We use the following Bertini property (B): if $\Gamma' \ge \Gamma$ is a reduced snc divisor on Y, $\mathcal{L}$ is an invertible sheaf on Y, and $V \subset H^0(Y, \mathcal{L})$ is a k'-vector subspace with $\dim_{k'} V < +\infty$ generating $\mathcal{L}$, then $\Gamma' + \sum_{j=1}^n (\sigma_j = 0)$ is reduced and snc for general $\sigma_1, \dots, \sigma_n \in V$, by Bertini's theorem on the strata of Γ' in $\pi^{-1}(\mathfrak{n}')$, since $C' \cap \pi^{-1}(\mathfrak{n}') \ne \emptyset$ for any non-empty closed subset $C' \subset Y$.

Let $V_\mathfrak{a} \subset \mathfrak{a}$ be the k'-span of a finite set of generators of $\mathfrak{a}$, and let $n > t$ be an integer. By (B) for $\Gamma' = \Gamma$ and $\mathcal{L} = \mathcal{O}_Y(-F)$, there exist $h_1, \dots, h_n \in V_\mathfrak{a}$ such that, letting $G_j := \pi^* \operatorname{div}(h_j) - F \ge 0$,

the divisor $\Gamma + \sum_j G_j$ is reduced and snc. Let

$$H := \frac{t}{n} \sum_{j=1}^{n} \operatorname{div}(h_j).$$

Then $K_Y + H_Y = \pi^*(K_X + H)$, where

$$H_Y = \sum_i \big(1 - a(E_i, X) + t \operatorname{mult}_{E_i} F\big) E_i + \frac{t}{n} \sum_{j=1}^{n} G_j.$$

Since $\pi^* \operatorname{div}(h_j) = F + G_j$, we have $\pi^* H = tF + \frac{t}{n} \sum_j G_j$. By [JM12, (5.1) and Corollary 5.4], t is the minimum of $a(E_i, X)/\operatorname{mult}_{E_i} F$ over the i with $\operatorname{mult}_{E_i} F > 0$. Hence

$$\mathcal{I} := \{i \mid a(E_i, X) = t \operatorname{mult}_{E_i} F\} \neq \emptyset,$$

and

$$\operatorname{mult}_{E_i}(\pi^* H) = t \operatorname{mult}_{E_i} F = a(E_i, X) > 0 \quad \text{for any } i \in \mathcal{I}.$$

Since π-ampleness of $-\sum_i \alpha_i E_i$ is an open condition on $(\alpha_i)_i$, possibly perturbing the α_i and replacing P with $-\sum_i \alpha_i E_i$, we may assume that the rational numbers $\operatorname{mult}_{E_i}(\pi^* H)/\alpha_i$, $i \in \mathcal{I}$, are pairwise distinct. Let

$$\lambda := \min_{i \in \mathcal{I}} \frac{\operatorname{mult}_{E_i}(\pi^* H)}{\alpha_i},$$

and let E be the unique E_i with $i \in \mathcal{I}$ and $\operatorname{mult}_{E_i}(\pi^* H) = \lambda \alpha_i$. Let δ be a rational number with $0 < \delta < 1$ and

$$\delta\big(\lambda \alpha_i - \operatorname{mult}_{E_i}(\pi^* H)\big) < a(E_i, X) - t \operatorname{mult}_{E_i} F \quad \text{for any } i \notin \mathcal{I},$$

and let $q \geq 2$ be an integer such that $q\delta\lambda P$ is Cartier and $\mathcal{O}_Y(q\delta\lambda P)$ is generated by $H^0(Y, \mathcal{O}_Y(q\delta\lambda P))$. Such q exists since P is π-ample and X is affine. By (B) for $\Gamma' = \Gamma + \sum_j G_j$, there exists $\sigma \in \operatorname{Frac}(B')^*$ with $\operatorname{div}_Y(\sigma) + q\delta\lambda P \geq 0$ such that

$$\Gamma + \sum_j G_j + \operatorname{div}_Y(\sigma) + q\delta\lambda P$$

is reduced and snc. Let

$$A_Y := \frac{1}{q}\big(\operatorname{div}_Y(\sigma) + q\delta\lambda P\big) \quad \text{and} \quad A := \pi_* A_Y = \frac{1}{q} \operatorname{div}_X(\sigma).$$

Then

$$\pi^* A = A_Y - \delta\lambda P = A_Y + \delta\lambda \sum_i \alpha_i E_i.$$

Let $\Delta' := (1 - \delta)H + A \geq 0$. Then $K_X + \Delta'$ is $\mathbb{Q}$-Cartier and $\pi^*(K_X + \Delta') = K_Y + \Delta'_Y$, where

$$\Delta'_Y = H_Y - \delta \pi^* H + A_Y + \delta\lambda \sum_i \alpha_i E_i.$$

For $i \in \mathcal{I}$, the coefficient of Δ'_Y along E_i is

$$1 - \delta\alpha_i\big(\operatorname{mult}_{E_i}(\pi^* H)/\alpha_i - \lambda\big),$$

which equals 1 if and only if $E_i = E$, and the coefficient of Δ'_Y along E_i is less than 1 for $i \notin \mathcal{I}$ by the choice of δ. The coefficients along G_j and the components of A_Y are $(1 - \delta)t/n < 1$ and $1/q < 1$. By (S2), (X, Δ') is lc and E is the only lc place of (X, Δ'). Let

$$v := \frac{\operatorname{ord}_E}{a(E, X)}.$$

Then v is a divisorial valuation centered at $\mathfrak{n}'$, $a(v, X) = 1$, and $v(\mathfrak{a}) = \operatorname{mult}_E F / a(E, X) = 1/t$, which is (1).

Let $0<\eta<1$ be a rational number with $\eta a(E,X)<1$, and let $\Delta_\eta := (1-\eta)\Delta'$. Since log discrepancies are affine in the boundary,

$$a(Q,X,\Delta_\eta) = (1-\eta)a(Q,X,\Delta') + \eta a(Q,X) > 0$$

for any prime divisor Q over X, so (X,Δ_η) is klt, and $a(E,X,\Delta_\eta) = \eta a(E,X) \in (0,1)$. By [LM22, Theorem 22.1], there exists a projective birational morphism $g_0 : Z_0 \to X$ with Z_0 $\mathbb{Q}$-factorial whose exceptional prime divisors are exactly the birational transform of E, which we also denote by E. Let $\Delta'_{Z_0} := (g_0)_*^{-1}\Delta'$. Since $a(E,X,\Delta')=0$,

$$K_{Z_0} + E + \Delta'_{Z_0} = g_0^*(K_X+\Delta') \sim_{\mathbb{Q},X} 0,$$

so, since E is the only lc place of (X,Δ'), $(Z_0,(1-\tau)E+\Delta'_{Z_0})$ is klt for any rational number $0<\tau<1$, and

$$K_{Z_0} + (1-\tau)E + \Delta'_{Z_0} \sim_{\mathbb{Q},X} -\tau E.$$

Fix such a τ. By [LM22, Theorem A], the MMP over X with scaling for $K_{Z_0}+(1-\tau)E+\Delta'_{Z_0}$ terminates with a model on which the birational transform of $-E$ is nef over X, and by [LM22, Theorem 11.1 (ii)], a multiple of it is generated over X, which gives the ample model $g : Z \to X$ over X of $-E$. Thus any prime divisor on Z is the birational transform of a prime divisor on Z_0, and, letting E_Z be the birational transform of E on Z ($E_Z := 0$ if E is contracted), $-E_Z$ is $\mathbb{Q}$-Cartier and g-ample, and for any sufficiently divisible positive integer j, the sheaf $\mathcal{O}_Z(-jE_Z)$ is invertible and generated by

$$H^0(Z,\mathcal{O}_Z(-jE_Z)) = H^0(Z_0,\mathcal{O}_{Z_0}(-jE)) = \mathfrak{a}_j(\mathrm{ord}_E).$$

Let $z \in g^{-1}(\mathfrak{n}')$. Since $\mathfrak{a}_j(\mathrm{ord}_E) \subset \mathfrak{n}'$, any element of $\mathfrak{a}_j(\mathrm{ord}_E)$ vanishes at z. If $z \notin \operatorname{Supp} E_Z$, then $\mathcal{O}_Z(-jE_Z) = \mathcal{O}_Z$ near z would be generated at z by elements vanishing at z, which is impossible. Thus $z \in \operatorname{Supp} E_Z$. Since the center of E on X is $\mathfrak{n}'$, $\operatorname{Supp} E_Z \subset g^{-1}(\mathfrak{n}')$. Hence $E_Z \neq 0$ and $g^{-1}(\mathfrak{n}') = \operatorname{Supp} E_Z$, and we write E for E_Z. Since $-E$ is g-ample and $g^{-1}(\mathfrak{n}')=E$, g is finite, hence an isomorphism, over $X\backslash\{\mathfrak{n}'\}$. Thus E is the only exceptional prime divisor of g. Let $\Delta'_Z := g_*^{-1}\Delta'$. Since $a(E,X,\Delta')=0$, $K_Z+E+\Delta'_Z = g^*(K_X+\Delta')$, so $(Z,E+\Delta'_Z)$ is lc and E is the only lc place of $(Z,E+\Delta'_Z)$. Since $K_Z+E = g^*K_X + a(E,X)E$ is $\mathbb{Q}$-Cartier and $\Delta'_Z \geq 0$, (Z,E) is plt, and

$$-(K_Z+E) \sim_{\mathbb{Q},X} -a(E,X)E \quad \text{is } g\text{-ample.} \tag{5.5}$$

Step 2. In this step we prove that E is normal, compute the coefficients of the different of (Z,E) on E, and prove an inversion of adjunction on Z.

(N) Normality. We show that for any separated normal integral excellent scheme U of characteristic zero admitting a dualizing complex and any reduced divisor S on U with K_U+S $\mathbb{Q}$-Cartier and (U,S) plt, S is normal. Let $\mu : W \to U$ be a log resolution of (U,S), let $S' := \mu_*^{-1}S$, and write $K_W + S' + \Delta_W = \mu^*(K_U+S)$. Since (U,S) is plt, Δ_W is μ-exceptional with coefficients less than 1. The components of S' are pairwise disjoint: otherwise, for a component C of the intersection of two of them, since $S'+\Delta_W$ has snc support and C has codimension 2, no other component of $S'+\Delta_W$ contains C, so by the equality case of (5.4) the exceptional prime divisor Q of the blow-up of W along C satisfies $a(Q,U,S)=0$, while Q is exceptional over U. Let $L := \lceil -\Delta_W \rceil$, a μ-exceptional divisor with $L \geq 0$. Then

$$L - S' = K_W + \{\Delta_W\} - \mu^*(K_U+S),$$

where $\{\Delta_W\}$ is a $\mathbb{Q}$-divisor with snc support and coefficients in $[0,1)$, and $-\mu^*(K_U+S)$ is μ-nef and μ-big. By [Mur25, Theorem A(i)], $R^1\mu_*\mathcal{O}_W(L-S')=0$, so the restriction

$$\mu_*\mathcal{O}_W(L) \to \mu_*\mathcal{O}_{S'}(L|_{S'})$$

is surjective. Since U is normal, $\mu_*\mathcal{O}_W(L) = \mathcal{O}_U$, whose image is $\mathcal{O}_S$. Since $L|_{S'} \geq 0$,

$$\mathcal{O}_S \subset \mu_*\mathcal{O}_{S'} \subset \mu_*\mathcal{O}_{S'}(L|_{S'}) = \mathcal{O}_S.$$

Since S' is regular and $S' \to S$ is proper birational, $\mu_*\mathcal{O}_{S'}$ is the normalization of $\mathcal{O}_S$. Hence S is normal.

(D) Different. For U as in (N), a normal prime divisor S on U with $K_U + S$ $\mathbb{Q}$-Cartier, and a $\mathbb{Q}$-divisor $\Delta \geq 0$ on U with $S \not\subset \operatorname{Supp}\Delta$ and $K_U + S + \Delta$ $\mathbb{Q}$-Cartier, we define $\operatorname{Diff}_S(\Delta)$ as follows. Take a log resolution $\mu : W \to U$ of $(U, S + \Delta)$, let $S' := \mu_*^{-1}S$, and write

$$K_W + S' + \Delta_W = \mu^*(K_U + S + \Delta).$$

Let $K_{S'} := (K_W + S')|_{S'}$ and $K_S := (\mu|_{S'})_*K_{S'}$, and let

$$\operatorname{Diff}_S(\Delta) := (\mu|_{S'})_*(\Delta_W|_{S'}).$$

Then $(K_U + S + \Delta)|_S = K_S + \operatorname{Diff}_S(\Delta)$. Since Δ is $\mathbb{Q}$-Cartier, $\operatorname{Diff}_S(\Delta) = \operatorname{Diff}_S(0) + \Delta|_S$, where $\Delta|_S = (\mu|_{S'})_*((\mu^*\Delta)|_{S'})$ may be computed on any log resolution of (U, S). In particular, Diff_S is affine in Δ. If (U, S) is plt and $\Delta = 0$, then $\Delta_W|_{S'}$ is snc with coefficients less than 1 and $K_{S'} + \Delta_W|_{S'} = (\mu|_{S'})^*(K_S + \operatorname{Diff}_S(0))$, so $(S, \operatorname{Diff}_S(0))$ is sub-klt by (S1).

By **Step 1** and (N), E is normal. Let

$$\Delta_E := \operatorname{Diff}_E(0).$$

By (D), (E, Δ_E) is sub-klt. It is klt since $\Delta_E \geq 0$ by (5.7) below.

(C) Cyclic covers. Let $z \in E$. Since E is $\mathbb{Q}$-Cartier, there exist an affine open neighborhood U of z in Z, a positive integer c, and $\psi \in \mathcal{O}_Z(U)$ with $cE|_U = \operatorname{div}(\psi)$. Let χ be a c-th root of ψ, and let $\rho : U' \to U$ be the normalization of U in $\operatorname{Frac}(B')(\chi)$. Since $\operatorname{mult}_E \operatorname{div}(\psi) = c$, ψ is a unit at any other codimension 1 point of U, and the characteristic is zero, ρ is étale over the codimension 1 points of U. Hence $K_{U'} = \rho^*K_U$, and

$$S := \rho^*(E|_U) = \operatorname{div}(\chi)$$

is a reduced Cartier divisor. For a $\mathbb{Q}$-divisor Δ on U with $K_U + \Delta$ $\mathbb{Q}$-Cartier,

$$\rho^*(K_U + \Delta) = K_{U'} + \rho^*\Delta.$$

Let Q' be a prime divisor over U' and Q the prime divisor over U with $\operatorname{ord}_{Q'}|_{\operatorname{Frac}(B')} = e \operatorname{ord}_Q$ for a positive integer e. Since ρ is étale over the codimension 1 points of U, $R(\rho) = 0$, and since $\operatorname{ord}_{Q'}$ restricts to $e \operatorname{ord}_Q$ on $\operatorname{Frac}(B')$, Lemma 3.2 gives $a(Q', U', \rho^*\Delta) = ea(Q, U, \Delta)$. Applying $a(Q', U', \rho^*\Delta) = ea(Q, U, \Delta)$ with $\Delta = E|_U$, for which $\rho^*\Delta = S$, and using that (Z, E) is plt and that Q is exceptional over U whenever Q' is exceptional over U', as ρ is finite, we see that (U', S) is plt, so S is normal by (N). Since $\rho|_S : S \to E \cap U$ is finite, if Δ_1 is a $\mathbb{Q}$-divisor on $E \cap U$ with $K_E + \Delta_1$ $\mathbb{Q}$-Cartier and $K_S + \Delta_S = (\rho|_S)^*(K_E + \Delta_1)$, then (S, Δ_S) is klt (resp. lc) if and only if $(E \cap U, \Delta_1)$ is klt (resp. lc). Let ζ_1 be a codimension 1 point of S. Then $\mathcal{O}_{S,\zeta_1} = \mathcal{O}_{U',\zeta_1}/(\chi)$ is a discrete valuation ring and χ is a nonzerodivisor, so $\mathcal{O}_{U',\zeta_1}$ is regular. Hence for any $\mathbb{Q}$-divisor $\Delta \geq 0$ on U with $E \not\subset \operatorname{Supp}\Delta$ and $K_U + E + \Delta$ $\mathbb{Q}$-Cartier, $\operatorname{Diff}_S(\rho^*\Delta) = (\rho^*\Delta)|_S$ as Weil divisors on S, and restricting $\rho^*(K_U + E + \Delta)$ to S,

$$K_S + (\rho^*\Delta)|_S = (\rho|_S)^*\big(K_E + \operatorname{Diff}_E(\Delta)\big) \quad \text{over } U. \tag{5.6}$$

Let ξ be a codimension 1 point of E, choose U above containing ξ, let $\zeta \in S$ be a point over ξ, and let e be the ramification index of $\rho|_S$ at ζ. Near ζ, $K_S = (\rho|_S)^*K_E + (e-1)\zeta$ by the Hurwitz formula, so (5.6) for $\Delta = 0$ implies that $e \operatorname{mult}_\xi \Delta_E = e - 1$, that is,

$$\operatorname{mult}_\xi \Delta_E = 1 - \frac{1}{e}. \tag{5.7}$$

In particular, $\Delta_E \geq 0$ and $\Delta_E \in \{1 - 1/e \mid e \in \mathbb{N}^+\}$.

(IA) Inversion of adjunction. Let $\Delta \geq 0$ be a $\mathbb{Q}$-divisor on Z such that $E \not\subset \operatorname{Supp}\Delta$, $K_Z + E + \Delta$ is $\mathbb{Q}$-Cartier, and $(E, \operatorname{Diff}_E(\Delta))$ is lc. We show that $(Z, E + \Delta)$ is lc. Since $K_Z + E$ is $\mathbb{Q}$-Cartier, so is Δ. Let $0 \leq s < 1$ be a rational number. By (D),

$$\operatorname{Diff}_E(s\Delta) = (1 - s)\Delta_E + s \operatorname{Diff}_E(\Delta),$$

and since log discrepancies are affine in the boundary, $(E, \mathrm{Diff}_E(s\Delta))$ is klt. Let $z \in E$, and let U, ρ, and S be as in (C). By (5.6) and (C), $(S, (s\rho^*\Delta)|_S)$ is klt. By [ST23, Theorem 3.2(2) and Remark 2.8] at the local rings of U' at the points of S, $(U', S + s\rho^*\Delta)$ is plt near S. Since log discrepancies are affine in s, letting s tend to 1, $(U', S + \rho^*\Delta)$ is lc near S, so $(U, E + \Delta)$ is lc near $E \cap U$ by (C). Since E is quasi-compact, $(Z, E + \Delta)$ is lc near E. By (5.4), the non-lc locus of $(Z, E + \Delta)$ is the image of the union of the components with coefficient larger than 1 of the crepant pullback of $E + \Delta$ to a log resolution, so it is closed. The non-lc locus is disjoint from $E = g^{-1}(\mathfrak{n}')$, so the image of the non-lc locus in X is a closed subset not containing $\mathfrak{n}'$, hence empty. Thus $(Z, E + \Delta)$ is lc.

Step 3. In this step we construct a complement of $K_Z + E$ of index N and push it down to X.

Since g is projective and $E = g^{-1}(\mathfrak{n}')$ set-theoretically, E is a normal projective variety of dimension $d - 1$ over k', and K_E is a canonical divisor of the k'-variety E [Sta, Tag 0A7L]. By (D), (5.7) and (5.5), (E, Δ_E) is klt and $-(K_E + \Delta_E) = -(K_Z + E)|_E$ is ample. In particular, E is of Fano type [Bir19, 2.10]. By (5.7), $\Delta_E \in \{1 - 1/e \mid e \in \mathbb{N}^+\}$, which is contained in the set of hyperstandard multiplicities associated to $\{1\}$ [Bir19, 2.2]. By [Bir19, Theorem 1.7], there exists a $\mathbb{Q}$-divisor Δ_E^+ on E satisfying the following.

(i) $\Delta_E^+ \geq \Delta_E$.
(ii) (E, Δ_E^+) is lc.
(iii) $N(K_E + \Delta_E^+) \sim 0$.

Let $R_E := \Delta_E^+ - \Delta_E \geq 0$, which is $\mathbb{Q}$-Cartier.

Let $\mu : W \to Z$ be a log resolution of (Z, E), let $E' := \mu_*^{-1}E$, and write

$$K_W + E' + \Delta_W = \mu^*(K_Z + E),$$

with canonical divisors satisfying $g_*\mu_*K_W = g_*K_Z = K_X$. By (N) and (D), Δ_W is μ-exceptional with coefficients less than 1, $\Delta_E = (\mu|_{E'})_*(\Delta_W|_{E'})$, and $K_{E'} + \Delta_W|_{E'} = (\mu|_{E'})^*(K_E + \Delta_E)$. Let $R_{E'} := (\mu|_{E'})^*R_E$ and

$$L_N := -NK_W - NE' - \lfloor (N+1)\Delta_W \rfloor.$$

Then L_N is Cartier and

$$L_N - E' = K_W + \{(N+1)\Delta_W\} - (N+1)\mu^*(K_Z + E),$$

where $\{(N+1)\Delta_W\}$ is a $\mathbb{Q}$-divisor with snc support and coefficients in $[0, 1)$, and $-(N+1)\mu^*(K_Z + E)$ is nef and big over X by (5.5). By [Mur25, Theorem A(i)] for $g \circ \mu : W \to X$, $R^1(g\circ\mu)_*\mathcal{O}_W(L_N - E') = 0$. Since X is affine, the restriction

$$H^0(W, \mathcal{O}_W(L_N)) \to H^0(E', \mathcal{O}_{E'}(L_N|_{E'}))$$

is surjective.

By (iii), there exists a rational function φ on E' with

$$N(K_{E'} + \Delta_W|_{E'} + R_{E'}) = (\mu|_{E'})^*N(K_E + \Delta_E^+) = \mathrm{div}(\varphi).$$

Since $\mathrm{Supp}\, \Delta_W + E'$ is snc,

$$\lfloor (N+1)\Delta_W \rfloor|_{E'} = \lfloor (N+1)\Delta_W|_{E'} \rfloor,$$

so

$$G_{E'} := L_N|_{E'} + \mathrm{div}(\varphi) = NR_{E'} + N\Delta_W|_{E'} - \lfloor (N+1)\Delta_W|_{E'} \rfloor.$$

The divisor $G_{E'}$ is integral, $R_{E'} \geq 0$, and

$$N\Delta_W|_{E'} - \lfloor (N+1)\Delta_W|_{E'} \rfloor = \{(N+1)\Delta_W|_{E'}\} - \Delta_W|_{E'}$$

has coefficients larger than -1. Hence $G_{E'} \geq 0$, that is, $\varphi \in H^0(E', \mathcal{O}_{E'}(L_N|_{E'}))$. Let σ_N be an element of $H^0(W, \mathcal{O}_W(L_N))$ restricting to φ, and let $G' := L_N + \mathrm{div}_W(\sigma_N) \geq 0$. Then $E' \not\subset \mathrm{Supp}\, G'$ and $G'|_{E'} = G_{E'}$. Let

$$\Delta_W^+ := \frac{1}{N}\big(\lfloor (N+1)\Delta_W \rfloor + G'\big) \quad \text{and} \quad \Delta_Z^+ := \mu_*\Delta_W^+ = \frac{1}{N}\mu_*G' \geq 0.$$

Then

$$N(K_W + E' + \Delta_W^+) = G' - L_N = \mathrm{div}_W(\sigma_N),$$

and since Δ_W is μ-exceptional,

$$N(K_Z + E + \Delta_Z^+) = \mathrm{div}_Z(\sigma_N).$$

Hence

$$\mu^*(K_Z + E + \Delta_Z^+) = K_W + E' + \Delta_W^+,$$

Δ_Z^+ is $\mathbb{Q}$-Cartier, $\mu^*\Delta_Z^+ = \Delta_W^+ - \Delta_W$, and $E \not\subset \mathrm{Supp}\,\Delta_Z^+$. By (D),

$$\begin{aligned}\mathrm{Diff}_E(\Delta_Z^+) &= \Delta_E + \Delta_Z^+|_E = (\mu|_{E'})_*\big((\Delta_W + \mu^*\Delta_Z^+)|_{E'}\big) = (\mu|_{E'})_*(\Delta_W^+|_{E'}) \\ &= \frac{1}{N}(\mu|_{E'})_*\big(\lfloor (N+1)\Delta_W|_{E'}\rfloor + G_{E'}\big) = (\mu|_{E'})_*(\Delta_W|_{E'} + R_{E'}) = \Delta_E^+.\end{aligned}$$

By (ii) and (IA), $(Z, E + \Delta_Z^+)$ is lc.

Let $\Delta^+ := g_*\Delta_Z^+$. Then $N(K_X + \Delta^+) = \mathrm{div}_X(\sigma_N)$, so $g^*(K_X + \Delta^+) = K_Z + E + \Delta_Z^+$. Hence (X, Δ^+) is lc and $a(E, X, \Delta^+) = 0$. Recall that $rK_X = \mathrm{div}(\kappa)$. Since $r \mid N$, $N\Delta^+ = \mathrm{div}_X(f_0)$ with $f_0 := \sigma_N\kappa^{-N/r}$. Since $N\Delta^+ \geq 0$ and B' is normal, $f_0 \in B'\backslash\{0\}$, and since $a(E, X, \Delta^+) = a(E, X) - \mathrm{ord}_E(f_0)/N$,

$$\Delta^+ = \frac{1}{N}\,\mathrm{div}(f_0), \qquad \Delta_Z^+ = g_*^{-1}\Delta^+, \quad \text{and} \quad \mathrm{ord}_E(f_0) = Na(E, X). \tag{5.8}$$

Step 4. In this step we replace f_0 with an element of a vector space depending only on ϵ and N. Let $\epsilon > 0$, and let J and V_ϵ be the integer with $J\epsilon > N$ and the subspace mapping isomorphically onto $B'/\mathfrak{n}'^J$ in the first paragraph of the proof.

Assume that $v(\mathfrak{n}') \geq \epsilon$, and let $f \in V_\epsilon$ with $f - f_0 \in \mathfrak{n}'^J$. Since

$$\mathrm{ord}_E(\mathfrak{n}') = a(E, X)v(\mathfrak{n}') \geq \epsilon a(E, X),$$

by (5.8),

$$\mathrm{ord}_E(f - f_0) \geq J\,\mathrm{ord}_E(\mathfrak{n}') \geq J\epsilon a(E, X) > Na(E, X) = \mathrm{ord}_E(f_0).$$

Hence $\mathrm{ord}_E(f) = \mathrm{ord}_E(f_0)$. In particular, $f \neq 0$. Let $\Delta_{Z,f} := g_*^{-1}(N^{-1}\,\mathrm{div}(f))$. Since

$$\begin{aligned}g^*K_X &= K_Z + (1 - a(E, X))E, \\ g^*\,\mathrm{div}(f) &= N\Delta_{Z,f} + \mathrm{ord}_E(f)E\end{aligned}$$

with $\mathrm{ord}_E(f) = Na(E, X)$,

$$K_Z + E + \Delta_{Z,f} = g^*\big(K_X + N^{-1}\,\mathrm{div}(f)\big),$$

so $\Delta_{Z,f}$ is $\mathbb{Q}$-Cartier.

We show that $\Delta_{Z,f}|_E = \Delta_Z^+|_E$. We may assume that $f \neq f_0$. Let $z \in E$, and let U, ρ, χ, and S be as in (C). Let

$$\widehat{f_0} := \frac{\rho^*g^*f_0}{\chi^{\mathrm{ord}_E(f_0)}} \quad \text{and} \quad \widehat{f} := \frac{\rho^*g^*f}{\chi^{\mathrm{ord}_E(f_0)}}.$$

Since $\rho^*(E|_U) = \mathrm{div}(\chi)$, we have $\mathrm{div}(\widehat{f_0}) = \rho^*(N\Delta_Z^+) \geq 0$ and $\mathrm{div}(\widehat{f}) = \rho^*(N\Delta_{Z,f}) \geq 0$ over U, so $\widehat{f_0}$ and $\widehat{f}$ are regular on U'. Since ρ is étale over the generic point of E, any component of S has coefficient $\mathrm{ord}_E(f - f_0) \geq \mathrm{ord}_E(f_0) + 1$ in $\rho^*g^*\,\mathrm{div}(f - f_0)$, so

$$\mathrm{div}\Big(\frac{\widehat{f} - \widehat{f_0}}{\chi}\Big) = \rho^*g^*\,\mathrm{div}(f - f_0) - (\mathrm{ord}_E(f_0) + 1)S \geq 0.$$

Hence $\widehat{f}|_S = \widehat{f_0}|_S$, which is non-zero since $S \not\subset \mathrm{Supp}\,\rho^*\Delta_Z^+$. Thus

$$N(\rho|_S)^*(\Delta_{Z,f}|_E) = \mathrm{div}(\widehat{f}|_S) = \mathrm{div}(\widehat{f_0}|_S) = N(\rho|_S)^*(\Delta_Z^+|_E) \quad \text{over } U,$$

so $\Delta_{Z,f}|_E = \Delta_Z^+|_E$ over U. Since z is arbitrary, $\Delta_{Z,f}|_E = \Delta_Z^+|_E$.

By (D),

$$\mathrm{Diff}_E(\Delta_{Z,f}) = \Delta_E + \Delta_{Z,f}|_E = \mathrm{Diff}_E(\Delta_Z^+) = \Delta_E^+,$$

so $(E, \mathrm{Diff}_E(\Delta_{Z,f}))$ is lc by (ii). By (IA), $(Z, E+\Delta_{Z,f})$ is lc. Thus $(X, N^{-1}\operatorname{div}(f))$ is lc and $a(E, X, N^{-1}\operatorname{div}(f)) = 0$, that is, E is an lc place of $(X, N^{-1}\operatorname{div}(f))$. Since $v = \operatorname{ord}_E / a(E, X)$ is a positive multiple of ord_E, (2) holds. The lemma follows. □

5.2. **Limits of lc places.**

Lemma 5.3. *Let $(B', \mathfrak{n}')$ be a normal excellent local domain of characteristic 0 admitting a dualizing complex, and let $k' \subset B'$ be a coefficient field. Assume that k' is algebraically closed, $K_{B'}$ is $\mathbb{Q}$-Cartier, and $\operatorname{Spec} B'$ is klt. Let N be a positive integer, let $V \subset B'$ be a k'-vector subspace with $\dim_{k'} V < +\infty$, and let ϵ be a positive real number. For any integer $i \geq 1$, let f_i be a non-zero element of V such that $(\operatorname{Spec} B', N^{-1}\operatorname{div}(f_i))$ is lc, and let v_i be a divisorial valuation of $\operatorname{Frac}(B')$ centered at $\mathfrak{n}'$ which is a positive multiple of the order of vanishing along an lc place of $(\operatorname{Spec} B', N^{-1}\operatorname{div}(f_i))$. Assume that*

$$a(v_i, \operatorname{Spec} B') = 1 \quad \text{and} \quad v_i(\mathfrak{n}') \geq \epsilon \quad \text{for any } i.$$

Then there exist integers $1 \leq i_1 < i_2 < \cdots$ and a quasi-monomial valuation v of $\operatorname{Frac}(B')$ centered at $\mathfrak{n}'$ such that

$$\lim_{j \to +\infty} v_{i_j}(x) = v(x) \text{ for any } x \in \operatorname{Frac}(B')^* \quad \text{and} \quad a(v, \operatorname{Spec} B') \leq 1.$$

Proof. The proof follows the strategy of [Xu20, Theorem 3.3 and Lemma 3.7]. We pass to subsequences of $(f_i, v_i)_{i\geq 1}$ several times and keep the notation $(f_i, v_i)_{i\geq 1}$. Let $d := \dim B'$.

We use the following notation of [JM12, Section 3.1]. Let Y be a separated regular excellent integral scheme of characteristic 0, and let $D = \sum_{k=1}^m D_k$ be a reduced snc divisor on Y. Let $J \subset \{1, \ldots, m\}$, let $\zeta \in Y$ be the generic point of an irreducible component of $\bigcap_{k\in J} D_k$, and let $\gamma = (\gamma_k)_{k\in J} \in \mathbb{R}^J_{\geq 0}$ be non-zero. Local equations $(e_k)_{k\in J}$ of $(D_k)_{k\in J}$ at ζ form a regular system of parameters of $\mathcal{O}_{Y,\zeta}$, and $\operatorname{val}_{\zeta,\gamma}$ is the monomial valuation [JM12, Proposition 3.1]

$$\operatorname{val}_{\zeta,\gamma}(x) = \min\{\langle \gamma, \beta\rangle \mid \beta \in \mathbb{N}^J,\ x_\beta \neq 0\} \quad \text{for} \quad x = \sum_{\beta\in\mathbb{N}^J} x_\beta e^\beta \in \widehat{\mathcal{O}}_{Y,\zeta}\backslash\{0\},$$

where the coefficients x_β belong to a coefficient field of $\widehat{\mathcal{O}}_{Y,\zeta}$. For a fixed x, the set of minimal elements of $\{\beta \mid x_\beta \neq 0\}$ is finite by Dickson's lemma, so $\gamma \to \operatorname{val}_{\zeta,\gamma}(x)$ is the minimum of finitely many linear functions on $\mathbb{R}^J_{\geq 0}$. In particular, it is continuous. The set $\mathrm{QM}(Y, D)$ of Definition-Lemma 2.7 consists of the valuations $\operatorname{val}_{\zeta,\gamma}$. For $u = \operatorname{val}_{\zeta,\gamma}$, we have $u(D_k) = \gamma_k$ for $k \in J$ and $u(D_k) = 0$ for $k \notin J$, since $\zeta \notin D_k$ for $k \notin J$. Moreover, with $J_u := \{k \in J \mid \gamma_k > 0\}$, the center $\operatorname{center}_Y u$ is the generic point ζ_u of the irreducible component of $\bigcap_{k\in J_u} D_k$ containing ζ, and $u = \operatorname{val}_{\zeta_u,(\gamma_k)_{k\in J_u}}$.

Step 1. In this step we fix a parameter variety for the elements f_i.

Let $\mathbb{P}(V)$ be the projective space of 1-dimensional k'-subspaces of V, and let $t_i := [f_i] \in \mathbb{P}(V)(k')$ be the point given by $k'f_i$. Let $T_0 \subset \mathbb{P}(V)$ be an irreducible closed subset containing t_i for infinitely many i, with $\dim T_0$ minimal. Possibly passing to a subsequence, we may assume that $t_i \in T_0$ for any i. Any proper closed subset of T_0 contains t_i for only finitely many i, since each irreducible component T_1 of it satisfies $\dim T_1 < \dim T_0$. Let T be the non-singular locus of T_0. Possibly passing to a subsequence, we may assume that $t_i \in T$ for any i, and

$$\text{for any nonempty open subset } T' \subset T,\ t_i \in T' \text{ for all but finitely many } i. \tag{5.9}$$

By [Tem18, Theorem 1.1.6], there exists a log resolution $\pi : W \to \operatorname{Spec} B'$. Write

$$K_W + \Delta_W = \pi^* K_{B'}.$$

By Definition 2.3(5),

$$a(u, \operatorname{Spec} B') = a(u, W) - u(\Delta_W) \tag{5.10}$$

for any real valuation u of $\operatorname{Frac}(B')$ which is non-negative on B'. Let

$$X_T := \operatorname{Spec} B' \times_{k'} T, \quad W_T := W \times_{k'} T, \quad \text{and} \quad \pi_T := \pi \times \mathrm{id}_T : W_T \to X_T,$$

and let Δ_{W_T} be the pullback of Δ_W to W_T. Since W is regular and T is smooth over k', W_T is regular. Let $\mathbb{K}$ be the function field of X_T. Then $\mathrm{Frac}(B') \subset \mathbb{K}$ and $\mathrm{trdeg}(\mathbb{K}/\mathrm{Frac}(B')) = \dim T$. For a scheme Y over T and $t \in T(k')$, let Y_t be the fiber of Y over t, and for a Cartier divisor G on Y, let $G_t := G|_{Y_t}$. Thus $(X_T)_t = \operatorname{Spec} B'$ and $(W_T)_t = W$. Let $g_1, \dots, g_n$ be a basis of V and $\lambda_1, \dots, \lambda_n$ the dual homogeneous coordinates on $\mathbb{P}(V)$. For any j, the function $\lambda_j^{-1} \sum_{k=1}^n \lambda_k g_k$ on $\operatorname{Spec} B' \times_{k'} (T \cap \{\lambda_j \neq 0\})$ is non-zero, and the functions $\lambda_j^{-1} \sum_{k=1}^n \lambda_k g_k$ for $1 \le j \le n$ define a Cartier divisor $D_T \ge 0$ on X_T with

$$(D_T)_t = \mathrm{div}(f) \quad \text{for any } t = [f] \in T(k').$$

Step 2. In this step we show that a log resolution of the family D_T restricts to log resolutions on the fibers over a non-empty open subset of T.

Let $\rho' : M' \to W_T$ be a projective birational morphism from a regular scheme M', let $\sigma' := \pi_T \circ \rho'$, and let $E' = \sum_{k=1}^{m'} E'_k$ be a reduced snc divisor on M'. Assume that $\mathfrak{n}'\mathcal{O}_{M'}$ is invertible and that E' contains

$$\rho'^{-1}(Z'), \quad \mathrm{Supp}(\mathfrak{n}'\mathcal{O}_{M'}), \quad \mathrm{Supp}\, \rho'^* \Delta_{W_T}, \quad \text{and} \quad \mathrm{Supp}\, \sigma'^* D_T,$$

where $Z' \subsetneq W_T$ is a closed subset such that ρ' is an isomorphism over $W_T \backslash Z'$. For a proper birational morphism $Y \to X$ of regular schemes, let $K_{Y/X}$ be the relative canonical divisor defined by $\mathrm{Fitt}_0(\Omega_{Y/X})$ [JM12, Section 1.3]. Define rational numbers b'_k and integers $o'_k, c'_k \ge 0$ by

$$\begin{aligned} &\sum_{k=1}^{m'} b'_k E'_k = \rho'^* \Delta_{W_T} - K_{M'/W_T}, \quad \sigma'^* D_T = \sum_{k=1}^{m'} o'_k E'_k, \\ &\text{and} \quad \mathfrak{n}'\mathcal{O}_{M'} = \mathcal{O}_{M'}\Big(- \sum_{k=1}^{m'} c'_k E'_k \Big), \end{aligned} \tag{5.11}$$

and let $F' := \mathrm{Supp}(\mathfrak{n}'\mathcal{O}_{M'}) = \bigcup_{c'_k > 0} E'_k$. We show that there exists a non-empty open subset $T' \subset T$ such that for any $t \in T'(k')$, the following hold.

(i) M'_t is regular and integral, $\rho'_t : M'_t \to W$ is birational, $E'_t := \sum_{k=1}^{m'} E'_{k,t}$ is a reduced snc divisor, and $E'_{k,t}$ is regular for any k.

(ii) For any real valuation u of $\mathrm{Frac}(B')$ which is non-negative on B' and any $f \in V \backslash \{0\}$ with $[f] = t$,

$$a(u, \operatorname{Spec} B') = a(u, M'_t) - \sum_{k=1}^{m'} b'_k u(E'_{k,t}), \quad u(f) = \sum_{k=1}^{m'} o'_k u(E'_{k,t}), \quad \text{and} \quad u(\mathfrak{n}') = \sum_{k=1}^{m'} c'_k u(E'_{k,t}).$$

(iii) If $t = t_i$ for some i, let $\zeta := \mathrm{center}_{M'_t} v_i$ and $J := \{k \mid \zeta \in E'_k\}$. Then ζ is the generic point of an irreducible component of $\bigcap_{k \in J} E'_{k,t}$, and with respect to (M'_t, E'_t),

$$v_i = \mathrm{val}_{\zeta, (v_i(E'_{k,t}))_{k \in J}}, \quad v_i(E'_{k,t}) > 0, \quad \text{and} \quad b'_k + \frac{o'_k}{N} = 1 \quad \text{for any } k \in J.$$

Let $\mathcal{S}$ be the finite set of the connected components of the schemes $\bigcap_{k \in J} E'_k$ for the subsets J of $\{1, \dots, m'\}$ containing some k with $c'_k > 0$. Any $S \in \mathcal{S}$ is regular and connected, hence integral, and

$$S \subset F' = \sigma'^{-1}(\operatorname{Spec}(B'/\mathfrak{n}') \times_{k'} T).$$

Since σ' is projective, S is projective over T. Thus S is a smooth variety over the perfect field k'. If $S \to T$ is not dominant, we remove the image of S, a proper closed subset, from T. If $S \to T$ is dominant, then by generic smoothness S is smooth over a non-empty open subset of T. Thus there exists a non-empty open subset $T' \subset T$ such that $S \times_T T'$ is smooth over T' for any $S \in \mathcal{S}$.

Let $t \in T'(k')$, and let $F'_t := F' \cap M'_t$. Let $\tau_1, \dots, \tau_{\dim T}$ be a regular system of parameters of $\mathcal{O}_{T,t}$, and let $z \in F'_t$. Let $J_z := \{k \mid z \in E'_k\}$, and let e_k be a local equation of E'_k at z for

$k \in J_z$. Since $z \in F'$, J_z contains some k with $c'_k > 0$. Let $S \in \mathcal{S}$ be the connected component of $\bigcap_{k\in J_z} E'_k$ containing z. Since $(e_k)_{k\in J_z}$ is part of a regular system of parameters of $\mathcal{O}_{M',z}$, $\mathcal{O}_{S,z} = \mathcal{O}_{M',z}/(e_k)_{k\in J_z}$, and the images of $\tau_1, \dots, \tau_{\dim T}$ in $\mathcal{O}_{S,z}$ are part of a regular system of parameters because S is smooth over T' at z, the elements $(e_k)_{k\in J_z}, \tau_1, \dots, \tau_{\dim T}$ form part of a regular system of parameters of $\mathcal{O}_{M',z}$. Thus $\mathcal{O}_{M'_t,z} = \mathcal{O}_{M',z}/(\tau_1, \dots, \tau_{\dim T})$ is regular and the images of $(e_k)_{k\in J_z}$ are part of a regular system of parameters of $\mathcal{O}_{M'_t,z}$. In particular, E'_t is an snc divisor at z, $E'_{k,t}$ is regular at z for any k, and the generic point of $\operatorname{Spec} \mathcal{O}_{M'_t,z}$ is not contained in E'.

Since

$$M'\backslash E' \subset M'\backslash \rho'^{-1}(Z') \cong W_T\backslash Z',$$

the open subset $M'_t\backslash E'$ of M'_t is isomorphic to an open subset of W, hence it is irreducible. By the previous paragraph, the irreducible component of M'_t through any $z \in F'_t$ intersects $M'_t\backslash E'$, hence it is the closure of $M'_t\backslash E'$. Let $\Sigma \subset M'_t$ be the union of the non-regular locus of M'_t, the locus where E'_t is not an snc divisor or some $E'_{k,t}$ is not regular, and the irreducible components of M'_t other than the closure of $M'_t\backslash E'$. Then Σ is closed and $\Sigma \cap F'_t = \emptyset$. Since $M'_t \to \operatorname{Spec} B'$ is proper and F'_t is the preimage of $\mathfrak{n}'$, the image of Σ is a closed subset of $\operatorname{Spec} B'$ not containing $\mathfrak{n}'$, so $\Sigma = \emptyset$. Since ρ'_t is surjective, $F'_t \neq \emptyset$, so M'_t is the closure of the non-empty set $M'_t\backslash E'$. This proves (i).

Since $M'_t = M' \times_{W_T} W$, $\Omega_{M'_t/W}$ is the pullback of Ω_{M'/W_T}, and taking Fitt_0 commutes with pullback. Since moreover $M'_t \not\subset E' \supset \operatorname{Supp} K_{M'/W_T}$, we get $K_{M'/W_T}|_{M'_t} = K_{M'_t/W}$. Moreover $(\rho'^*\Delta_{W_T})|_{M'_t} = \rho'^*_t\Delta_W$, $(\sigma'^* D_T)|_{M'_t} = \sigma'^*_t \operatorname{div}(f)$, and $\mathfrak{n}'\mathcal{O}_{M'_t} = \mathcal{O}_{M'_t}(-\sum_k c'_k E'_{k,t})$. By [JM12, Remark 5.6], $a(u, W) = a(u, M'_t) + u(K_{M'_t/W})$. Thus (ii) follows from (5.10) and (5.11).

Let $t = t_i$. For any prime divisor P on M'_t contained in $E'_{k,t}$, (ii) implies

$$a(P, \operatorname{Spec} B', N^{-1}\operatorname{div}(f_i)) = a(P, \operatorname{Spec} B') - N^{-1}\operatorname{mult}_P \operatorname{div}(f_i) = 1 - b'_k - \frac{o'_k}{N},$$

so $b'_k + o'_k/N \le 1$ whenever $E'_{k,t} \neq \emptyset$, because $(\operatorname{Spec} B', N^{-1}\operatorname{div}(f_i))$ is lc. Since v_i is a positive multiple of the order of vanishing along an lc place of $(\operatorname{Spec} B', N^{-1}\operatorname{div}(f_i))$, (ii) and Definition-Lemma 2.7 for (M'_t, E'_t) imply

$$0 = a(v_i, M'_t) - \sum_{k=1}^{m'} \Big(b'_k + \frac{o'_k}{N}\Big) v_i(E'_{k,t}) \ge a(v_i, M'_t) - \sum_{k=1}^{m'} v_i(E'_{k,t}) \ge 0.$$

Thus $v_i \in \operatorname{QM}(M'_t, E'_t)$, and $b'_k + o'_k/N = 1$ whenever $v_i(E'_{k,t}) > 0$, that is, whenever $\zeta \in E'_k$. By (i), the irreducible components of $E'_{k,t}$ are pairwise disjoint, so exactly one component of E'_t in $E'_{k,t}$ contains ζ for any $k \in J$. Thus (iii) follows from the description of $\operatorname{QM}(M'_t, E'_t)$ at the beginning of the proof.

Step 3. In this step we fix a regular scheme M over W_T on which the valuations v_i have convergent weights.

Let $M_1 \to W_T$ be the blow-up of $\mathfrak{n}'\mathcal{O}_{W_T}$, and let $\Sigma_1 \subset M_1$ be the union of the exceptional divisor of $M_1 \to W_T$ and the preimages of $\operatorname{Supp} \Delta_{W_T}$ and $\operatorname{Supp} \pi_T^* D_T$. By [Tem18, Theorem 1.1.6 and §2.2.6], there exists a projective birational morphism $M \to M_1$, a sequence of blow-ups, such that M is regular and the union $E_M = \sum_{j=1}^m E_j$ of the preimage of Σ_1 and the preimages of the centers of the blow-ups is a reduced snc divisor, and $\mathfrak{n}'\mathcal{O}_M$ is invertible. Let $\rho : M \to W_T$ be the composite. Then (M, ρ, E_M) satisfies the conditions of **Step 2**. Let b_j, o_j, c_j be the numbers (5.11) for (M, ρ, E_M), and let $a_j := 1 - b_j$. By (5.9), possibly replacing T with the open subset T' of **Step 2** for (M, ρ, E_M) and passing to a subsequence, we may assume that **Step 2**(i)–(iii) hold for any $t \in T(k')$.

For any i, let $\zeta_i := \mathrm{center}_{M_{t_i}}(v_i)$, $J_i := \{j \mid \zeta_i \in E_j\}$, and let S_i be the connected component of $\bigcap_{j\in J_i} E_j$ containing ζ_i. By **Step 2**(iii),

$$v_i = \mathrm{val}_{\zeta_i,\gamma_i} \text{ with } \gamma_i := (v_i(E_{j,t_i}))_{j\in J_i} \in \mathbb{R}^{J_i}_{>0}, \quad \text{and} \quad b_j + \frac{o_j}{N} = 1 \text{ for any } j \in J_i.$$

For $j \in J_i$, let P be the prime divisor on M_{t_i} contained in E_{j,t_i} and containing ζ_i. By **Step 2**(ii), $a(P, \operatorname{Spec} B') = 1 - b_j = a_j$, so $a_j > 0$ since $\operatorname{Spec} B'$ is klt. By **Step 2**(ii) and the equality case of Definition-Lemma 2.7 for (M_{t_i}, E_{M,t_i}),

$$1 = a(v_i, \operatorname{Spec} B') = \sum_{j\in J_i} a_j\gamma_{i,j} \quad \text{and} \quad v_i(\mathfrak{n}') = \sum_{j\in J_i} c_j\gamma_{i,j}.$$

Since the set of pairs (J_i, S_i) is finite, possibly passing to a subsequence, we may assume that $J_i = J$ and $S_i = S$ for any i. Then γ_i belongs to the compact simplex

$$\Big\{\gamma \in \mathbb{R}^J_{\geq 0} \;\Big|\; \sum_{j\in J} a_j\gamma_j = 1\Big\},$$

so possibly passing to a subsequence, we may assume that $\lim_{i\to+\infty} \gamma_i = \gamma$ for some γ in the simplex. Since E_M is an snc divisor, S is an irreducible component of $\bigcap_{j\in J} E_j$. Let ξ be the generic point of S. Define valuations of $\mathbb{K}$ by

$$w_i := \mathrm{val}_{\xi,\gamma_i} \quad \text{and} \quad w := \mathrm{val}_{\xi,\gamma} \quad \text{with respect to } (M, E_M).$$

By the continuity of $\gamma \to \mathrm{val}_{\xi,\gamma}(x)$,

$$\lim_{i\to+\infty} w_i(x) = w(x) \quad \text{for any } x \in \mathcal{O}_{M,\xi}\backslash\{0\}. \tag{5.12}$$

Since $\mathfrak{n}'\mathcal{O}_{M,\xi}$ is generated by $\prod_{j\in J} e_j^{c_j}$ for local equations e_j of E_j at ξ, we have

$$w(\mathfrak{n}') = \sum_{j\in J} c_j\gamma_j = \lim_{i\to+\infty} v_i(\mathfrak{n}') \geq \epsilon.$$

Step 4. In this step we show that for any $x \in B'\backslash\{0\}$, $v_i(x) = w_i(x)$ for all but finitely many i.

Fix $x \in B'\backslash\{0\}$. By [Tem18, Theorem 1.1.6], there exist a projective birational morphism $\rho_x : M_x \to M$ from a regular scheme M_x, a closed subset $Z_x \subsetneq M$ such that ρ_x is an isomorphism over $M\backslash Z_x$, and a reduced snc divisor $E' = \sum_{k=1}^{m'} E'_k$ on M_x containing $\rho_x^{-1}(E_M)$, $\rho_x^{-1}(\mathrm{Supp}\,\mathrm{div}_M(x))$, and $\rho_x^{-1}(Z_x)$. Then $(M_x, \rho\circ\rho_x, E')$ satisfies the conditions of **Step 2**. Let b'_k, o'_k, c'_k be the numbers (5.11) for $(M_x, \rho\circ\rho_x, E')$, and define integers $\mu_{jk}, \omega_k \geq 0$ by

$$\rho_x^*E_j = \sum_{k=1}^{m'} \mu_{jk}E'_k \quad \text{and} \quad \mathrm{div}_{M_x}(x) = \sum_{k=1}^{m'} \omega_k E'_k.$$

Let $\Gamma := \sum_{j=1}^m (b_j + o_j/N)E_j$. Since $K_{M_x/W_T} = \rho_x^*K_{M/W_T} + K_{M_x/M}$, (5.11) for M and M_x implies

$$\sum_{k=1}^{m'} \Big(b'_k + \frac{o'_k}{N}\Big)E'_k = \rho_x^*\Gamma - K_{M_x/M}. \tag{5.13}$$

By **Step 2** for $(M_x, \rho\circ\rho_x, E')$ and (5.9), **Step 2**(i)–(iii) for M_x hold at $t = t_i$ for all but finitely many i. Fix such an i and let $t := t_i$.

By **Step 2**(iii) for M_x, $\zeta'_i := \mathrm{center}_{M_{x,t}} v_i$ is the generic point of an irreducible component of $\bigcap_{k\in J'_i} E'_{k,t}$, where $J'_i := \{k \mid \zeta'_i \in E'_k\}$, and

$$v_i = \mathrm{val}_{\zeta'_i,\gamma'_i} \text{ with } \gamma'_i := (v_i(E'_{k,t}))_{k\in J'_i}, \quad b'_k + \frac{o'_k}{N} = 1 \text{ for any } k \in J'_i.$$

Let S_i' be the connected component of $\bigcap_{k\in J_i'} E_k'$ containing ζ_i', let ξ_i' be the generic point of S_i', and let $u_i := \mathrm{val}_{\xi_i',\gamma_i'}$ with respect to (M_x, E'), a valuation of $\mathbb{K}$. Restricting to $M_{x,t}$, we have $\rho_{x,t}^* E_{j,t} = \sum_k \mu_{jk} E_{k,t}'$ and $\mathrm{div}(x|_{M_{x,t}}) = \sum_k \omega_k E_{k,t}'$. Since $u_i = \mathrm{val}_{\xi_i',\gamma_i'}$ and $v_i = \mathrm{val}_{\zeta_i',\gamma_i'}$,

$$u_i(E_j) = \sum_{k\in J_i'} \mu_{jk}\gamma_{i,k}' = v_i(E_{j,t}) \text{ for any } j, \quad \text{and} \quad u_i(x) = \sum_{k\in J_i'} \omega_k \gamma_{i,k}' = v_i(x). \tag{5.14}$$

Since $\zeta_i \in E_j$ if and only if $j \in J$, (5.14) implies $u_i(E_j) = \gamma_{i,j}$ for $j \in J$ and $u_i(E_j) = 0$ for $j \notin J$.

We show that $u_i = w_i$. By [JM12, Remark 5.6], (5.13), and the equality case of Definition-Lemma 2.7 for $u_i \in \mathrm{QM}(M_x, E')$,

$$a(u_i, M) - u_i(\Gamma) = a(u_i, M_x) - \sum_{k=1}^{m'} \Big(b_k' + \frac{o_k'}{N}\Big) u_i(E_k') = \sum_{k\in J_i'} \gamma_{i,k}' - \sum_{k\in J_i'} \gamma_{i,k}' = 0.$$

Since $b_j + o_j/N = 1$ for $j \in J$, $u_i(\Gamma) = \sum_{j\in J} \gamma_{i,j} = \sum_{j=1}^m u_i(E_j)$. Thus $a(u_i, M) = \sum_{j=1}^m u_i(E_j)$, so $u_i \in \mathrm{QM}(M, E_M)$ by Definition-Lemma 2.7. Thus $\mathrm{center}_M u_i$ is the generic point of an irreducible component of $\bigcap_{j\in J} E_j$ and u_i is monomial there with weights γ_i. Moreover $\mathrm{center}_M u_i = \rho_x(\xi_i')$, and the closure $\rho_x(S_i')$ of $\rho_x(\xi_i')$ is irreducible, contained in $\bigcap_{j\in J} E_j$, and contains $\rho_x(\zeta_i')$, where

$$\rho_x(\zeta_i') = \mathrm{center}_{M_t} v_i = \zeta_i \in S.$$

Since S is a connected component of $\bigcap_{j\in J} E_j$, $\rho_x(S_i') \subset S$, so $\mathrm{center}_M u_i = \xi$. Thus $u_i = \mathrm{val}_{\xi,\gamma_i} = w_i$, and (5.14) implies $w_i(x) = v_i(x)$.

Step 5. In this step we conclude the proof.

By **Step 4** and (5.12), $\lim_{i\to+\infty} v_i(x) = w(x)$ for any $x \in B'\backslash\{0\}$, hence for any $x \in \mathrm{Frac}(B')^*$. Let v be the restriction of w to $\mathrm{Frac}(B')$. Then $v \geq 0$ on B' and $v(\mathfrak{n}') = w(\mathfrak{n}') \geq \epsilon$ by **Step 3**, so v is a real valuation of $\mathrm{Frac}(B')$ centered at $\mathfrak{n}'$.

Let $\xi_w := \mathrm{center}_M w$. Since $w(\mathfrak{n}') > 0$ and $w \geq 0$ on B', ξ_w maps to $\mathfrak{n}'$, whose residue field is k'. Since M is of finite type over B' and B' is excellent, the dimension formula [Mat89, Theorem 15.6] implies

$$\dim \mathcal{O}_{M,\xi_w} + \mathrm{trdeg}(\kappa(\xi_w)/k') = d + \dim T.$$

Since $w \in \mathrm{QM}(M, E_M)$, w is an Abhyankar valuation by [JM12, Proposition 3.7], that is,

$$\mathrm{ratrk}(w) + \mathrm{trdeg}(\kappa(w)/\kappa(\xi_w)) = \dim \mathcal{O}_{M,\xi_w}.$$

Thus

$$\mathrm{ratrk}(w) + \mathrm{trdeg}(\kappa(w)/k') = d + \dim T. \tag{5.15}$$

Let $\xi_v := \mathrm{center}_W v$. By [JM12, (3.4)] on the regular scheme W and the dimension formula [Mat89, Theorem 15.6],

$$\mathrm{ratrk}(v) + \mathrm{trdeg}(\kappa(v)/k') \leq \dim \mathcal{O}_{W,\xi_v} + \mathrm{trdeg}(\kappa(\xi_v)/k') = d, \tag{5.16}$$

and equality holds if and only if v is an Abhyankar valuation.

By the Abhyankar inequality for the extension $\mathrm{Frac}(B') \subset \mathbb{K}$ [Bou72, Chapter VI, Section 10, no. 3, Corollary 1],

$$\dim_{\mathbb{Q}}\big(\big(w(\mathbb{K}^*)/v(\mathrm{Frac}(B')^*)\big) \otimes_{\mathbb{Z}} \mathbb{Q}\big) + \mathrm{trdeg}(\kappa(w)/\kappa(v)) \leq \mathrm{trdeg}(\mathbb{K}/\mathrm{Frac}(B')) = \dim T. \tag{5.17}$$

Since

$$\mathrm{ratrk}(w) = \mathrm{ratrk}(v) + \dim_{\mathbb{Q}}\big(\big(w(\mathbb{K}^*)/v(\mathrm{Frac}(B')^*)\big) \otimes_{\mathbb{Z}} \mathbb{Q}\big)$$

and $k' \subset \kappa(v) \subset \kappa(w)$, (5.15), (5.17), and (5.16) imply

$$d + \dim T = \mathrm{ratrk}(w) + \mathrm{trdeg}(\kappa(w)/k') \leq \dim T + \mathrm{ratrk}(v) + \mathrm{trdeg}(\kappa(v)/k') \leq \dim T + d.$$

Thus equality holds in (5.16), so v is an Abhyankar valuation, and v is quasi-monomial by [JM12, Proposition 3.7] on W.

Finally, $\operatorname{Spec} B'$ and the zero divisor satisfy the hypotheses of Lemma 2.4, and $v_i \to v$ pointwise on $\operatorname{Frac}(B')^*$, so

$$a(v, \operatorname{Spec} B') \leq \liminf_{i \to +\infty} a(v_i, \operatorname{Spec} B') = 1.$$

The lemma follows. □

Theorem 5.4. *Let $(B, \mathfrak{n})$ be a normal local domain essentially of finite type over $\mathbb{C}\{z_1, \dots, z_n\}$ with $\dim B \geq 2$. Assume that K_B is $\mathbb{Q}$-Cartier and $\operatorname{Spec} B$ is klt. Let α, β be real numbers with $\beta \geq \alpha > 0$, and let $\mathfrak{a}_\bullet = (\mathfrak{a}_j)_{j \geq 1}$ be a graded sequence of $\mathfrak{n}$-primary ideals such that*

$$\mathfrak{n}^{\lceil \beta j \rceil} \subset \mathfrak{a}_j \subset \mathfrak{n}^{\lceil \alpha j \rceil} \quad \text{for any integer } j \geq 1.$$

Then $0 < \operatorname{lct}(B; \mathfrak{a}_\bullet) < +\infty$, and there exists a quasi-monomial valuation v of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$ such that

$$a(v, \operatorname{Spec} B) = 1 \quad \text{and} \quad v(\mathfrak{a}_\bullet) = \frac{1}{\operatorname{lct}(B; \mathfrak{a}_\bullet)}.$$

Proof of Theorem 5.4. Let $c := \operatorname{lct}(B; \mathfrak{a}_\bullet)$. By Lemma 4.5(2),

$$0 < c = \sup_{j \geq 1} j \operatorname{lct}(B; \mathfrak{a}_j) < +\infty. \tag{5.18}$$

Let B' be a strict henselization of B, and let $\mathfrak{n}' := \mathfrak{n}B'$. By Lemma 5.1(1) and $\dim B \geq 2$, B' and a coefficient field $k' \subset B'$ satisfy the assumptions of Lemmas 5.2 and 5.3. Let N be the positive integer of Lemma 5.2 for B' and k', let

$$\epsilon := \frac{1}{(\beta + 1)c},$$

and let $V := V_\epsilon \subset B'$ be the k'-vector subspace of Lemma 5.2.

For any integer $i \geq 1$, let $m_i := i!$. By Lemma 5.2 for the $\mathfrak{n}'$-primary ideal $\mathfrak{a}_{m_i}B'$, there exists a divisorial valuation v_i of $\operatorname{Frac}(B')$ centered at $\mathfrak{n}'$ with

$$a(v_i, \operatorname{Spec} B') = 1 \quad \text{and} \quad v_i(\mathfrak{a}_{m_i}B') = \frac{1}{\operatorname{lct}(B'; \mathfrak{a}_{m_i}B')}.$$

By Lemma 5.1(2) and (5.18),

$$v_i(\mathfrak{a}_{m_i}B') = \frac{1}{\operatorname{lct}(B; \mathfrak{a}_{m_i})} \geq \frac{m_i}{c}. \tag{5.19}$$

Since

$$\mathfrak{n}'^{\lceil \beta m_i \rceil} \subset \mathfrak{a}_{m_i}B',$$

(5.19) implies

$$v_i(\mathfrak{n}') \geq \frac{m_i}{c\lceil \beta m_i \rceil} \geq \frac{m_i}{c(\beta m_i + 1)} \geq \frac{1}{(\beta + 1)c} = \epsilon.$$

Thus by Lemma 5.2, there exists a non-zero $f_i \in V$ such that $(\operatorname{Spec} B', N^{-1} \operatorname{div}(f_i))$ is lc and v_i is a positive multiple of the order of vanishing along an lc place of $(\operatorname{Spec} B', N^{-1} \operatorname{div}(f_i))$. By Lemma 5.3, possibly passing to a subsequence of $(v_i)_{i \geq 1}$, there exists a quasi-monomial valuation w of $\operatorname{Frac}(B')$ centered at $\mathfrak{n}'$ such that

$$\lim_{i \to +\infty} v_i(x) = w(x) \text{ for any } x \in \operatorname{Frac}(B')^* \quad \text{and} \quad a(w, \operatorname{Spec} B') \leq 1.$$

Fix an integer $j \geq 1$. For any $i \geq j$ in the subsequence, j divides m_i, so $\mathfrak{a}_j^{m_i/j} \subset \mathfrak{a}_{m_i}$, and (5.19) implies

$$\frac{m_i}{j} v_i(\mathfrak{a}_j B') \geq v_i(\mathfrak{a}_{m_i}B') \geq \frac{m_i}{c}, \quad \text{that is,} \quad v_i(\mathfrak{a}_j B') \geq \frac{j}{c}.$$

Since $\mathfrak{a}_j B'$ is generated by finitely many generators of $\mathfrak{a}_j$,

$$w(\mathfrak{a}_j B') = \lim_{i \to +\infty} v_i(\mathfrak{a}_j B') \geq \frac{j}{c}.$$

Let v be the restriction of w to $\mathrm{Frac}(B)$. By Lemma 5.1(3), v is a quasi-monomial valuation of $\mathrm{Frac}(B)$ centered at $\mathfrak{n}$,

$$v(\mathfrak{a}_j) = w(\mathfrak{a}_j B') \geq \frac{j}{c} \text{ for any } j \geq 1, \quad \text{and} \quad a(v, \operatorname{Spec} B) \leq a(w, \operatorname{Spec} B') \leq 1.$$

Thus $v(\mathfrak{a}_\bullet) \geq 1/c > 0$, and by Definition 2.8(3),

$$c \leq \frac{a(v, \operatorname{Spec} B)}{v(\mathfrak{a}_\bullet)} \leq \frac{1}{v(\mathfrak{a}_\bullet)} \leq c.$$

Thus $a(v, \operatorname{Spec} B) = 1$ and $v(\mathfrak{a}_\bullet) = 1/c$. The theorem follows. □

6. Divisorial eigenvaluations on klt local rings

In this section we prove Theorem 6.5.

6.1. **Subadditivity and volume approximation.**

Lemma 6.1 (cf. [Tak13, Theorem 3.5]). *Let A_0, $\mathfrak{q}$, Y, and y satisfy Set-up 4.8(1), and let $B := \mathcal{O}_{Y,y}$. Assume that K_B is $\mathbb{Q}$-Cartier. Then there exists a non-zero ideal $\mathfrak{Q}_0$ of B such that*

$$\mathfrak{Q}_0 \mathcal{J}(B, \mathfrak{a}^\lambda \mathfrak{c}^\mu) \subset \overline{\mathcal{J}(B, \mathfrak{a}^\lambda)\mathcal{J}(B, \mathfrak{c}^\mu)}$$

for any non-zero ideals $\mathfrak{a}, \mathfrak{c}$ of B and any real numbers $\lambda, \mu > 0$, where $\mathcal{J}(B, \cdot)$ denotes the multiplier ideal of $\operatorname{Spec} B$ *(Definition 4.1) and $\overline{I}$ denotes the integral closure of an ideal I.*

Proof. Since A_0 is a quotient of $\mathbb{C}\{z_1, \dots, z_n\}$ for some n and Y is of finite type over $(A_0)_\mathfrak{q}$, B is a normal local domain essentially of finite type over $\mathbb{C}\{z_1, \dots, z_n\}$, so B satisfies the conditions of Definition 4.1 by Lemma 2.2. We divide the proof into four steps.

***Step 1**: the ideal $\mathfrak{Q}_0$.* Let $\pi : \widetilde{Y} \to \operatorname{Spec} B$ be a log resolution of the unit ideal B (Definition 4.1(1)), and let D_π be an snc divisor on $\widetilde{Y}$ whose support contains any π-exceptional prime divisor. For an $\mathcal{O}_{\widetilde{Y}}$-submodule $\mathcal{F}$ of the constant sheaf $\mathrm{Frac}(B)$ on $\widetilde{Y}$ and a divisor D on $\widetilde{Y}$, let $\mathcal{F}(D) := \mathcal{F} \cdot \mathcal{O}_{\widetilde{Y}}(D)$. Fix $K_{\widetilde{Y}}$ with $\pi_* K_{\widetilde{Y}} = K_B$ and a divisor $N_1 \geq 0$ on $\widetilde{Y}$ with

$$-N_1 \leq K_{\widetilde{Y}} - \pi^* K_B \leq N_1.$$

Since π is projective, $\widetilde{Y}$ is a closed subscheme of $\mathbb{P}^\nu_B$ for some $\nu \in \mathbb{N}$. Let $x_0, \dots, x_\nu$ be the homogeneous coordinates, numbered so that x_0 does not vanish identically on $\widetilde{Y}$. Since π is birational, $x_i/x_0 \in \mathrm{Frac}(B)$, so there exists $l \in B \setminus \{0\}$ with $l_i := l x_i/x_0 \in B$ for $0 \leq i \leq \nu$. Let $\mathfrak{l} := (l_0, \dots, l_\nu)$. Multiplication by l/x_0 maps $\mathcal{O}_{\widetilde{Y}}(1)$ isomorphically onto $\mathfrak{l}\mathcal{O}_{\widetilde{Y}}$, so $\mathfrak{l}\mathcal{O}_{\widetilde{Y}} = \mathcal{O}_{\widetilde{Y}}(-N_0)$ for a Cartier divisor $N_0 \geq 0$ on $\widetilde{Y}$, and $\mathcal{O}_{\widetilde{Y}}(-N_0)$ is very ample over $\operatorname{Spec} B$. Let

$$N := 3N_1 + 2\nu N_0 \quad \text{and} \quad \mathfrak{Q}_0 := \Gamma(\widetilde{Y}, \mathcal{O}_{\widetilde{Y}}(-N)).$$

Since B is normal and π is proper and birational, $\Gamma(\widetilde{Y}, \mathcal{O}_{\widetilde{Y}}) = B$, so $\mathfrak{Q}_0$ is an ideal of B, and $\mathfrak{Q}_0 \neq 0$ as $\mathcal{O}_{\widetilde{Y}}(-N)$ is invertible and π is birational.

Let $\mathfrak{a}, \mathfrak{c}$ be non-zero ideals of B and $\lambda, \mu > 0$ real numbers. Let $\rho : Y' \to \widetilde{Y}$ be a projective birational morphism from a regular integral scheme Y' with

$$\mathfrak{a}\mathcal{O}_{Y'} = \mathcal{O}_{Y'}(-F_\mathfrak{a}), \quad \mathfrak{c}\mathcal{O}_{Y'} = \mathcal{O}_{Y'}(-F_\mathfrak{c}), \quad \text{and} \quad \mathfrak{l}\mathcal{O}_{Y'} = \mathcal{O}_{Y'}(-F_\mathfrak{l})$$

for divisors $F_\mathfrak{a}, F_\mathfrak{c}, F_\mathfrak{l} \geq 0$ on Y', and with an snc divisor on Y' whose support contains $\operatorname{Supp} F_\mathfrak{a}$, $\operatorname{Supp} F_\mathfrak{c}$, $\operatorname{Supp} F_\mathfrak{l}$, $\operatorname{Supp} \rho^* D_\pi$, and any ρ-exceptional prime divisor [Tem18, Theorem 1.1.6]. Any $\pi \circ \rho$-exceptional prime divisor on Y' is ρ-exceptional or contained in $\operatorname{Supp} \rho^* D_\pi$, so $\pi \circ \rho$ is a log resolution of $\mathfrak{a}$, $\mathfrak{c}$, and $\mathfrak{l}$. Fix $K_{Y'}$ with $\rho_* K_{Y'} = K_{\widetilde{Y}}$, so that $(\pi \circ \rho)_* K_{Y'} = K_B$, and let $K' := K_{Y'} - (\pi \circ \rho)^* K_B$. For real numbers $t_1, t_2 \geq 0$ and integers $k \geq 0$ let

$$\mathcal{H}^k_{t_1,t_2} := \mathcal{O}_{Y'}(\lceil K' - t_1 F_\mathfrak{a} - t_2 F_\mathfrak{c} - k F_\mathfrak{l} \rceil) \quad \text{and} \quad \mathcal{J}'_{t_1,t_2} := \rho_* \mathcal{H}^0_{t_1,t_2}.$$

Since $F_{\mathfrak{l}} = \rho^* N_0$, the projection formula gives $\rho_* \mathcal{H}^k_{t_1,t_2} = \mathcal{J}'_{t_1,t_2}(-kN_0)$, and by Definition 4.1 for $\pi \circ \rho$,

$$\Gamma(Y', \mathcal{H}^k_{t_1,t_2}) = \mathcal{J}(B, \mathfrak{a}^{t_1}\mathfrak{c}^{t_2}\mathfrak{l}^k). \tag{6.1}$$

By (4.4) for $\pi \circ \rho$, and as $\operatorname{Spec} B$ is affine,

$$H^q(Y', \mathcal{H}^k_{t_1,t_2}) = 0 \quad \text{for any integer } q > 0. \tag{6.2}$$

Step 2: *subadditivity on $\widetilde{Y}$.* We show that

$$\mathcal{J}'_{\lambda,\mu}(-3N_1) \subset \mathcal{J}'_{\lambda,0} \cdot \mathcal{J}'_{0,\mu}. \tag{6.3}$$

Let $\mathfrak{a}' := \mathfrak{a}\mathcal{O}_{\widetilde{Y}}$ and $\mathfrak{c}' := \mathfrak{c}\mathcal{O}_{\widetilde{Y}}$, and let $\mathcal{J}(\widetilde{Y}, \cdot)$ denote the (mixed) multiplier ideals of the regular excellent scheme $\widetilde{Y}$, defined as in [JM12, Section 1.3]. They can be tested on the prime divisors on Y' [JM12, Section 1.3], and for any prime divisor E on Y', the log discrepancy over $\widetilde{Y}$ of the order of vanishing along E is $1 + \operatorname{mult}_E(K_{Y'} - \rho^* K_{\widetilde{Y}})$ by Definition 2.3(5). Hence (cf. Lemma 4.2), for real numbers $t_1, t_2 \geq 0$,

$$\mathcal{J}(\widetilde{Y}, \mathfrak{a}'^{t_1}\mathfrak{c}'^{t_2}) = \rho_* \mathcal{O}_{Y'}(\lceil K_{Y'} - \rho^* K_{\widetilde{Y}} - t_1 F_{\mathfrak{a}} - t_2 F_{\mathfrak{c}} \rceil).$$

Since

$$-\rho^* N_1 \leq K' - (K_{Y'} - \rho^* K_{\widetilde{Y}}) = \rho^*(K_{\widetilde{Y}} - \pi^* K_B) \leq \rho^* N_1,$$

the projection formula gives

$$\mathcal{J}'_{t_1,t_2}(-N_1) \subset \mathcal{J}(\widetilde{Y}, \mathfrak{a}'^{t_1}\mathfrak{c}'^{t_2}) \subset \mathcal{J}'_{t_1,t_2}(N_1).$$

By the subadditivity theorem for regular excellent schemes over $\mathbb{Q}$ [JM12, Theorem A.2],

$$\mathcal{J}'_{\lambda,\mu}(-N_1) \subset \mathcal{J}(\widetilde{Y}, \mathfrak{a}'^{\lambda}\mathfrak{c}'^{\mu}) \subset \mathcal{J}(\widetilde{Y}, \mathfrak{a}'^{\lambda}) \cdot \mathcal{J}(\widetilde{Y}, \mathfrak{c}'^{\mu}) \subset \mathcal{J}'_{\lambda,0}(N_1) \cdot \mathcal{J}'_{0,\mu}(N_1),$$

which gives (6.3).

Step 3: *global generation.* We show that

$$\mathcal{J}'_{\lambda,0}(-\nu N_0) \subset \mathcal{J}(B, \mathfrak{a}^{\lambda})\mathcal{O}_{\widetilde{Y}} \quad \text{and} \quad \mathcal{J}'_{0,\mu}(-\nu N_0) \subset \mathcal{J}(B, \mathfrak{c}^{\mu})\mathcal{O}_{\widetilde{Y}}. \tag{6.4}$$

The conditions on ρ in **Step 1** are symmetric in $\mathfrak{a}$ and $\mathfrak{c}$, so the second inclusion is the first one with $(\mathfrak{a}, \lambda)$ and $(\mathfrak{c}, \mu)$ interchanged. We prove the first inclusion. Let $\mathcal{H}^k := \mathcal{H}^k_{\lambda,0}$ for integers $k \geq 0$, and let $k \geq \nu + 1$ be an integer. Since $l_0, \ldots, l_\nu$ generate the invertible sheaf $\mathcal{O}_{Y'}(-F_{\mathfrak{l}})$, their Koszul complex gives an exact complex

$$0 \to \mathcal{H}^{k-\nu-1} \to \cdots \to (\mathcal{H}^{k-2})^{\oplus\binom{\nu+1}{2}} \to (\mathcal{H}^{k-1})^{\oplus(\nu+1)} \to \mathcal{H}^k \to 0$$

whose last map is $(g_i)_i \to \sum_i l_i g_i$. As $k - \nu - 1 \geq 0$, all terms other than $\mathcal{H}^k$ have vanishing higher cohomology by (6.2), so the last map is surjective on global sections. By (6.1),

$$\mathcal{J}(B, \mathfrak{a}^{\lambda}\mathfrak{l}^k) = \mathfrak{l}\mathcal{J}(B, \mathfrak{a}^{\lambda}\mathfrak{l}^{k-1}) \quad \text{for any integer } k \geq \nu + 1,$$

so

$$\mathcal{J}(B, \mathfrak{a}^{\lambda}\mathfrak{l}^k) = \mathfrak{l}^{k-\nu}\mathcal{J}(B, \mathfrak{a}^{\lambda}\mathfrak{l}^{\nu}) \quad \text{for any integer } k \geq \nu.$$

Since $\mathcal{O}_{\widetilde{Y}}(-N_0)$ is very ample over $\operatorname{Spec} B$, by Serre's theorem there exists an integer $k \geq \nu$ such that $\mathcal{J}'_{\lambda,0}(-kN_0) = \rho_* \mathcal{H}^k$ is generated by its global sections, which form $\mathcal{J}(B, \mathfrak{a}^{\lambda}\mathfrak{l}^k)$ by (6.1). Thus

$$\mathcal{J}'_{\lambda,0}(-kN_0) = \mathfrak{l}^{k-\nu}\mathcal{J}(B, \mathfrak{a}^{\lambda}\mathfrak{l}^{\nu})\mathcal{O}_{\widetilde{Y}} = \big(\mathcal{J}(B, \mathfrak{a}^{\lambda}\mathfrak{l}^{\nu})\mathcal{O}_{\widetilde{Y}}\big)(-(k-\nu)N_0).$$

As $\mathcal{O}_{\widetilde{Y}}(-(k-\nu)N_0)$ is invertible, and by (4.2),

$$\mathcal{J}'_{\lambda,0}(-\nu N_0) = \mathcal{J}(B, \mathfrak{a}^{\lambda}\mathfrak{l}^{\nu})\mathcal{O}_{\widetilde{Y}} \subset \mathcal{J}(B, \mathfrak{a}^{\lambda})\mathcal{O}_{\widetilde{Y}}.$$

This proves (6.4).

Step 4: *conclusion.* Let

$$I := \mathcal{J}(B, \mathfrak{a}^{\lambda})\mathcal{J}(B, \mathfrak{c}^{\mu}).$$

By (6.3) and (6.4),

$$\mathcal{J}'_{\lambda,\mu}(-N) \subset \mathcal{J}'_{\lambda,0}(-\nu N_0) \cdot \mathcal{J}'_{0,\mu}(-\nu N_0) \subset I\mathcal{O}_{\widetilde{Y}}.$$

Since $\mathcal{J}(B, \mathfrak{a}^\lambda \mathfrak{c}^\mu)$ is the module of global sections of $\mathcal{J}'_{\lambda,\mu}$ by (6.1) with $k = 0$,

$$\mathfrak{Q}_0 \mathcal{J}(B, \mathfrak{a}^\lambda \mathfrak{c}^\mu) \subset \Gamma(\widetilde{Y}, \mathcal{J}'_{\lambda,\mu}(-N)) \subset \Gamma(\widetilde{Y}, I\mathcal{O}_{\widetilde{Y}}).$$

Let $f \in \Gamma(\widetilde{Y}, I\mathcal{O}_{\widetilde{Y}})$, and let V be a discrete valuation ring of rank 1 with $B \subset V \subset \mathrm{Frac}(B)$ and valuation v. Then v has a center ξ on $\widetilde{Y}$ (Definition 2.3(1)), and $f \in I\mathcal{O}_{\widetilde{Y},\xi} \subset IV$. Hence $f \in \overline{I}$ by [HS06, Proposition 6.8.2]. □

The next lemma (cf. [Blu18, Proposition 3.7] for valuations on klt varieties) is the input for the existence of the minimizer in Section 6.3.

Lemma 6.2. *Notation and conditions as in Set-up 4.8 and Construction-Proposition 4.10. For any non-empty subset $\Sigma \subset \mathcal{W}$, let $\mathfrak{d}^\Sigma_\bullet$ be the family of the ideals $\mathfrak{d}^\Sigma_t := \bigcap_{v\in\Sigma} \mathfrak{a}_t(v)$, $t \in \mathbb{R}$. Then $\mathfrak{d}^\Sigma_\bullet$ is an $\mathfrak{n}$-filtration, and there exist real numbers $\epsilon_j \geq 0$, $j \in \mathbb{N}^+$, with $\lim_{j\to+\infty} \epsilon_j = 0$ such that*

$$0 \leq \frac{e(\mathfrak{d}^\Sigma_j)}{j^r} - e(\mathfrak{d}^\Sigma_\bullet) \leq \epsilon_j$$

for any $j \in \mathbb{N}^+$ and any non-empty subset $\Sigma \subset \mathcal{W}$.

Proof. Let $\Sigma \subset \mathcal{W}$ be a non-empty subset. For any $v \in \Sigma$, $v(\mathfrak{b}_\bullet) = 1$, so $\mathfrak{b}_t \subset \mathfrak{a}_t(v)$ for any $t > 0$. Hence, by Construction-Proposition 4.10,

$$\mathfrak{n}^{\lceil \beta t \rceil} \subset \mathfrak{b}_t \subset \mathfrak{d}^\Sigma_t \quad \text{for any real number } t > 0. \tag{6.5}$$

Since each $\mathfrak{a}_\bullet(v)$ satisfies the conditions of Definition 2.8(1) other than primarity, and $\mathfrak{d}^\Sigma_t \subset \mathfrak{n}$ for $t > 0$, (6.5) shows that $\mathfrak{d}^\Sigma_\bullet$ is an $\mathfrak{n}$-filtration. Since B is normal and excellent, B is analytically irreducible [Gro65, Scholie 7.8.3(v)]. Since $(\mathfrak{d}^\Sigma_j)_{j\geq 1}$ is an $\mathfrak{n}$-primary graded sequence and the function $\ell_B(B/\mathfrak{d}^\Sigma_t)$ of t is increasing, [BLQ24, Section 2, (2.1)] implies that $e(\mathfrak{d}^\Sigma_\bullet)$ exists and

$$e(\mathfrak{d}^\Sigma_\bullet) = \lim_{j\to+\infty} \frac{e(\mathfrak{d}^\Sigma_j)}{j^r} = \inf_{j\geq 1} \frac{e(\mathfrak{d}^\Sigma_j)}{j^r}. \tag{6.6}$$

This proves the left inequality of the lemma. By (6.5),

$$e(\mathfrak{d}^\Sigma_j) \leq e(\mathfrak{n}^{\lceil \beta j \rceil}) = \lceil \beta j \rceil^r e(\mathfrak{n}),$$

so (6.6) implies

$$e(\mathfrak{d}^\Sigma_\bullet) \leq \beta^r e(\mathfrak{n}). \tag{6.7}$$

Claim 6.3. *Let $\mathfrak{a}$ be an $\mathfrak{n}$-primary ideal of B, $c > 0$ a real number, $f \in \mathcal{J}(B, \mathfrak{a}^c)$, and v a real valuation of $\mathrm{Frac}(B)$ centered at $\mathfrak{n}$ with $a(v, \mathrm{Spec}\, B) < +\infty$. Then*

$$v(f) > cv(\mathfrak{a}) - a(v, \mathrm{Spec}\, B).$$

Proof. Let $g : Y' \to \mathrm{Spec}\, B$ be a log resolution of $\mathfrak{a}$ with Y' regular and g projective. Write $\mathfrak{a}\mathcal{O}_{Y'} = \mathcal{O}_{Y'}(-F_\mathfrak{a})$ and $K_{Y'/B} := K_{Y'} - g^*K_B$ with $g_*K_{Y'} = K_B$, and let $D = \sum_i D_i$ be an snc divisor on Y' containing $\mathrm{Exc}(g)$, $\mathrm{Supp}\, F_\mathfrak{a}$, and $\mathrm{Supp}\, K_{Y'/B}$. Let $s_i := \mathrm{mult}_{D_i} K_{Y'/B}$. By the definition of the multiplier ideal through the log resolution g (cf. [Laz04, Definition 9.2.3]),

$$\mathcal{J}(B, \mathfrak{a}^c) = g_*\mathcal{O}_{Y'}(\lceil K_{Y'/B} - cF_\mathfrak{a} \rceil),$$

so

$$\mathrm{mult}_{D_i} \mathrm{div}(f) \geq \lfloor c\, \mathrm{mult}_{D_i} F_\mathfrak{a} - s_i \rfloor \quad \text{for any } i.$$

First let $w \in \mathrm{QM}(Y', D)$ with weights $w_i := w(D_i) \geq 0$, not all zero, where $\mathrm{QM}(Y', D)$ and $r_{Y',D}$ are as in Definition-Lemma 2.7. Since $f \in B \subset \mathcal{O}_{Y',\mathrm{center}_{Y'} w}$, $w(f) \geq \sum_i w_i\, \mathrm{mult}_{D_i} \mathrm{div}(f)$. Since

$\lfloor x\rfloor > x-1$ for any real number x, $w(\mathfrak{a}) = \sum_i w_i \operatorname{mult}_{D_i} F_{\mathfrak{a}}$, $w(K_{Y'/B}) = \sum_i w_i s_i$, $a(w,Y') = \sum_i w_i$ [JM12, Proposition 5.1], and $a(w,\operatorname{Spec} B) = a(w,Y') + w(K_{Y'/B})$ by Definition 2.3(5),

$$w(f) \geq \sum_i w_i \lfloor c \operatorname{mult}_{D_i} F_{\mathfrak{a}} - s_i\rfloor > cw(\mathfrak{a}) - w(K_{Y'/B}) - a(w,Y') = cw(\mathfrak{a}) - a(w,\operatorname{Spec} B).$$

For general v, the center of v on Y' is contained in $g^{-1}(\mathfrak{n}) = \operatorname{Supp} F_{\mathfrak{a}}$ as $\mathfrak{a}$ is $\mathfrak{n}$-primary, so $w := r_{Y',D}(v)$ is a quasi-monomial valuation in $\mathrm{QM}(Y',D)$ with weights not all zero. By [JM12, §4.3, Lemma 4.7 and Corollary 5.4], $v \geq w$ on $\mathcal{O}_{Y',\operatorname{center}_{Y'} v}$, $w(D_i) = v(D_i)$ for any i, and $a(w,Y') \leq a(v,Y')$. Hence $w(\mathfrak{a}) = v(\mathfrak{a})$, $w(K_{Y'/B}) = v(K_{Y'/B})$, and

$$v(f) \geq w(f) > cv(\mathfrak{a}) - w(K_{Y'/B}) - a(w,Y') \geq cv(\mathfrak{a}) - v(K_{Y'/B}) - a(v,Y') = cv(\mathfrak{a}) - a(v,\operatorname{Spec} B).$$

This proves Claim 6.3. □

Proof of Lemma 6.2 continued. For $t>0$, let

$$\mathcal{J}(B, t\cdot\mathfrak{d}_\bullet^\Sigma) := \mathcal{J}(B, t\cdot(\mathfrak{d}_j^\Sigma)_{j\geq 1}),$$

the ideal defined in Section 4.1 for the graded sequence $(\mathfrak{d}_j^\Sigma)_{j\geq 1}$. By (4.2) and the definition of $\mathcal{J}(B, t\cdot(\mathfrak{d}_j^\Sigma)_{j\geq 1})$,

$$\mathcal{J}(B, t\cdot\mathfrak{d}_\bullet^\Sigma) = \mathcal{J}(B, (\mathfrak{d}_k^\Sigma)^{t/k}) \quad \text{for any sufficiently divisible } k.$$

Let $j\in\mathbb{N}^+$. For any $v\in\Sigma$, $a(v,\operatorname{Spec} B)=1$ and $v(\mathfrak{d}_k^\Sigma)\geq k$ as $\mathfrak{d}_k^\Sigma \subset \mathfrak{a}_k(v)$. By Claim 6.3 with $c=j/k$, any $f\in\mathcal{J}(B,(\mathfrak{d}_k^\Sigma)^{j/k})$ satisfies $v(f)>j-1$ for any $v\in\Sigma$, so

$$\mathcal{J}(B, j\cdot\mathfrak{d}_\bullet^\Sigma) \subset \mathfrak{d}_{j-1}^\Sigma. \tag{6.8}$$

Since $\operatorname{Spec} B$ is klt, for any $\mathfrak{n}$-primary ideal $\mathfrak{a}$, a projective log resolution $g: Y'\to \operatorname{Spec} B$ of $\mathfrak{a}$ with Y' regular, and $K_{Y'/B} := K_{Y'} - g^*K_B$ with $g_*K_{Y'} = K_B$, $\lceil K_{Y'/B}\rceil \geq 0$ is g-exceptional, so $\mathfrak{a}\subset\mathcal{J}(B,\mathfrak{a})$. In particular,

$$\mathfrak{d}_j^\Sigma \subset \mathcal{J}(B,\mathfrak{d}_j^\Sigma) \subset \mathcal{J}(B, j\cdot\mathfrak{d}_\bullet^\Sigma).$$

Since K_B is $\mathbb{Q}$-Cartier by Set-up 4.8(2), let $\mathfrak{Q}_0$ be the ideal of Lemma 6.1. Possibly replacing $\mathfrak{Q}_0$ with $\mathfrak{n}\mathfrak{Q}_0$, we may assume that $\mathfrak{Q}_0\subset\mathfrak{n}$. Let $m\in\mathbb{N}^+$, and let k be sufficiently divisible for both $t=j$ and $t=mj$. By Lemma 6.1 with $\mathfrak{a}=\mathfrak{c}=\mathfrak{d}_k^\Sigma$, used $m-1$ times, the equality

$$\mathcal{J}(B,\mathfrak{a}^\lambda\mathfrak{a}^\mu) = \mathcal{J}(B,\mathfrak{a}^{\lambda+\mu}) \quad \text{for any real numbers } \lambda,\mu>0,$$

and the relations $\overline{\overline{I}} = \overline{I}$ and $I\overline{J}\subset\overline{IJ}$ for ideals I,J of B [HS06, Corollary 1.3.1 and Remark 1.3.2(4)], we have

$$\mathfrak{Q}_0^{m-1}\mathcal{J}(B,(\mathfrak{d}_k^\Sigma)^{mj/k}) \subset \overline{\mathcal{J}(B,(\mathfrak{d}_k^\Sigma)^{j/k})^m}.$$

Together with (6.8),

$$\mathfrak{Q}_0^{m-1}\mathfrak{d}_{mj}^\Sigma \subset \mathfrak{Q}_0^{m-1}\mathcal{J}(B, mj\cdot\mathfrak{d}_\bullet^\Sigma) \subset \overline{\mathcal{J}(B, j\cdot\mathfrak{d}_\bullet^\Sigma)^m} \subset \overline{(\mathfrak{d}_{j-1}^\Sigma)^m}.$$

Replacing j with $j+1$ and using $\mathfrak{Q}_0^m\subset\mathfrak{Q}_0^{m-1}$, we obtain

$$\mathfrak{Q}_0^m\mathfrak{d}_{m(j+1)}^\Sigma \subset \overline{(\mathfrak{d}_j^\Sigma)^m} \quad \text{for any } j,m\in\mathbb{N}^+. \tag{6.9}$$

Let

$$\mathfrak{e}_j := \mathfrak{Q}_0 + \mathfrak{n}^{\lceil\beta j\rceil}, \quad \text{so that} \quad \mathfrak{e}_j^m = \sum_{i=0}^m \mathfrak{Q}_0^i\mathfrak{n}^{\lceil\beta j\rceil(m-i)}.$$

Let $0\leq i\leq m$. Since $i(j+1)\leq m(j+1)$, (6.9) (for $i\geq 1$), (6.5), and [HS06, Remark 1.3.2(4)] imply

$$\mathfrak{Q}_0^i\mathfrak{n}^{\lceil\beta j\rceil(m-i)}\mathfrak{d}_{m(j+1)}^\Sigma \subset \mathfrak{n}^{\lceil\beta j\rceil(m-i)}\mathfrak{Q}_0^i\mathfrak{d}_{i(j+1)}^\Sigma \subset (\mathfrak{d}_j^\Sigma)^{m-i}\overline{(\mathfrak{d}_j^\Sigma)^i} \subset \overline{(\mathfrak{d}_j^\Sigma)^m}.$$

Hence

$$\mathfrak{e}_j^m\mathfrak{d}_{m(j+1)}^\Sigma \subset \overline{(\mathfrak{d}_j^\Sigma)^m}. \tag{6.10}$$

All ideals in (6.10) are $\mathfrak{n}$-primary. The Minkowski inequality $e(IJ)^{1/r} \le e(I)^{1/r} + e(J)^{1/r}$ holds for $\mathfrak{n}$-primary ideals I, J of B [HS06, Corollary 17.7.3(3)], and $e(\bar{I}) = e(I)$ for any $\mathfrak{n}$-primary ideal I of B [HS06, Corollary 1.3.1 and Proposition 11.2.1]. By (6.10),

$$me(\mathfrak{d}_j^\Sigma)^{1/r} = e\big(\overline{(\mathfrak{d}_j^\Sigma)^m}\big)^{1/r} \le e(\mathfrak{e}_j^m \mathfrak{d}_{m(j+1)}^\Sigma)^{1/r} \le me(\mathfrak{e}_j)^{1/r} + e(\mathfrak{d}_{m(j+1)}^\Sigma)^{1/r}.$$

Dividing by mj and letting $m \to +\infty$, (6.6) implies

$$\Big(\frac{e(\mathfrak{d}_j^\Sigma)}{j^r}\Big)^{1/r} \le \delta_j + \Big(1+\frac{1}{j}\Big)e(\mathfrak{d}_\bullet^\Sigma)^{1/r}, \quad \text{where } \delta_j := \frac{e(\mathfrak{e}_j)^{1/r}}{j}. \tag{6.11}$$

We bound δ_j. Fix $0 \ne q_0 \in \mathfrak{Q}_0$. Since $q_0 B + \mathfrak{n}^{\lceil \beta j \rceil} \subset \mathfrak{e}_j$, Lech's inequality [Lec60, Theorem 3] implies

$$e(\mathfrak{e}_j) \le r! e(\mathfrak{n}) \ell_B(B/\mathfrak{e}_j) \le r! e(\mathfrak{n}) \ell_B(B/(q_0 B + \mathfrak{n}^{\lceil \beta j \rceil})).$$

Since B is a domain and $q_0 \in \mathfrak{n}\backslash\{0\}$, $\dim B/q_0 B \le r-1$, so the Hilbert–Samuel function of $B/q_0 B$ implies that there exists a real number $\gamma > 0$ with $\ell_B(B/(q_0 B + \mathfrak{n}^k)) \le \gamma k^{r-1}$ for any $k \in \mathbb{N}^+$. Hence

$$\delta_j \le \frac{(r! e(\mathfrak{n}) \gamma \lceil \beta j \rceil^{r-1})^{1/r}}{j},$$

so $\lim_{j\to+\infty} \delta_j = 0$, and δ_j depends only on $\mathfrak{Q}_0$, β, and j, not on Σ.

For $x \ge 0$, the function $x \to (\delta_j + (1+\frac{1}{j})x)^r - x^r$ is increasing, as the derivative of the function is

$$r\Big(1+\frac{1}{j}\Big)\Big(\delta_j + \Big(1+\frac{1}{j}\Big)x\Big)^{r-1} - rx^{r-1} \ge 0.$$

Let

$$\epsilon_j := \Big(\delta_j + \Big(1+\frac{1}{j}\Big)\beta e(\mathfrak{n})^{1/r}\Big)^r - \beta^r e(\mathfrak{n}).$$

Then $\epsilon_j \ge 0$ and $\lim_{j\to+\infty} \epsilon_j = 0$. By (6.11), and by the monotonicity of the function above at $x = e(\mathfrak{d}_\bullet^\Sigma)^{1/r}$, as $x \le \beta e(\mathfrak{n})^{1/r}$ by (6.7),

$$\frac{e(\mathfrak{d}_j^\Sigma)}{j^r} - e(\mathfrak{d}_\bullet^\Sigma) \le \Big(\delta_j + \Big(1+\frac{1}{j}\Big)e(\mathfrak{d}_\bullet^\Sigma)^{1/r}\Big)^r - e(\mathfrak{d}_\bullet^\Sigma) \le \epsilon_j.$$

The lemma follows. □

6.2. **Rational weights.**

Lemma 6.4. *Let $(A, \mathfrak{m}_A)$ be a local domain essentially of finite type over $\mathbb{C}\{z_1, \dots, z_n\}$, let $\varphi : A \to A$ be a finite injective local $\mathbb{C}$-algebra homomorphism, let $\Lambda > 0$ be a rational number, and let v be a quasi-monomial valuation of* $\mathrm{Frac}(A)$ *centered at $\mathfrak{m}_A$. Assume that $\varphi_* v = \Lambda v$. Then there exists a primitive divisorial valuation w of* $\mathrm{Frac}(A)$ *centered at $\mathfrak{m}_A$ such that*

$$\varphi_* w = \Lambda w.$$

Proof. The proof follows [Yos26, proof of Theorem 5.1]. We extend φ to $\mathrm{Frac}(A)$, and we let $f : \mathrm{Spec}\, A \to \mathrm{Spec}\, A$ be the finite morphism induced by φ. The ring A is excellent of characteristic zero [JM12, §1.1], and so is any scheme of finite type over $\mathrm{Spec}\, A$.

Step 1. By [JM12, Lemma 3.6(ii)] on the regular scheme in Definition 2.3(4), there exist a regular integral scheme Y_1 with a proper birational morphism $\pi_1 : Y_1 \to \mathrm{Spec}\, A$ and an snc divisor D_1 on Y_1 such that v is monomial in local equations $x_1, \dots, x_t$ of the components of D_1 at the generic point η of a stratum of D_1 and $v(x_1), \dots, v(x_t)$ are linearly independent over $\mathbb{Q}$. Here $x_1, \dots, x_t$ form a regular system of parameters of $\mathcal{O}_{Y_1,\eta}$ and $v(x_i) > 0$ for any i, so

$$\eta = \mathrm{center}_{Y_1} v.$$

Step 2. Let $Y_1 \times_{\mathrm{Spec}\, A} Y_1$ be the fiber product of $f \circ \pi_1$ and π_1, and let pr_1 and pr_2 be the projections of $Y_1 \times_{\mathrm{Spec}\, A} Y_1$ to the first and to the second factor. The morphism pr_1 is the base change of π_1 along $f \circ \pi_1$, so pr_1 is proper. Since φ is injective, $f \circ \pi_1$ maps the generic point of

Y_1 to the generic point of $\operatorname{Spec} A$, over which π_1 is an isomorphism. Thus the fiber of pr_1 over the generic point of Y_1 consists of one point, whose residue field is $\operatorname{Frac}(A)$ and whose image under pr_2 is the generic point of Y_1. Let Γ be the closure of the generic fiber of pr_1 with the reduced structure. Then $\mathrm{pr}_1|_\Gamma : \Gamma \to Y_1$ is proper birational and $\mathrm{pr}_2|_\Gamma : \Gamma \to Y_1$ is proper dominant.

Let $V \subset Y_1$ be the closed subset over which $\mathrm{pr}_1|_\Gamma$ is not an isomorphism, and let $\Gamma' \to \Gamma$ be the blow-up along $(\mathrm{pr}_1|_\Gamma)^{-1}(V)$. Let $\Sigma \subset \Gamma'$ be the union of the exceptional divisor of $\Gamma' \to \Gamma$ and the preimage of

$$\mathrm{pr}_1^{-1}(\operatorname{Supp} D_1) \cup \mathrm{pr}_2^{-1}(\operatorname{Supp} D_1).$$

By [Tem18, Theorem 1.1.6], there exist a regular integral scheme Y_2, a proper birational morphism $Y_2 \to \Gamma'$ and an snc divisor D_2 on Y_2 whose support contains the preimage of Σ, such that $Y_2 \to \Gamma'$ is an isomorphism over the regular locus of $\Gamma' \backslash \Sigma$. Let $\sigma : Y_2 \to \Gamma$ be the composite with $\Gamma' \to \Gamma$, and let

$$g_1 := \mathrm{pr}_1 \circ \sigma : Y_2 \to Y_1 \quad \text{and} \quad g_2 := \mathrm{pr}_2 \circ \sigma : Y_2 \to Y_1.$$

Then

$$\operatorname{Supp} D_2 \supset g_1^{-1}(\operatorname{Supp} D_1) \cup g_2^{-1}(\operatorname{Supp} D_1),$$

and Γ' is regular outside the exceptional divisor of $\Gamma' \to \Gamma$ since $\mathrm{pr}_1|_\Gamma$ is an isomorphism over $Y_1 \backslash V$, so g_1 is an open immersion on $Y_2 \backslash \operatorname{Supp} D_2$. The morphism g_1 is proper birational, g_2 is proper dominant, and

$$\pi_1 \circ g_2 = f \circ \pi_1 \circ g_1. \tag{6.12}$$

We regard $\operatorname{Frac}(A)$ as the function field of Y_1 via π_1^* and as the function field of Y_2 via $(\pi_1 \circ g_1)^*$. Then $g_1^* = \mathrm{id}_{\operatorname{Frac}(A)}$, and by (6.12),

$$g_2^* = \varphi \quad \text{on } \operatorname{Frac}(A).$$

Step 3. Since $\pi_1 \circ g_1 : Y_2 \to \operatorname{Spec} A$ is proper birational, $\zeta := \operatorname{center}_{Y_2} v$ exists (Definition 2.3(1)). For any $k \in \{1, 2\}$, g_k is dominant, so the homomorphism

$$g_k^\# : \mathcal{O}_{Y_1, g_k(\zeta)} \to \mathcal{O}_{Y_2, \zeta}$$

is injective and local, and $v \circ g_k^*$ is centered at $g_k(\zeta)$. Since $v \circ g_1^* = v$ and $v \circ g_2^* = \varphi_* v = \Lambda v$ are both centered at η and Y_1 is separated,

$$g_1(\zeta) = g_2(\zeta) = \eta.$$

Since g_1 is proper birational, $\operatorname{Supp} D_2 \supset g_1^{-1}(\operatorname{Supp} D_1)$ and g_1 is an open immersion on $Y_2 \backslash \operatorname{Supp} D_2$, by [JM12, Lemma 3.6(i)] over the regular scheme Y_1, ζ is the generic point of the intersection of exactly t components of D_2, so the local equations $x_1', \dots, x_t'$ at ζ of the components of D_2 through ζ form a regular system of parameters of $\mathcal{O}_{Y_2,\zeta}$, and $v(x_1'), \dots, v(x_t')$ are linearly independent over $\mathbb{Q}$. We let

$$v(x') := (v(x_1'), \dots, v(x_t'))^T \in \mathbb{R}^t_{>0}.$$

For any i and any $k \in \{1, 2\}$, $g_k^\#(x_i) \neq 0$ and the zero locus of $g_k^\#(x_i)$ near ζ is contained in $g_k^{-1}(\operatorname{Supp} D_1) \subset \operatorname{Supp} D_2$. Since $\mathcal{O}_{Y_2,\zeta}$ is a unique factorization domain, there exist $t \times t$ matrices $P = (P_{ij})$ and $P' = (P'_{ij})$ with entries in $\mathbb{N}$ and units $u_1, \dots, u_t, u_1', \dots, u_t'$ of $\mathcal{O}_{Y_2,\zeta}$ such that, for any $i \in \{1, \dots, t\}$,

$$g_1^\#(x_i) = u_i \prod_{j=1}^t (x_j')^{P_{ij}} \quad \text{and} \quad g_2^\#(x_i) = u_i' \prod_{j=1}^t (x_j')^{P'_{ij}}. \tag{6.13}$$

Since v is centered at ζ, $v(u_i) = v(u_i') = 0$. Since $g_1^\#(x_i) = x_i$ and $g_2^\#(x_i) = \varphi(x_i)$ in $\operatorname{Frac}(A)$, (6.13) and $\varphi_* v = \Lambda v$ imply

$$Pv(x') = (v(x_1), \dots, v(x_t))^T \quad \text{and} \quad P'v(x') = (v(\varphi(x_1)), \dots, v(\varphi(x_t)))^T = \Lambda P v(x').$$

Since $\Lambda \in \mathbb{Q}$, the matrix $P' - \Lambda P$ has rational entries, and since $v(x_1'), \dots, v(x_t')$ are linearly independent over $\mathbb{Q}$,

$$P' = \Lambda P.$$

Moreover, $\det P \neq 0$: if $\mathbf{d} \in \mathbb{Q}^t$ is a row vector with $\mathbf{d}P = 0$, then $\sum_{i=1}^t d_i v(x_i) = \mathbf{d}Pv(x') = 0$, so $\mathbf{d} = 0$ because $v(x_1), \dots, v(x_t)$ are linearly independent over $\mathbb{Q}$. Since $\Lambda > 0$, $\det P' = \Lambda^t \det P \neq 0$.

Step 4. For any $\mathbf{q} \in \mathbb{R}^t_{>0}$, let $\mu_{\mathbf{q}}$ be the monomial valuation at η in $x_1, \dots, x_t$ with weights $\mathbf{q}$, and for any $\mathbf{s} \in \mathbb{R}^t_{>0}$, let $v_{\mathbf{s}}$ be the monomial valuation at ζ in $x_1', \dots, x_t'$ with weights $\mathbf{s}$ [JM12, Section 3.1, Proposition 3.1]. Both $\mu_{\mathbf{q}}$ and $v_{\mathbf{s}}$ are real valuations of $\mathrm{Frac}(A)$. In this step we show that, for any $\mathbf{s} \in \mathbb{R}^t_{>0}$,

$$v_{\mathbf{s}} = \mu_{P\mathbf{s}} \quad \text{and} \quad \varphi_* v_{\mathbf{s}} = \Lambda v_{\mathbf{s}}. \tag{6.14}$$

Let (g, Q) be (g_1, P) or (g_2, P'), write $g^\#(x_i) = e_i \prod_{j=1}^t (x_j')^{Q_{ij}}$ with $e_i \in \{u_i, u_i'\}$ the unit in (6.13), and let $\mathbf{q} := Q\mathbf{s}$. Since Q has entries in $\mathbb{N}$ and $\det Q \neq 0$, no row of Q is zero, so $\mathbf{q} \in \mathbb{R}^t_{>0}$. By [Sta, Tag 0C0S], there exist coefficient fields k_1 of $\widehat{\mathcal{O}}_{Y_1,\eta}$ and k_2 of $\widehat{\mathcal{O}}_{Y_2,\zeta}$ with

$$\widehat{\mathcal{O}}_{Y_1,\eta} = k_1[[x_1, \dots, x_t]] \quad \text{and} \quad \widehat{\mathcal{O}}_{Y_2,\zeta} = k_2[[x_1', \dots, x_t']].$$

Write $x^\gamma := \prod_{i=1}^t x_i^{\gamma_i}$ and $x'^{\gamma'} := \prod_{j=1}^t (x_j')^{\gamma_j'}$ for $\gamma, \gamma' \in \mathbb{N}^t$. For any non-zero $\xi' \in k_2[[x_1', \dots, x_t']]$, write $\xi' = \sum_{\gamma'} c'_{\gamma'} x'^{\gamma'}$ with $c'_{\gamma'} \in k_2$, and let

$$\nu_{\mathbf{s}}(\xi') := \min\{\langle \mathbf{s}, \gamma' \rangle \mid c'_{\gamma'} \neq 0\}.$$

The function $\nu_{\mathbf{s}}$ is a valuation on $k_2[[x_1', \dots, x_t']] \backslash \{0\}$. By [JM12, Section 3.1, Proposition 3.1], $v_{\mathbf{s}}(\xi') = \nu_{\mathbf{s}}(\xi')$ for any non-zero $\xi' \in \mathcal{O}_{Y_2,\zeta}$, and for any non-zero $\xi \in \mathcal{O}_{Y_1,\eta}$, written as

$$\xi = \sum_{\gamma} c_\gamma x^\gamma \quad \text{with } c_\gamma \in k_1,$$

$$\mu_{\mathbf{q}}(\xi) = \min\{\langle \mathbf{q}, \gamma \rangle \mid c_\gamma \neq 0\}.$$

Let

$$\widehat{g} : \widehat{\mathcal{O}}_{Y_1,\eta} \to \widehat{\mathcal{O}}_{Y_2,\zeta}$$

be the local homomorphism induced by $g^\#$.

Fix a non-zero $\xi = \sum_\gamma c_\gamma x^\gamma \in \mathcal{O}_{Y_1,\eta}$, let $\lambda_0 := \mu_{\mathbf{q}}(\xi)$, and let

$$\xi_0 := \sum_{\langle \mathbf{q}, \gamma \rangle = \lambda_0} c_\gamma x^\gamma.$$

Since $\mathbf{q} \in \mathbb{R}^t_{>0}$, the set

$$\{\gamma \in \mathbb{N}^t \mid \langle \mathbf{q}, \gamma \rangle \leq \lambda_0 + 1\}$$

is finite, so ξ_0 is a polynomial and there exists a real number $\lambda_1 > \lambda_0$ with $\langle \mathbf{q}, \gamma \rangle \geq \lambda_1$ for any γ with $c_\gamma \neq 0$ and $\langle \mathbf{q}, \gamma \rangle \neq \lambda_0$. For any $\gamma \in \mathbb{N}^t$ with $c_\gamma \neq 0$,

$$\widehat{g}(c_\gamma x^\gamma) = \widehat{g}(c_\gamma) \prod_{i=1}^t e_i^{\gamma_i} \cdot x'^{Q^T\gamma} \quad \text{and} \quad \langle \mathbf{s}, Q^T\gamma \rangle = \langle Q\mathbf{s}, \gamma \rangle = \langle \mathbf{q}, \gamma \rangle,$$

where $\widehat{g}(c_\gamma) \prod_i e_i^{\gamma_i}$ is a unit. For any such γ, let $\overline{c}_\gamma \in k_2 \backslash \{0\}$ be the residue of the unit $\widehat{g}(c_\gamma) \prod_i e_i^{\gamma_i}$, and let $\overline{c}_\gamma := 0$ if $c_\gamma = 0$. Any element of the maximal ideal of $k_2[[x_1', \dots, x_t']]$ has $\nu_{\mathbf{s}} \geq \min_j s_j > 0$. Write

$$\widehat{g}(\xi) = \sum_{\langle \mathbf{q}, \gamma \rangle = \lambda_0} \overline{c}_\gamma x'^{Q^T\gamma} + \xi_1',$$

where

$$\xi_1' := \widehat{g}(\xi - \xi_0) + \sum_{\langle \mathbf{q}, \gamma \rangle = \lambda_0} \Big(\widehat{g}(c_\gamma) \prod_i e_i^{\gamma_i} - \overline{c}_\gamma \Big) x'^{Q^T\gamma}.$$

The second sum has $\nu_{\mathbf{s}} \geq \lambda_0 + \min_j s_j$ since each coefficient is an element of the maximal ideal, and $\nu_{\mathbf{s}}(\widehat{g}(\xi - \xi_0)) \geq \lambda_1$ because $\widehat{g}$ maps any term $c_\gamma x^\gamma$ with $\langle \mathbf{q}, \gamma \rangle \geq \lambda_1$ into the ideal $\{0\} \cup \{\xi' \mid \nu_{\mathbf{s}}(\xi') \geq \lambda_1\}$, which is closed since it is generated by finitely many monomials. Thus

$$\nu_{\mathbf{s}}(\xi_1') \geq \min\{\lambda_1, \lambda_0 + \min_j s_j\} > \lambda_0.$$

Since $\det Q \neq 0$, the map $\gamma \to Q^T\gamma$ is injective, so the monomials $x'^{Q^T\gamma}$ with $\langle \mathbf{q}, \gamma \rangle = \lambda_0$ and $c_\gamma \neq 0$ are pairwise distinct, and $\langle \mathbf{s}, Q^T\gamma \rangle = \lambda_0$ for each of them. Hence

$$v_{\mathbf{s}}(g^{\#}(\xi)) = \nu_{\mathbf{s}}(\widehat{g}(\xi)) = \lambda_0 = \mu_{Q\mathbf{s}}(\xi).$$

Since $\mathrm{Frac}(A)$ is the fraction field of $\mathcal{O}_{Y_1,\eta}$, $v_{\mathbf{s}} \circ g^* = \mu_{Q\mathbf{s}}$ on $\mathrm{Frac}(A)^*$.

For $(g, Q) = (g_1, P)$, $g_1^* = \mathrm{id}_{\mathrm{Frac}(A)}$ implies $v_{\mathbf{s}} = \mu_{P\mathbf{s}}$. For $(g, Q) = (g_2, P')$, $g_2^* = \varphi$ and $P' = \Lambda P$ imply

$$\varphi_* v_{\mathbf{s}} = v_{\mathbf{s}} \circ g_2^* = \mu_{\Lambda P\mathbf{s}} = \Lambda \mu_{P\mathbf{s}} = \Lambda v_{\mathbf{s}}.$$

This proves (6.14).

Step 5. Let $\mathbf{s} := (1, \dots, 1)$, so that $v_{\mathbf{s}}$ is the order of vanishing along the exceptional prime divisor E over ζ on the normalized blow-up of Y_2 along the closure of ζ. Thus $w := v_{\mathbf{s}}$ is a primitive divisorial valuation of $\mathrm{Frac}(A)$, centered at $\pi_1(g_1(\zeta)) = \pi_1(\eta) = \mathfrak{m}_A$, and $\varphi_* w = \Lambda w$ by (6.14). The lemma follows. □

6.3. **The volume minimizer.**

Theorem 6.5 (cf. [Yos26, Proposition 5.9 and Theorem 5.1])**.** *Notation and conditions as in Set-up 4.8. Then there exists a primitive divisorial valuation w of* $\mathrm{Frac}(B)$ *centered at $\mathfrak{n}$ with $h_* w = \Lambda w$. Moreover, w is a primitive divisorial valuation of* $\mathrm{Frac}(A_0) = \mathrm{Frac}(B)$ *centered at $\mathfrak{n} \cap A_0$.*

Instead of the parameter spaces of [Blu18, Section 5] used in [Yos26, Proposition 5.9], we construct the minimizer of vol_B on $\mathcal{W}$ from intersections of valuation ideals with Lemma 6.2 and (6.19), and we prove its uniqueness with (6.19) and Theorem 1.3. The proof below follows [Yos26, Proposition 5.9 and proof of Theorem 5.1] (see Theorem 1.2), with Theorem 5.4 in place of [Xu20, Theorem 1.1].

Proof of Theorem 6.5. We first show that any primitive divisorial valuation w of $\mathrm{Frac}(B)$ centered at $\mathfrak{n}$ is a primitive divisorial valuation of $\mathrm{Frac}(A_0) = \mathrm{Frac}(B)$ centered at $\mathfrak{n} \cap A_0$. The valuation ring $\mathcal{O}_w$ of w is the local ring at the generic point of a prime divisor on a normal scheme proper over $\mathrm{Spec}\, B$, and Y is of finite type over $\mathrm{Spec}(A_0)_{\mathfrak{q}}$, so $\mathcal{O}_w = A'_{\mathfrak{p}}$ for $A' := A_0[f_1/f_0, \dots, f_k/f_0]$ with $f_0, \dots, f_k \in A_0$, $f_0 \neq 0$, and a prime ideal $\mathfrak{p}$ of A'. Since A_0 is excellent [JM12, §1.1], the normalization X' of the blow-up of $\mathrm{Spec}\, A_0$ along $(f_0, \dots, f_k)$ is projective birational over $\mathrm{Spec}\, A_0$ [Gro65, Scholie 7.8.3(vi)] and has a point ξ with $\mathcal{O}_{X',\xi} = \mathcal{O}_w$, so w is the order of vanishing along the closure of ξ, a prime divisor on X'. Since $A_0 \subset B$, $w \geq 0$ on $A_0 \backslash \{0\}$, and $w > 0$ exactly on $(\mathfrak{n} \cap A_0) \backslash \{0\}$. Thus it suffices to find a primitive divisorial valuation w of $\mathrm{Frac}(B)$ centered at $\mathfrak{n}$ with $h_* w = \Lambda w$.

We divide the proof into six steps.

Step 1. In this step we prove Theorem 6.5 when $r = 1$. Since B is a normal Noetherian local domain with $\dim B = 1$, B is a discrete valuation ring. Let ϖ be a uniformizer of B, and let w be the normalized discrete valuation of $\mathrm{Frac}(B)$ given by B. Then $h(B)$ is a discrete valuation ring with uniformizer $h(\varpi)$ and fraction field $h(\mathrm{Frac}(B))$. Since B is normal and finite over $h(B)$, B is the integral closure of $h(B)$ in $\mathrm{Frac}(B)$, and $\mathfrak{n}$ is the only prime ideal of B over $h(\mathfrak{n})$. Thus [Ser79, Chapter I, §4, Proposition 10] implies

$$[\mathrm{Frac}(B) : h(\mathrm{Frac}(B))] = w(h(\varpi))[B/\mathfrak{n} : h_{B/\mathfrak{n}}(B/\mathfrak{n})],$$

so Set-up 4.8(3) with $r = 1$ gives $w(h(\varpi)) = \Lambda$. For any $f \in \mathrm{Frac}(B)^*$, write $f = g\varpi^i$ with $g \in B^*$ and $i = w(f)$. Since $h(g) \in B^*$,

$$w(h(f)) = i \cdot w(h(\varpi)) = \Lambda w(f),$$

that is, $h_* w = \Lambda w$. The closed point of $\operatorname{Spec} B$ is a prime divisor on $\operatorname{Spec} B$ and $w(\mathrm{Frac}(B)^*) = \mathbb{Z}$, so w is a primitive divisorial valuation of $\mathrm{Frac}(B)$ centered at $\mathfrak{n}$ (Definition 2.3(3) with $\mathrm{id}_{\operatorname{Spec} B}$). Thus w satisfies Theorem 6.5. From now on we assume that $r \geq 2$.

Step 2. In this step we prove the estimates (6.15)–(6.19) on $\mathcal{W}$. Let $\mathfrak{b}_\bullet$, β, and $\mathcal{W}$ be as in Construction-Proposition 4.10, and let

$$\tau := \inf_{v \in \mathcal{W}} \mathrm{vol}_B(v).$$

By Set-up 4.8(1), B is a normal local domain essentially of finite type over $\mathbb{C}\{z_1, \dots, z_n\}$ for some n, so it is excellent and analytically irreducible (Lemma 2.2). By Set-up 4.8(2) and $r \geq 2$, B satisfies the hypotheses of Lemma 4.7 and Theorem 5.4.

(a) *Linear bounds.* Let $v \in \mathcal{W}$. Since $a(v, \operatorname{Spec} B) = 1$, Lemma 4.6 gives a real number $\kappa_v > 0$ with

$$\mathfrak{a}_t(v) \subset \mathfrak{n}^{\lceil t/\kappa_v \rceil} \quad \text{for any real number } t > 0.$$

With (6.5) for $\Sigma = \{v\}$, we obtain

$$\mathfrak{n}^{\lceil \beta t \rceil} \subset \mathfrak{b}_t \subset \mathfrak{a}_t(v) \subset \mathfrak{n}^{\lceil t/\kappa_v \rceil} \quad \text{for any real number } t > 0. \tag{6.15}$$

Since $e(\mathfrak{n}) > 0$, (6.15) implies

$$0 < \kappa_v^{-r} e(\mathfrak{n}) \leq \mathrm{vol}_B(v) \leq \beta^r e(\mathfrak{n}). \tag{6.16}$$

(b) *Integer indices.* By Lemma 2.9 with $X = \operatorname{Spec} B$, $x = \mathfrak{n}$, and the boundary 0, for any $\mathfrak{n}$-filtration $\mathfrak{f}_\bullet$ of B and any real valuation v of $\mathrm{Frac}(B)$ centered at $\mathfrak{n}$,

$$v(\mathfrak{f}_\bullet) = v((\mathfrak{f}_j)_{j \geq 1}) \quad \text{and} \quad \mathrm{lct}(B; \mathfrak{f}_\bullet) = \mathrm{lct}(B; (\mathfrak{f}_j)_{j \geq 1}). \tag{6.17}$$

(c) *Geodesics.* Let $v, v' \in \mathcal{W}$, let $0 < \sigma < 1$ be a real number, and let $\mathfrak{g}_\bullet$ be the geodesic between $\mathfrak{a}_\bullet(v)$ and $\mathfrak{a}_\bullet(v')$ at σ. We show that there exists $v'' \in \mathcal{W}$ with $\mathfrak{g}_t \subset \mathfrak{a}_t(v'')$ for any real number t. In particular,

$$e(\mathfrak{g}_\bullet) \geq \mathrm{vol}_B(v'') \geq \tau. \tag{6.18}$$

Let $\alpha' := \min\{\beta, \kappa_v^{-1}, \kappa_{v'}^{-1}\}$. If $t > 0$ and $(1-\sigma)t_0 + \sigma t_1 = t$, then $\max\{t_0, t_1\} \geq t$, so (6.15) implies

$$\mathfrak{a}_{t_0}(v) \cap \mathfrak{a}_{t_1}(v') \subset \mathfrak{n}^{\lceil \alpha' t \rceil}.$$

The term with $t_0 = t_1 = t$ contains $\mathfrak{b}_t$ by (6.15). Thus

$$\mathfrak{n}^{\lceil \beta t \rceil} \subset \mathfrak{b}_t \subset \mathfrak{g}_t \subset \mathfrak{n}^{\lceil \alpha' t \rceil} \quad \text{for any real number } t > 0.$$

Since $a(v, \operatorname{Spec} B) = 1$ and $v(\mathfrak{a}_\bullet(v)) \geq 1$, we have $\mathrm{lct}(B; \mathfrak{a}_\bullet(v)) \leq 1$. Likewise, $\mathrm{lct}(B; \mathfrak{a}_\bullet(v')) \leq 1$. By (6.15),

$$\mathfrak{n}^{\lceil \beta t \rceil} \subset \mathfrak{a}_t(v) \subset \mathfrak{n}^{\lceil \alpha' t \rceil} \quad \text{and} \quad \mathfrak{n}^{\lceil \beta t \rceil} \subset \mathfrak{a}_t(v') \subset \mathfrak{n}^{\lceil \alpha' t \rceil} \quad \text{for any real number } t > 0,$$

so Lemma 4.7 with α' in place of α implies $\mathrm{lct}(B; \mathfrak{g}_\bullet) \leq 1$. Since $\mathfrak{b}_\bullet \subset \mathfrak{g}_\bullet$, we have

$$\mathrm{lct}(B; \mathfrak{g}_\bullet) \geq \mathrm{lct}(B; \mathfrak{b}_\bullet) = 1.$$

Hence (6.17) implies

$$\mathrm{lct}(B; (\mathfrak{g}_j)_{j \geq 1}) = \mathrm{lct}(B; \mathfrak{g}_\bullet) = 1.$$

By Theorem 5.4 with α' in place of α, there exists a quasi-monomial valuation v'' of $\mathrm{Frac}(B)$ centered at $\mathfrak{n}$ with

$$a(v'', \operatorname{Spec} B) = 1 \quad \text{and} \quad v''((\mathfrak{g}_j)_{j \geq 1}) = 1.$$

By (6.17), $v''(\mathfrak{g}_\bullet) = 1$, so $\mathfrak{g}_t \subset \mathfrak{a}_t(v'')$ for any real number t. Since $\mathfrak{b}_\bullet \subset \mathfrak{g}_\bullet$, $v''(\mathfrak{b}_\bullet) \geq 1$. Since $\mathrm{lct}(B; \mathfrak{b}_\bullet) = 1$,

$$1 \leq a(v'', \operatorname{Spec} B)/v''(\mathfrak{b}_\bullet) = 1/v''(\mathfrak{b}_\bullet).$$

Thus $v''(\mathfrak{b}_\bullet) = 1$ and $v'' \in \mathcal{W}$, which proves (6.18).

(d) *Intersections.* Let $v, v' \in \mathcal{W}$, let $\mathfrak{a}_\bullet := \mathfrak{a}_\bullet(v)$ and $\mathfrak{c}_\bullet := \mathfrak{a}_\bullet(v')$, and let $\mathfrak{a}_\bullet \cap \mathfrak{c}_\bullet$ be the $\mathfrak{n}$-filtration $(\mathfrak{a}_t \cap \mathfrak{c}_t)_{t\in\mathbb{R}}$. Let $e_0 := e(\mathfrak{a}_\bullet)$, $e_1 := e(\mathfrak{c}_\bullet)$, and $e_\sigma := e(\mathfrak{g}_\bullet^\sigma)$ for any real number $0 < \sigma < 1$, where $\mathfrak{g}_\bullet^\sigma$ is the geodesic between $\mathfrak{a}_\bullet$ and $\mathfrak{c}_\bullet$ at σ. Let

$$\delta(\mathfrak{a}_\bullet, \mathfrak{c}_\bullet) := 2e(\mathfrak{a}_\bullet \cap \mathfrak{c}_\bullet) - e_0 - e_1.$$

We show that

$$e_0 + e_1 - 2e_{1/2} \geq 0 \quad \text{and} \quad \delta(\mathfrak{a}_\bullet, \mathfrak{c}_\bullet)^2 \leq 2\big(e_0 + e_1\big)\big(e_0 + e_1 - 2e_{1/2}\big). \tag{6.19}$$

By (6.16), e_0 and e_1 are positive and finite. Let x, y be the coordinates on $\mathbb{R}^2$, and for a Borel subset $J' \subset [0,1]$ let $\mathrm{Cone}(J') := \mathbb{R}_{\geq 0} \cdot \{(z, 1-z) \mid z \in J'\}$. By [BLQ24, Section 4], there exist an integer $N \geq 2$ and a Borel measure μ on $\mathbb{R}^2$ attached to $(\mathfrak{a}_\bullet, \mathfrak{c}_\bullet)$ satisfying the following.

(i) By [BLQ24, Proposition 4.7(2)],
$$e(\mathfrak{a}_\bullet \cap \mathfrak{c}_\bullet) = \mu(\{x < 1\} \cup \{y < 1\}).$$
(ii) $\operatorname{supp}\mu \subset \{x/N \leq y \leq Nx\}$, and $\mu(\lambda\Omega) = \lambda^r \mu(\Omega)$ for any real number $\lambda > 0$ and any Borel subset $\Omega \subset \mathbb{R}^2$ [BLQ24, Proposition 4.9].
(iii) The Borel measure $\widetilde{\mu}$ on $[0,1]$ given by $\widetilde{\mu}(J') := \mu(\mathrm{Cone}(J') \cap \{x + y < 1\})$ satisfies $\operatorname{supp}\widetilde{\mu} \subset J := [1/(N+1), N/(N+1)]$, and $\mu(\{0\}) = 0$ [BLQ24, Section 4.4].
(iv) By [BLQ24, Lemma 4.11],
$$e_\sigma = \int_J \big((1-\sigma)z + \sigma(1-z)\big)^{-r} d\widetilde{\mu}(z) \quad \text{for any real number } 0 \leq \sigma \leq 1.$$

Since $N \geq 2$, $\{x/N \leq y \leq Nx\} = \mathrm{Cone}(J)$, so (ii) implies $\operatorname{supp}\mu \subset \mathrm{Cone}(J)$.

Claim 6.6. *We have*
$$e(\mathfrak{a}_\bullet \cap \mathfrak{c}_\bullet) = \int_J \min\{z, 1-z\}^{-r} d\widetilde{\mu}(z).$$

Proof. By the homogeneity in (ii) and the monotone convergence theorem, for any bounded Borel function $\chi : J \to [0, +\infty)$,

$$\mu(\{s(z, 1-z) \mid z \in J,\ 0 \leq s < \chi(z)\}) = \int_J \chi^r d\widetilde{\mu}.$$

Since $\operatorname{supp}\mu \subset \mathrm{Cone}(J)$, and $s(z, 1-z) \in \{x < 1\} \cup \{y < 1\}$ for $z \in J$ and $s \geq 0$ if and only if $s < \min\{z, 1-z\}^{-1}$, (i) implies the claim. □

For $z \in J$, let $\psi_0(z) := z^{-r}$ and $\psi_1(z) := (1-z)^{-r}$. Then $\min\{z, 1-z\}^{-r} = \max\{\psi_0(z), \psi_1(z)\}$. By (iv) with $\sigma \in \{0, \frac{1}{2}, 1\}$ and Claim 6.6,

$$e_0 + e_1 = \int_J (\psi_0 + \psi_1) d\widetilde{\mu}, \qquad e_{1/2} = 2^r \widetilde{\mu}(J), \quad \text{and}$$
$$\delta(\mathfrak{a}_\bullet, \mathfrak{c}_\bullet) = \int_J \big(2\max\{\psi_0, \psi_1\} - \psi_0 - \psi_1\big) d\widetilde{\mu} = \int_J |\psi_0 - \psi_1| d\widetilde{\mu},$$

and $\widetilde{\mu}(J) \leq e_0 < +\infty$ because $\psi_0 \geq 1$. For $z \in J$, we have $\psi_0(z)\psi_1(z) = (z(1-z))^{-r} \geq 4^r$, so $\psi_0 + \psi_1 \geq 2(\psi_0\psi_1)^{1/2} \geq 2^{r+1}$, and

$$\begin{aligned}\frac{(\psi_0 - \psi_1)^2}{\psi_0 + \psi_1} &= \psi_0 + \psi_1 - \frac{4\psi_0\psi_1}{\psi_0 + \psi_1} \leq \psi_0 + \psi_1 - \frac{4^{r+1}}{\psi_0 + \psi_1} \\ &= \frac{(\psi_0 + \psi_1 - 2^{r+1})(\psi_0 + \psi_1 + 2^{r+1})}{\psi_0 + \psi_1} \leq 2(\psi_0 + \psi_1 - 2^{r+1}).\end{aligned}$$

Thus

$$e_0 + e_1 - 2e_{1/2} = \int_J (\psi_0 + \psi_1 - 2^{r+1}) d\widetilde{\mu} \geq 0,$$

and the Cauchy–Schwarz inequality implies

$$\delta(\mathfrak{a}_\bullet, \mathfrak{c}_\bullet)^2 \leq \int_J \frac{(\psi_0 - \psi_1)^2}{\psi_0 + \psi_1} d\widetilde{\mu} \cdot \int_J (\psi_0 + \psi_1) d\widetilde{\mu} \leq 2\big(e_0 + e_1 - 2e_{1/2}\big)\big(e_0 + e_1\big).$$

This proves (6.19).

Step 3. In this step we construct a quasi-monomial valuation $v_0 \in \mathcal{W}$ with $\mathrm{vol}_B(v_0) = \tau$. For any integer $i \geq 1$, let v_i be an element of $\mathcal{W}$ with $\mathrm{vol}_B(v_i) \leq \tau + 4^{-i}$, and let $\mathfrak{a}_\bullet^i := \mathfrak{a}_\bullet(v_i)$. Let

$$\gamma := \big(8\beta^r e(\mathfrak{n})\big)^{1/2}.$$

With e_σ as in **Step 2**(d) for $(v, v') = (v_i, v_{i+1})$, we have $e_0 + e_1 \leq 2\beta^r e(\mathfrak{n})$ by (6.16), and (6.18) implies

$$e_0 + e_1 - 2e_{1/2} \leq (\tau + 4^{-i}) + (\tau + 4^{-i-1}) - 2\tau \leq 2 \cdot 4^{-i}.$$

Thus (6.19) implies

$$\delta(\mathfrak{a}_\bullet^i, \mathfrak{a}_\bullet^{i+1}) \leq \gamma 2^{-i} \quad \text{for any integer } i \geq 1. \tag{6.20}$$

For integers $m \geq k \geq 1$, let $\mathfrak{d}_\bullet^{k,m}$ and $\mathfrak{d}_\bullet^k$ be the $\mathfrak{n}$-filtrations with

$$\mathfrak{d}_t^{k,m} := \bigcap_{i=k}^{m} \mathfrak{a}_t^i \quad \text{and} \quad \mathfrak{d}_t^k := \bigcap_{i \geq k} \mathfrak{a}_t^i,$$

that is, $\mathfrak{d}_\bullet^\Sigma$ of Lemma 6.2 with $\Sigma = \{v_k, \dots, v_m\}$ and $\Sigma = \{v_i \mid i \geq k\}$ respectively. Let $m > k$ and $t > 0$. Since

$$\mathfrak{d}_t^{k,m} = \mathfrak{d}_t^{k,m-1} \cap \mathfrak{a}_t^m \quad \text{and} \quad \mathfrak{d}_t^{k,m-1} \subset \mathfrak{a}_t^{m-1},$$

the homomorphism

$$\mathfrak{d}_t^{k,m-1}/\mathfrak{d}_t^{k,m} \to \mathfrak{a}_t^{m-1}/(\mathfrak{a}_t^{m-1} \cap \mathfrak{a}_t^m)$$

is injective, so

$$\ell_B(B/\mathfrak{d}_t^{k,m}) - \ell_B(B/\mathfrak{d}_t^{k,m-1}) \leq \ell_B(B/(\mathfrak{a}_t^{m-1} \cap \mathfrak{a}_t^m)) - \ell_B(B/\mathfrak{a}_t^{m-1}).$$

Multiplying by $r!/t^r$ and letting $t \to +\infty$, we obtain

$$e(\mathfrak{d}_\bullet^{k,m}) - e(\mathfrak{d}_\bullet^{k,m-1}) \leq e(\mathfrak{a}_\bullet^{m-1} \cap \mathfrak{a}_\bullet^m) - e(\mathfrak{a}_\bullet^{m-1}) \leq \delta(\mathfrak{a}_\bullet^{m-1}, \mathfrak{a}_\bullet^m),$$

where the second inequality holds because

$$e(\mathfrak{a}_\bullet^{m-1} \cap \mathfrak{a}_\bullet^m) \geq e(\mathfrak{a}_\bullet^m).$$

Taking the sum over $m' = k+1, \dots, m$ in place of m, and using $\mathfrak{d}_\bullet^{k,k} = \mathfrak{a}_\bullet^k$, $e(\mathfrak{a}_\bullet^k) = \mathrm{vol}_B(v_k) \leq \tau + 4^{-k}$, and (6.20), we obtain

$$e(\mathfrak{d}_\bullet^{k,m}) \leq e(\mathfrak{a}_\bullet^k) + \sum_{i=k}^{m-1} \delta(\mathfrak{a}_\bullet^i, \mathfrak{a}_\bullet^{i+1}) \leq \tau + 4^{-k} + \gamma 2^{1-k}. \tag{6.21}$$

Fix integers $k, j \geq 1$. By (6.15), $\mathfrak{n}^{\lceil \beta j \rceil} \subset \mathfrak{d}_j^{k,m}$ for any $m \geq k$, and $\mathfrak{d}_j^{k,m+1} \subset \mathfrak{d}_j^{k,m}$. Since $B/\mathfrak{n}^{\lceil \beta j \rceil}$ is Artinian, $\mathfrak{d}_j^{k,m} = \mathfrak{d}_j^k$ for any $m \gg 0$. For such m, Lemma 6.2 and (6.21) imply

$$e(\mathfrak{d}_\bullet^k) \leq \frac{e(\mathfrak{d}_j^k)}{j^r} = \frac{e(\mathfrak{d}_j^{k,m})}{j^r} \leq e(\mathfrak{d}_\bullet^{k,m}) + \epsilon_j \leq \tau + 4^{-k} + \gamma 2^{1-k} + \epsilon_j.$$

Letting $j \to +\infty$, we obtain

$$e(\mathfrak{d}_\bullet^k) \leq \tau + 4^{-k} + \gamma 2^{1-k} \quad \text{for any integer } k \geq 1. \tag{6.22}$$

For any integer $j \geq 1$, let $\mathfrak{d}_j := \bigcup_{k\geq 1} \mathfrak{d}_j^k$. Since $\mathfrak{d}_j^k \subset \mathfrak{d}_j^{k+1}$ and B is Noetherian, there exists an integer $k_j \geq 1$ with $\mathfrak{d}_j = \mathfrak{d}_j^k$ for any $k \geq k_j$. For integers $j, j' \geq 1$ and $k \geq \max\{k_j, k_{j'}\}$, we have

$$\mathfrak{d}_j\mathfrak{d}_{j'} = \mathfrak{d}_j^k\mathfrak{d}_{j'}^k \subset \mathfrak{d}_{j+j'}^k \subset \mathfrak{d}_{j+j'}.$$

By (6.15),

$$\mathfrak{n}^{\lceil \beta j\rceil} \subset \mathfrak{b}_j \subset \mathfrak{d}_j.$$

Thus $(\mathfrak{d}_j)_{j\geq 1}$ is a graded sequence of $\mathfrak{n}$-primary ideals.

Since $\mathcal{W}$ is compact for the topology of pointwise convergence (Construction-Proposition 4.10(3)), the sequence $(v_i)_{i\geq 1}$ has a cluster point $v_\infty \in \mathcal{W}$. For any $f \in \mathfrak{d}_j$, $v_i(f) \geq j$ for any $i \geq k_j$, so $v_\infty(f) \geq j$. Thus (6.15) implies

$$\mathfrak{n}^{\lceil \beta j\rceil} \subset \mathfrak{b}_j \subset \mathfrak{d}_j \subset \mathfrak{a}_j(v_\infty) \subset \mathfrak{n}^{\lceil j/\kappa_{v_\infty}\rceil} \quad \text{for any integer } j \geq 1.$$

Since $\mathfrak{b}_j \subset \mathfrak{d}_j$, (6.17) implies

$$\operatorname{lct}(B; (\mathfrak{d}_j)_{j\geq 1}) \geq \operatorname{lct}(B; (\mathfrak{b}_j)_{j\geq 1}) = \operatorname{lct}(B; \mathfrak{b}_\bullet) = 1.$$

Since $v_\infty(\mathfrak{d}_j) \geq j$ and $a(v_\infty, \operatorname{Spec} B) = 1$, $\operatorname{lct}(B; (\mathfrak{d}_j)_{j\geq 1}) \leq 1$. Thus $\operatorname{lct}(B; (\mathfrak{d}_j)_{j\geq 1}) = 1$. By Theorem 5.4 with $\min\{\beta, \kappa_{v_\infty}^{-1}\}$ in place of α, there exists a quasi-monomial valuation v_0 of $\operatorname{Frac}(B)$ centered at $\mathfrak{n}$ with

$$a(v_0, \operatorname{Spec} B) = 1 \quad \text{and} \quad v_0((\mathfrak{d}_j)_{j\geq 1}) = 1.$$

Thus $\mathfrak{d}_j \subset \mathfrak{a}_j(v_0)$ for any integer $j \geq 1$. By (6.17),

$$v_0(\mathfrak{b}_\bullet) = v_0((\mathfrak{b}_j)_{j\geq 1}) \geq v_0((\mathfrak{d}_j)_{j\geq 1}) = 1,$$

and

$$v_0(\mathfrak{b}_\bullet) \leq a(v_0, \operatorname{Spec} B)/\operatorname{lct}(B; \mathfrak{b}_\bullet) = 1,$$

so $v_0 \in \mathcal{W}$. For any integer $k \geq 1$,

$$\mathfrak{d}_j^k \subset \mathfrak{d}_j \subset \mathfrak{a}_j(v_0) \quad \text{for any integer } j \geq 1,$$

so (6.22) implies

$$\operatorname{vol}_B(v_0) = \lim_{j\to+\infty} \frac{r!\ell_B(B/\mathfrak{a}_j(v_0))}{j^r} \leq \lim_{j\to+\infty} \frac{r!\ell_B(B/\mathfrak{d}_j^k)}{j^r} = e(\mathfrak{d}_\bullet^k) \leq \tau + 4^{-k} + \gamma 2^{1-k}.$$

Letting $k \to +\infty$, we obtain $\operatorname{vol}_B(v_0) \leq \tau$. Since $v_0 \in \mathcal{W}$, $\operatorname{vol}_B(v_0) = \tau$.

Step 4. In this step we show that v_0 is the only $v' \in \mathcal{W}$ with $\operatorname{vol}_B(v') = \tau$. Let $v' \in \mathcal{W}$ with $\operatorname{vol}_B(v') = \tau$, let $\mathfrak{a}_\bullet := \mathfrak{a}_\bullet(v_0)$ and $\mathfrak{c}_\bullet := \mathfrak{a}_\bullet(v')$, and let e_σ be as in **Step 2**(d). Then $e_0 = e_1 = \tau$, and $e_{1/2} \geq \tau$ by (6.18). By (6.19), $e_{1/2} \leq (e_0 + e_1)/2 = \tau$, so $e_0 + e_1 - 2e_{1/2} = 0$ and $\delta(\mathfrak{a}_\bullet, \mathfrak{c}_\bullet) = 0$. Thus

$$e(\mathfrak{a}_\bullet \cap \mathfrak{c}_\bullet) = e_0 = e_1 = \tau.$$

Since $\operatorname{vol}_B(v') > 0$ by (6.16) and

$$\mathfrak{a}_\bullet \cap \mathfrak{c}_\bullet \subset \mathfrak{a}_\bullet,$$

Theorem 1.3 implies

$$v'(\mathfrak{a}_\bullet) = v'(\mathfrak{a}_\bullet \cap \mathfrak{c}_\bullet) \geq v'(\mathfrak{c}_\bullet) \geq 1.$$

Thus $v'(\mathfrak{a}_t) \geq t$ for any real number $t > 0$. For any $f \in B\backslash\{0\}$, if $v_0(f) > 0$, then $f \in \mathfrak{a}_{v_0(f)}$, so $v'(f) \geq v_0(f)$. If $v_0(f) = 0$, then $v'(f) \geq 0 = v_0(f)$ since v' is centered at $\mathfrak{n}$. Hence $v' \geq v_0$ on B. Exchanging v_0 and v', and using $\operatorname{vol}_B(v_0) > 0$ and

$$\mathfrak{a}_\bullet \cap \mathfrak{c}_\bullet \subset \mathfrak{c}_\bullet,$$

we obtain $v_0 \geq v'$ on B. Thus $v' = v_0$ on B, hence on $\operatorname{Frac}(B)$.

Step 5. In this step we show that $h_*v_0 = \Lambda v_0$. By [EP05, Theorem 3.1.2], the valuation $h(f) \to v_0(f)$ of the subfield $h(\operatorname{Frac}(B))$ extends to a valuation u of $\operatorname{Frac}(B)$. Since $[\operatorname{Frac}(B) :$

$h(\mathrm{Frac}(B))] < +\infty$, the ramification index of u over $u|_{h(\mathrm{Frac}(B))}$ is finite [EP05, Theorem 3.3.4], so u is a real valuation. Then

$$h_*u = v_0,$$

and u is trivial on $\mathbb{C}^*$ since h is a $\mathbb{C}$-algebra homomorphism. Since B is local and integral over $h(B)$, u is centered at $\mathfrak{n}$. By Construction-Proposition 4.10(3), $\Lambda u \in \mathcal{W}$. By Set-up 4.8(3), Lemma 3.3(1) with $(S, \varphi, \lambda) = (B, h, \Lambda)$ for the real valuation Λu implies

$$\tau = \mathrm{vol}_B(v_0) = \mathrm{vol}_B(\Lambda^{-1}h_*(\Lambda u)) \geq \mathrm{vol}_B(\Lambda u) \geq \tau.$$

Thus $\mathrm{vol}_B(\Lambda u) = \tau$, and **Step 4** implies $\Lambda u = v_0$. Hence

$$h_*v_0 = h_*(\Lambda u) = \Lambda h_* u = \Lambda v_0.$$

Step 6. In this step we conclude. By **Step 3**, v_0 is a quasi-monomial valuation of $\mathrm{Frac}(B)$ centered at $\mathfrak{n}$, and $h_*v_0 = \Lambda v_0$ by **Step 5**, where Λ is an integer. By Set-up 4.8(3) and **Step 2**, B and h satisfy the hypotheses of Lemma 6.4, so Lemma 6.4 with $(A, \varphi, v) = (B, h, v_0)$ implies that there exists a primitive divisorial valuation w of $\mathrm{Frac}(B)$ centered at $\mathfrak{n}$ with $h_*w = \Lambda w$. □

7. A log discrepancy for local rings of klt type

A local ring A of klt type carries klt boundaries Δ, but for a finite endomorphism φ of A and a real valuation v of $\mathrm{Frac}(A)$ with $v \geq 0$ on $A\backslash\{0\}$ we have

$$a(\varphi_*v, \mathrm{Spec}\, A, \Delta) = a(v, \mathrm{Spec}\, A, \Delta) + v(R(\varphi) + \Delta - \varphi^*\Delta)$$

(Lemma 3.2), and the last term need not be non-negative. In this section we define a log discrepancy $a_{\mathrm{can}}(\cdot, A)$ which depends only on the ring A, changes under a finite endomorphism of A by a non-negative ramification term, and is bounded below by the log discrepancy of any klt boundary and above by a constant multiple of it (Definition-Theorem 7.3).

Set-up 7.1. Let A_0, ψ, $\mathfrak{q}$, A, k, d, φ, D, δ, and Δ be as follows.

(1) A_0 is a normal complex analytic local domain, $\psi : A_0 \to A_0$ is a finite injective local $\mathbb{C}$-algebra homomorphism, and $\mathfrak{q}$ is a prime ideal of A_0 with $\psi^{-1}(\mathfrak{q}) = \mathfrak{q}$ and $\sqrt{\psi(\mathfrak{q})A_0} = \mathfrak{q}$.
(2) $A := (A_0)_{\mathfrak{q}}$, $k := A/\mathfrak{m}_A$, and $d := \dim A \geq 1$.
(3) $\varphi : A \to A$ is the finite injective local homomorphism induced by ψ, $D := [\mathrm{Frac}(A) : \varphi(\mathrm{Frac}(A))]$, and $\delta := [k : \varphi_k(k)]$.
(4) $\Delta \geq 0$ is a $\mathbb{Q}$-divisor on $\mathrm{Spec}\, A$ such that $K_A + \Delta$ is $\mathbb{Q}$-Cartier and $(\mathrm{Spec}\, A, \Delta)$ is klt.

Let A be as in Set-up 7.1. By Lemma 2.2, A_0 and A are excellent and admit dualizing complexes, and the homomorphism φ of Set-up 7.1(3) is finite by Lemma 2.11 for ψ and $\mathfrak{q}$. We let K_A be a canonical divisor on $\mathrm{Spec}\, A$, and for a normal integral scheme Y with a proper birational morphism $\pi : Y \to \mathrm{Spec}\, A$ we let K_Y be the canonical divisor on Y with $\pi_*K_Y = K_A$.

Lemma 7.2. *Let Y be a separated regular integral excellent scheme over $\mathbb{Q}$, let $E_1, \ldots, E_n$ be distinct prime divisors on Y such that $\sum_{i=1}^n E_i$ is an snc divisor, and let $b_1, \ldots, b_n, c_1, \ldots, c_n$ be real numbers with $b_i < 1$ for any i. Let*

$$\gamma := \max\left\{1, \frac{1-c_1}{1-b_1}, \ldots, \frac{1-c_n}{1-b_n}\right\}.$$

Then any real valuation v of the function field of Y with a center on Y satisfies

$$a(v, Y) - \sum_{i=1}^n c_i v(E_i) \leq \gamma\Big(a(v, Y) - \sum_{i=1}^n b_i v(E_i)\Big),$$

where both sides may be $+\infty$.

Proof. By Definition-Lemma 2.7, $a(v,Y) \geq \sum_{i=1}^n v(E_i)$. Since $v(E_i) \geq 0$ and $1 - c_i \leq \gamma(1-b_i)$ for any i, and since $\gamma \geq 1$,

$$\begin{aligned} a(v,Y) - \sum_{i=1}^n c_i v(E_i) &= \Big(a(v,Y) - \sum_{i=1}^n v(E_i)\Big) + \sum_{i=1}^n (1-c_i)v(E_i) \\ &\leq \gamma\Big(a(v,Y) - \sum_{i=1}^n v(E_i)\Big) + \gamma \sum_{i=1}^n (1-b_i)v(E_i). \end{aligned}$$

The right-hand side is $\gamma(a(v,Y) - \sum_{i=1}^n b_i v(E_i))$. The lemma follows. □

In the rest of this section, v denotes a real valuation of $\mathrm{Frac}(A)$ with $v \geq 0$ on $A\backslash\{0\}$. The set of such valuations carries the topology of pointwise convergence on $\mathrm{Frac}(A)^*$, and any such v has a center on any normal integral scheme Y with a proper birational morphism $Y \to \operatorname{Spec} A$ (Definition 2.3(1)). For a non-zero finitely generated A-submodule M of $\mathrm{Frac}(A)$ we let

$$v(M) := \min\{v(f) \mid f \in M\backslash\{0\}\};$$

since $v \geq 0$ on A, the minimum is attained at one of finitely many generators of M, and $v(MM') = v(M) + v(M')$ for two such submodules M and M'. For a Weil divisor L on $\operatorname{Spec} A$ we let

$$\mathcal{O}_A(L) := \{f \in \mathrm{Frac}(A)^* \mid \operatorname{div}(f) + L \geq 0\} \cup \{0\},$$

a non-zero finitely generated A-submodule of $\mathrm{Frac}(A)$.

Following [BdFF12, Section 1], a $\mathbb{Q}$-*Cartier* $\boldsymbol{b}$-*divisor* $\mathbf{L}$ over $\operatorname{Spec} A$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $\mathbf{L}_Y$ on a normal integral scheme Y with a proper birational morphism $Y \to \operatorname{Spec} A$, together with the pullbacks $\mathbf{L}_{Y'}$ of $\mathbf{L}_Y$ to all normal integral schemes Y' with a proper birational morphism $Y' \to Y$. We say that $\mathbf{L}$ is *determined on* Y, and we call $\mathbf{L}_{Y'}$ the *trace* of $\mathbf{L}$ on Y'. The value $v(\mathbf{L}) := v(\mathbf{L}_Y)$ (Definition 2.3(1)) does not depend on Y. For $\mathbb{Q}$-Cartier $\boldsymbol{b}$-divisors $\mathbf{L}$ and $\mathbf{L}'$ we write $\mathbf{L} \leq \mathbf{L}'$ if $v(\mathbf{L}) \leq v(\mathbf{L}')$ for any v, and we regard a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on $\operatorname{Spec} A$ as the $\mathbb{Q}$-Cartier $\boldsymbol{b}$-divisor determined on $\operatorname{Spec} A$. If $\mathbf{L}$ and $\mathbf{L}'$ are determined on Y_1 and Y_2, then $\mathbf{L} \pm \mathbf{L}'$ is the $\mathbb{Q}$-Cartier $\boldsymbol{b}$-divisor determined by $\mathbf{L}_W \pm \mathbf{L}'_W$ for a normal integral scheme W with proper birational morphisms $W \to Y_1$ and $W \to Y_2$ over $\operatorname{Spec} A$, and $v(\mathbf{L} \pm \mathbf{L}') = v(\mathbf{L}) \pm v(\mathbf{L}')$.

Definition-Theorem 7.3. Notation and conditions as in Set-up 7.1.

(1) For any Weil divisor L on $\operatorname{Spec} A$, the A-algebra $\bigoplus_{j\geq 0} \mathcal{O}_A(jL)$ is finitely generated, and there exists a $\mathbb{Q}$-Cartier $\boldsymbol{b}$-divisor $\mathrm{Env}_A(L)$ over $\operatorname{Spec} A$ whose value at any real valuation v of $\mathrm{Frac}(A)$ with $v \geq 0$ on $A\backslash\{0\}$ is

$$\mathrm{Env}_A(L)(v) := -\inf_{j\geq 1} \frac{v(\mathcal{O}_A(jL))}{j}.$$

(2) For a real valuation v of $\mathrm{Frac}(A)$ with $v \geq 0$ on $A\backslash\{0\}$, let

$$a_{\mathrm{can}}(v,A) := a(v,Y) + v(K_Y + \mathrm{Env}_A(-K_A)_Y),$$

where $Y \to \operatorname{Spec} A$ is a log resolution of $\operatorname{Spec} A$ on which $\mathrm{Env}_A(-K_A)$ is determined, with trace $\mathrm{Env}_A(-K_A)_Y$ on Y. Then $a_{\mathrm{can}}(v,A)$ does not depend on Y, the function $a_{\mathrm{can}}(\cdot,A)$ is lower semicontinuous for the topology of pointwise convergence, and for any $\mathbb{Q}$-divisor $\Delta' \geq 0$ on $\operatorname{Spec} A$ such that $K_A + \Delta'$ is $\mathbb{Q}$-Cartier and $(\operatorname{Spec} A, \Delta')$ is klt, there exists a real number $\gamma \geq 1$ with

$$a(v, \operatorname{Spec} A, \Delta') \leq a_{\mathrm{can}}(v,A) \leq \gamma a(v, \operatorname{Spec} A, \Delta')$$

for any such v.

(3) Let $R(\varphi)$ be the ramification divisor of $\operatorname{Spec} \varphi$, and let

$$\mathbf{R}_\varphi := \mathrm{Env}_A(-K_A + R(\varphi)) - \mathrm{Env}_A(-K_A).$$

Then $\mathbf{R}_\varphi$ is a $\mathbb{Q}$-Cartier $\boldsymbol{b}$-divisor, there exists an integer $m_\varphi \geq 1$ with

$$0 \leq v(\mathbf{R}_\varphi) \leq m_\varphi^{-1} v(\mathcal{O}_A(-m_\varphi R(\varphi)))$$

for any real valuation v of $\mathrm{Frac}(A)$ with $v \geq 0$ on $A\backslash\{0\}$, and

$$a_{\mathrm{can}}(\varphi_* v, A) = a_{\mathrm{can}}(v, A) + v(\mathbf{R}_\varphi) \quad \text{in } (-\infty, +\infty]$$

for any such v.

Proof. We divide the proof into six steps.

Step 1. Let L and L' be Weil divisors on $\operatorname{Spec} A$. Let $x \in A\backslash\{0\}$ with $\operatorname{div}(x) \geq L$ and $\operatorname{div}(x) \geq -L$. For any $j \geq 1$ we have $x^j \in \mathcal{O}_A(jL)$, and $x^j f \in A$ for any $f \in \mathcal{O}_A(jL)$ since A is normal. Thus

$$-jv(x) \leq v(\mathcal{O}_A(jL)) \leq jv(x).$$

Since $\mathcal{O}_A(iL)\mathcal{O}_A(jL) \subset \mathcal{O}_A((i+j)L)$ for any $i, j \geq 1$, the sequence $(v(\mathcal{O}_A(jL)))_{j\geq 1}$ is subadditive, so by Fekete's lemma

$$\mathrm{Env}_A(L)(v) = -\lim_{j\to+\infty} \frac{v(\mathcal{O}_A(jL))}{j} \in [-v(x), v(x)].$$

For any $j \geq 1$, any $g \in \mathrm{Frac}(A)^*$ and any integer $N \geq 1$, we have

$$\mathcal{O}_A(jL)\mathcal{O}_A(jL') \subset \mathcal{O}_A(j(L+L')), \qquad \mathcal{O}_A(j(L+\operatorname{div}(g))) = g^{-j}\mathcal{O}_A(jL),$$

and $1 \in \mathcal{O}_A(jL)$ if $L \geq 0$. Taking the limit along $(jN)_{j\geq 1}$ and along $(j)_{j\geq 1}$, we obtain, for any v,

$$\begin{aligned} &\mathrm{Env}_A(L+L')(v) \geq \mathrm{Env}_A(L)(v) + \mathrm{Env}_A(L')(v), \qquad \mathrm{Env}_A(NL)(v) = N\,\mathrm{Env}_A(L)(v), \\ &\mathrm{Env}_A(L+\operatorname{div}(g))(v) = \mathrm{Env}_A(L)(v) + v(g), \qquad \mathrm{Env}_A(L)(v) \geq 0 \text{ if } L \geq 0. \end{aligned} \tag{7.1}$$

Step 2. We prove that $\bigoplus_{j\geq 0} \mathcal{O}_A(jL)$ is finitely generated. For $g \in \mathrm{Frac}(A)^*$, multiplication by g^j in degree j is an isomorphism of graded A-algebras from $\bigoplus_{j\geq 0} \mathcal{O}_A(j(L+\operatorname{div}(g)))$ onto $\bigoplus_{j\geq 0} \mathcal{O}_A(jL)$. Possibly replacing L with $L + \operatorname{div}(x)$ for x as in **Step 1**, we may assume that $L \geq 0$.

By [LM22, Corollary 22.3], there exists a small birational morphism $\pi : W \to \operatorname{Spec} A$ from a normal $\mathbb{Q}$-factorial integral scheme W. Let $\Delta_W := \pi_*^{-1}\Delta$ and $L_W := \pi_*^{-1}L$. Since π is small, $K_W + \Delta_W = \pi^*(K_A + \Delta)$, so (W, Δ_W) is klt, and $L_W \geq 0$ is $\mathbb{Q}$-Cartier. Let $\epsilon > 0$ be a rational number such that $(W, \Delta_W + \epsilon L_W)$ is klt, and let $N_\Delta \geq 1$ be an integer such that $N_\Delta(K_A + \Delta)$ is Cartier and $m := N_\Delta \epsilon$ is an integer. Since π is birational, $\Delta_W + \epsilon L_W$ is big over $\operatorname{Spec} A$, so the A-algebra

$$\bigoplus_{j\geq 0} \pi_* \mathcal{O}_W(jN_\Delta(K_W + \Delta_W + \epsilon L_W))$$

is finitely generated [LM22, Theorem B]. Since A is local, $N_\Delta(K_A + \Delta) = \operatorname{div}(g_\Delta)$ for some $g_\Delta \in \mathrm{Frac}(A)^*$, and since π is small,

$$\pi_* \mathcal{O}_W(jN_\Delta(K_W + \Delta_W + \epsilon L_W)) = \pi_* \mathcal{O}_W(\pi^* \operatorname{div}(g_\Delta^j) + jmL_W) = g_\Delta^{-j}\mathcal{O}_A(jmL)$$

for any $j \geq 0$. Thus the A-algebra $\bigoplus_{j\geq 0} \mathcal{O}_A(jmL)$ is finitely generated.

Let τ be an indeterminate, and let

$$\mathcal{R}_L := \bigoplus_{j\geq 0} \mathcal{O}_A(jL)\tau^j \subset \mathrm{Frac}(A)[\tau], \qquad \mathcal{R}_L^{(m)} := \bigoplus_{j\geq 0} \mathcal{O}_A(jmL)\tau^{jm}.$$

Since $y^m \in \mathcal{R}_L^{(m)}$ for any homogeneous $y \in \mathcal{R}_L$, the ring $\mathcal{R}_L$ is integral over $\mathcal{R}_L^{(m)}$. Since $L \geq 0$, the fraction field of $\mathcal{R}_L^{(m)}$ is $\mathrm{Frac}(A)(\tau^m)$, over which $\mathrm{Frac}(A)(\tau)$ is finite. The finitely generated A-algebra $\mathcal{R}_L^{(m)}$ is excellent, hence universally Japanese [Gro65, Scholie 7.8.3(ii)(vi)], so the integral closure of $\mathcal{R}_L^{(m)}$ in $\mathrm{Frac}(A)(\tau)$ is a finite $\mathcal{R}_L^{(m)}$-module. Since $\mathcal{R}_L$ is contained in the

integral closure and $\mathcal{R}_L^{(m)}$ is Noetherian, $\mathcal{R}_L$ is a finite $\mathcal{R}_L^{(m)}$-module, hence a finitely generated A-algebra.

Step 3. We prove that $\mathrm{Env}_A(L)$ is a $\mathbb{Q}$-Cartier $\boldsymbol{b}$-divisor. By **Step 2** there exists an integer $n_L \geq 1$ such that the n_L-th Veronese subalgebra of $\bigoplus_{j\geq 0}\mathcal{O}_A(jL)$ is generated in degree 1, that is, $\mathcal{O}_A(jn_LL) = \mathcal{O}_A(n_LL)^j$ for any $j \geq 1$. Thus $v(\mathcal{O}_A(jn_LL)) = jv(\mathcal{O}_A(n_LL))$, and by **Step 1**,

$$\mathrm{Env}_A(L)(v) = -\frac{v(\mathcal{O}_A(n_LL))}{n_L} \quad \text{for any } v. \tag{7.2}$$

Let $x_L \in A\backslash\{0\}$ with $I := x_L\mathcal{O}_A(n_LL) \subset A$. Let $\pi_L : Y_L \to \operatorname{Spec} A$ be the normalization of the blow-up of $\operatorname{Spec} A$ along I, which is finite over the blow-up since A is excellent [Gro65, Scholie 7.8.3(vi)], and let $F \geq 0$ be the Cartier divisor on Y_L with $I\mathcal{O}_{Y_L} = \mathcal{O}_{Y_L}(-F)$. For any v with center y on Y_L, the ideal $I\mathcal{O}_{Y_L,y}$ is generated by a local equation of F at y, and also by the images of the generators of I, so $v(I) = v(F)$. By (7.2),

$$\mathrm{Env}_A(L)(v) = \frac{v(x_L) - v(I)}{n_L} = v\big(\tfrac{1}{n_L}(\pi_L^*\operatorname{div}(x_L) - F)\big).$$

Thus $\mathrm{Env}_A(L)$ is the $\mathbb{Q}$-Cartier $\boldsymbol{b}$-divisor determined on Y_L by $\frac{1}{n_L}(\pi_L^*\operatorname{div}(x_L) - F)$. The $\mathbb{Q}$-Cartier $\boldsymbol{b}$-divisor $\mathrm{Env}_A(L)$ is the nef envelope of L in the sense of [BdFF12, Definition 2.3]. This proves (1).

Step 4. We prove (2). For a regular integral scheme Y with a proper birational morphism $Y \to \operatorname{Spec} A$, let

$$\beta_Y(v) := a(v, Y) + v(K_Y) \in (-\infty, +\infty].$$

By [JM12, Remark 5.6], β_Y does not depend on Y. We write $\beta := \beta_Y$. Let Y be a normal integral scheme with a proper birational morphism $Y \to \operatorname{Spec} A$, let B be a $\mathbb{Q}$-divisor on Y with $K_Y + B$ $\mathbb{Q}$-Cartier, let $\rho : Y_1 \to Y$ be a log resolution of (Y, B), and let $K_{Y_1} + B_{Y_1} := \rho^*(K_Y + B)$. By Definition 2.3(5),

$$a(v, Y, B) = a(v, Y_1) - v(B_{Y_1}) = \beta(v) - v(K_Y + B). \tag{7.3}$$

Let Y be as in (2). Then

$$a_{\mathrm{can}}(v, A) = \beta(v) + \mathrm{Env}_A(-K_A)(v),$$

so $a_{\mathrm{can}}(v, A)$ does not depend on Y.

Write $K_Y + \mathrm{Env}_A(-K_A)_Y = \sum_i \lambda_i P_i$ with $\lambda_i \in \mathbb{Q}$ and prime divisors P_i on Y. Since Y is regular, each P_i is Cartier and $v(P_i) = v(\mathcal{O}_Y(-P_i))$. Since $Y \to \operatorname{Spec} A$ is proper, the valuations v are exactly the real valuations of $\mathrm{Frac}(A)$ with a center on Y, so $v \to v(P_i)$ is continuous [JM12, Lemma 4.1] and $a(\cdot, Y)$ is lower semicontinuous [JM12, Lemma 5.7]. Thus $a_{\mathrm{can}}(\cdot, A)$ is lower semicontinuous.

Let Δ' be as in (2), and let $N' \geq 1$ be an integer such that $N'\Delta'$ is a Weil divisor and $N'(K_A + \Delta')$ is Cartier. Since A is local, $N'(K_A + \Delta') = \operatorname{div}(g')$ for some $g' \in \mathrm{Frac}(A)^*$. By (7.1),

$$N'\,\mathrm{Env}_A(-K_A)(v) = \mathrm{Env}_A(N'\Delta' - \operatorname{div}(g'))(v) = \mathrm{Env}_A(N'\Delta')(v) - v(g').$$

By (7.3) with $(\operatorname{Spec} A, \Delta')$ in place of (Y, B), $\beta(v) - v(g')/N' = a(v, \operatorname{Spec} A, \Delta')$, so

$$a_{\mathrm{can}}(v, A) = a(v, \operatorname{Spec} A, \Delta') + v(\mathbf{G}'), \qquad \mathbf{G}' := \frac{1}{N'}\mathrm{Env}_A(N'\Delta'). \tag{7.4}$$

By (1) and (7.1), $\mathbf{G}' \geq 0$ is a $\mathbb{Q}$-Cartier $\boldsymbol{b}$-divisor, and (7.4) implies $a(v, \operatorname{Spec} A, \Delta') \leq a_{\mathrm{can}}(v, A)$. Let $\pi : Y \to \operatorname{Spec} A$ be a log resolution of $(\operatorname{Spec} A, \Delta')$ on which $\mathbf{G}'$ is determined, let $K_Y + \Delta'_Y := \pi^*(K_A + \Delta')$, and let $E_1, \dots, E_n$ be distinct prime divisors on Y such that $\sum_{i=1}^n E_i$ is an snc

divisor whose support contains $\mathrm{Exc}(\pi)$, $\operatorname{Supp}\Delta'_Y$, and $\operatorname{Supp}\mathbf{G}'_Y$. Write $\Delta'_Y = \sum_{i=1}^n b_i E_i$ and $\mathbf{G}'_Y = \sum_{i=1}^n g_i E_i$. Then $g_i \geq 0$ since $\mathbf{G}' \geq 0$, and $b_i < 1$ since $(\operatorname{Spec} A, \Delta')$ is klt. By (7.4),

$$a_{\mathrm{can}}(v, A) = a(v, Y) - \sum_{i=1}^n (b_i - g_i) v(E_i), \qquad a(v, \operatorname{Spec} A, \Delta') = a(v, Y) - \sum_{i=1}^n b_i v(E_i),$$

so Lemma 7.2 with $c_i := b_i - g_i$ gives $\gamma \geq 1$ with $a_{\mathrm{can}}(v, A) \leq \gamma a(v, \operatorname{Spec} A, \Delta')$ for any v. This proves (2).

Step 5. Let $X := \operatorname{Spec} A$, $X' := \operatorname{Spec} A$, and let $f := \operatorname{Spec} \varphi : X' \to X$, a finite surjective morphism. Since $\varphi(A) \subset A$, $\varphi_* v \geq 0$ on $A \backslash \{0\}$ for any v. We prove, for any Weil divisor L on X and any v (cf. [BdFF12, Proposition 2.19]),

$$\mathrm{Env}_A(f^*L)(v) = \mathrm{Env}_A(L)(\varphi_* v). \tag{7.5}$$

For $s \in \mathcal{O}_A(jL)$ we have

$$\operatorname{div}(\varphi(s)) + j f^* L = f^*(\operatorname{div}(s) + jL) \geq 0,$$

so $\varphi(\mathcal{O}_A(jL)) \subset \mathcal{O}_A(jf^*L)$ and $v(\mathcal{O}_A(jf^*L)) \leq \varphi_* v(\mathcal{O}_A(jL))$ for any $j \geq 1$. Thus the left-hand side of (7.5) is at least the right-hand side.

Conversely, let $\alpha := -\mathrm{Env}_A(L)(\varphi_* v)$, so that $\varphi_* v(\mathcal{O}_A(jL)) \geq j\alpha$ for any $j \geq 1$ by **Step 1**. Let $j \geq 1$, let $t \in \mathcal{O}_A(jf^*L) \backslash \{0\}$, and let

$$T^D + \sum_{i=1}^D \varphi(h_i) T^{D-i}, \qquad h_1, \dots, h_D \in \mathrm{Frac}(A),$$

be the characteristic polynomial of multiplication by t on the $\varphi(\mathrm{Frac}(A))$-vector space $\mathrm{Frac}(A)$. We show $h_i \in \mathcal{O}_A(ijL)$ for any i. Let P be a prime divisor on X, let $\mathcal{M}$ be a normal closure of $\mathrm{Frac}(A)$ over $\varphi(\mathrm{Frac}(A))$, and let u be a valuation of $\mathcal{M}$ extending $\operatorname{ord}_P \circ \varphi^{-1}$. In $\mathcal{M}[T]$ the characteristic polynomial is $\prod_\sigma (T - \sigma(t))$, the product over the D embeddings σ of $\mathrm{Frac}(A)$ into $\mathcal{M}$ over $\varphi(\mathrm{Frac}(A))$. Since A is the integral closure of $\varphi(A)$ in $\mathrm{Frac}(A)$, for any such σ we have $u \circ \sigma = e_{P'}^{-1} \operatorname{ord}_{P'}$ for a prime divisor P' on X' with $f(P') = P$, where $e_{P'}$ is the coefficient of P' in f^*P. Thus

$$u(\sigma(t)) = e_{P'}^{-1} \operatorname{mult}_{P'} \operatorname{div}(t) \geq -j e_{P'}^{-1} \operatorname{mult}_{P'}(f^*L) = -j \operatorname{mult}_P L.$$

Since $\varphi(h_i)$ is, up to sign, the i-th elementary symmetric polynomial in the roots $\sigma(t)$, $\operatorname{mult}_P \operatorname{div}(h_i) \geq -ij \operatorname{mult}_P L$. Since P is arbitrary and A is normal, $h_i \in \mathcal{O}_A(ijL)$, so $v(\varphi(h_i)) \geq ij\alpha$ if $h_i \neq 0$. If $v(t) < j\alpha$, then $v(\varphi(h_i) t^{D-i}) \geq ij\alpha + (D-i)v(t) > Dv(t)$ for any $i \geq 1$ with $h_i \neq 0$, which contradicts $t^D = -\sum_{i=1}^D \varphi(h_i) t^{D-i}$. Thus $v(t) \geq j\alpha$, so $v(\mathcal{O}_A(jf^*L)) \geq j\alpha$ for any $j \geq 1$, and

$$\mathrm{Env}_A(f^*L)(v) \leq -\alpha = \mathrm{Env}_A(L)(\varphi_* v).$$

This proves (7.5).

Step 6. We prove (3) (cf. [BdFF12, Lemma 3.3]). By (1), $\mathbf{R}_\varphi$ is a $\mathbb{Q}$-Cartier $\boldsymbol{b}$-divisor. Since A contains $\mathbb{Q}$, the coefficient of $R(\varphi) = R(f)$ along a prime divisor P' on X' is $e_{P'} - 1 \geq 0$, where $e_{P'}$ is the ramification index of f at the generic point of P'. Thus $R(\varphi) \geq 0$. By (7.1),

$$\mathbf{R}_\varphi \geq \mathrm{Env}_A(R(\varphi)) \geq 0, \qquad \mathrm{Env}_A(-K_A) \geq \mathrm{Env}_A(-K_A + R(\varphi)) + \mathrm{Env}_A(-R(\varphi)),$$

so $\mathbf{R}_\varphi \leq -\mathrm{Env}_A(-R(\varphi))$. Let $m_\varphi := n_{-R(\varphi)}$ be the integer of **Step 3** for $L = -R(\varphi)$. By (7.2),

$$-\mathrm{Env}_A(-R(\varphi))(v) = m_\varphi^{-1} v(\mathcal{O}_A(-m_\varphi R(\varphi))),$$

and $\mathcal{O}_A(-m_\varphi R(\varphi)) \subset A$ is a non-zero ideal. This proves the bounds on $v(\mathbf{R}_\varphi)$.

We use the canonical divisor K_A on X and $K_{X'} := f^*K_A + R(\varphi)$ on X', and write

$$K_{X'} = K_A + \operatorname{div}(g_{X'}), \qquad g_{X'} \in \mathrm{Frac}(A)^*.$$

Since

$$-K_A + R(\varphi) = -K_{X'} + R(\varphi) + \operatorname{div}(g_{X'}), \qquad -K_A = -K_{X'} + \operatorname{div}(g_{X'}),$$

(7.1) gives

$$\mathbf{R}_\varphi = \operatorname{Env}_A(-K_{X'} + R(\varphi)) - \operatorname{Env}_A(-K_{X'}).$$

Since $f^*(-K_A) = -K_{X'} + R(\varphi)$, (7.5) implies

$$v(\mathbf{R}_\varphi) = \operatorname{Env}_A(-K_A)(\varphi_* v) - \operatorname{Env}_A(-K_{X'})(v). \tag{7.6}$$

Let β' be the function of **Step 4** on X', computed with $K_{X'}$. Then $\beta'(v) = \beta(v) + v(g_{X'})$ and, by (7.1),

$$\operatorname{Env}_A(-K_{X'})(v) = \operatorname{Env}_A(-K_A)(v) - v(g_{X'}),$$

so the formula for $a_{\mathrm{can}}(v, A)$ in **Step 4** gives

$$a_{\mathrm{can}}(v, A) = \beta'(v) + \operatorname{Env}_A(-K_{X'})(v). \tag{7.7}$$

Let $Y \to X$ be a log resolution on which $\operatorname{Env}_A(-K_A)$ is determined, and let

$$B_Y := -K_Y - \operatorname{Env}_A(-K_A)_Y.$$

By (7.3),

$$a(w, Y, B_Y) = \beta(w) + \operatorname{Env}_A(-K_A)(w) = a_{\mathrm{can}}(w, A)$$

for any real valuation w of $\operatorname{Frac}(A)$ with $w \geq 0$ on $A\backslash\{0\}$. Let Y' be the normalization of the irreducible component of $Y \times_X X'$ dominating X', and let $f_Y : Y' \to Y$ and $Y' \to X'$ be the induced morphisms. Since A is excellent, f_Y is finite and surjective, and $Y' \to X'$ is proper and birational. Let $K_{Y'} := f_Y^* K_Y + R(f_Y)$. Let $U \subset X$ be an open subset with $\operatorname{codim}_X(X\backslash U) \geq 2$ over which $Y \to X$ is an isomorphism. Then $\operatorname{codim}_{X'}(X'\backslash f^{-1}(U)) \geq 2$, and over $f^{-1}(U)$ the morphism $Y' \to X'$ is an isomorphism and $f_Y = f$. Thus the pushforward of $K_{Y'}$ to X' is $K_{X'}$. The restriction of v to the function field of Y is $\varphi_* v$, so Lemma 3.2 implies

$$a_{\mathrm{can}}(\varphi_* v, A) = a(\varphi_* v, Y, B_Y) = a(v, Y', f_Y^* B_Y - R(f_Y)).$$

We have

$$K_{Y'} + f_Y^* B_Y - R(f_Y) = f_Y^*(K_Y + B_Y) = -f_Y^* \operatorname{Env}_A(-K_A)_Y,$$

and

$$v(f_Y^* \operatorname{Env}_A(-K_A)_Y) = \varphi_* v(\operatorname{Env}_A(-K_A)_Y).$$

Thus (7.3) on X' implies

$$a(v, Y', f_Y^* B_Y - R(f_Y)) = \beta'(v) + \operatorname{Env}_A(-K_A)(\varphi_* v).$$

By (7.6) and (7.7), we obtain

$$a_{\mathrm{can}}(\varphi_* v, A) = a_{\mathrm{can}}(v, A) + v(\mathbf{R}_\varphi) \quad \text{in } (-\infty, +\infty]$$

for any v. This proves (3). □

Lemma 7.4 (cf. [Li18, Theorem 3.1] and [Kim26, Lemma 2.3]). *Let A_0 be a normal complex analytic local domain, let $\mathfrak{q}$ be a prime ideal of A_0, let $A := (A_0)_{\mathfrak{q}}$, and let $\Delta \geq 0$ be a $\mathbb{Q}$-divisor on $\operatorname{Spec} A$ such that $K_A + \Delta$ is $\mathbb{Q}$-Cartier and $(\operatorname{Spec} A, \Delta)$ is klt. Then for any non-zero prime ideal P of A there exists a real number $\gamma_P > 0$ such that any real valuation v of $\operatorname{Frac}(A)$ centered at P with $a(v, \operatorname{Spec} A, \Delta) < +\infty$ satisfies*

$$\mathfrak{a}_t(v) \subset (PA_P)^{\lceil t/(\gamma_P a(v, \operatorname{Spec} A, \Delta))\rceil} \quad \text{in } A_P$$

for any real number $t > 0$.

Proof. Let P be a non-zero prime ideal of A, let $P_0 := P \cap A_0$, let $S := A_P = (A_0)_{P_0}$, let $r := \dim S \geq 1$, and let $n := \dim A_0$. Let Δ_S be the pullback of Δ to $\operatorname{Spec} S$. A log resolution of $(\operatorname{Spec} A, \Delta)$ restricts over $\operatorname{Spec} S$ to a log resolution of $(\operatorname{Spec} S, \Delta_S)$, so $(\operatorname{Spec} S, \Delta_S)$ is klt and $a(v, \operatorname{Spec} A, \Delta) = a(v, \operatorname{Spec} S, \Delta_S)$ for any real valuation v of $\operatorname{Frac}(A)$ centered at P. It suffices to find $\gamma_P > 0$ with

$$v(f) \leq \gamma_P a(v, \operatorname{Spec} S, \Delta_S) \operatorname{ord}_{\mathfrak{m}_S}(f) \tag{7.8}$$

for any $f \in S\backslash\{0\}$ and any real valuation v of $\operatorname{Frac}(A)$ centered at $\mathfrak{m}_S = PA_P$ with $a(v, \operatorname{Spec} S, \Delta_S) < +\infty$. Applied to some $f \in \mathfrak{m}_S\backslash\{0\}$, (7.8) gives $a(v, \operatorname{Spec} S, \Delta_S) > 0$, and since $\operatorname{ord}_{\mathfrak{m}_S}$ takes integer values, (7.8) implies the inclusion in the lemma.

Step 1. We construct a finite injective homomorphism from a regular henselian local ring to the henselization of S. By prime avoidance there exist $x_1, \ldots, x_r \in P_0$ such that any minimal prime ideal of $(x_1, \ldots, x_r)A_0$ has height r. Since A_0 is a catenary local domain [Mat89, Theorem 17.9], $\dim A_0/(x_1, \ldots, x_r)A_0 = n - r$ [Mat89, Theorem 31.4], so there exist $y_1, \ldots, y_{n-r} \in \mathfrak{m}_{A_0}$ such that $x_1, \ldots, x_r, y_1, \ldots, y_{n-r}$ is a system of parameters of A_0. Since $A_0/(x_1, \ldots, x_r, y_1, \ldots, y_{n-r})A_0$ is finite-dimensional over $\mathbb{C}$, the $\mathbb{C}$-algebra homomorphism $T_0 := \mathbb{C}\{z_1, \ldots, z_n\} \to A_0$ sending z_i to x_i for $i \leq r$ and z_{r+i} to y_i for $i \leq n - r$ is finite [dJP00, Theorem 3.4.24], and it is injective since A_0 is a domain of dimension n [dJP00, Lemma 4.1.4]. The prime ideal $\mathfrak{p} := P_0 \cap T_0$ contains the prime ideal $(z_1, \ldots, z_r)T_0$ of height r, and $\operatorname{ht} \mathfrak{p} = \operatorname{ht} P_0 = r$ by going-down for the normal ring T_0. Thus $\mathfrak{p} = (z_1, \ldots, z_r)T_0$. Let $T := (T_0)_{\mathfrak{p}}$, a regular local ring with $\dim T = r$, and let A_1 be the localization of A_0 at $T_0\backslash\mathfrak{p}$, a finite T-algebra. Then P_0A_1 is a maximal ideal of A_1 over $\mathfrak{p}T$, and $S = (A_1)_{P_0A_1}$.

Let T^h be the henselization of T. The finite T^h-algebra $A_1 \otimes_T T^h$ is the henselization of the semi-local ring A_1, that is, the product of the henselizations of the localizations of A_1 at its maximal ideals, so its factor over P_0A_1 is the henselization S^h of S [Gro67, (18.6.7), Proposition 18.6.8]. Moreover, $\mathfrak{m}_S S^h = \mathfrak{m}_{S^h}$, $S \to S^h$ is faithfully flat and induces an isomorphism of completions, so $S/\mathfrak{m}_S^j \cong S^h/\mathfrak{m}_{S^h}^j$ for any j [Gro67, Théorème 18.6.6(iii)(iv)]. The ring S^h is normal and the fibers of $\operatorname{Spec} S^h \to \operatorname{Spec} S$ are geometrically regular, so $S \to S^h$ is a regular homomorphism [Gro67, Théorème 18.6.9]. The ring T^h is regular [Gro67, Corollaire 18.6.10], and T^h and S^h are excellent [Gro67, Corollaire 18.7.6]. Since T^h is regular, it is Gorenstein and $\operatorname{Spec} T^h$ admits a dualizing complex [Sta, Tag 0AWX], and since S^h is finite over T^h, $\operatorname{Spec} S^h$ admits a dualizing complex [Sta, Tag 0A7K]. Since S^h is a domain finite over T^h and $\dim S^h = \dim S = \dim T^h$, the homomorphism $T^h \to S^h$ is injective. Let $\pi : \operatorname{Spec} S^h \to \operatorname{Spec} T^h$ be the induced finite surjective morphism. The rings T^h and S^h are analytically irreducible [Gro65, Scholie 7.8.3(v)]. For $f \in S^h\backslash\{0\}$ let $\operatorname{Nm}(f)$ be the norm of f from $\operatorname{Frac}(S^h)$ to $\operatorname{Frac}(T^h)$. Since f is integral over the normal ring T^h, $\operatorname{Nm}(f) \in T^h$. By [RS14, Lemma 2.8], there exists a real number $\gamma_2 > 0$ with

$$\operatorname{ord}_{\mathfrak{m}_{T^h}}(\operatorname{Nm}(f)) \leq \gamma_2 \operatorname{ord}_{\mathfrak{m}_{S^h}}(f) \quad \text{for any } f \in S^h\backslash\{0\}. \tag{7.9}$$

Let Δ^h be the pullback of Δ_S to $\operatorname{Spec} S^h$. Since $S \to S^h$ is regular, the base change to $\operatorname{Spec} S^h$ of a log resolution $Y_S \to \operatorname{Spec} S$ of $(\operatorname{Spec} S, \Delta_S)$ is a log resolution of $(\operatorname{Spec} S^h, \Delta^h)$, and the crepant pullback of Δ^h is the pullback of that of Δ_S. Thus $K_{S^h} + \Delta^h$ is $\mathbb{Q}$-Cartier and $(\operatorname{Spec} S^h, \Delta^h)$ is klt.

Step 2. We prove that there exists $\gamma_P > 0$ such that any real valuation w of $\operatorname{Frac}(S^h)$ centered at $\mathfrak{m}_{S^h}$ with $a(w, \operatorname{Spec} S^h, \Delta^h) < +\infty$ satisfies

$$w(f) \leq \gamma_P a(w, \operatorname{Spec} S^h, \Delta^h) \operatorname{ord}_{\mathfrak{m}_{S^h}}(f) \quad \text{for any } f \in S^h\backslash\{0\}. \tag{7.10}$$

Let u be the restriction of w to $\operatorname{Frac}(T^h)$. Since $\operatorname{Frac}(S^h)$ is algebraic over $\operatorname{Frac}(T^h)$ and S^h is integral over T^h, u is a real valuation centered at $\mathfrak{m}_{T^h}$. With $K_{S^h} := \pi^* K_{T^h} + R(\pi)$, Lemma 3.2 implies $a(u, \operatorname{Spec} T^h) = a(w, \operatorname{Spec} S^h, -R(\pi))$. Let $Y_1 \to \operatorname{Spec} S^h$ be a log resolution of $(\operatorname{Spec} S^h, \Delta^h)$ and of $(\operatorname{Spec} S^h, -R(\pi))$, with distinct prime divisors $E_1, \ldots, E_N$ on Y_1 such that $\sum_{i=1}^N E_i$ is an snc divisor containing the exceptional locus and the supports of the crepant

pullbacks $\sum_{i=1}^N b_i E_i$ of Δ^h and $\sum_{i=1}^N c_i E_i$ of $-R(\pi)$. Since $(\operatorname{Spec} S^h, \Delta^h)$ is klt, $b_i < 1$ for any i. Let $\gamma_1 \geq 1$ be the number γ of Lemma 7.2 for $b_1, \dots, b_N$ and $c_1, \dots, c_N$. Then Lemma 7.2 implies

$$a(u, \operatorname{Spec} T^h) = a(w, \operatorname{Spec} S^h, -R(\pi)) \leq \gamma_1 a(w, \operatorname{Spec} S^h, \Delta^h) < +\infty.$$

By Lemma 2.5(1) for $\operatorname{Spec} T^h$,

$$u(g) \leq a(u, \operatorname{Spec} T^h) \operatorname{ord}_{\mathfrak{m}_{T^h}}(g) \quad \text{for any } g \in T^h \backslash \{0\}.$$

Let $f \in S^h \backslash \{0\}$. The element $\operatorname{Nm}(f)/f \in \operatorname{Frac}(S^h)$ is a product of conjugates of f over $\operatorname{Frac}(T^h)$, hence integral over T^h, so $\operatorname{Nm}(f)/f \in S^h$ and $w(f) \leq w(\operatorname{Nm}(f)) = u(\operatorname{Nm}(f))$. By (7.9),

$$w(f) \leq a(u, \operatorname{Spec} T^h) \operatorname{ord}_{\mathfrak{m}_{T^h}}(\operatorname{Nm}(f)) \leq \gamma_1 \gamma_2 a(w, \operatorname{Spec} S^h, \Delta^h) \operatorname{ord}_{\mathfrak{m}_{S^h}}(f),$$

which is (7.10) with $\gamma_P := \gamma_1 \gamma_2$.

Step 3. We prove (7.8) for divisorial valuations. Let $v = c \operatorname{ord}_E$ be a divisorial valuation centered at $\mathfrak{m}_S$, where $c > 0$ and E is a prime divisor on a log resolution $Y_S \to \operatorname{Spec} S$ of $(\operatorname{Spec} S, \Delta_S)$. Let $Y^h := Y_S \times_{\operatorname{Spec} S} \operatorname{Spec} S^h$, a log resolution of $(\operatorname{Spec} S^h, \Delta^h)$ by **Step 1**. Since E is contained in the fiber over $\mathfrak{m}_S$ and $S/\mathfrak{m}_S^j \cong S^h/\mathfrak{m}_{S^h}^j$ for any j, the preimage E^h of E in Y^h maps isomorphically onto E, and a uniformizer of $\mathcal{O}_{Y_S,E}$ is a uniformizer of $\mathcal{O}_{Y^h,E^h}$. Thus $w := c \operatorname{ord}_{E^h}$ restricts to v on $\operatorname{Frac}(S)$, w is centered at $\mathfrak{m}_{S^h}$, and the coefficient of E^h in the crepant pullback of Δ^h equals the coefficient of E in that of Δ_S, so $a(w, \operatorname{Spec} S^h, \Delta^h) = a(v, \operatorname{Spec} S, \Delta_S)$ (cf. [JM12, Proposition 5.13]). Since $S \to S^h$ is faithfully flat and $\mathfrak{m}_S S^h = \mathfrak{m}_{S^h}$, we have $\mathfrak{m}_{S^h}^j \cap S = \mathfrak{m}_S^j$ for any j, so $\operatorname{ord}_{\mathfrak{m}_{S^h}}(f) = \operatorname{ord}_{\mathfrak{m}_S}(f)$ for $f \in S \backslash \{0\}$. Thus (7.10) for w implies (7.8) for v.

Step 4. We prove (7.8) in general. Let v be a real valuation of $\operatorname{Frac}(A)$ centered at $\mathfrak{m}_S$ with $a(v, \operatorname{Spec} S, \Delta_S) < +\infty$, and let $f \in S \backslash \{0\}$. Let $\rho_0 : Y_0 \to \operatorname{Spec} S$ be a log resolution of $(\operatorname{Spec} S, \Delta_S)$ with $\mathfrak{m}_S \mathcal{O}_{Y_0} = \mathcal{O}_{Y_0}(-F_0)$ for a Cartier divisor $F_0 \geq 0$, and let $K_{Y_0} + \Delta_{Y_0} := \rho_0^*(K_S + \Delta_S)$. Possibly replacing Y_0 with a log resolution of $(Y_0, \Delta_{Y_0} + F_0)$, we may assume that the union of $\operatorname{Exc}(\rho_0)$, $\operatorname{Supp} \Delta_{Y_0}$, and $\operatorname{Supp} F_0$ is the support of an snc divisor G_0. Then $a(v, \operatorname{Spec} S, \Delta_S) = a(v, Y_0) - v(\Delta_{Y_0})$ and $v(F_0) = v(\mathfrak{m}_S) > 0$.

For a log resolution $Y \to Y_0$ and an snc divisor G on Y containing the exceptional locus and the preimage of G_0, let $r_{Y,G}(v)$ be the retraction of v onto $\operatorname{QM}(Y, G)$ (Definition-Lemma 2.7). Since the pullbacks of F_0 and Δ_{Y_0} to Y are supported on G, we have $r_{Y,G}(v)(F_0) = v(F_0) > 0$ and $r_{Y,G}(v)(\Delta_{Y_0}) = v(\Delta_{Y_0})$. In particular, $r_{Y,G}(v)$ is centered at $\mathfrak{m}_S$. The pairs (Y, G) form a directed set, where $(Y', G') \geq (Y, G)$ if $Y' \to Y_0$ factors through Y and G' contains the preimage of G, and $r_{Y,G}(v)$ converges to v pointwise and $a(r_{Y,G}(v), Y_0)$ converges to $a(v, Y_0)$ along the directed set of pairs (Y, G) [JM12, Lemma 4.6, Corollary 4.8, (5.2) and Corollary 5.4]. Thus it suffices to prove (7.8) for the valuations $r_{Y,G}(v)$.

Let K_{Y/Y_0} be the relative canonical divisor of $Y \to Y_0$. Let w_α be the quasi-monomial valuation which is monomial in local equations of components $G_1, \dots, G_m$ of G at the generic point of an irreducible component of $G_1 \cap \cdots \cap G_m$, with weights $\alpha \in \mathbb{R}_{>0}^m$. The function $\alpha \to w_\alpha(f)$ is continuous [JM12, Lemma 4.5]. Since K_{Y/Y_0} and the pullbacks of F_0 and Δ_{Y_0} to Y are supported on G, the functions $\alpha \to w_\alpha(F_0)$, $\alpha \to w_\alpha(\Delta_{Y_0})$, and $\alpha \to a(w_\alpha, Y_0)$ are linear, where

$$a(w_\alpha, Y_0) = a(w_\alpha, Y) + w_\alpha(K_{Y/Y_0})$$

[JM12, Proposition 5.1 and Remark 5.6]. Moreover, w_α is divisorial for $\alpha \in \mathbb{Q}_{>0}^m$ [JM12, Remark 3.9]. Thus if $w_\alpha(F_0) > 0$, then w_α is a limit of divisorial valuations $w_{\alpha'}$ with $\alpha' \in \mathbb{Q}_{>0}^m$, $w_{\alpha'}(F_0) > 0$, $w_{\alpha'}(f) \to w_\alpha(f)$, and

$$a(w_{\alpha'}, \operatorname{Spec} S, \Delta_S) \to a(w_\alpha, \operatorname{Spec} S, \Delta_S).$$

By **Step 3**, (7.8) holds for each $w_{\alpha'}$, hence for w_α, hence for any $r_{Y,G}(v)$, and hence for v. The lemma follows. □

8. Selecting a totally invariant prime

In this section, for A and φ as in Set-up 7.1 such that $Q := c_\infty(\varphi) \geq 2$ is an integer with $D = \delta Q^d$, we study the failure of the uniform bound $c(\varphi^j) \geq bQ^j$. If the bound fails for every $b > 0$, then there exist an integer $a \geq 1$ and a prime ideal P of A with $0 \neq P \neq \mathfrak{m}_A$, totally invariant under φ^a, such that

$$[\mathrm{Frac}(A/P) : (\varphi^a)_{A/P}(\mathrm{Frac}(A/P))] = \delta^a Q^{a\dim(A/P)}$$

and $c((\varphi^a)^j_{A_P}) \geq bQ^{aj}$ for a real number $b > 0$ and any integer $j \geq 0$ (Theorem 8.2).

Lemma 8.1 (cf. [Gig14, Theorem 1.2 and Proposition 6.4])**.** *Let Y be a Noetherian scheme, $g : Y \to Y$ a continuous map, $\mathcal{K}$ a compact Hausdorff space, $f : \mathcal{K} \to \mathcal{K}$ a continuous map, and $\zeta : \mathcal{K} \to Y$ a map with $\zeta \circ f = g \circ \zeta$. Assume that $\zeta^{-1}(Z)$ is open for any closed subset $Z \subset Y$. Let μ be an f-invariant Radon probability measure on $\mathcal{K}$. For an irreducible closed subset $V \subset Y$ with generic point η_V, let $\mathcal{K}_V := \zeta^{-1}(\eta_V)$. Then:*

(1) $\mathcal{K}_V$ is a Borel set for any V.
(2) The set of V with $\mu(\mathcal{K}_V) > 0$ is countable, and $\sum_V \mu(\mathcal{K}_V) = 1$.
(3) If $\mu(\mathcal{K}_V) > 0$, then there exists an integer $N \geq 1$ with $g^N(\eta_V) = \eta_V$, and for the least such N the measure $\mu(\mathcal{K}_V)^{-1}\mu|_{\mathcal{K}_V}$ is f^N-invariant.

Proof. Since $\zeta^{-1}(Z)$ is open for any closed subset $Z \subset Y$, the preimage under ζ of any open subset of Y is closed. Thus ζ is Borel and $\zeta_*\mu$ is a Borel probability measure on Y. Since $\zeta \circ f = g \circ \zeta$ and μ is f-invariant,

$$\zeta_*\mu(g^{-1}(E)) = \mu(f^{-1}(\zeta^{-1}(E))) = \zeta_*\mu(E)$$

for any Borel set $E \subset Y$. The underlying space of Y is a Zariski space [Gig14, Definition 1.1]. By [Gig14, Theorem 1.2], there exists a unique family $(c_y)_{y\in Y}$ of real numbers with $\sum_{y\in Y}|c_y| < +\infty$ and

$$\zeta_*\mu(E) = \sum_{y\in E} c_y \quad \text{for any Borel set } E \subset Y. \tag{8.1}$$

By [Gig14, Proposition 6.4] and the uniqueness of $(c_y)_{y\in Y}$, if $c_y \neq 0$, then $g^N(y) = y$ for some integer $N \geq 1$.

Let $V \subset Y$ be an irreducible closed subset. A point $y \in V$ is different from η_V if and only if $\overline{\{y\}} \subsetneq V$. Thus

$$\mathcal{K}_V = \zeta^{-1}(V)\backslash U_V, \qquad U_V := \bigcup_{Z\subsetneq V,\ Z \text{ closed in } Y} \zeta^{-1}(Z).$$

Both $\zeta^{-1}(V)$ and U_V are open, so $\mathcal{K}_V$ is Borel. We show that

$$\mu(\mathcal{K}_V) = c_{\eta_V}. \tag{8.2}$$

By (8.1), $\mu(\zeta^{-1}(V)) = \sum_{y\in V} c_y$. Since μ is Radon, $\mu(U_V)$ is the supremum of $\mu(C)$ over the compact subsets $C \subset U_V$. Any such C is covered by finitely many $\zeta^{-1}(Z)$, and a finite union of closed subsets $Z \subsetneq V$ is again a closed subset $Z \subsetneq V$ because V is irreducible. Hence $\mu(U_V) = \sup_{Z\subsetneq V} \mu(\zeta^{-1}(Z))$. For a finite subset $F \subset V\backslash\{\eta_V\}$, the closed subset $Z_F := \bigcup_{y\in F} \overline{\{y\}}$ satisfies $Z_F \subsetneq V$, and by (8.1), for any closed subset Z with $Z_F \subset Z \subsetneq V$,

$$\left|\mu(\zeta^{-1}(Z)) - \sum_{y\in V\backslash\{\eta_V\}} c_y\right| \leq \sum_{y\in V\backslash(F\cup\{\eta_V\})} |c_y|. \tag{8.3}$$

Since μ is positive, $\mu(\zeta^{-1}(Z)) \leq \mu(\zeta^{-1}(Z \cup Z_F))$ for any closed $Z \subsetneq V$, so the supremum defining $\mu(U_V)$ may be taken over the closed Z with $Z_F \subset Z \subsetneq V$, and (8.3) gives

$$\left|\mu(U_V) - \sum_{y\in V\backslash\{\eta_V\}} c_y\right| \leq \sum_{y\in V\backslash(F\cup\{\eta_V\})} |c_y|$$

for any finite F. Letting F grow, we obtain $\mu(U_V) = \sum_{y\in V\setminus\{\eta_V\}} c_y$, which implies (8.2).

Since $y \to \overline{\{y\}}$ is a bijection from Y onto the set of irreducible closed subsets of Y, with inverse $V \to \eta_V$, (8.2) and $\sum_y |c_y| < +\infty$ imply that the set of V with $\mu(\mathcal{K}_V) > 0$ is countable and

$$\sum_V \mu(\mathcal{K}_V) = \sum_{y\in Y} c_y = \zeta_*\mu(Y) = 1.$$

If $\mu(\mathcal{K}_V) > 0$, then $c_{\eta_V} \neq 0$, so $g^N(\eta_V) = \eta_V$ for some integer $N \geq 1$.

Let N be the least such integer, and let $\mathcal{K}^+$ be the union of the fibers $\mathcal{K}_W$ with $\mu(\mathcal{K}_W) > 0$. Since $\sum_W \mu(\mathcal{K}_W) = 1$, we have $\mu(\mathcal{K}\setminus\mathcal{K}^+) = 0$. We show that

$$f^{-N}(\mathcal{K}_V) \cap \mathcal{K}^+ = \mathcal{K}_V.$$

Since $\zeta(f^N(u)) = g^N(\eta_V) = \eta_V$ for $u \in \mathcal{K}_V$, we have $\mathcal{K}_V \subset f^{-N}(\mathcal{K}_V)$. Conversely, let $u \in \mathcal{K}_W \cap f^{-N}(\mathcal{K}_V)$ with $\mu(\mathcal{K}_W) > 0$. Then $g^N(\eta_W) = \eta_V$ and $g^{N'}(\eta_W) = \eta_W$ for some integer $N' \geq 1$, so $\eta_W = g^{NN'}(\eta_W) = g^{N(N'-1)}(\eta_V) = \eta_V$ and $W = V$. Thus, for any Borel set $E \subset \mathcal{K}$, the sets $f^{-N}(E\cap\mathcal{K}_V)$ and $f^{-N}(E)\cap\mathcal{K}_V$ differ by a subset of $\mathcal{K}\setminus\mathcal{K}^+$, and since μ is f^N-invariant,

$$\mu(f^{-N}(E)\cap\mathcal{K}_V) = \mu(f^{-N}(E\cap\mathcal{K}_V)) = \mu(E\cap\mathcal{K}_V).$$

The lemma follows. □

Theorem 8.2. *Notation and conditions as in Set-up 7.1. Assume that there exists an integer $Q \geq 2$ with $D = \delta Q^d$, $c_\infty(\varphi) = Q$, and $\inf_{j\geq 0} c(\varphi^j)/Q^j = 0$. Then there exist an integer $a \geq 1$ and a prime ideal P of A with $0 \neq P \neq \mathfrak{m}_A$ such that:*

(1) P is totally invariant under φ^a.

(2) The equality

$$[\mathrm{Frac}(A/P) : (\varphi^a)_{A/P}(\mathrm{Frac}(A/P))] = \delta^a Q^{a\dim(A/P)}$$

holds.

(3) There exists a real number $b > 0$ with $c((\varphi^a)^j_{A_P}) \geq bQ^{aj}$ for any integer $j \geq 0$.

Proof. Since $c_\infty(\varphi) = Q \geq 2$, we have $c(\varphi^j) \to +\infty$, and we fix an integer $a_0 \geq 1$ with $c(\varphi^{a_0}) \geq 2$. Let $x_1, \dots, x_{n_0}$ generate $\mathfrak{m}_A$, so that $u(\mathfrak{m}_A) = \min_k u(x_k)$ for any real valuation u with $u \geq 0$ on $A\setminus\{0\}$, and $c(\varphi^j) = \min_k \mathrm{ord}_{\mathfrak{m}_A}(\varphi^j(x_k))$ for any $j \geq 0$. By Lemma 2.13,

$$c(\varphi^{j+j'}) \geq c(\varphi^j)c(\varphi^{j'}) \quad \text{for any integers } j, j' \geq 0. \tag{8.4}$$

We divide the rest of the proof into six steps.

Step 1. In this step we construct valuations w_j centered at $\mathfrak{m}_A$ and show that the ratios $a_{\mathrm{can}}(w_j, A)/w_j(\mathfrak{m}_A)$ are unbounded and grow at most geometrically.

First, we show that

$$\inf_{j\geq 0} \frac{c(\varphi^{a_0 j})}{Q^{a_0 j}} = 0. \tag{8.5}$$

If $c(\varphi^{a_0 j}) \geq b'Q^{a_0 j}$ for a real number $b' > 0$ and any $j \geq 0$, then by (8.4),

$$c(\varphi^{a_0 j+i}) \geq c(\varphi^{a_0 j}) \geq b'Q^{-a_0}Q^{a_0 j+i} \quad \text{for any integers } j \geq 0 \text{ and } 0 \leq i < a_0,$$

contrary to $\inf_{j\geq 0} c(\varphi^j)/Q^j = 0$.

For any real valuation w of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$ with $a(w, \mathrm{Spec}\, A, \Delta) < +\infty$, Lemma 7.4 with $P = \mathfrak{m}_A$ and $t = w(x)$, and Definition-Theorem 7.3(2) with $\Delta' = \Delta$ imply

$$w(x) \leq \gamma_{\mathfrak{m}_A} a(w, \mathrm{Spec}\, A, \Delta)\,\mathrm{ord}_{\mathfrak{m}_A}(x) \leq \gamma_{\mathfrak{m}_A} a_{\mathrm{can}}(w, A)\,\mathrm{ord}_{\mathfrak{m}_A}(x) \quad \text{for any } x \in A\setminus\{0\}. \tag{8.6}$$

Fix a divisorial valuation w_0 of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$, and let

$$\alpha := w_0(\mathfrak{m}_A), \qquad \beta := \gamma_{\mathfrak{m}_A} a(w_0, \mathrm{Spec}\, A, \Delta).$$

Since $x \in \mathfrak{m}_A^{\operatorname{ord}_{\mathfrak{m}_A}(x)}$ for any $x \in A\backslash\{0\}$, and by (8.6),

$$\alpha \operatorname{ord}_{\mathfrak{m}_A} \le w_0 \le \beta \operatorname{ord}_{\mathfrak{m}_A} \quad \text{on } A\backslash\{0\}.$$

For $j \ge 0$ let $w_j := (\varphi^{a_0 j})_* w_0$ and $\rho_j := w_j(\mathfrak{m}_A)$. Since φ is local, w_j is centered at $\mathfrak{m}_A$. Since

$$\varphi^{a_0 j}(\mathfrak{m}_A)A \subset \mathfrak{m}_A^{c(\varphi^{a_0 j})},$$

and since $\operatorname{ord}_{\mathfrak{m}_A}(\varphi^{a_0 j}(x_k)) = c(\varphi^{a_0 j})$ for some k, we have, for any $j, j' \ge 0$,

$$\alpha c(\varphi^{a_0 j}) \le \rho_j \le \beta c(\varphi^{a_0 j}), \qquad \rho_{j+j'} = w_j(\varphi^{a_0 j'}(\mathfrak{m}_A)A) \ge c(\varphi^{a_0 j'})\rho_j. \tag{8.7}$$

In particular $\rho_{j+1} \ge 2\rho_j$. Set-up 7.1 holds with ψ^{a_0} and φ^{a_0} in place of ψ and φ. By Definition-Theorem 7.3(2), $a_{\mathrm{can}}(w_0, A) < +\infty$, and by Definition-Theorem 7.3(3) with φ^{a_0} in place of φ and induction on j, $a_{\mathrm{can}}(w_j, A) < +\infty$ and

$$a_{\mathrm{can}}(w_{j+1}, A) = a_{\mathrm{can}}(w_j, A) + w_j(\mathbf{R}_{\varphi^{a_0}}). \tag{8.8}$$

Since $(\operatorname{Spec} A, \Delta)$ is klt, $a_{\mathrm{can}}(w_j, A) \ge a(w_j, \operatorname{Spec} A, \Delta) > 0$. Let $z_j := a_{\mathrm{can}}(w_j, A)/\rho_j$.

Let x_R be a non-zero element of the ideal $\mathcal{O}_A(-m_{\varphi^{a_0}} R(\varphi^{a_0}))$, and let

$$\kappa_1 := m_{\varphi^{a_0}}^{-1} \gamma_{\mathfrak{m}_A} \operatorname{ord}_{\mathfrak{m}_A}(x_R).$$

By Definition-Theorem 7.3(3) and (8.6),

$$0 \le w_j(\mathbf{R}_{\varphi^{a_0}}) \le m_{\varphi^{a_0}}^{-1} w_j(x_R) \le \kappa_1 a_{\mathrm{can}}(w_j, A).$$

With (8.8) and $\rho_{j+1} \ge 2\rho_j$,

$$z_{j+1} = z_j\Big(1 + \frac{w_j(\mathbf{R}_{\varphi^{a_0}})}{a_{\mathrm{can}}(w_j, A)}\Big)\frac{\rho_j}{\rho_{j+1}} \le \kappa_2 z_j, \qquad \kappa_2 := 2 + \kappa_1. \tag{8.9}$$

We show that $\sup_j z_j = +\infty$. Let $j \ge 0$ and $m \ge 1$ be integers, and let x be one of $x_1, \dots, x_{n_0}$ with

$$\operatorname{ord}_{\mathfrak{m}_A}(\varphi^{a_0(m-1)j}(x)) = c(\varphi^{a_0(m-1)j}).$$

Since $\varphi^{a_0 mj}(x) \in \mathfrak{m}_A^{c(\varphi^{a_0 mj})}$ and $\alpha \operatorname{ord}_{\mathfrak{m}_A} \le w_0$, and by (8.6) for w_j,

$$\alpha c(\varphi^{a_0 mj}) \le w_0(\varphi^{a_0 mj}(x)) = w_j(\varphi^{a_0(m-1)j}(x)) \le \gamma_{\mathfrak{m}_A} a_{\mathrm{can}}(w_j, A) c(\varphi^{a_0(m-1)j}).$$

Since $c(\varphi^0) = 1$, induction on m implies

$$c(\varphi^{a_0 mj}) \le \Big(\frac{\gamma_{\mathfrak{m}_A} a_{\mathrm{can}}(w_j, A)}{\alpha}\Big)^m.$$

Since $c(\varphi^{a_0 mj})^{1/m} \to Q^{a_0 j}$ as $m \to +\infty$, we obtain

$$Q^{a_0 j} \le \frac{\gamma_{\mathfrak{m}_A} a_{\mathrm{can}}(w_j, A)}{\alpha}.$$

By (8.7),

$$z_j \ge \frac{\alpha Q^{a_0 j}}{\gamma_{\mathfrak{m}_A}\rho_j} \ge \frac{\alpha}{\beta\gamma_{\mathfrak{m}_A}} \cdot \frac{Q^{a_0 j}}{c(\varphi^{a_0 j})},$$

and by (8.5), $\sup_j z_j = +\infty$.

For any integer $i \ge 1$, let

$$\lambda_i := i + \kappa_2 + z_0, \qquad j_i' := \min\{j \ge 0 \mid z_j \ge \lambda_i^2\},$$
$$j_i := 1 + \max\{j < j_i' \mid z_j < \lambda_i\}, \quad \text{and} \quad L_i := j_i' - j_i.$$

The set in the definition of j_i contains 0 because $z_0 < \lambda_i \le \lambda_i^2$. Then

$$z_j \ge \lambda_i \ \text{ for } j_i \le j \le j_i', \qquad z_{j_i} < \kappa_2 \lambda_i, \quad \text{and} \quad z_{j_i'} \ge \lambda_i^2. \tag{8.10}$$

By (8.9),

$$\lambda_i^2 \le \kappa_2^{L_i} z_{j_i} < \kappa_2^{L_i+1}\lambda_i.$$

Since $\lambda_i > \kappa_2$, we have $L_i \ge 1$, and $L_i \to +\infty$ as $i \to +\infty$.

Step 2. In this step we construct a compact space $\mathcal{K}_0$ of valuations, a continuous map $f : \mathcal{K}_0 \to \mathcal{K}_0$, and an f-invariant Radon probability measure μ on $\mathcal{K}_0$ with $\mu(\{u \mid u(\mathfrak{m}_A) > 0\}) = 0$ and

$$\int \log(1 + u(\mathbf{R}_{\varphi^{a_0}}))d\mu(u) \geq a_0 \log Q.$$

Let $u_j := w_j/a_{\mathrm{can}}(w_j, A)$. Since, for Y as in Definition-Theorem 7.3(2), $a(\cdot, Y)$ and $v \to v(K_Y + \mathrm{Env}_A(-K_A)_Y)$ are homogeneous of degree 1, the definition there gives $a_{\mathrm{can}}(u_j, A) = 1$, and $u_j(\mathfrak{m}_A) = z_j^{-1}$. By (8.6),

$$0 \leq u_j(x) \leq \gamma_{\mathfrak{m}_A} \operatorname{ord}_{\mathfrak{m}_A}(x) \quad \text{for any } x \in A\backslash\{0\}.$$

Let $\mathcal{K}_0$ be the closure of $\{u_j \mid j \geq 0\}$ in

$$\prod_{x \in A\backslash\{0\}} [0, \gamma_{\mathfrak{m}_A} \operatorname{ord}_{\mathfrak{m}_A}(x)],$$

a compact Hausdorff space whose topology is the topology of pointwise convergence. The conditions $u(xy) = u(x) + u(y)$, $u(x + y) \geq \min\{u(x), u(y)\}$ for $x + y \neq 0$, and $u(\mathbb{C}^*) = \{0\}$ are closed and hold for any u_j. Thus any $u \in \mathcal{K}_0$, extended by $u(x/y) := u(x) - u(y)$, is either a real valuation of $\mathrm{Frac}(A)$ with $u \geq 0$ on $A\backslash\{0\}$, or the trivial valuation v_0, for which we set $a_{\mathrm{can}}(v_0, A) := 0$, $v_0(\mathbf{R}_{\varphi^{a_0}}) := 0$, and $\mathrm{center}_{\mathrm{Spec}\, A}\, v_0 :=$ the generic point of $\mathrm{Spec}\, A$.

The function $a_{\mathrm{can}}(\cdot, A)$ is lower semicontinuous on $\mathcal{K}_0$: at v_0 because $a_{\mathrm{can}} \geq 0$, and at $u \neq v_0$ by Definition-Theorem 7.3(2), since $\mathcal{K}_0\backslash\{v_0\}$ is an open neighborhood of u in the Hausdorff space $\mathcal{K}_0$. Since $a_{\mathrm{can}}(u_j, A) = 1$ for any j,

$$a_{\mathrm{can}}(u, A) \leq 1 \quad \text{for any } u \in \mathcal{K}_0. \tag{8.11}$$

We show that the function $u \to u(\mathbf{R}_{\varphi^{a_0}})$ is continuous on $\mathcal{K}_0$. Let $Y \to \mathrm{Spec}\, A$ be a projective log resolution on which $\mathbf{R}_{\varphi^{a_0}}$ is determined, with trace R_Y on Y, let $N_0 \geq 1$ be an integer with $N_0 R_Y$ Cartier, let H be an ample Cartier divisor on Y, and let $k_0 \geq 1$ be an integer with $\mathcal{O}_Y(k_0 H)$ and $\mathcal{O}_Y(k_0 H - N_0 R_Y)$ globally generated. Let $g_1, \ldots, g_{m_1}$ and $g'_1, \ldots, g'_{m_2}$ be rational functions on Y which are global sections generating $\mathcal{O}_Y(k_0 H)$ and $\mathcal{O}_Y(k_0 H - N_0 R_Y)$ respectively. For $u \in \mathcal{K}_0\backslash\{v_0\}$, since $Y \to \mathrm{Spec}\, A$ is proper and $u \geq 0$ on A, u has a center $\xi \in Y$. If t_1 and t_2 are local equations of $k_0 H$ and $k_0 H - N_0 R_Y$ at ξ, then $g_k t_1, g'_k t_2 \in \mathcal{O}_{Y,\xi}$ for any k, with at least one unit among the $g_k t_1$ and among the $g'_k t_2$. Thus $\min_k u(g_k) = -u(t_1)$, $\min_k u(g'_k) = -u(t_2)$, and

$$N_0 u(\mathbf{R}_{\varphi^{a_0}}) = u(t_1) - u(t_2) = \min_k u(g'_k) - \min_k u(g_k).$$

Both sides vanish at $u = v_0$. The right side is a difference of minima of finitely many coordinates, hence continuous on $\mathcal{K}_0$. Moreover, by Definition-Theorem 7.3(3) and the coordinate bounds of $\mathcal{K}_0$,

$$0 \leq u(\mathbf{R}_{\varphi^{a_0}}) \leq m_{\varphi^{a_0}}^{-1} u(x_R) \leq \kappa_1 \quad \text{for any } u \in \mathcal{K}_0. \tag{8.12}$$

Let

$$f(u) := \frac{(\varphi^{a_0})_* u}{1 + u(\mathbf{R}_{\varphi^{a_0}})}, \qquad \sigma(u) := \log(1 + u(\mathbf{R}_{\varphi^{a_0}})) \qquad (u \in \mathcal{K}_0).$$

Both are continuous on $\mathcal{K}_0$, and $0 \leq \sigma \leq \log(1 + \kappa_1)$ by (8.12). By (8.8),

$$1 + u_j(\mathbf{R}_{\varphi^{a_0}}) = \frac{a_{\mathrm{can}}(w_{j+1}, A)}{a_{\mathrm{can}}(w_j, A)},$$

so $f(u_j) = u_{j+1}$, $f(\mathcal{K}_0) \subset \mathcal{K}_0$, and

$$\sigma(u_j) = \log a_{\mathrm{can}}(w_{j+1}, A) - \log a_{\mathrm{can}}(w_j, A). \tag{8.13}$$

For $i \geq 1$, let μ_i be the probability measure on $\mathcal{K}_0$ with mass L_i^{-1} at each u_j, $j_i \leq j < j_i'$. For any continuous function F on $\mathcal{K}_0$,

$$\left| \int F \circ f d\mu_i - \int F d\mu_i \right| = L_i^{-1} |F(u_{j_i'}) - F(u_{j_i})| \leq 2L_i^{-1} \sup |F| \to 0.$$

Since the Radon probability measures on $\mathcal{K}_0$ form a weak-$*$ compact set, a subnet of $(\mu_i)_{i\geq 1}$ converges in the weak-$*$ topology to a Radon probability measure μ on $\mathcal{K}_0$. Since f is continuous, $\int F \circ f d\mu = \int F d\mu$ for any continuous function F on $\mathcal{K}_0$, so μ is f-invariant.

The function $u \to u(\mathfrak{m}_A) = \min_k u(x_k)$ is continuous on $\mathcal{K}_0$. For $\tau > 0$, the function $F_\tau(u) := \min\{1, u(\mathfrak{m}_A)/\tau\}$ is continuous, and by (8.10), $\int F_\tau d\mu_i \leq (\tau\lambda_i)^{-1}$. Thus $\int F_\tau d\mu = 0$ for any $\tau > 0$, and

$$\mu(\{u \in \mathcal{K}_0 \mid u(\mathfrak{m}_A) > 0\}) = 0. \tag{8.14}$$

By (8.13), (8.10), and (8.7),

$$\int \sigma d\mu_i = \frac{1}{L_i} \log \frac{z_{j_i'} \rho_{j_i'}}{z_{j_i} \rho_{j_i}} \geq \frac{1}{L_i} \log c(\varphi^{a_0 L_i}).$$

Since $L_i \to +\infty$ along the subnet, $c(\varphi^{a_0 L})^{1/L} \to Q^{a_0}$ as $L \to +\infty$, and σ is continuous,

$$\int \sigma d\mu \geq a_0 \log Q > 0. \tag{8.15}$$

Step 3. In this step we decompose μ along the centers of the valuations. For $u \in \mathcal{K}_0$ let

$$\zeta(u) := \operatorname{center}_{\operatorname{Spec} A} u = \{0\} \cup \{x \in A \backslash \{0\} \mid u(x) > 0\}.$$

Then

$$\zeta(f(u)) = \{0\} \cup \{x \in A \backslash \{0\} \mid u(\varphi^{a_0}(x)) > 0\} = (\varphi^{a_0})^{-1}(\zeta(u)),$$

that is, $\zeta \circ f = \operatorname{Spec}(\varphi^{a_0}) \circ \zeta$. For a non-zero ideal $I = (y_1, \ldots, y_m)$ of A, $\zeta^{-1}(V(I)) = \{u \mid \min_k u(y_k) > 0\}$ is open in $\mathcal{K}_0$, and $\zeta^{-1}(\operatorname{Spec} A) = \mathcal{K}_0$. For a prime ideal P of A write $\mathcal{K}_{V(P)} := \zeta^{-1}(P)$. Lemma 8.1, with $\operatorname{Spec} A$, $\operatorname{Spec}(\varphi^{a_0})$, and $\mathcal{K}_0$ in place of Y, g, and $\mathcal{K}$, implies:

(1) $\sum_{P \in \operatorname{Spec} A} \mu(\mathcal{K}_{V(P)}) = 1$, and the set of P with $\mu(\mathcal{K}_{V(P)}) > 0$ is countable.
(2) If $\mu(\mathcal{K}_{V(P)}) > 0$, then $(\varphi^{a_0 N})^{-1}(P) = P$ for some integer $N \geq 1$, and for the least such N the probability measure $\mu_P := \mu(\mathcal{K}_{V(P)})^{-1} \mu|_{\mathcal{K}_{V(P)}}$ is f^N-invariant.

Moreover, $\mathcal{K}_{V(0)} = \{v_0\}$, and $\mathcal{K}_{V(\mathfrak{m}_A)} = \{u \mid u(\mathfrak{m}_A) > 0\}$ is μ-null by (8.14). Let $\mathcal{P}$ be the set of prime ideals P with $0 \neq P \neq \mathfrak{m}_A$ and $\mu(\mathcal{K}_{V(P)}) > 0$. Then

$$\mu(\{v_0\}) + \sum_{P \in \mathcal{P}} \mu(\mathcal{K}_{V(P)}) = 1. \tag{8.16}$$

Step 4. In this step we let $P \in \mathcal{P}$ be arbitrary and prove a lower bound of the residue degree of φ^a on A/P and an inequality between volumes on A_P. Let $N \geq 1$ be the least integer with $(\varphi^{a_0 N})^{-1}(P) = P$, let $a := a_0 N$, and let $r := \dim A_P$. Since A is a localization of a quotient of a convergent power series ring, A is a catenary local domain [Mat89, Theorem 17.9], so $\dim(A/P) = d - r$ [Mat89, Theorem 31.4] and $1 \leq r \leq d - 1$.

Since $(\varphi^a)^{-1}(P) = P$, φ^a induces a finite injective local homomorphism $A/P \to A/P$, which we denote by $(\varphi^a)_{A/P}$. The residue field of A/P is k, and the induced homomorphism of k is φ_k^a, so $[k : \varphi_k^a(k)] = \delta^a$. Let

$$\delta_1 := [\operatorname{Frac}(A/P) : (\varphi^a)_{A/P}(\operatorname{Frac}(A/P))].$$

Since

$$\varphi^{aj}(\mathfrak{m}_A)(A/P) \subset \mathfrak{m}_{A/P}^{c(\varphi^{aj})},$$

Lemma 2.12 with A/P and $(\varphi^a)_{A/P}$ in place of S and φ implies

$$c(\varphi^{aj}) \leq c\big((\varphi^a)_{A/P}^j\big) \leq (\delta_1/\delta^a)^{j/(d-r)}$$

for any $j \geq 0$. Taking j-th roots and letting $j \to +\infty$,

$$\delta_1 \geq \delta^a Q^{a(d-r)}. \tag{8.17}$$

Let $u \in \mathcal{K}_{V(P)}$. For $x \in A\backslash\{0\}$, $u(x) > 0$ if and only if $x \in P$, so u is a real valuation of $\mathrm{Frac}(A)$ non-negative on A_P and centered at PA_P. By (8.11) and Definition-Theorem 7.3(2), $a(u, \mathrm{Spec}\, A, \Delta) \leq a_{\mathrm{can}}(u, A) \leq 1$, so, since $u((PA_P)^m) = mu(PA_P)$ for any integer $m \geq 1$ and by Lemma 7.4,

$$(PA_P)^{\lceil t/u(PA_P)\rceil} \subset \mathfrak{a}_t(u) \subset (PA_P)^{\lceil t/\gamma_P\rceil} \quad \text{for any real number } t > 0, \tag{8.18}$$

with $\mathfrak{a}_t(u)$ taken in A_P. Since A_P is a normal excellent local domain, it is analytically irreducible [Gro65, Scholie 7.8.3 (ii), (v)]. By [Cut14, Theorem 6.5], the volume $\mathrm{vol}_{A_P}(u)$ of Definition 2.8(2), with A_P in place of A, exists and

$$\mathrm{vol}_{A_P}(u) = \lim_{t\to+\infty} \frac{e(\mathfrak{a}_t(u))}{t^r}. \tag{8.19}$$

By (8.18),

$$e(PA_P)\gamma_P^{-r} \leq \mathrm{vol}_{A_P}(u) \leq e(PA_P)u(PA_P)^{-r} < +\infty.$$

Moreover,

$$\mathrm{vol}_{A_P}(u) = \inf_I \frac{e(I)}{u(I)^r},$$

where I runs over the PA_P-primary ideals of A_P: the inequality $\leq$ holds because $I^m \subset \mathfrak{a}_{mu(I)}(u)$ for any $m \geq 1$, and the inequality $\geq$ follows from $u(\mathfrak{a}_j(u)) \geq j$ and (8.19). Since any such I is generated by finitely many elements of A, $u \to u(I)$ is continuous on $\mathcal{K}_{V(P)}$. Thus vol_{A_P} is upper semicontinuous, hence Borel, on $\mathcal{K}_{V(P)}$.

For $u \in \mathcal{K}_0$ let

$$\Gamma(u) := \prod_{i=0}^{N-1} \left(1 + f^i(u)(\mathbf{R}_{\varphi^{a_0}})\right),$$

so $\log\Gamma = \sum_{i=0}^{N-1} \sigma \circ f^i$ is continuous and $1 \leq \Gamma \leq (1+\kappa_1)^N$ by (8.12). By the definition of f and induction on i,

$$f^i(u) = \frac{(\varphi^{a_0 i})_* u}{\prod_{i'<i}(1 + f^{i'}(u)(\mathbf{R}_{\varphi^{a_0}}))} \quad \text{for } 0 \leq i \leq N.$$

For $i = N$ this is

$$f^N(u) = \frac{(\varphi^a)_* u}{\Gamma(u)}. \tag{8.20}$$

Let $P_1 = P, P_2, \ldots, P_n$ be the prime ideals P' of A with $(\varphi^a)^{-1}(P') = P$. Since A is normal, going down holds for the integral extension $\varphi^a(A) \subset A$ [Mat89, Theorem 9.4(ii)]. By going down and [Mat89, Exercise 9.8], $\dim A_{P_i} = r$ for any i. For $2 \leq i \leq n$, let

$$\delta_i := [A_{P_i}/P_iA_{P_i} : \varphi^a(\mathrm{Frac}(A/P))],$$

the degree of the residue field extension induced by φ^a. Since $A_P/PA_P = \mathrm{Frac}(A/P)$, δ_1 is the degree of the residue field extension at $P_1 = P$. Similar to [Yos26, Proof of Lemma 8.2], we show that for any PA_P-primary ideal I of A_P,

$$D^a e(I) = \sum_{i=1}^{n} \delta_i e(\varphi^a(I)A_{P_i}). \tag{8.21}$$

Since φ^a makes the localization M of A at $\varphi^a(A\backslash P)$ a finite torsion-free A_P-module of rank D^a, $M/I^mM \cong \prod_{i=1}^n A_{P_i}/(\varphi^a(I)A_{P_i})^m$ for any integer $m \geq 1$, an A_{P_i}-module of finite length ℓ has A_P-length $\delta_i\ell$, and $\dim A_{P_i} = r$ for any i, the associativity formula [Mat89, Theorem 14.7] for the A_P-module M gives (8.21).

Let $u \in \mathcal{K}_{V(P)}$ and $v := f^N(u)$. Since $\zeta(v) = (\varphi^a)^{-1}(P) = P$, $v \in \mathcal{K}_{V(P)}$. Let $j \geq 1$ be an integer. By (8.20), $u(\varphi^a(x)) = \Gamma(u)v(x)$ for $x \in A_P$, so

$$\varphi^a(\mathfrak{a}_j(v))A_P \subset \mathfrak{a}_{\Gamma(u)j}(u).$$

For $i \geq 2$, (8.18) for v and $\varphi^a(P) \subset P_i$ imply

$$\varphi^a(\mathfrak{a}_j(v))A_{P_i} \subset (P_iA_{P_i})^{\lceil j/\gamma_P \rceil}, \qquad \text{so} \qquad e(\varphi^a(\mathfrak{a}_j(v))A_{P_i}) \geq (j/\gamma_P)^r e(P_iA_{P_i}).$$

By (8.21) with $I = \mathfrak{a}_j(v)$,

$$D^a \frac{e(\mathfrak{a}_j(v))}{j^r} \geq \delta_1 \frac{e(\mathfrak{a}_{\Gamma(u)j}(u))}{j^r} + \kappa_0, \qquad \kappa_0 := \gamma_P^{-r} \sum_{i=2}^{n} \delta_i e(P_iA_{P_i}).$$

Letting $j \to +\infty$, by (8.19),

$$D^a \operatorname{vol}_{A_P}(f^N(u)) \geq \delta_1 \Gamma(u)^r \operatorname{vol}_{A_P}(u) + \kappa_0 \quad \text{for any } u \in \mathcal{K}_{V(P)}. \tag{8.22}$$

Here $\kappa_0 > 0$ if $n \geq 2$, and $\kappa_0 = 0$ if $n = 1$.

Step 5. In this step we show that $\mathcal{P} \neq \emptyset$ and that any $P \in \mathcal{P}$ satisfies (1) and (2) with a as in **Step 4**.

Let $P \in \mathcal{P}$, with N, a, r, δ_1, n, and κ_0 as in **Step 4**. For $u \in \mathcal{K}_{V(P)}$ let

$$\omega(u) := \log\left(\gamma_P^r \operatorname{vol}_{A_P}(u)/e(PA_P)\right),$$

a finite non-negative Borel function on $\mathcal{K}_{V(P)}$, and let

$$\Xi(u) := \omega(f^N(u)) - \omega(u) - r\log\Gamma(u) - \log\delta_1 + \log D^a.$$

By (8.22),

$$\Xi(u) \geq \log\left(1 + \frac{\kappa_0}{\delta_1 \Gamma(u)^r \operatorname{vol}_{A_P}(u)}\right) \geq 0, \tag{8.23}$$

so $\omega \circ f^N - \omega \geq \log\delta_1 - \log D^a$ on $\mathcal{K}_{V(P)}$. We show that $\omega \circ f^N - \omega$ is μ_P-integrable with integral 0. For $\tau > 0$, the function $\min\{\omega, \tau\}$ is bounded, so, by the f^N-invariance of μ_P,

$$\int (\min\{\omega,\tau\} \circ f^N - \min\{\omega,\tau\}) d\mu_P = 0.$$

As $\tau \to +\infty$, the positive and negative parts of $\min\{\omega,\tau\} \circ f^N - \min\{\omega,\tau\}$ are increasing and converge pointwise to those of $\omega \circ f^N - \omega$, and the negative parts are bounded by $\log D^a - \log\delta_1$. By monotone convergence, the positive part and the negative part of $\omega \circ f^N - \omega$ have finite and equal integrals. Since $\log\Gamma$ is bounded, integrating the definition of Ξ against μ_P implies

$$0 = \int \Xi d\mu_P + r\int \log\Gamma d\mu_P + \log\delta_1 - \log D^a. \tag{8.24}$$

Since $\Xi \geq 0$, $D^a = \delta^a Q^{ad}$, and (8.17) holds,

$$\int \log\Gamma d\mu_P \leq \frac{\log D^a - \log\delta_1}{r} \leq a\log Q. \tag{8.25}$$

For any $P_\circ \in \mathcal{P}$, let $P_\circ^- := (\varphi^{a_0})^{-1}(P_\circ)$. The fibers of ζ of positive mass are over periodic points by **Step 3**(2). If $y, y' \in \operatorname{Spec} A$ satisfy $\operatorname{Spec}(\varphi^{a_0})(y) = \operatorname{Spec}(\varphi^{a_0})(y')$, $\operatorname{Spec}(\varphi^{a_0N_1})(y) = y$, and $\operatorname{Spec}(\varphi^{a_0N_2})(y') = y'$ for integers $N_1, N_2 \geq 1$, then

$$y = \operatorname{Spec}(\varphi^{a_0N_1N_2})(y) = \operatorname{Spec}(\varphi^{a_0N_1N_2})(y') = y'.$$

Since $\zeta \circ f = \operatorname{Spec}(\varphi^{a_0}) \circ \zeta$, we have

$$\mathcal{K}_{V(P_\circ)} \subset f^{-1}(\mathcal{K}_{V(P_\circ^-)}),$$

and since $P_\circ$ is periodic, the set

$$f^{-1}(\mathcal{K}_{V(P_\circ^-)}) \backslash \mathcal{K}_{V(P_\circ)}$$

is contained in the union of the fibers of ζ of mass 0, which is μ-null by **Step 3**(1). Hence

$$f_*(\mu|_{\mathcal{K}_{V(P_\circ)}}) = \mu|_{\mathcal{K}_{V(P_\circ^-)}}.$$

In particular

$$\mu(\mathcal{K}_{V(P_\circ^-)}) = \mu(\mathcal{K}_{V(P_\circ)}) > 0,$$

so $P_\circ^- \neq \mathfrak{m}_A$ by (8.14). If $N' \geq 1$ is an integer with $(\varphi^{a_0 N'})^{-1}(P_\circ) = P_\circ$ and $P_\circ^- = 0$, then $P_\circ = (\varphi^{a_0(N'-1)})^{-1}(P_\circ^-) = 0$, a contradiction. Thus $P_\circ^- \neq 0$, so $P_\circ^- \in \mathcal{P}$, and $\mathcal{P}$ is a disjoint union of cycles under $(\varphi^{a_0})^{-1}$. Let $P^{(i)} := (\varphi^{a_0 i})^{-1}(P)$ for $0 \leq i \leq N-1$. The primes $P^{(0)}, \dots, P^{(N-1)}$ are distinct by the minimality of N, and $\mu(\mathcal{K}_{V(P^{(i)})}) = \mu(\mathcal{K}_{V(P)})$. Since $\log \Gamma = \sum_{i=0}^{N-1} \sigma \circ f^i$, (8.25) implies

$$\sum_{i=0}^{N-1} \int_{\mathcal{K}_{V(P^{(i)})}} \sigma d\mu = \mu(\mathcal{K}_{V(P)}) \int \log \Gamma d\mu_P \leq a_0 \log Q \sum_{i=0}^{N-1} \mu(\mathcal{K}_{V(P^{(i)})}). \tag{8.26}$$

Since $\sigma(v_0) = 0$, (8.15), (8.16), and (8.26) for all the cycles in $\mathcal{P}$ imply

$$a_0 \log Q \leq \int \sigma d\mu = \sum_{P \in \mathcal{P}} \int_{\mathcal{K}_{V(P)}} \sigma d\mu \leq a_0 \log Q \sum_{P \in \mathcal{P}} \mu(\mathcal{K}_{V(P)}) = a_0 \log Q (1 - \mu(\{v_0\})).$$

Since $a_0 \log Q > 0$, we have $\mu(\{v_0\}) = 0$, $\mathcal{P} \neq \emptyset$ by (8.16), and equality holds in (8.26) for any $P \in \mathcal{P}$. Thus $\int \log \Gamma d\mu_P = a \log Q$ for any $P \in \mathcal{P}$.

Fix $P \in \mathcal{P}$. Since $D^a = \delta^a Q^{ad}$, $\dim(A/P) = d - r$, and $\int \log \Gamma d\mu_P = a \log Q$, (8.24) becomes

$$0 = \int \Xi d\mu_P + \log \delta_1 - \log(\delta^a Q^{a \dim(A/P)}),$$

where $\int \Xi d\mu_P \geq 0$ by (8.23) and $\log \delta_1 - \log(\delta^a Q^{a \dim(A/P)}) \geq 0$ by (8.17). Thus

$$\delta_1 = \delta^a Q^{a \dim(A/P)},$$

and $\Xi = 0$ μ_P-almost everywhere. If $n \geq 2$, then $\kappa_0 > 0$ and (8.23) implies $\Xi(u) > 0$ for any $u \in \mathcal{K}_{V(P)}$, a contradiction. Thus $n = 1$: P is the only prime ideal P' of A with $(\varphi^a)^{-1}(P') = P$. By going down for $\varphi^a(A) \subset A$ [Mat89, Theorem 9.4(ii)], any minimal prime ideal P' of $\varphi^a(P)A$ satisfies $(\varphi^a)^{-1}(P') = P$, so $P' = P$ since $n = 1$. Since $\varphi^a(P) \subset P$,

$$(\varphi^a)^{-1}(P) = P \quad \text{and} \quad \sqrt{\varphi^a(P)A} = P,$$

that is, P is totally invariant under φ^a. This proves (1) and (2).

Step 6. In this step we prove (3) for $P \in \mathcal{P}$ and a as in **Step 5**. By **Step 5**, $\Xi = 0$ μ_P-almost everywhere and $\delta_1 = \delta^a Q^{a(d-r)}$, so the set $G_0 := \{u \in \mathcal{K}_{V(P)} \mid \Xi(u) = 0\}$ is Borel, $\mu_P(G_0) = 1$, and

$$G_0 = \{u \in \mathcal{K}_{V(P)} \mid \mathrm{vol}_{A_P}(f^N(u)) = (\Gamma(u)/Q^a)^r \mathrm{vol}_{A_P}(u)\}.$$

For $u \in \mathcal{K}_{V(P)}$ let $h(u) := \mathrm{vol}_{A_P}(u)^{1/r} u$, a real valuation centered at PA_P with $\mathrm{vol}_{A_P}(h(u)) = 1$. By (8.20), for $u \in G_0$,

$$h(f^N(u)) = \frac{\Gamma(u)}{Q^a} \mathrm{vol}_{A_P}(u)^{1/r} \frac{(\varphi^a)_* u}{\Gamma(u)} = Q^{-a} (\varphi^a)_* h(u).$$

Let $G := \bigcap_{j \geq 0} f^{-Nj}(G_0) \cap \mathcal{K}_{V(P)}$. Since μ_P is f^N-invariant, $\mu_P(G) = 1$, and by induction on j,

$$h(f^{Nj}(u)) = Q^{-aj} (\varphi^{aj})_* h(u) \quad \text{for any } u \in G \text{ and } j \geq 0. \tag{8.27}$$

Since $\mathrm{vol}_{A_P}(u) < +\infty$ and $h(u)(PA_P) > 0$ for any $u \in \mathcal{K}_{V(P)}$, there exist real numbers $\kappa_3 > 0$ and $\epsilon > 0$ with $\mu_P(E_1) > 2/3$ and $\mu_P(E_2) > 2/3$, where

$$E_1 := \{u \in \mathcal{K}_{V(P)} \mid \mathrm{vol}_{A_P}(u)^{1/r} \leq \kappa_3\}, \qquad E_2 := \{u \in \mathcal{K}_{V(P)} \mid h(u)(PA_P) \geq \epsilon\}.$$

The sets E_1 and E_2 are Borel. Let $j \geq 0$ be an integer. Since $\mu_P(f^{-Nj}(E_2)) = \mu_P(E_2)$, the set $E_1 \cap f^{-Nj}(E_2) \cap G$ has positive μ_P-measure. Let u be an element of $E_1 \cap f^{-Nj}(E_2) \cap G$. By (8.11), Definition-Theorem 7.3(2), and the homogeneity of degree 1 of $a(\cdot, Y)$ and $v \to v(K_Y + \mathrm{Env}_A(-K_A)_Y)$ for Y as in Definition-Theorem 7.3(2),

$$a(h(u), \operatorname{Spec} A, \Delta) \leq a_{\mathrm{can}}(h(u), A) = \mathrm{vol}_{A_P}(u)^{1/r} a_{\mathrm{can}}(u, A) \leq \kappa_3,$$

so Lemma 7.4 with $t = h(u)(x)$ gives $h(u)(x) \leq \gamma_P \kappa_3 \operatorname{ord}_{PA_P}(x)$ for any $x \in A_P \backslash \{0\}$. Since $f^{Nj}(u) \in E_2$, (8.27) implies

$$\begin{aligned} \epsilon Q^{aj} &\leq Q^{aj} h(f^{Nj}(u))(PA_P) = h(u)(\varphi^{aj}(PA_P)A_P) \\ &\leq \gamma_P \kappa_3 \operatorname{ord}_{PA_P}(\varphi^{aj}(PA_P)A_P) = \gamma_P \kappa_3 c((\varphi^a)^j_{A_P}). \end{aligned}$$

Thus (3) holds with $b := \epsilon/(\gamma_P \kappa_3)$. The theorem follows. □

9. Equivariant extraction and minimal lc centers

In this section, we start from a primitive divisorial eigenvaluation w centered at a totally invariant prime ideal. We show that a Veronese of $\bigoplus_{j \geq 0} \mathfrak{a}_j(w)$ is generated in degree 1 (Proposition 9.3), extract the divisor of w equivariantly on a $\mathbb{Q}$-factorial scheme (Construction-Proposition 9.4), and take the minimal lc center of the ideal pair given by the extraction (Section 9.2).

9.1. Valuation ideals and equivariant extraction.

Lemma 9.1 (cf. [Yos26, Lemmas 3.2 and 3.3]). *Notation and conditions as in Set-up 7.1. Let $m \geq 1$ be an integer, let $f := \operatorname{Spec}(\varphi^m) : \operatorname{Spec} A \to \operatorname{Spec} A$, and let*

$$B := D^{-m} f_*(R(f) + \Delta).$$

Then:

1. *$B \geq 0$ and $K_A + B$ is $\mathbb{Q}$-Cartier.*
2. *$(\operatorname{Spec} A, B)$ is klt.*
3. *Let v be a divisorial valuation of $\mathrm{Frac}(A)$ with a center on $\operatorname{Spec} A$. If u is the only real valuation of $\mathrm{Frac}(A)$ with $f_* u = v$, then*

$$a(v, \operatorname{Spec} A, B) = a(u, \operatorname{Spec} A, \Delta).$$

Proof. We let $X := \operatorname{Spec} A$ be the codomain and $X' := \operatorname{Spec} A$ the domain of f, so that $f^* = \varphi^m$ on function fields and

$$[\mathrm{Frac}(A) : \varphi^m(\mathrm{Frac}(A))] = D^m.$$

Let K_X be a canonical divisor of X and let $K_{X'} := f^* K_X + R(f)$. Since the characteristic is zero, $R(f) \geq 0$, so $B \geq 0$.

Let T be a prime divisor on X, let T' run over the prime divisors on X' with $f(T') = T$, and let $e_{T'}$ and $\delta_{T'}$ be the ramification index and the residue degree of $\mathcal{O}_{X',T'}$ over $\mathcal{O}_{X,T}$. Then $f^* T = \sum_{T'} e_{T'} T'$, $f_* T' = \delta_{T'} T$, and $\sum_{T'} e_{T'} \delta_{T'} = D^m$ [Ser79, Chapter I, §4, Proposition 10]. Thus $f_* f^* L = D^m L$ for any $\mathbb{Q}$-divisor L on X, and

$$f_*(K_{X'} + \Delta) = D^m K_X + f_* R(f) + f_* \Delta = D^m (K_X + B).$$

For $g \in \mathrm{Frac}(A)^*$, let $\mathrm{Nm}_{\varphi^m}(g) \in \mathrm{Frac}(A)$ be the element with

$$\varphi^m(\mathrm{Nm}_{\varphi^m}(g)) = \mathrm{Nm}_{\mathrm{Frac}(A)/\varphi^m(\mathrm{Frac}(A))}(g).$$

By [Ser79, Chapter I, §5, Proposition 14] at any prime divisor on X,

$$f_* \operatorname{div}_{X'}(g) = \operatorname{div}_X(\mathrm{Nm}_{\varphi^m}(g)).$$

Since $K_{X'}$ is a canonical divisor of X', $K_{X'} + \Delta$ is $\mathbb{Q}$-Cartier, and X' is local, there exist an integer $N \geq 1$ and $g \in \mathrm{Frac}(A)^*$ with $N(K_{X'} + \Delta) = \operatorname{div}_{X'}(g)$. Thus

$$N D^m (K_X + B) = f_* \operatorname{div}_{X'}(g) = \operatorname{div}_X(\mathrm{Nm}_{\varphi^m}(g)),$$

which implies (1).

Let v be a divisorial valuation of $\mathrm{Frac}(A)$ with a center on X, and let $c > 0$ be the real number with $v(\mathrm{Frac}(A)^*) = c\mathbb{Z}$. Let $\mathcal{O}'$ be the integral closure of $\varphi^m(\mathcal{O}_v)$ in $\mathrm{Frac}(A)$, where $\mathcal{O}_v$ is the valuation ring of v. By [Ser79, Chapter I, §4, Propositions 8, 9 and Corollary to Proposition 10], $\mathcal{O}'$ is a semilocal Dedekind domain. Let $z_1, \dots, z_n$ be the valuations with value group $\mathbb{Z}$ of the localizations of $\mathcal{O}'$ at the maximal ideals, let e_i and δ_i be the ramification index and the residue degree of z_i over v, and let $u_i := ce_i^{-1}z_i$. Then $f_*u_i = v$ for any i, $u_1, \dots, u_n$ are pairwise distinct, and

$$\sum_{i=1}^{n} e_i\delta_i = D^m, \qquad v(\mathrm{Nm}_{\varphi^m}(g)) = c\sum_{i=1}^{n} \delta_i z_i(g) = \sum_{i=1}^{n} e_i\delta_i u_i(g) \tag{9.1}$$

[Ser79, Chapter I, §4, Proposition 10 and §5, Proposition 14]. Moreover, $u_i \geq 0$ on A since A is integral over $\varphi^m(A) \subset \varphi^m(\mathcal{O}_v)$. Thus u_i has a center on X', each z_i is divisorial by Lemma 3.1 for f, and $a(u_i, X', \Delta) > 0$ since (X', Δ) is klt.

Let

$$B_0 := B - (ND^m)^{-1}\,\mathrm{div}_X(\mathrm{Nm}_{\varphi^m}(g)),$$

so that $K_X + B_0 = 0$. The $\mathbb{Q}$-divisor $B_0' := f^*B_0 - R(f)$ satisfies

$$K_{X'} + B_0' = f^*(K_X + B_0) = 0 = K_{X'} + \Delta - N^{-1}\,\mathrm{div}_{X'}(g),$$

so $B_0' = \Delta - N^{-1}\,\mathrm{div}_{X'}(g)$. By Lemma 3.2, $a(v, X, B_0) = a(u_i, X', B_0')$ for any i, that is,

$$a(v, X, B) + \frac{v(\mathrm{Nm}_{\varphi^m}(g))}{ND^m} = a(u_i, X', \Delta) + \frac{u_i(g)}{N}.$$

Multiplying by $e_i\delta_i$ and adding over $i \in \{1, \dots, n\}$, (9.1) implies

$$a(v, X, B) = D^{-m}\sum_{i=1}^{n} e_i\delta_i a(u_i, X', \Delta). \tag{9.2}$$

The right-hand side of (9.2) is positive, which implies (2) by (1). If u is the only real valuation with $f_*u = v$, then $n = 1$, $u_1 = u$, and $e_1\delta_1 = D^m$, so (9.2) implies (3). □

Lemma 9.2. *Let $(A, \mathfrak{m}_A)$ be a normal Noetherian local domain, let $\pi : Y \to \mathrm{Spec}\,A$ be a proper birational morphism from a normal integral scheme Y, let E be a prime divisor on Y, let w be the order of vanishing along E, and let $F \geq 0$ be a π-exceptional Weil divisor on Y with $E \not\subset \mathrm{Supp}\,F$. Then $H^0(Y, \mathcal{O}_Y(F - jE)) = \mathfrak{a}_j(w)$ for any integer $j \geq 0$.*

Proof. Let $g \in \mathrm{Frac}(A)^*$ with $\mathrm{div}_Y(g) + F - jE \geq 0$. Since F is π-exceptional and π_*E is 0 or a prime divisor,

$$\mathrm{div}_A(g) = \pi_*(\mathrm{div}_Y(g) + F - jE) + j\pi_*E \geq 0,$$

so $g \in A$ since A is normal. Since $E \not\subset \mathrm{Supp}\,F$, $w(g) = \mathrm{mult}_E\,\mathrm{div}_Y(g) \geq j$. Conversely, if $g \in A$ and $w(g) \geq j$, then $\mathrm{div}_Y(g) \geq 0$ and $\mathrm{mult}_E\,\mathrm{div}_Y(g) = w(g) \geq j$, so $\mathrm{div}_Y(g) + F - jE \geq 0$ since $F \geq 0$. □

Proposition 9.3 (cf. [Yos26, Lemma 5.7 and Proposition 6.4])**.** *Notation and conditions as in Set-up 7.1. Let P be a prime ideal of A totally invariant under φ, let $Q \geq 2$ be an integer with*

$$D = Q^{\mathrm{ht}\,P}[\mathrm{Frac}(A/P) : \varphi_{A/P}(\mathrm{Frac}(A/P))],$$

and let w be a primitive divisorial valuation of $\mathrm{Frac}(A)$ centered at P with $\varphi_*w = Qw$. Then:

1. *w/Q is the only real valuation u of $\mathrm{Frac}(A)$ with $\varphi_*u = w$.*
2. *There exists an integer $N \geq 1$ such that $\mathfrak{a}_N(w)$ is P-primary, $\mathfrak{a}_{jN}(w) = \mathfrak{a}_N(w)^j$ for any integer $j \geq 1$, and $\varphi(\mathfrak{a}_N(w))A \subset \mathfrak{a}_N(w)^Q$.*
3. *There exists a $\mathbb{Q}$-divisor $B \geq 0$ on $\mathrm{Spec}\,A$ such that $K_A + B$ is $\mathbb{Q}$-Cartier, $(\mathrm{Spec}\,A, B)$ is klt, and $0 < a(w, \mathrm{Spec}\,A, B) < 1$.*

Proof. (1) Let $S := A_P$ and $\kappa := \mathrm{Frac}(A/P)$, the residue field of S. Since P is totally invariant under φ, φ induces a finite injective local $\mathbb{C}$-algebra homomorphism $\varphi_{A_P} : S \to S$ (Lemma 2.11), equal to φ on $\mathrm{Frac}(S) = \mathrm{Frac}(A)$, whose residue field homomorphism $\varphi_\kappa : \kappa \to \kappa$ is induced by $\varphi_{A/P}$. Since w is non-trivial and centered at P, $\dim S = \mathrm{ht}\, P \geq 1$. The ring S is a localization of the excellent ring A_0, so S is an excellent normal local domain, and the completion of S is normal [Gro65, Scholie 7.8.3 (ii), (v)], hence a domain. Thus S is analytically irreducible, and by Lemma 2.5(2), there exists a real number $\beta > 0$ with $\mathfrak{a}_t(w) \subset \mathfrak{m}_S^{\lceil t/\beta \rceil}$ in S for any real number $t > 0$.

Let $u := w/Q$, a real valuation of $\mathrm{Frac}(S)$ centered at $\mathfrak{m}_S$. Then

$$Q^{-1}(\varphi_{A_P})_* u = Q^{-2}\varphi_* w = u, \qquad \mathfrak{a}_t(u) = \mathfrak{a}_{Qt}(w) \subset \mathfrak{m}_S^{\lceil t/(\beta/Q) \rceil}$$

for any real number $t > 0$. Moreover, by assumption,

$$[\mathrm{Frac}(S) : \varphi_{A_P}(\mathrm{Frac}(S))] = D = Q^{\dim S}[\kappa : \varphi_\kappa(\kappa)].$$

By Lemma 3.3(2) with $\lambda := Q$, u is the only real valuation z of $\mathrm{Frac}(A)$ with $\varphi_* z = \varphi_* u = w$.

(3) By (1) and induction on m, for any integer $m \geq 1$,

$$w/Q^m \text{ is the only real valuation } u \text{ of } \mathrm{Frac}(A) \text{ with } (\varphi^m)_* u = w. \tag{9.3}$$

If $(\varphi^{m+1})_* u = w$, then $\varphi_*((\varphi^m)_* u) = w$, so $(\varphi^m)_*(Qu) = w$ by (1), and $Qu = w/Q^m$ by induction.

Since $(\mathrm{Spec}\, A, \Delta)$ is klt and w is divisorial, $0 < a(w, \mathrm{Spec}\, A, \Delta) < +\infty$. We let $m \geq 1$ be an integer with

$$\alpha := Q^{-m} a(w, \mathrm{Spec}\, A, \Delta) < 1,$$

let $f := \mathrm{Spec}(\varphi^m) : \mathrm{Spec}\, A \to \mathrm{Spec}\, A$, and let $B := D^{-m} f_*(R(f) + \Delta)$. By Lemma 9.1(1)(2), $B \geq 0$, $K_A + B$ is $\mathbb{Q}$-Cartier, and $(\mathrm{Spec}\, A, B)$ is klt. By Lemma 9.1(3) and (9.3),

$$a(w, \mathrm{Spec}\, A, B) = a(Q^{-m} w, \mathrm{Spec}\, A, \Delta) = \alpha \in (0, 1).$$

This proves (3).

(2) Let B be as in (3), and let $\alpha := a(w, \mathrm{Spec}\, A, B) \in (0, 1)$. Since w is primitive divisorial, by Chow's lemma and [Tem18, Theorem 1.1.6] there exist a log resolution $\pi : Y \to \mathrm{Spec}\, A$ of $(\mathrm{Spec}\, A, B)$ with π projective and a prime divisor E on Y such that w is the order of vanishing along E.

Write

$$K_Y + B_Y = \pi^*(K_A + B), \qquad B_Y = B_Y^+ - F_Y,$$

where $B_Y^+ \geq 0$ and $F_Y \geq 0$ have no common components. Since π is a log resolution of $(\mathrm{Spec}\, A, B)$, $\mathrm{Supp}\, B_Y$ is snc, and since $(\mathrm{Spec}\, A, B)$ is klt, every coefficient of B_Y^+ is less than 1. Moreover,

$$\mathrm{mult}_E B_Y = 1 - \alpha > 0,$$

so $E \not\subset \mathrm{Supp}\, F_Y$. Since $\pi_* B_Y = B \geq 0$, any component of F_Y is π-exceptional. We let $N_1 \geq 1$ be an integer with $1/N_1 < 1 - \alpha$, $N_1 F_Y$ integral, and $N_1(K_A + B) = \mathrm{div}_A(\tau)$ for some $\tau \in \mathrm{Frac}(A)^*$, and let $B_Y' := B_Y^+ - N_1^{-1} E$. Then $B_Y' \geq 0$ has snc support and coefficients in $[0, 1)$, so (Y, B_Y') is klt, and

$$N_1(K_Y + B_Y') = \mathrm{div}_Y(\tau) + N_1 F_Y - E.$$

Since $\mathrm{Spec}\, A$ is excellent and admits a dualizing complex, π is projective and birational, and $K_Y + B_Y'$ is π-big (π is birational), the A-algebra

$$\mathcal{S} := \bigoplus_{j \geq 0} \mathcal{S}_j, \qquad \mathcal{S}_j := H^0(Y, \mathcal{O}_Y(\lfloor j(K_Y + B_Y') \rfloor)),$$

is finitely generated [LM22, Theorem B]. For any integer $j \geq 0$, multiplication by τ^j is a bijection

$$\mathcal{S}_{jN_1} \to H^0(Y, \mathcal{O}_Y(jN_1 F_Y - jE)),$$

and the bijections for $j \geq 0$ are compatible with products. By Lemma 9.2 with jN_1F_Y in place of F,

$$H^0(Y, \mathcal{O}_Y(jN_1F_Y - jE)) = \mathfrak{a}_j(w) \quad \text{for any integer } j \geq 0. \tag{9.4}$$

By [Bou72, Chapter III, Section 1, no. 3, Proposition 3], there exists an integer $N \geq 1$ with $\mathcal{S}_{jN} = \mathcal{S}_N^j$ for any integer $j \geq 1$. Then $\mathcal{S}_{jNN_1} = \mathcal{S}_{NN_1}^j$, and (9.4) implies

$$\mathfrak{a}_{jN}(w) = \mathfrak{a}_N(w)^j \quad \text{for any integer } j \geq 1.$$

Since $w \geq 0$ on A with center P, $\mathfrak{a}_N(w)$ is the contraction to A of the ideal $\{x \in \mathcal{O}_w \mid w(x) \geq N\}$, which is primary to the maximal ideal $\mathfrak{m}_w$ of the rank 1 valuation ring $\mathcal{O}_w$. Thus $\mathfrak{a}_N(w)$ is primary with radical $\mathfrak{m}_w \cap A = P$. Finally, for $g \in \mathfrak{a}_N(w)$, $w(\varphi(g)) = Qw(g) \geq QN$, so

$$\varphi(\mathfrak{a}_N(w))A \subset \mathfrak{a}_{QN}(w) = \mathfrak{a}_N(w)^Q.$$

The proposition follows. □

Construction-Proposition 9.4 (cf. [Yos26, Proposition 6.4])**.** Let $R := \mathbb{C}\{z_1, \dots, z_n\}$, let $\varphi : R \to R$ be a finite injective local $\mathbb{C}$-algebra homomorphism, let $J \neq 0$ be a prime ideal of R totally invariant under φ, let $Q \geq 2$ be an integer with

$$[\mathrm{Frac}(R) : \varphi(\mathrm{Frac}(R))] = Q^{\mathrm{ht}\, J}[\mathrm{Frac}(R/J) : \varphi_{R/J}(\mathrm{Frac}(R/J))],$$

and let w be a primitive divisorial valuation of $\mathrm{Frac}(R)$ centered at J with $\varphi_* w = Qw$. Then there exist an integer $N \geq 1$, a normal $\mathbb{Q}$-factorial integral scheme W with a projective birational morphism $\pi : W \to \operatorname{Spec} R$, a prime divisor E on W, and a finite surjective morphism $g : W \to W$ such that:

(1) w is the order of vanishing along E, $\mathfrak{a}_{jN}(w) = \mathfrak{a}_N(w)^j$ for any integer $j \geq 1$, $\mathfrak{a}_N(w)\mathcal{O}_W = \mathcal{O}_W(-NE)$, and NE is Cartier.
(2) π is an isomorphism over $\operatorname{Spec} R \backslash V(J)$, and $\pi^{-1}(V(J)) = \operatorname{Supp} E$.
(3) $\pi \circ g = \operatorname{Spec}(\varphi) \circ \pi$ and $g^*E = QE$.
(4) W is klt, (W, E) is lc, and $R(g) - (Q-1)E \geq 0$.

Proof. We divide the proof into several steps.

Step 1. The data $A_0 := R$, $\psi := \varphi$, $\mathfrak{q} := \mathfrak{m}_R$, and $\Delta := 0$ satisfy Set-up 7.1: $\sqrt{\varphi(\mathfrak{m}_R)R} = \mathfrak{m}_R$ since R is finite over $\varphi(R)$, $\dim R = n \geq \mathrm{ht}\, J \geq 1$, and $(\operatorname{Spec} R, 0)$ is klt since R is regular. Then $A = R$, and the hypotheses of Proposition 9.3 hold with $P := J$. By Proposition 9.3(1), w/Q is the only real valuation u of $\mathrm{Frac}(R)$ with $\varphi_* u = w$.

Step 2. By Proposition 9.3(3) with $P := J$, there exists a $\mathbb{Q}$-divisor $B \geq 0$ on $\operatorname{Spec} R$ such that $(\operatorname{Spec} R, B)$ is klt and $\alpha := a(w, \operatorname{Spec} R, B) \in (0, 1)$.

Step 3. We construct π, W, E, and N, and show (1), (2), that W is klt, that $-NE$ is a π-ample Cartier divisor, and

$$\begin{gathered} H^0(W, \mathcal{O}_W(-jNE)) = \mathfrak{a}_{jN}(w) \text{ for any integer } j \geq 0, \\ W \cong \operatorname{Proj}_R \bigoplus_{j \geq 0} \mathfrak{a}_N(w)^j \text{ over } \operatorname{Spec} R. \end{gathered} \tag{9.5}$$

First assume that $\mathrm{ht}\, J = 1$. Since R is a unique factorization domain, $J = tR$ for some $t \in R$. The valuation ring of w dominates the discrete valuation ring R_J, so it equals R_J, and w is the order of vanishing along $V(J)$ since w is primitive. We let $W := \operatorname{Spec} R$, $\pi := \mathrm{id}_W$, $E := V(J)$, and $N := 1$. Then $\mathfrak{a}_j(w) = t^j R = J^j$ for any integer $j \geq 1$, and $-NE = -\operatorname{div}_R(t)$ is Cartier and π-ample since π is the identity. This implies (1), (2), and (9.5), and W is klt since R is regular.

Now assume that $\mathrm{ht}\, J \geq 2$. By Chow's lemma and [Tem18, Theorem 1.1.6], there exist a log resolution $\pi_Y : Y \to \operatorname{Spec} R$ of $(\operatorname{Spec} R, B)$ with π_Y projective and a prime divisor E_Y on Y such that w is the order of vanishing along E_Y. Since $\mathrm{ht}\, J \geq 2$, E_Y is π_Y-exceptional. Since $\operatorname{Spec} R$ is regular, it admits a dualizing complex. Since $(\operatorname{Spec} R, B)$ is klt and $a(w, \operatorname{Spec} R, B) = \alpha < 1$,

by [LM22, Theorem 22.1], there exists a projective birational morphism $\pi : W \to \operatorname{Spec} R$ from a normal $\mathbb{Q}$-factorial integral scheme W whose only exceptional prime divisor is the birational transform E of E_Y. In particular, w is the order of vanishing along E.

Since E is the only exceptional prime divisor,

$$K_W + \pi_*^{-1}B + (1-\alpha)E = \pi^*(K_R + B).$$

Thus $(W, \pi_*^{-1}B + (1-\alpha)E)$ is klt. Since W is $\mathbb{Q}$-factorial and $\pi_*^{-1}B + (1-\alpha)E \geq 0$, W is klt.

Since R is factorial and E is the only π-exceptional prime divisor, $-E$ is π-ample by the negativity lemma [LM22, Lemma 5.15].

Since $-E$ is π-ample and $\mathbb{Q}$-Cartier, there exists an integer $N \geq 1$ such that $-NE$ is Cartier, $\mathcal{O}_W(-NE)$ is globally generated, the R-algebra $\bigoplus_{j\geq 0} H^0(W, \mathcal{O}_W(-jNE))$ is generated in degree 1 [Bou72, Chapter III, Section 1, no. 3, Proposition 3], and $W \cong \operatorname{Proj}_R \bigoplus_{j\geq 0} H^0(W, \mathcal{O}_W(-jNE))$ over $\operatorname{Spec} R$. By Lemma 9.2 with W, 0, and jN in place of Y, F, and j, $H^0(W, \mathcal{O}_W(-jNE)) = \mathfrak{a}_{jN}(w)$ for any integer $j \geq 0$, so $\mathfrak{a}_{jN}(w) = \mathfrak{a}_N(w)^j$ for any integer $j \geq 1$, (9.5) holds, and $\mathfrak{a}_N(w)\mathcal{O}_W = \mathcal{O}_W(-NE)$ since $\mathcal{O}_W(-NE)$ is globally generated. This proves (1). By (9.5), W is the blow-up of $\operatorname{Spec} R$ along $\mathfrak{a}_N(w)$. Since w is centered at J, $\sqrt{\mathfrak{a}_N(w)} = J$. Thus π is an isomorphism over $\operatorname{Spec} R \backslash V(J)$, and

$$\pi^{-1}(V(J)) = V(\mathfrak{a}_N(w)\mathcal{O}_W) = \operatorname{Supp} E,$$

which proves (2).

Step 4. We construct g and prove (3). Let $\nu : W' \to W$ be the normalization of W in the finite extension

$$\varphi : \operatorname{Frac}(R) \to \operatorname{Frac}(R).$$

Thus W' is a normal integral scheme with function field $\operatorname{Frac}(R)$, ν is finite since W is excellent, and $\nu^* = \varphi$ on function fields. Since R is integral over $\varphi(R)$, there exists a proper birational morphism $\pi' : W' \to \operatorname{Spec} R$ with

$$\pi \circ \nu = \operatorname{Spec}(\varphi) \circ \pi'.$$

Since R is normal and finite over $\varphi(R)$, and $\sqrt{\varphi(J)R} = J$, π' is an isomorphism over $\operatorname{Spec} R \backslash V(J)$, and $\pi'^{-1}(V(J)) = \nu^{-1}(\operatorname{Supp} E)$.

Let E' be a prime divisor on W' with $\nu(E') = E$, let e' be the ramification index of ν along E', and let w' be the order of vanishing along E'. Since w is the order of vanishing along E, $\varphi_*(w'/e') = w$, so $w'/e' = w/Q$ by **Step 1**. Since both w' and w have value group $\mathbb{Z}$, $e' = Q$ and $w' = w$. Since distinct prime divisors on W' have distinct valuations, E' is the only prime divisor on W' with $\nu(E') = E$. Thus

$$\nu^* E = QE', \qquad \pi'^{-1}(V(J)) = \operatorname{Supp} E'.$$

Let $\mathcal{L}' := \nu^*\mathcal{O}_W(-NE) = \mathcal{O}_{W'}(-QNE')$, which is π'-ample since ν is finite. By Lemma 9.2 with W', π', E', 0, and jQN in place of Y, π, E, F, and j, and by **Step 3**,

$$\bigoplus_{j\geq 0} H^0(W', \mathcal{L}'^{\otimes j}) = \bigoplus_{j\geq 0} \mathfrak{a}_{jQN}(w) = \bigoplus_{j\geq 0} \mathfrak{a}_N(w)^{jQ},$$

the Q-th Veronese of $\bigoplus_{j\geq 0} \mathfrak{a}_N(w)^j$. Since $\mathcal{L}'$ is π'-ample,

$$W' \cong \operatorname{Proj}_R \bigoplus_{j\geq 0} H^0(W', \mathcal{L}'^{\otimes j}) \text{ over } \operatorname{Spec} R.$$

Since the Proj of a Veronese subalgebra is the Proj of the algebra, (9.5) implies that there exists an isomorphism $\iota : W' \to W$ with $\pi \circ \iota = \pi'$ which is the identity on $\operatorname{Frac}(R)$. We let

$$g := \nu \circ \iota^{-1} : W \to W.$$

Then g is finite and surjective, and

$$\pi \circ g = \pi \circ \nu \circ \iota^{-1} = \operatorname{Spec}(\varphi) \circ \pi' \circ \iota^{-1} = \operatorname{Spec}(\varphi) \circ \pi.$$

Since ι is the identity on $\mathrm{Frac}(R)$, the order of vanishing along $\iota(E')$ is $w' = w$, which is also the order of vanishing along E. Thus $\iota(E') = E$, so $g^*E = (\iota^{-1})^*(QE') = QE$. This proves (3).

Step 5. We prove (4). By **Step 3**, W is klt. Let $k \geq 1$ be an integer. By (3), $(g^k)^*E = Q^kE$, so E is the only prime divisor on W mapped onto E by g^k, and the ramification index of g^k along E is Q^k. Since the characteristic is zero, the coefficient of E in $R(g^k)$ is $Q^k - 1$, so

$$R(g^k) - (Q^k - 1)E \geq 0.$$

For $k = 1$, this proves the last inequality of (4). Since W is $\mathbb{Q}$-factorial, $K_W + E$ and $R(g^k)$ are $\mathbb{Q}$-Cartier. Let

$$B'_k := (g^k)^*E - R(g^k) = E - (R(g^k) - (Q^k - 1)E) \leq E,$$

so $K_W + B'_k = (g^k)^*(K_W + E)$. The scheme W is of finite type over $\operatorname{Spec} R$, hence excellent and admits a dualizing complex.

Let v be a divisorial valuation of $\mathrm{Frac}(R)$ with a center on W. By [EP05, Theorem 3.1.2], v extends to a valuation u_k of $\mathrm{Frac}(R)$ through g^k, which is real since its ramification index over v is finite [EP05, Theorem 3.3.4], and $(g^k)_*u_k = v$. Since g^k is finite, u_k has a center on W, and u_k is divisorial by Lemma 3.1 for g^k. Since $(g^k)^*E = Q^kE$, $u_k(E) = Q^{-k}v(E)$. By Lemma 3.2, since $E - B'_k \geq 0$ and W is klt,

$$a(v, W, E) = a(u_k, W, B'_k) \geq a(u_k, W, E) = a(u_k, W) - u_k(E) > -Q^{-k}v(E).$$

Letting $k \to +\infty$, $a(v, W, E) \geq 0$. Thus (W, E) is lc, which proves (4). □

9.2. **The minimal lc center.**

Definition 9.5. Let X be a regular integral excellent scheme of characteristic zero, let $I \subset \mathcal{O}_X$ be a non-zero ideal, and let $t \geq 0$ be a real number. For any prime divisor E over X, let

$$a(E, X, I^t) := a(E, X) - t \operatorname{mult}_E I, \quad \text{where } \operatorname{mult}_E I := \min\{\operatorname{mult}_E \operatorname{div}(f) \mid f \in I \backslash \{0\}\}$$

is computed on a higher model of X on which E is a divisor. We call (X, I^t) an *ideal pair*. It is *lc* (resp. *klt*) if $a(E, X, I^t) \geq 0$ (resp. $a(E, X, I^t) > 0$) for any prime divisor E over X. *Lc places* and *lc centers* of (X, I^t) are defined as in Definition 2.1. If $X = \operatorname{Spec} S$ for a local ring S, an lc center of (X, I^t) is *minimal* if it contains no other lc center of (X, I^t).

Lemma 9.6 (cf. [FG12, Theorem 7.2])**.** *Let $R := \mathbb{C}\{z_1, \dots, z_n\}$, let P be a prime ideal of R, let $S := R_P$, let I be a non-zero proper ideal of S, and let $t > 0$ be a rational number. Assume that $(\operatorname{Spec} S, I^t)$ is lc and not klt. Then $(\operatorname{Spec} S, I^t)$ has a unique minimal lc center $V(\mathfrak{c})$, $V(\mathfrak{c})$ is contained in any lc center of $(\operatorname{Spec} S, I^t)$, and the local ring $S/\mathfrak{c}$ is of klt type.*

Proof. **Step 1.** Let $J := I \cap R$, so that $JS = I$. Let $X \subset \mathbb{C}^n$ be an open polydisc with center 0 on which finitely many generators of J converge. Thus $R = \mathcal{O}_{X,0}$. For a prime ideal $\mathfrak{p}$ of R, we also denote by $V(\mathfrak{p})$ the closed analytic subset of X defined by $\mathfrak{p}$. For a projective morphism $Y' \to X$ of complex spaces, we let Y'_S be the base change to $\operatorname{Spec} S$ of the corresponding projective scheme over $\operatorname{Spec} R$ [LM22, Theorem 24.3]. Possibly shrinking X to a neighborhood of 0 finitely many times, we may assume that the irreducible components of the finitely many closed analytic subsets of X (resp. of spaces projective over X) considered below are the irreducible components of their germs at 0 (resp. along the fibers over 0). In particular, for an irreducible closed analytic subset Σ_0 of a space Y' projective over X, we have $\Sigma_{0,S} \neq \emptyset$ if and only if the image of Σ_0 in X contains $V(P)$. We let $\mathcal{D}_P(Y')$ be the set of prime divisors E over Y' such that the image in X of the center of E on Y' contains $V(P)$. For $E \in \mathcal{D}_P(Y')$ with Y' normal and any $\mathbb{Q}$-divisor B' on Y' with $K_{Y'} + B'$ $\mathbb{Q}$-Cartier, $a(E, Y', B') = a(E, Y'_S, B'_S)$, where B'_S is the pullback of B' to Y'_S [LM22, Theorem 24.4, Lemma 25.10 and Remark 25.11], and any prime divisor over Y'_S is E_S for some $E \in \mathcal{D}_P(Y')$, since it is a prime divisor on the normalization of a blow-up of Y'_S, which is the base change of a projective scheme over $\operatorname{Spec} R$ [LM22, Theorem 24.3].

Let $\mu : Y_0 \to X$ be a log resolution of J, let M be the divisor on Y_0 with $J\mathcal{O}_{Y_0} = \mathcal{O}_{Y_0}(-M)$, and let

$$B^J_{Y_0} := tM - K_{Y_0/X}, \quad \text{where } K_{Y_0/X} := K_{Y_0} - \mu^* K_X.$$

For any prime divisor E over $Y_{0,S}$, $\operatorname{mult}_E I = \operatorname{mult}_E M_S$, so

$$a(E, \operatorname{Spec} S, I^t) = a(E, Y_{0,S}, B^J_{Y_0,S}).$$

Let $m > t$ be an integer, let $\xi_1, \dots, \xi_m$ be general $\mathbb{C}$-linear combinations of the generators of J converging on X, and let $B := \frac{t}{m}\sum_{i=1}^m \operatorname{div}(\xi_i) \geq 0$. Since $J\mathcal{O}_{Y_0} = \mathcal{O}_{Y_0}(-M)$, by Bertini, $\mu^* \operatorname{div}(\xi_i) = M + G_i$ for any i, where $G_1, \dots, G_m$ are smooth divisors such that $\operatorname{Exc}(\mu) \cup \operatorname{Supp} M \cup \bigcup_{i=1}^m G_i$ is an snc divisor and no G_i contains a stratum of $\operatorname{Exc}(\mu) \cup \operatorname{Supp} M$. Thus

$$K_{Y_0} + B_{Y_0} = \mu^*(K_X + B), \quad \text{where } B_{Y_0} := B^J_{Y_0} + \frac{t}{m}\sum_{i=1}^m G_i.$$

Write $B_{Y_0} = \sum_k b_k D_k$, where the D_k are distinct prime divisors. Since $(\operatorname{Spec} S, I^t)$ is lc, $b_k \leq 1$ for any k with $D_{k,S} \neq \emptyset$. Moreover $b_k = t/m < 1$ if $D_k \subset \bigcup_i G_i$. Thus $B_{Y_0,S}$ and $B^J_{Y_0,S}$ are snc sub-boundaries with coefficients at most 1, the components of coefficient 1 of $B_{Y_0,S}$ are those of $B^J_{Y_0,S}$, and no stratum of the components of coefficient 1 is contained in any G_i. Since

$$a(E, \operatorname{Spec} S, B_S) = a(E, \operatorname{Spec} S, I^t) - \frac{t}{m}\sum_{i=1}^m \operatorname{mult}_E G_{i,S}$$

for any prime divisor E over $Y_{0,S}$, where B_S is the pullback of B to $\operatorname{Spec} S$, and $\operatorname{mult}_E G_{i,S} = 0$ for any i when the center of E on $Y_{0,S}$ is a stratum of $\sum_{b_k=1} D_{k,S}$, we deduce that $(\operatorname{Spec} S, B_S)$ is lc, that the lc places of $(\operatorname{Spec} S, B_S)$ are those of $(\operatorname{Spec} S, I^t)$, and that their lc centers are the images in $\operatorname{Spec} S$ of the strata of $\sum_{b_k=1} D_{k,S}$. The closed analytic subset $\mu(\operatorname{Supp} B^{>1}_{Y_0})$ of X does not contain $V(P)$.

Step 2. For an irreducible closed analytic subset V of X considered below, let $\mathfrak{p}_V$ be the prime ideal of R with $V = V(\mathfrak{p}_V)$. Let $\mathcal{V}$ be the set of images $\mu(\Sigma)$, where Σ runs through the strata of $\sum_{b_k=1} D_k$, and let $\mathcal{V}_P := \{V \in \mathcal{V} \mid V(P) \subset V\}$. Each $V \in \mathcal{V}$ is irreducible. By **Step 1**, the lc centers of $(\operatorname{Spec} S, I^t)$ are the closed subsets $V(\mathfrak{p}_V S)$ with $V \in \mathcal{V}_P$, and, for any $V, V' \in \mathcal{V}_P$, $V(\mathfrak{p}_V S) \subset V(\mathfrak{p}_{V'} S)$ if and only if $V \subset V'$. Since $(\operatorname{Spec} S, I^t)$ is not klt, $\mathcal{V}_P \neq \emptyset$. Thus it suffices to show that the finite set $\mathcal{V}_P$ has a unique minimal element with respect to inclusion. We then let $V(C)$ be the minimal element and $\mathfrak{c} := CS$, so that $C = \mathfrak{c} \cap R$.

The subsets $\mu(\operatorname{Supp} B^{>1}_{Y_0}) \cap V(P)$ and $V \cap V(P)$, $V \in \mathcal{V}\backslash\mathcal{V}_P$, are proper closed analytic subsets of the irreducible set $V(P)$. Let x be a smooth point of $V(P)$ outside $\mu(\operatorname{Supp} B^{>1}_{Y_0})$ and outside any $V \in \mathcal{V}\backslash\mathcal{V}_P$, and let U_x be an open ball around x such that $V(P) \cap U_x$ is a connected submanifold of U_x, hence irreducible, $U_x \cap \mu(\operatorname{Supp} B^{>1}_{Y_0}) = \emptyset$, and $U_x \cap V = \emptyset$ for any $V \in \mathcal{V}\backslash\mathcal{V}_P$. Then $(U_x, B|_{U_x})$ is lc, and the lc centers of $(U_x, B|_{U_x})$ are the images $\mu(\Sigma')$ of the irreducible components Σ' of $\Sigma \cap \mu^{-1}(U_x)$, where Σ runs through the strata of $\sum_{b_k=1} D_k$. Since Σ' contains a non-empty open subset of Σ, the general fibers of $\mu|_{\Sigma'}$ and of $\mu|_\Sigma$ have dimension $\dim \Sigma - \dim \mu(\Sigma)$. Since moreover $\emptyset \neq \mu(\Sigma') \subset U_x$ and $U_x \cap V = \emptyset$ for any $V \in \mathcal{V}\backslash\mathcal{V}_P$, we obtain

$$\mu(\Sigma') \subset \mu(\Sigma) \in \mathcal{V}_P \quad \text{and} \quad \dim \mu(\Sigma') = \dim \mu(\Sigma). \tag{9.6}$$

Let V_1, V_2 be minimal elements of $\mathcal{V}_P$. Since $V(P) \cap U_x$ is irreducible and $V_1 \cap U_x$ is a finite union of lc centers of $(U_x, B|_{U_x})$, there exists an lc center $V_1' \subset V_1$ of $(U_x, B|_{U_x})$ containing $V(P) \cap U_x$. Similarly there exists an lc center $V_2' \subset V_2$ of $(U_x, B|_{U_x})$ containing $V(P) \cap U_x$. By [Fuj24, Theorem 1.1.12], $V_1' \cap V_2'$ is a union of lc centers of $(U_x, B|_{U_x})$, so $V(P) \cap U_x$ is contained in an lc center $V' \subset V_1' \cap V_2'$. By (9.6), $V' \subset V_3$ for some $V_3 \in \mathcal{V}_P$ with $\dim V' = \dim V_3$. Since V_3 is irreducible and $V_3 \cap V_1 \supset V'$ with $\dim V' = \dim V_3$, we have $V_3 \subset V_1$. Similarly $V_3 \subset V_2$. By the minimality of V_1 and V_2, $V_1 = V_3 = V_2$. Hence $\mathcal{V}_P$ has a unique minimal element $V(C)$, and

$$V(C) \subset V \quad \text{for any } V \in \mathcal{V}_P,$$

that is, by **Step 1**, $V(\mathfrak{c})$ is contained in any lc center of $(\operatorname{Spec} S, I^t)$.

We show that $V(C)$ is normal at x. Let V'_x be the minimal lc center of $(U_x, B|_{U_x})$ through x, which is normal at x [Fuj24, Theorem 1.1.12]. Possibly replacing U_x with a smaller ball around x, $V(C) \cap U_x$ is the union of finitely many lc centers $V'^{(i)}$ of $(U_x, B|_{U_x})$ through x, each of dimension $\dim V(C)$ by (9.6), and each containing V'_x. By (9.6), $V'_x \subset V_4$ for some $V_4 \in \mathcal{V}_P$ with $\dim V'_x = \dim V_4$, and $V(C) \subset V_4$. Thus

$$\dim V(C) \le \dim V_4 = \dim V'_x \le \dim V'^{(i)} = \dim V(C),$$

so $V'_x = V'^{(i)}$ for any i, and $V(C) \cap U_x = V'_x$ is normal at x.

Let $\nu : V(C)^\nu \to V(C)$ be the normalization. The stalk of $\nu_*\mathcal{O}_{V(C)^\nu}$ at 0 is the integral closure $(R/C)^\nu$ of R/C in $\operatorname{Frac}(R/C)$ [GR84, Chapter 6, §4.1 and Chapter 8, §3.3], a finite R/C-module since R/C is excellent, and a local ring since R/C is Henselian [Nag62, (45.5)(1) and (43.12)]. The coherent sheaf $\nu_*\mathcal{O}_{V(C)^\nu}/\mathcal{O}_{V(C)}$ vanishes at any point where $V(C)$ is normal, and the stalk of $\nu_*\mathcal{O}_{V(C)^\nu}/\mathcal{O}_{V(C)}$ at 0 is $(R/C)^\nu/(R/C)$. If $\big((R/C)^\nu/(R/C)\big) \otimes_R S \ne 0$, then the support of $\nu_*\mathcal{O}_{V(C)^\nu}/\mathcal{O}_{V(C)}$ contains $V(P)$, hence contains x, a contradiction. Thus

$$S/\mathfrak{c} = (R/C) \otimes_R S = (R/C)^\nu \otimes_R S$$

is normal. Since $(R/C)^\nu \otimes_R S$ is the localization of $(R/C)^\nu$ at the image of $R \setminus P$ and $S/\mathfrak{c}$ is local, $(R/C)^\nu$ has a unique prime ideal lying over P/C, and $S/\mathfrak{c}$ is the localization of $(R/C)^\nu$ at the prime ideal lying over P/C. We let $A := S/\mathfrak{c}$.

Step 3. By [Fuj22, Theorem 1.21], which does not require (X, B) to be lc (see the proof on pp. 73–74 of [Fuj22]), possibly shrinking X to a neighborhood of 0, there exists a projective bimeromorphic morphism $f : Y \to X$ such that Y is $\mathbb{Q}$-factorial over a Stein compact subset $W \ni 0$ of X, $a(E, X, B) \le 0$ for any f-exceptional prime divisor E, and

$$(Y, B_Y^{\mathrm{tr}}) \text{ is dlt}, \quad \text{where } K_Y + B_Y := f^*(K_X + B) \text{ and } B_Y^{\mathrm{tr}} := B_Y^{<1} + \operatorname{Supp} B_Y^{\ge 1}.$$

Since $B \ge 0$ and $a(E, X, B) \le 0$ for any f-exceptional prime divisor E, $B_Y \ge 0$. Let $F := B_Y - B_Y^{\mathrm{tr}} \ge 0$. Any component of F has coefficient > 1 in B_Y. Since $(\operatorname{Spec} S, B_S)$ is lc by **Step 1**, the image in X of any component of F does not contain $V(P)$. Thus $F_S = 0$, and $(Y_S, B_{Y,S}) = (Y_S, B^{\mathrm{tr}}_{Y,S})$ is a dlt pair with $K_{Y_S} + B_{Y,S} = f_S^*(K_{\operatorname{Spec} S} + B_S)$.

Possibly shrinking X to an open neighborhood of 0 contained in W, the finitely many prime divisors on Y used below (the components of F and of $\lfloor B_Y^{\mathrm{tr}} \rfloor$) are $\mathbb{Q}$-Cartier. Possibly shrinking X further, let $h : \widetilde{Y} \to Y$ be a composite of blow-ups such that $\operatorname{Exc}(h) \cup \operatorname{Supp} h_*^{-1} B_Y^{\mathrm{tr}}$ is an snc divisor and $a(E, Y, B_Y^{\mathrm{tr}}) > 0$ for any h-exceptional prime divisor E [Fuj22, Definition 3.7, Lemma 3.9], and let

$$K_{\widetilde{Y}} + B_{\widetilde{Y}} := h^*(K_Y + B_Y^{\mathrm{tr}}).$$

Then $B_{\widetilde{Y}}$ is an snc sub-boundary with coefficients ≤ 1 whose components of coefficient 1 are the components of $h_*^{-1}\lfloor B_Y^{\mathrm{tr}} \rfloor$. Since $\operatorname{Exc}(h) \cup \operatorname{Supp} h_*^{-1} B_Y^{\mathrm{tr}}$ is snc, no stratum Σ of $h_*^{-1}\lfloor B_Y^{\mathrm{tr}} \rfloor$ is contained in $\operatorname{Exc}(h)$, so $h(\Sigma)$ is an irreducible component of dimension $\dim \Sigma$ of an intersection of components of $\lfloor B_Y^{\mathrm{tr}} \rfloor$. In this proof, a *stratum* of $\lfloor B_Y^{\mathrm{tr}} \rfloor$ means such an image $h(\Sigma)$. It is an lc center of (Y, B_Y^{tr}), since the exceptional divisor of the blow-up of Σ has log discrepancy 0 with respect to $(\widetilde{Y}, B_{\widetilde{Y}})$.

Since $(\widetilde{Y}, B_{\widetilde{Y}})$ is an snc sub-pair with coefficients ≤ 1, the center on Y of any prime divisor E over Y with $a(E, Y, B_Y^{\mathrm{tr}}) = 0$ is a stratum of $\lfloor B_Y^{\mathrm{tr}} \rfloor$. Let E be an lc place of $(\operatorname{Spec} S, B_S)$ with center $V(\mathfrak{c})$. The image in X of the center of E on Y contains $V(P)$, so $E \in \mathcal{D}_P(Y)$ and $\operatorname{mult}_E F = 0$, hence $a(E, Y, B_Y^{\mathrm{tr}}) = a(E, Y, B_Y) = 0$ by **Step 1**. Thus some stratum of $\lfloor B_Y^{\mathrm{tr}} \rfloor$ has image $V(C)$ in X. Let T be a stratum of $\lfloor B_Y^{\mathrm{tr}} \rfloor$ which is minimal with respect to inclusion among those with $f(T) = V(C)$. If a stratum $T' \subsetneq T$ of $\lfloor B_Y^{\mathrm{tr}} \rfloor$ satisfied $V(P) \subset f(T')$, then $V(\mathfrak{p}_{f(T')}S)$ would be an lc center of $(\operatorname{Spec} S, I^t)$ contained in $V(\mathfrak{c})$, so $f(T') = V(C)$ by **Step 2**, contradicting the choice of T. Hence

$$V(P) \not\subset f(T') \quad \text{for any stratum } T' \subsetneq T \text{ of } \lfloor B_Y^{\mathrm{tr}} \rfloor. \tag{9.7}$$

Let $y \in T$ be a point outside the strata $T' \subsetneq T$. For any lc center T'' of (Y, B_Y^{tr}) through y, $T \cap T''$ is a union of lc centers [Fuj24, Theorem 1.1.12], that is, of strata contained in T. A stratum contained in T and containing y equals T, so $T \subset T''$. Thus T is the minimal lc center of (Y, B_Y^{tr}) through y, and T is normal at y [Fuj24, Theorem 1.1.12]. Let $T^\nu \to T$ be the normalization, which is an isomorphism outside the preimage of $\bigcup_{T' \subsetneq T} T'$.

Let D_i' be the components of $\lfloor B_Y^{\mathrm{tr}} \rfloor$ containing T, so that T is an irreducible component of $\bigcap_i D_i'$. Since T is a stratum of the dlt pair (Y, B_Y^{tr}) and the D_i' are $\mathbb{Q}$-Cartier, adjunction along the D_i' defines a $\mathbb{Q}$-divisor $B_T^{\mathrm{tr}} \geq 0$ on T^ν with

$$(K_Y + B_Y^{\mathrm{tr}})|_{T^\nu} = K_{T^\nu} + B_T^{\mathrm{tr}}.$$

Let $\widetilde{T}$ be the stratum of $h_*^{-1}\lfloor B_Y^{\mathrm{tr}} \rfloor$ with $h(\widetilde{T}) = T$. Then $h|_{\widetilde{T}}$ factors through a bimeromorphic morphism $\widetilde{T} \to T^\nu$, and

$$(K_{\widetilde{Y}} + B_{\widetilde{Y}})|_{\widetilde{T}} = K_{\widetilde{T}} + B_{\widetilde{T}}, \quad \text{where } B_{\widetilde{T}} := \Big(B_{\widetilde{Y}} - \sum_i h_*^{-1} D_i'\Big)\Big|_{\widetilde{T}}$$

is an snc sub-boundary with coefficients ≤ 1 whose components of coefficient 1 are the components of $\widetilde{T} \cap h_*^{-1} D''$ for the components $D'' \not\supset T$ of $\lfloor B_Y^{\mathrm{tr}} \rfloor$. Since $K_{\widetilde{T}} + B_{\widetilde{T}}$ is the pullback of $K_{T^\nu} + B_T^{\mathrm{tr}}$, the pair $(T^\nu, B_T^{\mathrm{tr}})$ is lc and an lc place of $(T^\nu, B_T^{\mathrm{tr}})$ has center on $\widetilde{T}$ a stratum of the components $\widetilde{T} \cap h_*^{-1} D''$, whose image is contained in a stratum $T' \subsetneq T$.

Since $T_S \neq \emptyset$ and $F_S = 0$, $T \not\subset \operatorname{Supp} F$, so $F|_{T^\nu} \geq 0$. Let $B_T := B_T^{\mathrm{tr}} + F|_{T^\nu} \geq 0$. Since $K_X = \operatorname{div}(dz_1 \wedge \cdots \wedge dz_n) = 0$ and $t \in \mathbb{Q}$,

$$K_X + B = \frac{t}{m} \operatorname{div}(\xi_1 \cdots \xi_m) \sim_{\mathbb{Q}} 0, \quad \text{so} \quad K_{T^\nu} + B_T = \big(f^*(K_X + B)\big)|_{T^\nu} \sim_{\mathbb{Q}} 0.$$

By (9.7) and since the image of $\operatorname{Supp} F$ in X does not contain $V(P)$,

$$a(E, T^\nu, B_T) > 0 \quad \text{for any } E \in \mathcal{D}_P(T^\nu). \tag{9.8}$$

The morphism $T^\nu \to V(C)$ is projective and surjective, and it factors through $V(C)^\nu$ since T^ν is normal. Let

$$T^\nu \xrightarrow{\tau} Z' \xrightarrow{\alpha} V(C)^\nu$$

be the Stein factorization, so that τ is a projective contraction, Z' is normal, and α is finite. The generic point of Z' maps to the generic point of $V(C) \supset V(P)$, so by (9.8) the general fiber of τ is klt with the restriction of B_T. By [BFMT25, Definition 6.18, Remarks 6.19 and 6.31], $\tau : (T^\nu, B_T) \to Z'$ is an lc-trivial fibration with $K_{T^\nu} + B_T \sim_{\mathbb{Q}} \tau^* 0$. Let $B_{Z'}$ be the boundary divisor of τ, and let $\mathbf{M}$ be Shokurov's moduli $\boldsymbol{b}$-divisor of τ, defined with $K_{T^\nu} + B_T \sim_{\mathbb{Q}} \tau^* 0$ and a fixed rational function as in [BFMT25, Definitions 6.20, 6.21 and Remark 6.22]. By [BFMT25, Remarks 6.22 and 6.31], for any higher model $\pi_1 : Z_1 \to Z'$,

$$K_{Z_1} + B_{Z_1} + \mathbf{M}_{Z_1} = \pi_1^*(K_{Z'} + B_{Z'} + \mathbf{M}_{Z'}) \sim_{\mathbb{Q}} 0,$$

where B_{Z_1} is the trace on Z_1 of the boundary $\boldsymbol{b}$-divisor, and $\mathbf{M}$ is $\boldsymbol{b}$-semi-ample by [BFMT25, Theorem 7.3 and Remark 6.30]. The coefficient of a prime divisor E' on Z' in $B_{Z'}$ is $1 - s$, where s is the lc threshold of $\tau^* E'$ with respect to (T^ν, B_T) over the generic point of E'. Since a component of $\tau^* E'$ has multiplicity at least 1 and non-negative coefficient in B_T, we have $s \leq 1$, so $B_{Z'} \geq 0$. For a prime divisor E' over Z' whose center has image in X containing $V(P)$, the coefficient of E' in the boundary $\boldsymbol{b}$-divisor is < 1, since the corresponding lc threshold is positive by (9.8).

Possibly shrinking X to a neighborhood of 0, let $\pi'' : Z'' \to Z'$ be a resolution such that $\mathbf{M}$ descends to Z'', $\operatorname{Supp} B_{Z''} \cup \operatorname{Exc}(\pi'')$ is an snc divisor, and $m_0 \mathbf{M}_{Z''}$ is base point free for an integer $m_0 \geq 2$. Let $H'' \in |m_0 \mathbf{M}_{Z''}|$ be general, and let ψ be the meromorphic function with $m_0 \mathbf{M}_{Z''} = H'' + \operatorname{div}(\psi)$. Let

$$\Delta_{Z'} := B_{Z'} + \frac{1}{m_0} \pi''_* H'' \geq 0.$$

Then

$$K_{Z''} + B_{Z''} + \frac{1}{m_0}H'' = \pi''^*(K_{Z'} + \Delta_{Z'}), \quad K_{Z'} + \Delta_{Z'} = K_{Z'} + B_{Z'} + \mathbf{M}_{Z'} - \frac{1}{m_0}\operatorname{div}(\psi) \sim_{\mathbb{Q}} 0.$$

The coefficients of $B_{Z''} + \frac{1}{m_0}H''$ along the prime divisors in $\mathcal{D}_P(Z')$ that are divisors on Z'' are < 1, and $\operatorname{Supp}(B_{Z''} + H'') \cup \operatorname{Exc}(\pi'')$ is an snc divisor, so

$$a(E, Z', \Delta_{Z'}) > 0 \quad \text{for any } E \in \mathcal{D}_P(Z').$$

We descend $\Delta_{Z'}$ to $\operatorname{Spec} A$ (cf. [FG12, Lemma 1.1]). By **Step 2**, A is the localization of

$$\mathcal{O}_{V(C)^\nu,0} = (R/C)^\nu$$

at the image of $R\backslash P$. Let z' be a point of Z' over 0, let $A' := \mathcal{O}_{Z',z'} \otimes_R S$, and let $\alpha_A : \operatorname{Spec} A' \to \operatorname{Spec} A$ be the morphism induced by α. Since $\mathcal{O}_{Z',z'}$ is a normal domain finite over $(R/C)^\nu$, A' is a normal domain finite over A, and A, A', and the finite A-algebra $A^\dagger$ below are excellent and admit dualizing complexes, being essentially of finite type over R, which is excellent and admits a dualizing complex (Lemma 2.2) [Gro65, Scholie 7.8.3 (ii)], [Sta, Tag 0A7K]. Let Δ' be the restriction of $\Delta_{Z'}$ to $\operatorname{Spec} A'$, and let $K_{A'} := \alpha_A^* K_A + R(\alpha_A)$. Since $K_{Z'} + \Delta_{Z'} \sim_{\mathbb{Q}} 0$ and the restriction of $K_{Z'}$ to $\operatorname{Spec} A'$ is linearly equivalent to $K_{A'}$, there exist an integer $m_1 \geq 1$ and $\chi \in \operatorname{Frac}(A')$ with $m_1(K_{A'} + \Delta') = \operatorname{div}(\chi)$. Since Z' is finite over X and R is Henselian, $\operatorname{Spec} A'$ is an open subscheme of Z'_S, so by **Step 1** $(\operatorname{Spec} A', \Delta')$ is klt, since the prime divisors over $\operatorname{Spec} A'$ belong to $\mathcal{D}_P(Z')$. Let $r := [\operatorname{Frac}(A') : \operatorname{Frac}(A)]$ and

$$\Delta := \frac{1}{r}\alpha_{A*}\big(R(\alpha_A) + \Delta'\big) \geq 0.$$

Then $K_A + \Delta = \frac{1}{r}\alpha_{A*}(K_{A'} + \Delta')$, so $rm_1(K_A + \Delta)$ is the divisor of the norm of χ from $\operatorname{Frac}(A')$ to $\operatorname{Frac}(A)$, and $K_A + \Delta$ is $\mathbb{Q}$-Cartier.

Let L' be a finite Galois extension of $\operatorname{Frac}(A)$ containing $\operatorname{Frac}(A')$, let $\Gamma := \operatorname{Gal}(L'/\operatorname{Frac}(A))$, let $A^\dagger$ be the integral closure of A in L', a finite A-algebra since A is excellent, and let

$$\beta : \operatorname{Spec} A^\dagger \to \operatorname{Spec} A, \qquad \gamma : \operatorname{Spec} A^\dagger \to \operatorname{Spec} A'$$

be the induced morphisms. Let $K_{A^\dagger} := \beta^* K_A + R(\beta)$, which is Γ-invariant, let $\Delta^\dagger := \gamma^*\Delta' - R(\gamma)$, so that $K_{A^\dagger} + \Delta^\dagger = \gamma^*(K_{A'} + \Delta')$, and let

$$\Delta^\ddagger := \frac{1}{\#\Gamma}\sum_{\sigma\in\Gamma} \sigma^*\Delta^\dagger.$$

Since $R(\beta) = \gamma^* R(\alpha_A) + R(\gamma)$ is Γ-invariant, since $\beta_*\gamma^* = [L' : \operatorname{Frac}(A')]\alpha_{A*}$ on $\mathbb{Q}$-divisors on $\operatorname{Spec} A'$, and since

$$\sum_{\sigma\in\Gamma} \sigma^* D^\dagger = \beta^*\beta_* D^\dagger \quad \text{for any } \mathbb{Q}\text{-divisor } D^\dagger \text{ on } \operatorname{Spec} A^\dagger,$$

we obtain

$$\Delta^\ddagger = \frac{1}{\#\Gamma}\beta^*\beta_*\gamma^*\big(R(\alpha_A) + \Delta'\big) - R(\beta) = \beta^*\Delta - R(\beta), \quad \text{that is,} \quad K_{A^\dagger} + \Delta^\ddagger = \beta^*(K_A + \Delta).$$

Let v be a divisorial valuation of $\operatorname{Frac}(A)$ with a center on $\operatorname{Spec} A$, and let u be an extension of v to L'. The valuation u has a center on $\operatorname{Spec} A^\dagger$. For any $\sigma \in \Gamma$, $(u \circ \sigma^*)|_{\operatorname{Frac}(A')}$ extends v, so it has a center on $\operatorname{Spec} A'$, and it is divisorial by Lemma 3.1 for α_A. For any $\sigma \in \Gamma$, $R(\sigma) = 0$ since $K_{A^\dagger}$ is Γ-invariant, so by Lemma 3.2 for σ and then for γ,

$$a(u, \operatorname{Spec} A^\dagger, \sigma^*\Delta^\dagger) = a(u \circ \sigma^*, \operatorname{Spec} A^\dagger, \Delta^\dagger) = a\big((u \circ \sigma^*)|_{\operatorname{Frac}(A')}, \operatorname{Spec} A', \Delta'\big) > 0.$$

Since $\sigma^*\Delta^\dagger - \Delta^\dagger$ is $\mathbb{Q}$-Cartier for any $\sigma \in \Gamma$, $a(u, \operatorname{Spec} A^\dagger, \Delta^\ddagger)$ is the mean of the numbers $a(u, \operatorname{Spec} A^\dagger, \sigma^*\Delta^\dagger)$, $\sigma \in \Gamma$, so it is positive. By Lemma 3.2 for β,

$$a(v, \operatorname{Spec} A, \Delta) = a(u, \operatorname{Spec} A^\dagger, \Delta^\ddagger) > 0.$$

Thus $(\operatorname{Spec} A, \Delta)$ is klt. Since A is normal by **Step 2**, excellent, and admits a dualizing complex, A is of klt type. □

Set-up 9.7. Let n, R, φ, Q, P, d, S, and φ_S be as follows.

(1) $n \geq 1$ is an integer, $R := \mathbb{C}\{z_1, \dots, z_n\}$, $\varphi : R \to R$ is a finite injective local $\mathbb{C}$-algebra homomorphism, and $Q \geq 2$ is an integer with
$$[\operatorname{Frac}(R) : \varphi(\operatorname{Frac}(R))] = Q^n.$$
(2) P is a prime ideal of R totally invariant under φ with $d := \operatorname{ht} P \geq 2$ and
$$[\operatorname{Frac}(R/P) : \varphi_{R/P}(\operatorname{Frac}(R/P))] = Q^{n-d}.$$
(3) $S := R_P$, $\varphi_S := \varphi_{R_P}$, $c_\infty(\varphi_S) = Q$, and $\inf_{j\geq 0} c(\varphi_S^j)/Q^j = 0$.

Construction-Proposition 9.8. Notation and conditions as in Set-up 9.7. Let $J \subset P$ be a non-zero prime ideal of R totally invariant under φ with
$$[\operatorname{Frac}(R/J) : \varphi_{R/J}(\operatorname{Frac}(R/J))] = Q^{n-\operatorname{ht} J},$$
let w be a primitive divisorial valuation of $\operatorname{Frac}(R)$ centered at J with $\varphi_* w = Qw$, and let $N \geq 1$ be an integer with $\mathfrak{a}_{jN}(w) = \mathfrak{a}_N(w)^j$ for any integer $j \geq 1$. Let $I := \mathfrak{a}_N(w)S$ and $t := a(w, \operatorname{Spec} R)/N$. Then $(\operatorname{Spec} S, I^t)$ is lc and not klt, and the minimal lc center $V(\mathfrak{c})$ of $(\operatorname{Spec} S, I^t)$ (see Lemma 9.6) satisfies the following, where $C := \mathfrak{c} \cap R$ and $A := S/\mathfrak{c}$.

(1) $0 \neq \mathfrak{c} \neq PS$, and C is totally invariant under φ.
(2) We have
$$[\operatorname{Frac}(R/C) : \varphi_{R/C}(\operatorname{Frac}(R/C))] = Q^{n-d+\dim A},$$
and the local homomorphism $h : A \to A$ induced by φ satisfies $c_\infty(h) = Q$.
(3) There exists a real number $\beta > 0$ with $\varphi^j(C)R \subset C^{(\lceil \beta Q^j \rceil)}$ for any integer $j \geq 0$, where $C^{(i)}$ is the i-th symbolic power of C.

Proof. **Step 1.** Since $Q^n = Q^{\operatorname{ht} J} Q^{n-\operatorname{ht} J}$, by Construction-Proposition 9.4 there exist an integer $N_0 \geq 1$, a normal $\mathbb{Q}$-factorial integral scheme W with a projective birational morphism $\pi : W \to \operatorname{Spec} R$, a prime divisor E on W, and a finite surjective morphism $g : W \to W$ satisfying (1)–(4) of Construction-Proposition 9.4 with N_0 in place of N. By (2) there, any π-exceptional prime divisor equals E. By (1) there, $w = \operatorname{ord}_E$, and $a(w, \operatorname{Spec} R) = 1$ if E is not π-exceptional. Thus
$$K_W + E = \pi^* K_R + a(w, \operatorname{Spec} R)E.$$
Since E is $\mathbb{Q}$-Cartier, $a(v, \operatorname{Spec} R) = a(v, W, E) + a(w, \operatorname{Spec} R)v(E)$ for any divisorial valuation v of $\operatorname{Frac}(R)$ with a center on $\operatorname{Spec} R$. Such a v has a center on W since π is projective. By the choice of N and by (1) there, where $N_0 E$ is Cartier,
$$\mathfrak{a}_N(w)^{N_0} = \mathfrak{a}_{NN_0}(w) = \mathfrak{a}_{N_0}(w)^N \quad \text{and} \quad \mathfrak{a}_{N_0}(w)\mathcal{O}_W = \mathcal{O}_W(-N_0 E),$$
so $N_0 v(\mathfrak{a}_N(w)) = NN_0 v(E)$, that is, $v(\mathfrak{a}_N(w)) = Nv(E)$. Since $t = a(w, \operatorname{Spec} R)/N$, we obtain
$$a(v, \operatorname{Spec} R) - tv(\mathfrak{a}_N(w)) = a(v, W, E) \tag{9.9}$$
for any divisorial valuation v of $\operatorname{Frac}(R)$ with a center on $\operatorname{Spec} R$.

Step 2. For any prime divisor F over $\operatorname{Spec} S$,
$$\operatorname{mult}_F I = \operatorname{ord}_F(\mathfrak{a}_N(w)) \quad \text{and} \quad a(F, \operatorname{Spec} S) = a(\operatorname{ord}_F, \operatorname{Spec} R),$$
so by (9.9),
$$a(F, \operatorname{Spec} S, I^t) = a(\operatorname{ord}_F, W, E) \geq 0,$$
since (W, E) is lc by Construction-Proposition 9.4(4). Since $J \subset P$, $w = \operatorname{ord}_E$ has a center on $\operatorname{Spec} S$, and $a(w, W, E) = 0$. Thus $(\operatorname{Spec} S, I^t)$ is lc and not klt. Moreover $0 \neq I \subset JS \subsetneq S$, and t is a positive rational number since $a(w, \operatorname{Spec} R)$ is a positive integer. By Lemma 9.6, $(\operatorname{Spec} S, I^t)$ has a unique minimal lc center $V(\mathfrak{c})$. Since $\mathfrak{c}$ is the center of a divisorial valuation of $\operatorname{Frac}(S)$, $\mathfrak{c} \neq 0$. Since $\mathfrak{c} \subset PS$, $C \subset P$.

Step 3. By Construction-Proposition 9.4(3), g induces φ on the function field $\mathrm{Frac}(R)$ of W, so $g_*u = \varphi_*u$ for any real valuation u of $\mathrm{Frac}(R)$. Let

$$G_g := R(g) - (Q-1)E.$$

Since W is $\mathbb{Q}$-factorial, G_g is $\mathbb{Q}$-Cartier, and $G_g \geq 0$ by Construction-Proposition 9.4(4). Since $g^*E = QE$, by Lemma 3.2 for g and the $\mathbb{Q}$-divisor E, $a(\varphi_*u, W, E) = a(u, W, QE - R(g))$, that is,

$$a(\varphi_*u, W, E) = a(u, W, E) + u(G_g) \geq a(u, W, E) \tag{9.10}$$

for any divisorial valuation u of $\mathrm{Frac}(R)$ with a center on $\operatorname{Spec} R$.

Step 4. We show that C is totally invariant under φ. Let L be a finite Galois extension of $\varphi(\mathrm{Frac}(R))$ containing $\mathrm{Frac}(R)$, let $\Gamma := \mathrm{Gal}(L/\varphi(\mathrm{Frac}(R)))$, and let $R^\dagger$ be the integral closure of R in L. Since R is integral over $\varphi(R)$ and $\varphi(R)$ is normal, $R^\dagger$ is the integral closure of $\varphi(R)$ in L and $(R^\dagger)^\Gamma = \varphi(R)$. Let v be a divisorial valuation of $\mathrm{Frac}(R)$ with center C on $\operatorname{Spec} R$. Since R is integral over $\varphi(R)$, there exists a prime ideal C' of R with $\varphi^{-1}(C') = C$ (lying over). Let C' be any such prime ideal. Let $\widetilde{u}$ be an extension of $v \circ \varphi^{-1}$ to L, with center $\mathfrak{q}$ on $\operatorname{Spec} R^\dagger$, and let $\mathfrak{q}'$ be a prime ideal of $R^\dagger$ lying over C'. Since both $\mathfrak{q} \cap \varphi(R)$ and $\mathfrak{q}' \cap \varphi(R)$ equal $\varphi(C)$, there exists $\sigma \in \Gamma$ with $\sigma(\mathfrak{q}) = \mathfrak{q}'$ [Sta, Tag 0BRI]. Let u be the restriction of $\widetilde{u} \circ \sigma^{-1}$ to $\mathrm{Frac}(R)$. Then

$$\varphi_*u = v, \quad \text{and the center of } u \text{ on } \operatorname{Spec} R \text{ is } C'.$$

By Lemma 3.1 for $\operatorname{Spec} \varphi$, u is divisorial.

By going-up for the integral extension $\varphi(R) \subset R$, there exists a prime ideal $P' \supset C'$ of R with $\varphi^{-1}(P') = P$. By Lemma 2.11, $P' = P$. Thus $C' \subset P$, and u has a center on $\operatorname{Spec} S$. Now we take $v = \mathrm{ord}_{F_0}$, where F_0 is an lc place of $(\operatorname{Spec} S, I^t)$ with center $V(\mathfrak{c})$, and let u be a divisorial valuation of $\mathrm{Frac}(R)$ with $\varphi_*u = v$ and center C' on $\operatorname{Spec} R$, constructed in the first paragraph of **Step 4**. By (9.9), (9.10) and **Step 2**,

$$0 = a(v, W, E) = a(u, W, E) + u(G_g) \geq a(u, W, E) \geq 0,$$

so $a(u, W, E) = 0$ and $V(C'S)$ is an lc center of $(\operatorname{Spec} S, I^t)$. By Lemma 9.6, $V(\mathfrak{c}) \subset V(C'S)$, that is, $C' \subset C$. Since $\varphi(R)$ is normal and R is a domain integral over $\varphi(R)$, going-down holds, so

$$\operatorname{ht} C' = \operatorname{ht} \varphi(C) = \operatorname{ht} C.$$

Hence $C' = C$. Since a prime ideal C' with $\varphi^{-1}(C') = C$ exists, $\varphi^{-1}(C) = C$. Since, by going-down, any minimal prime ideal of $\varphi(C)R$ is such a C', $\sqrt{\varphi(C)R} = C$. This proves that C is totally invariant under φ.

Step 5. Let $u_0 := \mathrm{ord}_{F_0}$. Since C is totally invariant, the first paragraph of **Step 4**, applied to $v := Qu_j$ and $C' := C$, implies inductively that there exist divisorial valuations $u_1, u_2, \dots$ of $\mathrm{Frac}(R)$ with center C on $\operatorname{Spec} R$ and

$$\varphi_*u_{j+1} = Qu_j \quad \text{for any } j \geq 0.$$

By (9.10), $0 \leq a(u_{j+1}, W, E) \leq Qa(u_j, W, E)$, so $a(u_j, W, E) = 0$ for any j. Since $g^*E = QE$,

$$Qu_{j+1}(E) = u_{j+1}(g^*E) = (\varphi_*u_{j+1})(E) = Qu_j(E),$$

so $u_j(E) = u_0(E)$ for any j. By **Step 1**, $a(u_j, \operatorname{Spec} R) = a(u_j, W, E) + a(w, \operatorname{Spec} R)u_j(E)$, hence

$$a(u_j, \operatorname{Spec} R) = a(w, \operatorname{Spec} R)u_0(E) = a(u_0, \operatorname{Spec} R) =: a_0 < +\infty \quad \text{for any } j \geq 0.$$

Moreover $\varphi_*^j u_j = Q^j u_0$, that is, $u_j(\varphi^j(f)) = Q^j u_0(f)$ for any $f \in R$. By Lemma 2.5(1) for $X = \operatorname{Spec} R$ and u_j, whose center is C, $\mathfrak{a}_\rho(u_j) \subset (CR_C)^{\lceil \rho/a_0 \rceil}$ in R_C for any real number $\rho > 0$. Since $u_j(\varphi^j(f)) \geq Q^j u_0(CR_C)$ for any $f \in C$, we obtain

$$\varphi^j(C)R_C \subset (CR_C)^{\lceil \beta Q^j \rceil} \quad \text{for any } j \geq 0, \quad \text{where } \beta := u_0(CR_C)/a_0 > 0. \tag{9.11}$$

Since $C^{(i)} = (CR_C)^i \cap R$, (9.11) implies (3). If $\mathfrak{c} = PS$, then $C = P$, $R_C = S$, and (9.11) implies $c(\varphi_S^j) \geq \beta Q^j$ for any $j \geq 0$, which contradicts $\inf_{j\geq 0} c(\varphi_S^j)/Q^j = 0$. Thus $\mathfrak{c} \neq PS$, which, together with **Steps 2 and 4**, proves (1).

Step 6. Let

$$D_C := [\mathrm{Frac}(R/C) : \varphi_{R/C}(\mathrm{Frac}(R/C))].$$

Since $\varphi^{-1}(C) = C$, φ_S induces a finite injective local homomorphism $h : A \to A$ with $[\mathrm{Frac}(A) : h(\mathrm{Frac}(A))] = D_C$. The residue field of A is $\mathrm{Frac}(R/P)$, on which h induces $\varphi_{R/P}$ of degree Q^{n-d}. By (1), $\dim A \geq 1$ and $\mathrm{ht}\, C \geq 1$. Since R is catenary, $\mathrm{ht}\, C = d - \dim A$. Since A is a quotient of S, $c(h^j) \geq c(\varphi_S^j)$ for any j, and by Lemma 2.12 for h, $c(h^j) \leq (D_C/Q^{n-d})^{j/\dim A}$. Thus

$$Q = c_\infty(\varphi_S) \leq c_\infty(h) \leq (D_C/Q^{n-d})^{1/\dim A}, \quad \text{so} \quad D_C \geq Q^{n-d+\dim A}.$$

Since C is totally invariant, $\varphi_{R_C} : R_C \to R_C$ is a finite injective local homomorphism by Lemma 2.11, whose fraction field extension has degree Q^n by Set-up 9.7(1) and whose residue field extension has degree D_C. By Lemma 2.12 for φ_{R_C} and induction on j, and by (9.11),

$$(Q^n/D_C)^j = e\big(\varphi^j(C)R_C\big) \geq e\big((CR_C)^{\lceil \beta Q^j \rceil}\big) \geq (\beta Q^j)^{\mathrm{ht}\, C}$$

for any $j \geq 0$, where $e(CR_C) = 1$ since R_C is regular. Taking j-th roots and letting $j \to +\infty$, $Q^n/D_C \geq Q^{\mathrm{ht}\, C}$, that is, $D_C \leq Q^{n-\mathrm{ht}\, C} = Q^{n-d+\dim A}$. Hence $D_C = Q^{n-d+\dim A}$ and $c_\infty(h) = Q$, which proves (2). □

10. Reduction to a local ring of klt type

In this section we reduce the failure of the uniform bound $\inf_{j\geq 0} c(\varphi^j)/q^j > 0$ to a local ring A of klt type with an endomorphism h and an integer $Q \geq 2$ such that $c(h^j) \geq bQ^j$ for a real number $b > 0$ and any integer $j \geq 0$, while $h^i(I)A \not\subset \overline{I^{Q^i}}$ for any integer $i \geq 1$ and any $\mathfrak{m}_A$-primary ideal I of A (Construction-Theorem 10.6).

Lemma 10.1. *Let $(S, \mathfrak{n})$ be a regular local ring, $\varphi : S \to S$ an injective local homomorphism, $Q \geq 2$ an integer, and $\mathfrak{c} \subsetneq \mathfrak{n}$ a prime ideal with $\varphi(\mathfrak{c}) \subset \mathfrak{c}$, and let $\varphi_{S/\mathfrak{c}}$ be the endomorphism of $S/\mathfrak{c}$ induced by φ. Assume that there exists a real number $\beta > 0$ with*

$$\varphi^j(\mathfrak{c})S \subset \mathfrak{n}^{\lceil \beta Q^j \rceil} \quad \textit{for any integer } j \geq 0,$$

and that there exists an $(\mathfrak{n}/\mathfrak{c})$-primary ideal I of $S/\mathfrak{c}$ with $\varphi_{S/\mathfrak{c}}(I)(S/\mathfrak{c}) \subset \overline{I^Q}$. Then there exists a real number $b > 0$ with $c(\varphi^j) \geq bQ^j$ for any integer $j \geq 0$.

Proof. Since S is regular, $\mathrm{ord}_\mathfrak{n}$ is a valuation, so $\mathrm{ord}_\mathfrak{n}$ is additive on products of non-zero ideals of S, and $c(\varphi^k) = \mathrm{ord}_\mathfrak{n}(\varphi^k(\mathfrak{n})S)$ for any integer $k \geq 0$.

Let $f_1, \ldots, f_m \in \mathfrak{n}$ be elements whose images in $S/\mathfrak{c}$ generate I, and let $\mathfrak{b} := (f_1, \ldots, f_m)S$. Since I is $(\mathfrak{n}/\mathfrak{c})$-primary, there exists an integer $N \geq 1$ with

$$\mathfrak{n}^N \subset \mathfrak{c} + \mathfrak{b}. \tag{10.1}$$

Since $\varphi_{S/\mathfrak{c}}(I)(S/\mathfrak{c}) \subset \overline{I^Q}$ and $\mathfrak{b}^{Qj}$ maps onto I^{Qj} for any $j \geq 1$, for any $i \in \{1, \ldots, m\}$ there exist an integer $e_i \geq 1$ and elements $a_{i,j} \in \mathfrak{b}^{Qj}$, $1 \leq j \leq e_i$, with

$$g_i := \varphi(f_i)^{e_i} + \sum_{j=1}^{e_i} a_{i,j}\varphi(f_i)^{e_i - j} \in \mathfrak{c}. \tag{10.2}$$

Let $e := \max_i e_i$, and let $x_k := \mathrm{ord}_\mathfrak{n}(\varphi^k(\mathfrak{b})S)$ for any integer $k \geq 0$. Then $x_0 \geq 1$. We show that

$$x_{k+1} \geq \min\{Qx_k, \beta Q^k/e\} \quad \text{for any integer } k \geq 0. \tag{10.3}$$

Fix i and k, let $y := \mathrm{ord}_{\mathfrak{n}}(\varphi^{k+1}(f_i))$, and suppose that $y < \min\{Qx_k, \beta Q^k/e_i\}$. Applying φ^k to (10.2), we obtain

$$\varphi^{k+1}(f_i)^{e_i} = \varphi^k(g_i) - \sum_{j=1}^{e_i} \varphi^k(a_{i,j})\varphi^{k+1}(f_i)^{e_i-j}.$$

The left side has order $e_i y$. Since $\varphi^k(a_{i,j}) \in \varphi^k(\mathfrak{b})^{Qj}S$ and $Qx_k > y$, the j-th term of the sum has order at least $Qjx_k + (e_i - j)y > e_i y$. Since $g_i \in \mathfrak{c}$, $\mathrm{ord}_{\mathfrak{n}}(\varphi^k(g_i)) \geq \beta Q^k > e_i y$. Since $\mathrm{ord}_{\mathfrak{n}}$ is a valuation, the right side has order greater than $e_i y$, a contradiction. Thus $y \geq \min\{Qx_k, \beta Q^k/e\}$ for any i, and (10.3) follows because $\varphi^{k+1}(\mathfrak{b})S$ is generated by $\varphi^{k+1}(f_1), \dots, \varphi^{k+1}(f_m)$.

Let $b_0 := \min\{1, \beta/(eQ)\}$. Since $x_0 \geq 1 \geq b_0$ and $\beta Q^k/e \geq b_0 Q^{k+1}$, induction on k in (10.3) implies $x_k \geq b_0 Q^k$ for any $k \geq 0$. Applying φ^k to (10.1), we obtain

$$\varphi^k(\mathfrak{n})^N S \subset \varphi^k(\mathfrak{c})S + \varphi^k(\mathfrak{b})S.$$

Since $b_0 \leq \beta$,

$$Nc(\varphi^k) = \mathrm{ord}_{\mathfrak{n}}(\varphi^k(\mathfrak{n})^N S) \geq \min\{\beta Q^k, x_k\} \geq b_0 Q^k.$$

The lemma follows with $b := b_0/N$. □

Lemma 10.2. *Let $R := \mathbb{C}\{z_1, \dots, z_n\}$, and let $C \subset P$ be prime ideals of R. Then $C^{(i)}R_P \subset P^i R_P$ for any integer $i \geq 1$.*

Proof. Let $f \in C^{(i)}$. Since R_P is a regular local ring, [Nag62, (38.3)] for R_P and its prime ideal CR_P gives $\mathrm{ord}_{PR_P}(f) \geq \mathrm{ord}_{CR_C}(f) \geq i$, so $f \in P^i R_P$. □

Lemma 10.3. *Let $R := \mathbb{C}\{z_1, \dots, z_n\}$, let $\varphi : R \to R$ be a finite injective local $\mathbb{C}$-algebra homomorphism, let $Q \geq 2$ be an integer with*

$$[\mathrm{Frac}(R) : \varphi(\mathrm{Frac}(R))] = Q^n,$$

and let $C \subsetneq P$ be prime ideals of R totally invariant under φ with

$$[\mathrm{Frac}(R/P) : \varphi_{R/P}(\mathrm{Frac}(R/P))] = Q^{n-\mathrm{ht}\,P}.$$

Assume that there exists a real number $\beta > 0$ with $\varphi^j(C)R \subset C^{(\lceil \beta Q^j \rceil)}$ for any integer $j \geq 0$, and that the local homomorphism h of $(R/C)_{P/C}$ induced by φ satisfies $c(h^j) \geq bQ^j$ for some real number $b > 0$ and any integer $j \geq 0$. Then $c_\infty(\varphi_{R_P}) = Q$.

Proof. Let $S := R_P$, $\mathfrak{n} := PS$, and $\varphi_S := \varphi_{R_P}$. By Lemma 2.11, φ_S is finite. Since $\dim S = \mathrm{ht}\,P \geq 1$, the residue field of S is $\mathrm{Frac}(R/P)$, and $Q^n/Q^{n-\mathrm{ht}\,P} = Q^{\dim S}$, Lemma 2.12 implies $c(\varphi_S^j) \leq Q^j$ for any $j \geq 0$. Thus $c_\infty(\varphi_S) \leq Q$.

By the hypothesis on β and Lemma 10.2,

$$\varphi_S^j(CS)S = \varphi^j(C)S \subset \mathfrak{n}^{\lceil \beta Q^j \rceil} \quad \text{for any integer } j \geq 0. \tag{10.4}$$

Let $x_j := c(\varphi_S^j)$ for any integer $j \geq 0$. Then $x_{j+j'} \geq x_j x_{j'}$ and $x_j \geq 1$ for any integers $j, j' \geq 0$. Fix a real number ρ with $1 < \rho < Q$. Since $b(Q/\rho)^N \to +\infty$ as $N \to +\infty$, there exist integers $N \geq 1$ and m with $\rho^N < m \leq bQ^N$. Since $b \leq c(h^0) = 1$, $m \leq Q^N$. Since $S/CS = (R/C)_{P/C}$, h is induced by φ_S, and $c(h^N) \geq m$, we have $\varphi_S^N(\mathfrak{n})S \subset \mathfrak{n}^m + CS$. Applying φ_S^{iN} and using (10.4), we obtain, for any integer $i \geq 0$,

$$\varphi_S^{(i+1)N}(\mathfrak{n})S \subset (\varphi_S^{iN}(\mathfrak{n})S)^m + \varphi_S^{iN}(CS)S \subset \mathfrak{n}^{mx_{iN}} + \mathfrak{n}^{\lceil \beta Q^{iN} \rceil},$$

so

$$x_{(i+1)N} \geq \min\{mx_{iN}, \beta Q^{iN}\}. \tag{10.5}$$

Let $b_0 := \min\{1, \beta/m\}$. Since $x_0 = 1$ and $b_0 m^{i+1} \leq \beta m^i \leq \beta Q^{iN}$, induction on i in (10.5) implies $x_{iN} \geq b_0 m^i$ for any $i \geq 0$. For an integer $j \geq N$, let $i := \lfloor j/N \rfloor$. Then $x_j \geq x_{iN} x_{j-iN} \geq b_0 m^{j/N-1}$. Thus $c_\infty(\varphi_S) \geq m^{1/N} > \rho$. Since $\rho \in (1, Q)$ is arbitrary, $c_\infty(\varphi_S) \geq Q$. The lemma follows. □

Lemma 10.4. *Let $R := \mathbb{C}\{z_1, \dots, z_n\}$, let $\varphi : R \to R$ be a finite injective local $\mathbb{C}$-algebra homomorphism, and let $C \subsetneq P$ be prime ideals of R totally invariant under φ such that $(R/C)_{P/C}$ is normal. Let A_0 be the normalization of R/C. Then A_0 is a normal complex analytic local domain, φ induces a finite injective local $\mathbb{C}$-algebra homomorphism $\psi : A_0 \to A_0$, and there exists a unique prime ideal $\mathfrak{q}$ of A_0 over P/C. Moreover, $\psi^{-1}(\mathfrak{q}) = \mathfrak{q}$, $\sqrt{\psi(\mathfrak{q})A_0} = \mathfrak{q}$, and $(A_0)_{\mathfrak{q}} = (R/C)_{P/C}$.*

Proof. By [Nag62, (45.5)(2)], R/C is a finite module over a convergent power series ring $\mathbb{C}\{y_1, \dots, y_N\}$. By [Nag62, (45.6)], R/C is a Weierstrass ring. In particular, R/C is Henselian and pseudo-geometric, so A_0 is a finite R/C-module. By [Nag62, (43.12)], A_0 is local. Since A_0 is a finite local $\mathbb{C}\{y_1, \dots, y_N\}$-algebra, Weierstrass division [Nag62, (45.3)] shows that A_0 is a quotient of a convergent power series ring $\mathbb{C}\{x_1, \dots, x_M\}$. Thus A_0 is a normal complex analytic local domain.

Since $\varphi^{-1}(C) = C$, φ induces an injective homomorphism $\varphi_{R/C} : R/C \to R/C$, and R/C is finite over $\varphi_{R/C}(R/C)$ because R is finite over $\varphi(R)$. The extension of $\varphi_{R/C}$ to $\mathrm{Frac}(R/C)$ maps elements integral over R/C to elements integral over R/C, so it restricts to an injective $\mathbb{C}$-algebra homomorphism $\psi : A_0 \to A_0$. Since A_0 is finite over R/C and R/C is finite over $\varphi_{R/C}(R/C) \subset \psi(A_0)$, ψ is finite. Since A_0 is integral over $\psi(A_0)$, the maximal ideal of A_0 contracts to the maximal ideal of $\psi(A_0)$, so ψ is local.

Since normalization commutes with localization and $(R/C)_{P/C}$ is normal,

$$A_0 \otimes_{R/C} (R/C)_{P/C} = (R/C)_{P/C}.$$

Thus A_0 has a unique prime ideal $\mathfrak{q}$ over P/C, and $(A_0)_{\mathfrak{q}} = (R/C)_{P/C}$. Since $\varphi^{-1}(P) = P$, $\psi^{-1}(\mathfrak{q}) \cap (R/C) = \varphi_{R/C}^{-1}(P/C) = P/C$. By the uniqueness of $\mathfrak{q}$, $\psi^{-1}(\mathfrak{q}) = \mathfrak{q}$. In particular $\psi(\mathfrak{q})A_0 \subset \mathfrak{q}$.

Let $\mathfrak{p}$ be a minimal prime ideal of $\psi(\mathfrak{q})A_0$. Since A_0 is a domain integral over the normal domain $\psi(A_0)$, going-down [Mat89, Theorem 9.4(ii)] and the minimality of $\mathfrak{p}$ imply $\psi^{-1}(\mathfrak{p}) = \mathfrak{q}$. Let $P_1 \supset C$ be the prime ideal of R with $P_1/C = \mathfrak{p} \cap (R/C)$. Then $\varphi^{-1}(P_1) = P$, so $P_1 = P$ by Lemma 2.11. Thus $\mathfrak{p}$ is over P/C, so $\mathfrak{p} = \mathfrak{q}$. Hence $\sqrt{\psi(\mathfrak{q})A_0} = \mathfrak{q}$. □

Lemma 10.5. *Let $R := \mathbb{C}\{z_1, \dots, z_n\}$, let $\varphi : R \to R$ be a finite injective local homomorphism, and let $C \subset P_1 \subset P$ be prime ideals of R with C and P totally invariant under φ. Assume that $P_1(R/C)_{P/C}$ is totally invariant under the local homomorphism of $(R/C)_{P/C}$ induced by φ. Then P_1 is totally invariant under φ.*

Proof. Let $A := (R/C)_{P/C}$, let h be the local homomorphism of A induced by φ, and let $\mathfrak{p} := P_1 A$. Since $C \subset P_1 \subset P$, $\mathfrak{p} \cap (R/C) = P_1/C$, so $h^{-1}(\mathfrak{p}) = \mathfrak{p}$ implies $\varphi^{-1}(P_1) = P_1$. Let J'' be a minimal prime ideal of $\varphi(P_1)R$. Then J'' contains $\sqrt{\varphi(C)R} = C$. Since R is a domain integral over the normal domain $\varphi(R)$, going-down [Mat89, Theorem 9.4(ii)] and the minimality of J'' imply $J'' \cap \varphi(R) = \varphi(P_1)$. Since $P_1 \subset P$, going-up [Mat89, Theorem 9.4(i)] gives a prime ideal of R containing J'' and lying over $\varphi(P)$, which is P by Lemma 2.11. Thus $C \subset J'' \subset P$, so $J''A$ is a minimal prime ideal of $h(\mathfrak{p})A = \varphi(P_1)A$, and $J''A = \sqrt{h(\mathfrak{p})A} = \mathfrak{p}$. Hence $J'' = P_1$, and $\sqrt{\varphi(P_1)R} = P_1$. The lemma follows. □

Construction-Theorem 10.6. Let $n \geq 1$ and $q \geq 2$ be integers, let $R := \mathbb{C}\{z_1, \dots, z_n\}$, and let $\varphi : R \to R$ be a finite injective local $\mathbb{C}$-algebra homomorphism with $c_\infty(\varphi) = q$, $\inf_{j \geq 0} c(\varphi^j)/q^j = 0$, and

$$[\mathrm{Frac}(R) : \varphi(\mathrm{Frac}(R))] = q^n.$$

Then there exist an integer $a \geq 1$ and prime ideals $0 \neq C \subsetneq P$ of R, both totally invariant under φ^a, with the following properties, where $Q := q^a$, $A := (R/C)_{P/C}$, $\mathfrak{m}_A := (P/C)A$, $k := A/\mathfrak{m}_A$, and $h : A \to A$ is the local homomorphism induced by φ^a:

(1) $\dim A \geq 2$ and

$$[\mathrm{Frac}(A) : h(\mathrm{Frac}(A))] = Q^{\dim A}[k : h_k(k)].$$

(2) A is of klt type.
(3) There exists a real number $b > 0$ with $c(h^j) \geq bQ^j$ for any integer $j \geq 0$.
(4) There do not exist an integer $i \geq 1$ and an $\mathfrak{m}_A$-primary ideal I of A with $h^i(I)A \subset \overline{I^{Q^i}}$.

Proof. **Step 1.** Let $\mathcal{F}$ be the set of pairs (e, P) of an integer $e \geq 1$ and a prime ideal P of R such that Set-up 9.7 holds with φ^e, q^e in place of φ, Q.

We show that $(1, \mathfrak{m}_R) \in \mathcal{F}$. If $n = 1$, then R is a discrete valuation ring, and for any $j \geq 0$ the field degree of φ^j is q^j and its residue degree is 1, so Lemma 2.12 gives $\ell_R(R/\varphi^j(\mathfrak{m}_R)R) = q^j$, that is, $\varphi^j(\mathfrak{m}_R)R = \mathfrak{m}_R^{q^j}$ and $c(\varphi^j) = q^j$, contrary to $\inf_{j\geq 0} c(\varphi^j)/q^j = 0$. Thus $n \geq 2$. Since φ is local, $\varphi^{-1}(\mathfrak{m}_R) = \mathfrak{m}_R$. Since $R/\varphi(\mathfrak{m}_R)R$ is a finite module over $\varphi(R)/\varphi(\mathfrak{m}_R) \cong \mathbb{C}$, $\varphi(\mathfrak{m}_R)R$ is $\mathfrak{m}_R$-primary. Thus $\mathfrak{m}_R$ is totally invariant under φ, $\operatorname{ht} \mathfrak{m}_R = n \geq 2$, $R_{\mathfrak{m}_R} = R$, and

$$[R/\mathfrak{m}_R : \varphi_{R/\mathfrak{m}_R}(R/\mathfrak{m}_R)] = 1 = q^{n-n}.$$

By the hypotheses on φ, $(1, \mathfrak{m}_R) \in \mathcal{F}$.

Fix $(e, P) \in \mathcal{F}$ with $\operatorname{ht} P$ minimal, and let $S := R_P$, $\mathfrak{n} := PS$, and $\varphi_S := (\varphi^e)_{R_P}$ as in Set-up 9.7.

Let $e_1, m_0 \geq 1$ be integers, and let P_1 be a prime ideal of R totally invariant under φ^{e_1}. Then $(\varphi^{e_1 m_0})^{-1}(P_1) = P_1$. We show by induction on m_0 that any prime ideal P_2 of R containing $\varphi^{e_1 m_0}(P_1)$ contains P_1. Since $(\varphi^{e_1})^{-1}(P_2)$ contains $\varphi^{e_1(m_0-1)}(P_1)$, it contains P_1 by induction, so $P_2 \supset \sqrt{\varphi^{e_1}(P_1)R} = P_1$. Thus:

(i) P_1 is totally invariant under $\varphi^{e_1 m_0}$.
(ii) P_1 is the only prime ideal of R lying over $\varphi^{e_1 m_0}(P_1)$, and $(\varphi^{e_1 m_0})_{R_{P_1}}$ is finite, by (i) and Lemma 2.11 for $\varphi^{e_1 m_0}$.

We show that $(em_0, P) \in \mathcal{F}$ for any integer $m_0 \geq 1$. By (i), P is totally invariant under φ^{em_0}. Since field degrees are multiplicative in towers,

$$[\operatorname{Frac}(R) : \varphi^{em_0}(\operatorname{Frac}(R))] = q^{em_0 n}, \quad [\operatorname{Frac}(R/P) : (\varphi^{em_0})_{R/P}(\operatorname{Frac}(R/P))] = q^{em_0(n-\operatorname{ht} P)}.$$

Moreover, $(\varphi^{em_0})_{R_P} = \varphi_S^{m_0}$ and

$$c_\infty(\varphi_S^{m_0}) = c_\infty(\varphi_S)^{m_0} = q^{em_0}.$$

Since

$$c(\varphi_S^{i+i'}) \geq c(\varphi_S^i)c(\varphi_S^{i'}) \geq c(\varphi_S^i) \quad \text{for any integers } i, i' \geq 0,$$

a real number $b_0 > 0$ with $c(\varphi_S^{m_0 j}) \geq b_0 q^{em_0 j}$ for any $j \geq 0$ would imply, with $j := \lfloor i/m_0 \rfloor$,

$$c(\varphi_S^i) \geq c(\varphi_S^{m_0 j}) \geq b_0 q^{em_0 j} \geq b_0 q^{-em_0} q^{ei} \quad \text{for any integer } i \geq 0,$$

contrary to $\inf_{i\geq 0} c(\varphi_S^i)/q^{ei} = 0$. Thus $\inf_{j\geq 0} c(\varphi_S^{m_0 j})/q^{em_0 j} = 0$, and $(em_0, P) \in \mathcal{F}$.

Step 2. Let $\mathcal{G}$ be the set of quadruples (m, J, w, N) satisfying the following:

(a) $m \geq 1$ is an integer.
(b) $J \subsetneq P$ is a non-zero prime ideal of R totally invariant under φ^{em} with

$$[\operatorname{Frac}(R/J) : (\varphi^{em})_{R/J}(\operatorname{Frac}(R/J))] = q^{em(n-\operatorname{ht} J)}.$$

(c) w is a primitive divisorial valuation of $\operatorname{Frac}(R)$ centered at J with $(\varphi^{em})_* w = q^{em} w$.
(d) $N \geq 1$ is an integer with $\mathfrak{a}_{jN}(w) = \mathfrak{a}_N(w)^j$ for any integer $j \geq 1$.

Let $\gamma = (m, J, w, N) \in \mathcal{G}$, and let $\mathfrak{b}_\gamma := \mathfrak{a}_N(w)S$ and $t_\gamma := a(w, \operatorname{Spec} R)/N$. Since $(em, P) \in \mathcal{F}$ by **Step 1** with $m_0 := m$, Construction-Proposition 9.8 with φ^{em}, q^{em} in place of φ, Q shows that $(\operatorname{Spec} S, \mathfrak{b}_\gamma^{t_\gamma})$ is lc and not klt. Let $V(\mathfrak{c}_\gamma)$ be its minimal lc center.

We construct an element of $\mathcal{G}$. By **Step 1**(ii), φ_S is finite. Since S is regular, K_S is Cartier and $\operatorname{Spec} S$ is klt. Thus Set-up 7.1 holds with R, φ^e, P, φ_S, 0 in place of A_0, ψ, $\mathfrak{q}$, φ, Δ, with $D = q^{en}$, $\delta = q^{e(n-\operatorname{ht} P)}$, and $\dim S = \operatorname{ht} P \geq 2$. By Set-up 9.7(3) with φ^e, q^e in place of φ, Q, Theorem 8.2 with φ_S, q^e in place of φ, Q yields m_1, $\mathfrak{p}_1$, b_1 in place of a, P, b: an integer $m_1 \geq 1$,

a prime ideal $\mathfrak{p}_1$ of S with $0 \neq \mathfrak{p}_1 \neq \mathfrak{n}$ totally invariant under $\varphi_S^{m_1}$, and a real number $b_1 > 0$. Let $J_1 := \mathfrak{p}_1 \cap R$. Then $0 \neq J_1 \subsetneq P$, $S/\mathfrak{p}_1 = (R/J_1)_{P/J_1}$, $S_{\mathfrak{p}_1} = R_{J_1}$, and, by [Mat89, Theorem 17.4],

$$\dim(S/\mathfrak{p}_1) = \operatorname{ht} P - \operatorname{ht} J_1.$$

Since P is totally invariant under φ^{em_1} by **Step 1**(i) and $J_1 S = \mathfrak{p}_1$, Lemma 10.5 with φ^{em_1}, 0, and J_1 in place of φ, C, and P_1 shows that J_1 is totally invariant under φ^{em_1}, so $(\varphi^{em_1})_{R_{J_1}}$ is finite by **Step 1**(ii). Thus Theorem 8.2(2) and (3) read

$$[\operatorname{Frac}(R/J_1) : (\varphi^{em_1})_{R/J_1}(\operatorname{Frac}(R/J_1))] = q^{em_1(n-\operatorname{ht} P)} q^{em_1(\operatorname{ht} P - \operatorname{ht} J_1)} = q^{em_1(n-\operatorname{ht} J_1)},$$

$$c\big(((\varphi^{em_1})_{R_{J_1}})^j\big) \geq b_1 q^{em_1 j} \quad \text{for any integer } j \geq 0.$$

Set-up 4.8 holds with R, J_1, $\operatorname{Spec} R_{J_1}$, the closed point of $\operatorname{Spec} R_{J_1}$, R_{J_1}, $(\varphi^{em_1})_{R_{J_1}}$, q^{em_1}, b_1 in place of A_0, $\mathfrak{q}$, Y, y, B, h, Λ, b: R_{J_1} is regular, $\dim R_{J_1} = \operatorname{ht} J_1 \geq 1$, the residue field of R_{J_1} is $\operatorname{Frac}(R/J_1)$, and

$$q^{em_1 n} = q^{em_1 \operatorname{ht} J_1} \cdot q^{em_1(n-\operatorname{ht} J_1)}.$$

By Theorem 6.5, there exists a primitive divisorial valuation w_1 of $\operatorname{Frac}(R)$ centered at J_1 with $(\varphi^{em_1})_* w_1 = q^{em_1} w_1$. By Construction-Proposition 9.4(1) with φ^{em_1}, J_1, q^{em_1}, w_1 in place of φ, J, Q, w, there exists an integer $N_1 \geq 1$ with $\mathfrak{a}_{jN_1}(w_1) = \mathfrak{a}_{N_1}(w_1)^j$ for any integer $j \geq 1$. Thus $(m_1, J_1, w_1, N_1) \in \mathcal{G}$.

Choose $\gamma = (m, J, w, N) \in \mathcal{G}$ with $\dim(S/\mathfrak{c}_\gamma)$ minimal, let $\mathfrak{c} := \mathfrak{c}_\gamma$, let $C := \mathfrak{c} \cap R$, so that $\mathfrak{c} = CS$, let $A := S/\mathfrak{c}$, and let $h : A \to A$ be the homomorphism induced by φ^{em}. By Construction-Proposition 9.8(1)–(3) with φ^{em}, q^{em} in place of φ, Q:

(A) $0 \neq \mathfrak{c} \neq \mathfrak{n}$, so $0 \neq C \subsetneq P$, $\dim A \geq 1$, and $A = (R/C)_{P/C}$. Moreover, C is totally invariant under φ^{em}.

(B) $c_\infty(h) = q^{em}$ and

$$[\operatorname{Frac}(R/C) : (\varphi^{em})_{R/C}(\operatorname{Frac}(R/C))] = q^{em(n-\operatorname{ht} P + \dim A)}.$$

(C) There exists a real number $\beta > 0$ with $\varphi^{emj}(C)R \subset C^{(\lceil \beta q^{emj} \rceil)}$ for any integer $j \geq 0$.

Step 3. By Lemma 9.6 for $(\operatorname{Spec} S, \mathfrak{b}_\gamma^{t_\gamma})$, A is of klt type. Let $\Delta \geq 0$ be a $\mathbb{Q}$-divisor on $\operatorname{Spec} A$ such that $K_A + \Delta$ is $\mathbb{Q}$-Cartier and $(\operatorname{Spec} A, \Delta)$ is klt. Let A_0 be the normalization of R/C. By (A), $C \subsetneq P$ are totally invariant under φ^{em} (for P by **Step 1**(i)). By Lemma 10.4 with φ^{em} in place of φ, A_0, the homomorphism $\psi : A_0 \to A_0$ induced by φ^{em}, and the prime ideal $\mathfrak{q}$ of A_0 over P/C satisfy Set-up 7.1(1), and $(A_0)_{\mathfrak{q}} = A$. The homomorphism of A induced by ψ is h. By Lemma 2.11 for ψ and $\mathfrak{q}$, h is finite. Thus Set-up 7.1 holds with h in place of φ. Since $k = S/\mathfrak{n} = \operatorname{Frac}(R/P)$, (B) and **Step 1** imply

$$[\operatorname{Frac}(A) : h(\operatorname{Frac}(A))] = q^{em(n-\operatorname{ht} P + \dim A)} = q^{em \dim A}[k : h_k(k)]. \tag{10.6}$$

Suppose that $\inf_{j \geq 0} c(h^j)/q^{emj} = 0$. By (10.6) and (B), Theorem 8.2 with h, q^{em} in place of φ, Q yields m', $\mathfrak{p}'$, b' in place of a, P, b: an integer $m' \geq 1$, a prime ideal $\mathfrak{p}'$ of A with $0 \neq \mathfrak{p}' \neq \mathfrak{m}_A$ totally invariant under $h^{m'}$, and a real number $b' > 0$ such that

$$[\operatorname{Frac}(A/\mathfrak{p}') : (h^{m'})_{A/\mathfrak{p}'}(\operatorname{Frac}(A/\mathfrak{p}'))] = q^{emm'(n-\operatorname{ht} P)} q^{emm' \dim(A/\mathfrak{p}')},$$

$$c\big(((h^{m'})_{A_{\mathfrak{p}'}})^j\big) \geq b' q^{emm' j} \quad \text{for any integer } j \geq 0.$$

Let P' be the inverse image of $\mathfrak{p}'$ in R. Then $C \subsetneq P' \subsetneq P$, $A/\mathfrak{p}' = (R/P')_{P/P'}$, $A_{\mathfrak{p}'} = (R/C)_{P'/C}$, and $\dim(A/\mathfrak{p}') = \operatorname{ht} P - \operatorname{ht} P'$ [Mat89, Theorem 17.4]. By **Step 1**(i), C and P are totally invariant under $\varphi^{emm'}$. By Lemma 10.5 with $\varphi^{emm'}$ and P' in place of φ and P_1, P' is totally invariant under $\varphi^{emm'}$, so $(\varphi^{emm'})_{R_{P'}}$ is finite by **Step 1**(ii).

Thus Set-up 9.7(1) and (2) hold with $\varphi^{emm'}$, $q^{emm'}$, P' in place of φ, Q, P: indeed

$$\operatorname{ht} P' > \operatorname{ht} C \geq 1,$$

and

$$[\mathrm{Frac}(R/P') : (\varphi^{emm'})_{R/P'}(\mathrm{Frac}(R/P'))] = q^{emm'(n-\mathrm{ht}\,P)}q^{emm'(\mathrm{ht}\,P-\mathrm{ht}\,P')} = q^{emm'(n-\mathrm{ht}\,P')}.$$

By (C) for $j \in m'\mathbb{N}$, Lemma 10.3 with $\varphi^{emm'}$, $q^{emm'}$, P', $(h^{m'})_{A_{\mathfrak{p}'}}$, b' in place of φ, Q, P, h, b implies $c_\infty((\varphi^{emm'})_{R_{P'}}) = q^{emm'}$. Since $\mathrm{ht}\,P' < \mathrm{ht}\,P$, $(emm', P') \notin \mathcal{F}$ by the minimality of $\mathrm{ht}\,P$ in **Step 1**. Hence there exists a real number $b'' > 0$ with

$$c\big(((\varphi^{emm'})_{R_{P'}})^j\big) \geq b''q^{emm'j} \quad \text{for any integer } j \geq 0.$$

Thus Set-up 4.8 holds with R, P', $\mathrm{Spec}\,R_{P'}$, the closed point of $\mathrm{Spec}\,R_{P'}$, $R_{P'}$, $(\varphi^{emm'})_{R_{P'}}$, $q^{emm'}$, b'' in place of A_0, $\mathfrak{q}$, Y, y, B, h, Λ, b, since $R_{P'}$ is regular with residue field $\mathrm{Frac}(R/P')$. By Theorem 6.5 and Construction-Proposition 9.4(1), there exist a primitive divisorial valuation w' of $\mathrm{Frac}(R)$ centered at P' with $(\varphi^{emm'})_*w' = q^{emm'}w'$ and an integer $N' \geq 1$ with

$$\mathfrak{a}_{jN'}(w') = \mathfrak{a}_{N'}(w')^j \quad \text{for any integer } j \geq 1.$$

Thus $\gamma' := (mm', P', w', N') \in \mathcal{G}$.

Since w' is centered at P', $\sqrt{\mathfrak{a}_{N'}(w')} = P'$. Since S is regular, $(\mathrm{Spec}\,S, \mathfrak{b}_{\gamma'}^{t_{\gamma'}})$ is klt outside $V(P'S)$, so $P'S \subset \mathfrak{c}_{\gamma'}$. Since $\mathfrak{c} \subsetneq P'S$,

$$\dim(S/\mathfrak{c}_{\gamma'}) \leq \dim(S/P'S) < \dim A, \tag{10.7}$$

contrary to the minimality of $\dim A$ in **Step 2**. Thus there exists a real number $b > 0$ with $c(h^j) \geq bq^{emj}$ for any integer $j \geq 0$.

Step 4. Suppose that there exist an integer $i \geq 1$ and an $\mathfrak{m}_A$-primary ideal I of A with $h^i(I)A \subset \overline{I^{q^{emi}}}$. By (C) and Lemma 10.2,

$$\varphi_S^{mij}(\mathfrak{c})S = \varphi^{emij}(C)S \subset \mathfrak{n}^{\lceil \beta q^{emij} \rceil} \quad \text{for any integer } j \geq 0.$$

By (A), $\varphi_S^{mi}(\mathfrak{c}) \subset \mathfrak{c}$, and the homomorphism of A induced by φ_S^{mi} is h^i. By Lemma 10.1 with φ_S^{mi}, q^{emi} in place of φ, Q, there exists a real number $b_2 > 0$ with $c(\varphi_S^{mij}) \geq b_2q^{emij}$ for any integer $j \geq 0$, contrary to $\inf_{j\geq 0} c(\varphi_S^{mij})/q^{emij} = 0$, which holds by **Step 1** with $m_0 := mi$. Thus such i and I do not exist. If $\dim A = 1$, then A is a discrete valuation ring since it is normal, and Lemma 2.12 with (10.6) gives $\ell_A(A/h(\mathfrak{m}_A)A) = q^{em}$, so $h(\mathfrak{m}_A)A = \mathfrak{m}_A^{q^{em}}$, which is impossible with $i = 1$ and $I = \mathfrak{m}_A$. Since $\dim A \geq 1$ by (A), $\dim A \geq 2$.

Let $a := em$ and $Q := q^a$. By **Step 1**(i) and (A), P and C are totally invariant under φ^a, $0 \neq C \subsetneq P$, $A = (R/C)_{P/C}$, and h is induced by φ^a. Conclusion (1) holds by **Step 4** and (10.6), conclusions (2) and (3) hold by **Step 3**, and conclusion (4) holds by **Step 4**. □

11. An equivariant small model

Construction-Theorem 11.1. Notation and conditions as in Set-up 7.1. Then there exist an integer $m \geq 1$, a normal integral scheme Y, a small projective birational morphism $\pi : Y \to \mathrm{Spec}\,A$, and a finite surjective morphism $g : Y \to Y$ with

$$\pi \circ g = \mathrm{Spec}(\varphi^m) \circ \pi$$

satisfying the following, where $T := \pi^{-1}(\mathfrak{m}_A)$ and $\mathfrak{m}_A$ is regarded as the closed point of $\mathrm{Spec}\,A$.

(1) Y is $\mathbb{Q}$-factorial and klt, $g^{-1}(T) = T$, and $g_*v = (\varphi^m)_*v$ for any real valuation v of $\mathrm{Frac}(A)$.

(2) For any real valuation v of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$ with $a(v, Y) < +\infty$,

$$a(g_*v, Y) = a(v, Y) + v(R(g)).$$

(3) There exists a real number $\gamma \geq 1$ such that for any real valuation v of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$,

$$0 < \gamma^{-1}a_{\mathrm{can}}(v, A) \leq a(v, Y) \leq \gamma a_{\mathrm{can}}(v, A),$$

where $a_{\mathrm{can}}(v, A)$ and $a(v, Y)$ may be $+\infty$.

Proof. Log resolutions of pairs over $\operatorname{Spec} A$ exist [Tem18, Theorem 1.1.6].
***Step 1**: small $\mathbb{Q}$-factorial models are klt.* A *small $\mathbb{Q}$-factorial model* of $\operatorname{Spec} A$ is a normal integral scheme Y' with a small birational morphism $\pi' : Y' \to \operatorname{Spec} A$ such that π' is projective and Y' is $\mathbb{Q}$-factorial. For such Y', let $\Delta_{Y'} := (\pi')_*^{-1}\Delta$. Since π' is small, the $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors $K_{Y'} + \Delta_{Y'}$ and $\pi'^*(K_A + \Delta)$ are equal at the generic point of any prime divisor on Y', so

$$K_{Y'} + \Delta_{Y'} = \pi'^*(K_A + \Delta).$$

Thus $(Y', \Delta_{Y'})$ is klt, hence so is Y'. By [LM22, Corollary 22.3], there exists a small $\mathbb{Q}$-factorial model $\pi_0 : Y_0 \to \operatorname{Spec} A$. Let $\Delta_0 := \Delta_{Y_0}$.
***Step 2**: the divisor class space.* Let

$$V := \operatorname{Cl}(A) \otimes_{\mathbb{Z}} \mathbb{Q} \quad \text{and} \quad \rho := \dim_{\mathbb{Q}} V.$$

We show that $\rho < +\infty$. For any small $\mathbb{Q}$-factorial model $\pi' : Y' \to \operatorname{Spec} A$, the push-forward π'_* is a bijection from the prime divisors on Y' to the prime divisors on $\operatorname{Spec} A$ and maps $\operatorname{div}_{Y'}(u)$ to $\operatorname{div}_A(u)$ for any $u \in \operatorname{Frac}(A)^*$, so

$$\operatorname{Pic}(Y') \otimes_{\mathbb{Z}} \mathbb{Q} = \operatorname{Cl}(Y') \otimes_{\mathbb{Z}} \mathbb{Q} \xrightarrow{\pi'_*} V \tag{11.1}$$

is an isomorphism. By [LM22, Theorem 4.11], $\dim_{\mathbb{Q}} N^1(Y_0/\operatorname{Spec} A) < +\infty$, so it suffices to show that the natural surjection

$$\operatorname{Pic}(Y_0) \otimes_{\mathbb{Z}} \mathbb{Q} \to N^1(Y_0/\operatorname{Spec} A)$$

is injective. Let M be a Cartier divisor on Y_0 which is numerically trivial over $\operatorname{Spec} A$. Since $K_{Y_0} + \Delta_0 = \pi_0^*(K_A + \Delta)$, the $\mathbb{Q}$-divisor $M - (K_{Y_0} + \Delta_0)$ is numerically trivial over $\operatorname{Spec} A$, hence π_0-nef, and it is π_0-big since π_0 is birational. By [LM22, Theorem 11.1(ii)], nM is π_0-generated for some positive integer n. Let $Y_0 \to X \to \operatorname{Spec} A$ be the Stein factorization of the morphism over $\operatorname{Spec} A$ defined by nM. Then X is normal, and $\mathcal{O}_{Y_0}(nM)$ is the pull-back of a line bundle $\mathcal{L}$ on X which is ample over $\operatorname{Spec} A$. If a fiber of $X \to \operatorname{Spec} A$ contained a curve C_X, then a curve $C \subset Y_0$ mapping onto C_X would satisfy $nM \cdot C = \deg(C \to C_X)(\mathcal{L} \cdot C_X) > 0$, which is impossible. Thus $X \to \operatorname{Spec} A$ is proper with finite fibers, hence finite, and it is birational, so it is an isomorphism as A is normal. Hence $\mathcal{O}_{Y_0}(nM)$ is the pull-back of a line bundle on $\operatorname{Spec} A$, which is trivial as A is local. Thus the class of M in $\operatorname{Pic}(Y_0) \otimes_{\mathbb{Z}} \mathbb{Q}$ is 0, and $\rho < +\infty$.
***Step 3**: there exist finitely many small $\mathbb{Q}$-factorial models.* Let $L_1, \dots, L_\rho$ be Weil divisors on $\operatorname{Spec} A$ whose classes form a basis of V. Since $\mathcal{O}_A(L_i) \neq 0$ and $\mathcal{O}_A(-L_i) \neq 0$, there exist Weil divisors $L_i^+ \geq 0$ and $L_i^- \geq 0$ on $\operatorname{Spec} A$ with $L_i^+ \sim L_i$ and $L_i^- \sim -L_i$ for any i. By **Step 1**, $(\pi_0)_*^{-1}L_i^\pm$ is $\mathbb{Q}$-Cartier, so computing on a log resolution of $(Y_0, \Delta_0 + \sum_i (\pi_0)_*^{-1}(L_i^+ + L_i^-))$, there exists a rational number $\epsilon > 0$ such that the pairs $(Y_0, \Delta_0 + \epsilon(\pi_0)_*^{-1}L_i^+)$ and $(Y_0, \Delta_0 + \epsilon(\pi_0)_*^{-1}L_i^-)$ are klt for any i. Let $G_i^\pm := K_{Y_0} + \Delta_0 + \epsilon(\pi_0)_*^{-1}L_i^\pm$ for any i, and let

$$G(\mathbf{n}^+, \mathbf{n}^-) := \sum_{i=1}^{\rho} (n_i^+ G_i^+ + n_i^- G_i^-) \quad \text{for } (\mathbf{n}^+, \mathbf{n}^-) \in \mathbb{N}^\rho \times \mathbb{N}^\rho.$$

Since the $G_i^\pm$ are π_0-big as π_0 is birational, the ring

$$\mathcal{A} := \bigoplus_{(\mathbf{n}^+, \mathbf{n}^-) \in \mathbb{N}^\rho \times \mathbb{N}^\rho} H^0\big(Y_0, \mathcal{O}_{Y_0}(\lfloor G(\mathbf{n}^+, \mathbf{n}^-) \rfloor)\big)$$

is finitely generated over A [LM22, Theorem 17.3].

Let n_0 be a positive integer such that $n_0\epsilon \in \mathbb{Z}$ and $n_0(K_A + \Delta)$ is Cartier, hence principal as A is local. Let $w \in \operatorname{Frac}(A)^*$ with $n_0(K_A + \Delta) = \operatorname{div}_A(w)$, and let $u_i^\pm \in \operatorname{Frac}(A)^*$ with $L_i^\pm = \pm L_i + \operatorname{div}_A(u_i^\pm)$ for any i. Since π_0 is small,

$$n_0 G_i^\pm = \pm n_0\epsilon(\pi_0)_*^{-1}L_i + \operatorname{div}_{Y_0}\left(w(u_i^\pm)^{n_0\epsilon}\right)$$

and $\pi_{0*}\mathcal{O}_{Y_0}(L') = \mathcal{O}_A(\pi_{0*}L')$ for any Weil divisor L' on Y_0. Thus, with

$$\theta_{\mathbf{n}^+,\mathbf{n}^-} := \prod_i \big(w(u_i^+)^{n_0\epsilon}\big)^{n_i^+}\big(w(u_i^-)^{n_0\epsilon}\big)^{n_i^-},$$

multiplication by $\theta_{\mathbf{n}^+,\mathbf{n}^-}$ followed by π_{0*} is an isomorphism

$$\Phi_{\mathbf{n}^+,\mathbf{n}^-} : H^0\big(Y_0, \mathcal{O}_{Y_0}(n_0G(\mathbf{n}^+,\mathbf{n}^-))\big) \to \mathcal{O}_A\Big(\sum_i (n_i^+ - n_i^-)n_0\epsilon L_i\Big).$$

Since $\theta_{\mathbf{n}^+,\mathbf{n}^-}$ is multiplicative in $(\mathbf{n}^+,\mathbf{n}^-)$, the $\Phi_{\mathbf{n}^+,\mathbf{n}^-}$ form a ring homomorphism from the Veronese subring of $\mathcal{A}$ in degrees $(n_0\mathbb{N})^{2\rho}$, which is finitely generated, to

$$\bigoplus_{\mathbf{j}\in\mathbb{Z}^\rho} \mathcal{O}_A\Big(\sum_i j_i n_0\epsilon L_i\Big),$$

and it is surjective since any $\mathbf{j}\in\mathbb{Z}^\rho$ is $\mathbf{n}^+ - \mathbf{n}^-$ for some $\mathbf{n}^\pm \in \mathbb{N}^\rho$. Possibly replacing each L_i with $n_0\epsilon L_i$, we may assume that the $\mathbb{Z}^\rho$-graded ring

$$S := \bigoplus_{\mathbf{j}\in\mathbb{Z}^\rho} S_{\mathbf{j}}, \quad S_{\mathbf{j}} := \mathcal{O}_A\Big(\sum_{i=1}^{\rho} j_i L_i\Big) \subset \operatorname{Frac}(A),$$

with multiplication induced by $\operatorname{Frac}(A)$, is finitely generated over $A = S_{\mathbf{0}}$. For $\rho = 0$ we let $\mathbb{Z}^0 = \{\mathbf{0}\}$ and $S = A$.

Let $x_1,\dots,x_N$ be homogeneous generators of the A-algebra S, of degrees $\mathbf{d}_1,\dots,\mathbf{d}_N \in \mathbb{Z}^\rho$. For $\mathbf{a}\in\mathbb{Z}^\rho$ let

$$S^{(\mathbf{a})} := \bigoplus_{n\ge 0} S_{n\mathbf{a}} \quad\text{and}\quad \mathcal{I}_{\mathbf{a}} := \Big\{ I\subset\{1,\dots,N\} \;\Big|\; \mathbf{a}\in\sum_{i\in I}\mathbb{Q}_{>0}\mathbf{d}_i\Big\},$$

where the empty sum is $\{\mathbf{0}\}$. We show that the isomorphism class over A of $\operatorname{Proj}_A S^{(\mathbf{a})}$ only depends on $\mathcal{I}_{\mathbf{a}}$. For $\mathbf{e}\in\mathbb{N}^N$ let $x^{\mathbf{e}} := \prod_i x_i^{e_i}$, and for $I\subset\{1,\dots,N\}$ let $x_I := \prod_{i\in I} x_i$. For any $n\ge 1$, $S_{n\mathbf{a}}$ is spanned over A by the monomials $x^{\mathbf{e}}$ of degree $n\mathbf{a}$, so $\operatorname{Proj}_A S^{(\mathbf{a})}$ is covered by the open subsets $D_+(x^{\mathbf{e}})$ for such n and $\mathbf{e}$. If $\mathbf{e}$ has support I, then $S_{x^{\mathbf{e}}} = S_{x_I}$ and

$$D_+(x^{\mathbf{e}}) = \operatorname{Spec}\big(S_{x_I}\big)_{\mathbf{0}},$$

where $(\cdot)_{\mathbf{0}}$ is the degree-$\mathbf{0}$ part. The supports I which occur are exactly the elements of $\mathcal{I}_{\mathbf{a}}$: clearing denominators in $\mathbf{a} = \sum_{i\in I} q_i\mathbf{d}_i$ with $q_i\in\mathbb{Q}_{>0}$ yields $n\mathbf{a} = \sum_{i\in I} e_i\mathbf{d}_i$ with n, e_i positive integers. For $I, I'\in\mathcal{I}_{\mathbf{a}}$ we have $I\cup I'\in\mathcal{I}_{\mathbf{a}}$, the intersection of $\operatorname{Spec}(S_{x_I})_{\mathbf{0}}$ and $\operatorname{Spec}(S_{x_{I'}})_{\mathbf{0}}$ is $\operatorname{Spec}(S_{x_{I\cup I'}})_{\mathbf{0}}$, and the inclusions are induced by the localization maps

$$(S_{x_I})_{\mathbf{0}} \to (S_{x_{I\cup I'}})_{\mathbf{0}},$$

which do not depend on $\mathbf{a}$. Thus $\operatorname{Proj}_A S^{(\mathbf{a})}$ is glued from data determined by $\mathcal{I}_{\mathbf{a}}$, and

$$\#\big\{ \operatorname{Proj}_A S^{(\mathbf{a})} \;\big|\; \mathbf{a}\in\mathbb{Z}^\rho\big\}/\cong_A \le 2^{2^N}.$$

Let $\pi' : Y' \to \operatorname{Spec} A$ be a small $\mathbb{Q}$-factorial model, and let H' be a Cartier divisor on Y' which is ample over $\operatorname{Spec} A$. By (11.1), possibly replacing H' with a positive multiple, we may assume that $\pi'_*H' \sim \sum_i a_iL_i$ for some $\mathbf{a}\in\mathbb{Z}^\rho$. Since π' is small, $\pi'_*\mathcal{O}_{Y'}(nH') = \mathcal{O}_A(n\pi'_*H')$ for any $n\ge 0$, so

$$Y' \cong_A \operatorname{Proj}_A \bigoplus_{n\ge 0} H^0(Y', \mathcal{O}_{Y'}(nH')) \cong_A \operatorname{Proj}_A S^{(\mathbf{a})}.$$

Thus

$$\#\{\text{small } \mathbb{Q}\text{-factorial models of } \operatorname{Spec} A\}/\cong_A \le 2^{2^N}.$$

***Step 4**: finite pull-back of divisor classes.* Since $\operatorname{Spec}\varphi$ is a finite surjective morphism of normal integral schemes, Weil divisors on $\operatorname{Spec} A$ pull back to Weil divisors, and the pull-back of Weil divisors induces a linear map $(\operatorname{Spec}\varphi)^* : V \to V$. For $u\in\operatorname{Frac}(A)^*$ let $\operatorname{Nm}(u)\in\varphi(\operatorname{Frac}(A))$ be

the norm of u for the field extension $\mathrm{Frac}(A)/\varphi(\mathrm{Frac}(A))$ of degree D. Computing in the discrete valuation rings at the generic points of prime divisors,

$$(\operatorname{Spec}\varphi)_*(\operatorname{Spec}\varphi)^*L = DL \quad \text{and} \quad (\operatorname{Spec}\varphi)_*\operatorname{div}(u) = \operatorname{div}\left(\varphi^{-1}(\mathrm{Nm}(u))\right)$$

for any Weil divisor L on $\operatorname{Spec} A$. Thus if $n(\operatorname{Spec}\varphi)^*L = \operatorname{div}(u)$ for a positive integer n, then nDL is principal. Hence $(\operatorname{Spec}\varphi)^*$ is injective on V, and since $\rho < +\infty$, $(\operatorname{Spec}\varphi)^*$ is bijective on V.

***Step 5**: the normalized pull-back of a small $\mathbb{Q}$-factorial model.* Let $\pi' : Y' \to \operatorname{Spec} A$ be a small $\mathbb{Q}$-factorial model, and let $U \subset \operatorname{Spec} A$ be the largest open subset over which π' is an isomorphism. Then $\operatorname{codim}(\operatorname{Spec} A \backslash U) \geq 2$. Let W be the fiber product of $\operatorname{Spec}\varphi : \operatorname{Spec} A \to \operatorname{Spec} A$ and π'. The projection $W \to \operatorname{Spec} A$ to the first factor is an isomorphism over $U' := (\operatorname{Spec}\varphi)^{-1}(U)$, so W has a unique irreducible component W_1 dominating the first factor. Let Z be the normalization of W_1 with the reduced structure, and let $p : Z \to \operatorname{Spec} A$ and $h : Z \to Y'$ be the induced morphisms, so that

$$\pi' \circ h = \operatorname{Spec}\varphi \circ p.$$

Then p is projective and birational, and p is an isomorphism over U'. Since $W \to Y'$ is the base change of the finite morphism $\operatorname{Spec}\varphi$ and normalization is finite for excellent schemes, h is finite, and h is surjective as $\dim Z = \dim Y'$. For a finite surjective morphism of normal integral schemes, incomparability and going-down [Mat89, Theorems 9.3 (ii), 9.4 (ii)] imply that the codimension of any point ξ equals the codimension of the image of ξ. In particular, $\operatorname{codim}\xi = \operatorname{codim} h(\xi)$ for any point $\xi \in Z$, and $\operatorname{codim}\eta = \operatorname{codim}\operatorname{Spec}\varphi(\eta)$ for any point $\eta \in \operatorname{Spec} A$.

We show that p is small. Let E be a prime divisor on Z with $\operatorname{codim} p(E) \geq 2$. Then $h(E)$ is a prime divisor on Y', and

$$\operatorname{codim}\pi'(h(E)) = \operatorname{codim}\operatorname{Spec}\varphi(p(E)) = \operatorname{codim} p(E) \geq 2,$$

so $h(E)$ is exceptional over $\operatorname{Spec} A$, which is impossible as π' is small. In particular, $Z\backslash p^{-1}(U')$ contains no prime divisor, since $\operatorname{codim}(\operatorname{Spec} A\backslash U') \geq 2$.

We show that Z is $\mathbb{Q}$-factorial. For any Weil divisor L on $\operatorname{Spec} A$, the divisor $(\pi')_*^{-1}L$ is $\mathbb{Q}$-Cartier, hence so is $h^*(\pi')_*^{-1}L$, and over U' it is the pull-back of L by $\operatorname{Spec}\varphi$. As $Z\backslash p^{-1}(U')$ contains no prime divisor,

$$h^*(\pi')_*^{-1}L = p_*^{-1}(\operatorname{Spec}\varphi)^*L.$$

Let E be a prime divisor on Z. By **Step 4**, there exists a $\mathbb{Q}$-divisor L on $\operatorname{Spec} A$ with $p_*E \sim_{\mathbb{Q}} (\operatorname{Spec}\varphi)^*L$. Since p is small, $E = p_*^{-1}p_*E$ and $p_*^{-1}\operatorname{div}_A(u) = \operatorname{div}_Z(u)$ for any $u \in \mathrm{Frac}(A)^*$, so

$$E \sim_{\mathbb{Q}} p_*^{-1}(\operatorname{Spec}\varphi)^*L = h^*(\pi')_*^{-1}L,$$

which is $\mathbb{Q}$-Cartier. Thus $p : Z \to \operatorname{Spec} A$ is a small $\mathbb{Q}$-factorial model.

***Step 6**: a cycle, and the proof of (1).* Starting from Y_0 and applying **Step 5** repeatedly, we obtain small $\mathbb{Q}$-factorial models $\pi_j : Y_j \to \operatorname{Spec} A$ and finite surjective morphisms $h_j : Y_{j+1} \to Y_j$ with $\pi_j \circ h_j = \operatorname{Spec}\varphi \circ \pi_{j+1}$ for any $j \geq 0$. By **Step 3**, there exist integers $j_0 \geq 0$ and $m \geq 1$ and an isomorphism $\iota : Y_{j_0} \to Y_{j_0+m}$ with $\pi_{j_0+m} \circ \iota = \pi_{j_0}$. Let

$$Y := Y_{j_0}, \quad \pi := \pi_{j_0}, \quad \text{and} \quad g := h_{j_0} \circ h_{j_0+1} \circ \cdots \circ h_{j_0+m-1} \circ \iota.$$

Then g is finite and surjective, and

$$\pi \circ g = (\operatorname{Spec}\varphi)^m \circ \pi_{j_0+m} \circ \iota = \operatorname{Spec}(\varphi^m) \circ \pi.$$

By **Step 1**, Y is $\mathbb{Q}$-factorial and klt. Since φ^m is finite and local, $\operatorname{Spec}(\varphi^m)^{-1}(\mathfrak{m}_A) = \{\mathfrak{m}_A\}$, so $g^{-1}(T) = \pi^{-1}(\operatorname{Spec}(\varphi^m)^{-1}(\mathfrak{m}_A)) = T$. Since π is birational, we regard $\mathrm{Frac}(A)$ as the function field of Y via π^*, and then $g^* = \varphi^m$ since $\pi \circ g = \operatorname{Spec}(\varphi^m) \circ \pi$. Thus $g_*v = v \circ g^* = (\varphi^m)_*v$ for any real valuation v of $\mathrm{Frac}(A)$. This proves (1).

***Step 7**: proof of (2).* By (1), K_Y and $R(g)$ are $\mathbb{Q}$-Cartier. Let v be a real valuation of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$ with $a(v, Y) < +\infty$. Since π is proper, v has a center on Y, contained in T. By

Lemma 3.2 for $g : Y \to Y$ and $B = 0$, so that $B' = -R(g)$,

$$a(g_*v, Y) = a(v, Y, -R(g)) = a(v, Y) + v(R(g)).$$

This proves (2).

***Step 8**: proof of (3).* Let $\Delta_Y := \pi_*^{-1}\Delta$. By (1), Δ_Y is $\mathbb{Q}$-Cartier, and (Y, Δ_Y) is klt by **Step 1**. Let $f_Y : Y'' \to Y$ be a log resolution of (Y, Δ_Y), and write $K_{Y''} + B_0 = f_Y^* K_Y$ and $K_{Y''} + B_1 = f_Y^*(K_Y + \Delta_Y)$. Any real valuation of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$ has a center on Y'' since $\pi \circ f_Y$ is proper. By Lemma 7.2 for Y'' and the prime divisors in $\operatorname{Supp} B_0 \cup \operatorname{Supp} B_1$, with their coefficients in B_1 (less than 1 since (Y, Δ_Y) is klt) and in B_0 as the b_i and the c_i, there exists a real number $\gamma_1 \geq 1$ with $a(v, Y) \leq \gamma_1 a(v, Y, \Delta_Y)$ for any real valuation v of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$.

Let v be a real valuation of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$, and let ξ be the center of v on Y''. Let $z_1, \dots, z_n$ be a regular system of parameters of $\mathcal{O}_{Y'',\xi}$ such that any prime divisor in $\operatorname{Supp} B_1$ containing ξ is defined by some z_i near ξ, so that $\sum_i (z_i = 0)$ is snc on an open neighborhood of ξ, and let $\tau_i < 1$ be the coefficient in B_1 of the divisor $(z_i = 0)$ (with $\tau_i = 0$ if it is not in $\operatorname{Supp} B_1$). By [JM12, Proposition 5.1, (5.1), Corollary 5.4 and Remark 5.5], $a(v, Y'') \geq \sum_{i=1}^n v(z_i)$, so

$$a(v, Y, \Delta_Y) = a(v, Y'') - v(B_1) \geq \sum_{i=1}^{n} (1 - \tau_i) v(z_i) > 0,$$

where the last inequality holds since $v(z_i) \geq 0$ for any i and $v(z_i) > 0$ for some i. Since $\Delta_Y \geq 0$,

$$0 < a(v, Y, \Delta_Y) \leq a(v, Y) \leq \gamma_1 a(v, Y, \Delta_Y),$$

where the terms may be $+\infty$. Since $K_Y + \Delta_Y = \pi^*(K_A + \Delta)$, we have $a(v, Y, \Delta_Y) = a(v, \operatorname{Spec} A, \Delta)$. By Definition-Theorem 7.3(2) with $\Delta' = \Delta$, there exists a real number $\gamma_0 \geq 1$, independent of v, with

$$a(v, \operatorname{Spec} A, \Delta) \leq a_{\mathrm{can}}(v, A) \leq \gamma_0 a(v, \operatorname{Spec} A, \Delta).$$

Thus

$$\gamma_0^{-1} a_{\mathrm{can}}(v, A) \leq a(v, \operatorname{Spec} A, \Delta) = a(v, Y, \Delta_Y) \leq a(v, Y) \leq \gamma_1 a(v, Y, \Delta_Y) \leq \gamma_1 a_{\mathrm{can}}(v, A).$$

Let $\gamma := \max\{\gamma_0, \gamma_1\}$. Since $a(v, Y) > 0$, also $a_{\mathrm{can}}(v, A) > 0$. This proves (3). □

Construction-Proposition 11.2. Notation and conditions as in Set-up 7.1 and Construction-Theorem 11.1. Assume that there exist an integer $Q \geq 2$ and a real number $b > 0$ such that

$$D = \delta Q^d \quad \text{and} \quad c(\varphi^j) \geq b Q^j \text{ for any integer } j \geq 0,$$

and let $\lambda := Q^m$. Then there exist a non-empty set $\mathcal{K}$ of real valuations of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$, compact for the topology of pointwise convergence on $\mathrm{Frac}(A)^*$, real numbers $\alpha, \beta > 0$, and a Radon probability measure μ on $\mathcal{K}$ with the following properties.

(1) For any $v \in \mathcal{K}$ and any real number $t > 0$,

$$\mathfrak{m}_A^{\lceil \beta t \rceil} \subset \mathfrak{a}_t(v) \subset \mathfrak{m}_A^{\lceil \alpha t \rceil}.$$

(2) $\lambda^{-1} g_* v \in \mathcal{K}$ for any $v \in \mathcal{K}$, and μ is invariant under the map $\mathcal{K} \to \mathcal{K}$, $v \to \lambda^{-1} g_* v$.

(3) The function $v \to a(v, Y)$ is bounded on $\mathcal{K}$, and

$$\int_{\mathcal{K}} a(v, Y) d\mu(v) = 1 \quad \text{and} \quad \int_{\mathcal{K}} v(R(g)) d\mu(v) = \lambda - 1.$$

Proof. For any integer $j \geq 0$ let $c_j := c(\varphi^j)$, so that $\varphi^j(\mathfrak{m}_A) A \subset \mathfrak{m}_A^{c_j}$ and $c_j \geq b Q^j$.

***Step 1**: the order estimate.* Since A is analytically irreducible (Lemma 2.2) and $D = \delta Q^d$, Lemma 4.9 with A, $\mathfrak{m}_A$, φ, Q, and d in place of S, $\mathfrak{n}$, φ, Q, and r gives a real number $C_1 > 0$ with

$$b Q^j \operatorname{ord}_{\mathfrak{m}_A}(f) \leq \operatorname{ord}_{\mathfrak{m}_A}(\varphi^j(f)) \leq C_1 Q^j \operatorname{ord}_{\mathfrak{m}_A}(f) \tag{11.2}$$

for any $f \in \mathfrak{m}_A \backslash \{0\}$ and any integer $j \geq 0$, and also for any unit f of A, where all three terms are 0. The left inequality holds since $c_j \geq bQ^j$ and

$$\varphi^j(f) \in (\varphi^j(\mathfrak{m}_A)A)^{\operatorname{ord}_{\mathfrak{m}_A}(f)} \subset \mathfrak{m}_A^{c_j \operatorname{ord}_{\mathfrak{m}_A}(f)},$$

and the right one is Lemma 4.9 with $\operatorname{ord}_{\mathfrak{m}_A}(\varphi^j(f))$ in place of k.

Step 2: *an orbit with bounded log discrepancies.* Let $v_0 := \operatorname{ord}_E$ for an irreducible component E of the inverse image of $\mathfrak{m}_A$ in the normalized blow-up of $\mathfrak{m}_A$. Then v_0 is a divisorial valuation of $\operatorname{Frac}(A)$ centered at $\mathfrak{m}_A$. Let $\alpha_0 := v_0(\mathfrak{m}_A) > 0$. By Lemma 2.5(2), there exists a real number $\beta_0 > 0$ with

$$\mathfrak{a}_t(v_0) \subset \mathfrak{m}_A^{\lceil t/\beta_0 \rceil} \quad \text{for any real number } t > 0.$$

Since $f \in \mathfrak{m}_A^{\operatorname{ord}_{\mathfrak{m}_A}(f)}$, and taking $t = v_0(f)$ in the last inclusion, we obtain

$$\alpha_0 \operatorname{ord}_{\mathfrak{m}_A}(f) \leq v_0(f) \leq \beta_0 \operatorname{ord}_{\mathfrak{m}_A}(f) \quad \text{for any } f \in A\backslash\{0\}.$$

For any integer $j \geq 0$ let

$$v_j := Q^{-j}(\varphi^j)_* v_0.$$

Applying the last inequalities to $\varphi^j(f)$ and then (11.2), we get

$$\alpha_0 b \operatorname{ord}_{\mathfrak{m}_A}(f) \leq v_j(f) \leq \beta_0 C_1 \operatorname{ord}_{\mathfrak{m}_A}(f) \quad \text{for any } f \in A\backslash\{0\} \text{ and any } j \geq 0. \tag{11.3}$$

Let $\mathbf{R}_\varphi$ and m_φ be as in Definition-Theorem 7.3(3). Since $R(\varphi) \geq 0$, $\mathcal{O}_A(-m_\varphi R(\varphi))$ is a non-zero ideal of A. Let u be a non-zero element of $\mathcal{O}_A(-m_\varphi R(\varphi))$. By Definition-Theorem 7.3(3) and (11.3), for any $j \geq 0$,

$$0 \leq v_j(\mathbf{R}_\varphi) \leq m_\varphi^{-1} v_j(u) \leq \kappa := m_\varphi^{-1} \beta_0 C_1 \operatorname{ord}_{\mathfrak{m}_A}(u).$$

Since $(\varphi^{j+1})_* v_0 = \varphi_*\big((\varphi^j)_* v_0\big)$, we have $Q v_{j+1} = \varphi_* v_j$. Since v_0 is divisorial, $a(v_0, \operatorname{Spec} A, \Delta) < +\infty$, so $a_{\mathrm{can}}(v_0, A) < +\infty$ by Definition-Theorem 7.3(2). If $a_{\mathrm{can}}(v_j, A) < +\infty$, then, since $a_{\mathrm{can}}(sv, A) = s a_{\mathrm{can}}(v, A)$ for any real number $s > 0$ (by the definition in Definition-Theorem 7.3(2) and [JM12, Section 4.3, (5.1), (5.2)]), Definition-Theorem 7.3(3) implies

$$Q a_{\mathrm{can}}(v_{j+1}, A) = a_{\mathrm{can}}(\varphi_* v_j, A) = a_{\mathrm{can}}(v_j, A) + v_j(\mathbf{R}_\varphi) \leq a_{\mathrm{can}}(v_j, A) + \kappa < +\infty.$$

Let $\kappa_1 := a_{\mathrm{can}}(v_0, A) + \kappa/(Q-1)$. If $a_{\mathrm{can}}(v_j, A) \leq \kappa_1$, then, since $\kappa \leq (Q-1)\kappa_1$,

$$a_{\mathrm{can}}(v_{j+1}, A) \leq (\kappa_1 + \kappa)/Q \leq \kappa_1.$$

By induction on j,

$$a_{\mathrm{can}}(v_j, A) \leq \kappa_1 \quad \text{for any integer } j \geq 0. \tag{11.4}$$

Step 3: *the compact set.* Let $\omega(f) := \beta_0 C_1 \operatorname{ord}_{\mathfrak{m}_A}(f)$ for $f \in A\backslash\{0\}$. By (11.3), $v_{mj} \in \mathcal{V}(\alpha_0 b, \omega)$ for any $j \geq 0$. Let $\mathcal{K}_0$ be the closure of $\{v_{mj} \mid j \geq 0\}$ in $\mathcal{V}(\alpha_0 b, \omega)$, which is compact by Definition-Lemma 4.4. Thus $\mathcal{K}_0$ is a non-empty compact Hausdorff space of real valuations of $\operatorname{Frac}(A)$ centered at $\mathfrak{m}_A$, with the topology of pointwise convergence on $\operatorname{Frac}(A)^*$. Since $\mathcal{K}_0 \subset \mathcal{V}(\alpha_0 b, \omega)$ and $v(f) \geq v(\mathfrak{m}_A) \operatorname{ord}_{\mathfrak{m}_A}(f)$ for any $v \in \mathcal{K}_0$ and any $f \in A\backslash\{0\}$,

$$\alpha_0 b \operatorname{ord}_{\mathfrak{m}_A}(f) \leq v(f) \leq \beta_0 C_1 \operatorname{ord}_{\mathfrak{m}_A}(f) \quad \text{for any } v \in \mathcal{K}_0 \text{ and any } f \in A\backslash\{0\}. \tag{11.5}$$

Since $a_{\mathrm{can}}(\cdot, A)$ is lower semicontinuous (Definition-Theorem 7.3(2)), (11.4) implies $a_{\mathrm{can}}(v, A) \leq \kappa_1$ for any $v \in \mathcal{K}_0$, so by Construction-Theorem 11.1(3),

$$0 < a(v, Y) \leq \gamma \kappa_1 \quad \text{for any } v \in \mathcal{K}_0. \tag{11.6}$$

By Construction-Theorem 11.1(1), $\lambda^{-1} g_* v = Q^{-m}(\varphi^m)_* v$, so the value of $\lambda^{-1} g_* v$ at $f \in A\backslash\{0\}$ is $Q^{-m} v(\varphi^m(f))$. Thus $v \to \lambda^{-1} g_* v$ is continuous on $\mathcal{K}_0$, and it maps v_{mj} to $v_{m(j+1)}$ for any $j \geq 0$. Hence it maps $\mathcal{K}_0$ into $\mathcal{K}_0$.

Step 4: *measurability.* Since Y is normal and projective over $\operatorname{Spec} A$, it is separated, excellent, and admits a dualizing complex, and K_Y is $\mathbb{Q}$-Cartier by Construction-Theorem 11.1(1). Any $v \in \mathcal{K}_0$ has a center on Y since π is proper. By Lemma 2.4, $v \to a(v, Y)$ is lower semicontinuous,

and in particular Borel, on $\mathcal{K}_0$. Since $a(\cdot, Y)$ is homogeneous of degree 1 and $a(v, Y) < +\infty$ for any $v \in \mathcal{K}_0$ by (11.6), Construction-Theorem 11.1(2) implies

$$v(R(g)) = a(g_*v, Y) - a(v, Y) = \lambda a(\lambda^{-1} g_* v, Y) - a(v, Y) \quad \text{for any } v \in \mathcal{K}_0,$$

so by (11.6) and **Step 3**, $v \to v(R(g))$ is the difference of two bounded Borel functions on $\mathcal{K}_0$.

***Step 5**: the invariant measure.* By Lemma 2.15 for $\mathcal{K}_0$, the map $v \to \lambda^{-1} g_* v$, and the point v_0, there exists a Radon probability measure μ_0 on $\mathcal{K}_0$ with $\mu_0(E) = \mu_0(\{v \mid \lambda^{-1} g_* v \in E\})$ for any Borel subset $E \subset \mathcal{K}_0$. Thus

$$\int_{\mathcal{K}_0} \psi(\lambda^{-1} g_* v) d\mu_0(v) = \int_{\mathcal{K}_0} \psi(v) d\mu_0(v)$$

for any bounded Borel function ψ on $\mathcal{K}_0$. Let

$$t_0 := \int_{\mathcal{K}_0} a(v, Y) d\mu_0(v).$$

By (11.6), $0 < t_0 < +\infty$. By **Step 4** and the invariance of μ_0,

$$\int_{\mathcal{K}_0} v(R(g)) d\mu_0(v) = \lambda \int_{\mathcal{K}_0} a(\lambda^{-1} g_* v, Y) d\mu_0(v) - t_0 = (\lambda - 1) t_0.$$

Let

$$\mathcal{K} := \{t_0^{-1} v \mid v \in \mathcal{K}_0\}, \quad \alpha := t_0/(\beta_0 C_1), \quad \text{and} \quad \beta := t_0/(\alpha_0 b),$$

and let μ be the push-forward of μ_0 under the homeomorphism $\mathcal{K}_0 \to \mathcal{K}$, $v \to t_0^{-1} v$. Then $\mathcal{K}$ is a non-empty compact set of real valuations of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$. By (11.5),

$$\beta^{-1} \operatorname{ord}_{\mathfrak{m}_A}(f) \le v(f) \le \alpha^{-1} \operatorname{ord}_{\mathfrak{m}_A}(f) \quad \text{for any } v \in \mathcal{K} \text{ and any } f \in A \backslash \{0\}.$$

For $t > 0$, if $f \in \mathfrak{m}_A^{\lceil \beta t \rceil}$ then $v(f) \ge \beta^{-1} \lceil \beta t \rceil \ge t$, and if $v(f) \ge t$ then $\operatorname{ord}_{\mathfrak{m}_A}(f) \ge \alpha t$, hence $\operatorname{ord}_{\mathfrak{m}_A}(f) \ge \lceil \alpha t \rceil$. This proves (1). Since $\lambda^{-1} g_*(t_0^{-1} v) = t_0^{-1} \lambda^{-1} g_* v$, **Step 3** and the invariance of μ_0 imply (2). Since $a(t_0^{-1} v, Y) = t_0^{-1} a(v, Y)$ and $(t_0^{-1} v)(R(g)) = t_0^{-1} v(R(g))$, (11.6) implies that $v \to a(v, Y)$ is bounded on $\mathcal{K}$, and

$$\int_{\mathcal{K}} a(v, Y) d\mu(v) = t_0^{-1} t_0 = 1 \quad \text{and} \quad \int_{\mathcal{K}} v(R(g)) d\mu(v) = t_0^{-1}(\lambda - 1) t_0 = \lambda - 1.$$

This proves (3). □

12. A totally invariant center on the small model

In this section we find, on the small model Y of Construction-Theorem 11.1, a local ring to which Set-up 4.8 can be applied. The local ring is that of Y at the generic point of an irreducible closed subset $V \subset T$ with $(g^N)^{-1}(V) = V$ for an integer $N \ge 1$ (Construction-Theorem 12.2, with the small morphism $\pi : Y \to \operatorname{Spec} A$ as the morphism $Y \to \operatorname{Spec}(A_0)_{\mathfrak{q}}$ of Set-up 4.8(1)).

Lemma 12.1. *Let $(A_1, \mathfrak{m}_1) \subset (A_2, \mathfrak{m}_2)$ be a finite extension of Noetherian local domains containing $\mathbb{C}$ with $\dim A_2 \ge 1$, let $L_1 \subset L_2$ be the fraction fields, and let $D_0 := [L_2 : L_1]$ and $\delta_0 := [A_2/\mathfrak{m}_2 : A_1/\mathfrak{m}_1]$. Let w be a real valuation of L_2 centered at $\mathfrak{m}_2$ such that w is the only real valuation of L_2 extending $w|_{L_1}$. Assume that $\operatorname{vol}_{A_1}(w|_{L_1})$ exists and $\operatorname{vol}_{A_1}(w|_{L_1}) < +\infty$. Then $\operatorname{vol}_{A_2}(w)$ exists and*

$$\operatorname{vol}_{A_2}(w) = \frac{D_0}{\delta_0} \operatorname{vol}_{A_1}(w|_{L_1}).$$

Proof. Let $r := \dim A_2$ and $w_1 := w|_{L_1}$. Since A_2 is finite over A_1, we have $\dim A_1 = r$, $\mathfrak{m}_2 \cap A_1 = \mathfrak{m}_1$, and $\delta_0 < +\infty$. In particular, w_1 is centered at $\mathfrak{m}_1$. Any composition factor $A_2/\mathfrak{m}_2$ has A_1-length δ_0, so

$$\ell_{A_1}(N') = \delta_0 \ell_{A_2}(N') \quad \text{for any } A_2\text{-module } N' \text{ of finite length.}$$

Below $\mathfrak{a}_t(w_1)$ is taken in A_1 and $\mathfrak{a}_t(w)$ in A_2, and $\mathfrak{a}_t(w_1) = A_1$ for $t \le 0$.

Let $\mathrm{Tr} : L_2 \to L_1$ be the trace map. We show

$$w_1(\mathrm{Tr}(y)) \geq w(y) \quad \text{for any } y \in L_2, \tag{12.1}$$

where $w_1(0) = +\infty$. Since $\operatorname{char} L_1 = 0$, a normal closure L' of L_2 over L_1 is Galois over L_1. Let w' be a valuation of L' extending w. Since $[L' : L_2] < +\infty$, the value group of w' contains that of w with finite index [Sta, Tag 0ASH], so w' is a real valuation. For any L_1-embedding $\sigma : L_2 \to L'$, $w' \circ \sigma$ is a real valuation of L_2 extending w_1, so $w' \circ \sigma = w$ by assumption. Since $\mathrm{Tr}(y) = \sum_\sigma \sigma(y)$, the sum running over the D_0 embeddings σ, we obtain

$$w_1(\mathrm{Tr}(y)) = w'\Big(\sum_\sigma \sigma(y)\Big) \geq \min_\sigma w'(\sigma(y)) = w(y).$$

Since any element of L_2 is of the form y/x with $y \in A_2$ and $x \in A_1 \backslash \{0\}$, there exists an L_1-basis $e_1, \dots, e_{D_0}$ of L_2 with $e_i \in A_2$ for any i. Since L_2/L_1 is separable, the trace form is non-degenerate, and we let $e_1^*, \dots, e_{D_0}^* \in L_2$ be the dual basis: $\mathrm{Tr}(e_i^* e_j)$ is 1 if $i = j$ and 0 otherwise. Thus

$$y = \sum_{i=1}^{D_0} \mathrm{Tr}(e_i^* y) e_i \quad \text{for any } y \in L_2.$$

Let $M := \bigoplus_{i=1}^{D_0} A_1 e_i \subset A_2$ and $\kappa := \max\{0, -w(e_1^*), \dots, -w(e_{D_0}^*)\}$. We show

$$\bigoplus_{i=1}^{D_0} \mathfrak{a}_t(w_1) e_i \subset M \cap \mathfrak{a}_t(w) \subset \bigoplus_{i=1}^{D_0} \mathfrak{a}_{t-\kappa}(w_1) e_i \quad \text{for any } t \in \mathbb{R}. \tag{12.2}$$

If $a_i \in \mathfrak{a}_t(w_1)$ for any i, then $w(\sum_i a_i e_i) \geq \min_i(w_1(a_i) + w(e_i)) \geq t$, since $w(e_i) \geq 0$. If $y = \sum_i a_i e_i \in M$ with $a_i \in A_1$ and $w(y) \geq t$, then $a_i = \mathrm{Tr}(e_i^* y)$, so by (12.1),

$$w_1(a_i) \geq w(e_i^*) + w(y) \geq t - \kappa.$$

This proves (12.2).

We have

$$M \Big/ \bigoplus_{i=1}^{D_0} \mathfrak{a}_t(w_1) e_i \cong (A_1/\mathfrak{a}_t(w_1))^{\oplus D_0}.$$

Together with (12.2), this implies

$$D_0 \ell_{A_1}(A_1/\mathfrak{a}_{t-\kappa}(w_1)) \leq \ell_{A_1}(M/(M \cap \mathfrak{a}_t(w))) \leq D_0 \ell_{A_1}(A_1/\mathfrak{a}_t(w_1)).$$

Since the limit

$$\mathrm{vol}_{A_1}(w_1) = \lim_{t \to +\infty} \frac{r! \ell_{A_1}(A_1/\mathfrak{a}_t(w_1))}{t^r}$$

exists and $(t - \kappa)/t \to 1$, we obtain

$$\lim_{t \to +\infty} \frac{r! \ell_{A_1}(M/(M \cap \mathfrak{a}_t(w)))}{t^r} = D_0 \,\mathrm{vol}_{A_1}(w_1). \tag{12.3}$$

The A_1-module A_2/M is finitely generated and $(A_2/M) \otimes_{A_1} L_1 = 0$, so there exists $x \in A_1 \backslash \{0\}$ with $xA_2 \subset M$. Let $n_t := \lceil t/w(\mathfrak{m}_2) \rceil$. Since $\mathfrak{m}_2^{n_t} \subset \mathfrak{a}_t(w)$, the module $A_2/(M + \mathfrak{a}_t(w))$ is a quotient of $A_2/(xA_2 + \mathfrak{m}_2^{n_t})$, so

$$\ell_{A_1}(A_2/(M + \mathfrak{a}_t(w))) \leq \delta_0 \ell_{A_2}(A_2/(xA_2 + \mathfrak{m}_2^{n_t})).$$

Since A_2 is a domain and $x \neq 0$, $\dim(A_2/xA_2) \leq r - 1$, so by the Hilbert–Samuel polynomial of A_2/xA_2 the right-hand side is $O(t^{r-1})$ as $t \to +\infty$. The exact sequence of A_1-modules

$$0 \to M/(M \cap \mathfrak{a}_t(w)) \to A_2/\mathfrak{a}_t(w) \to A_2/(M + \mathfrak{a}_t(w)) \to 0$$

and (12.3) imply

$$\lim_{t \to +\infty} \frac{r! \ell_{A_1}(A_2/\mathfrak{a}_t(w))}{t^r} = D_0 \,\mathrm{vol}_{A_1}(w_1).$$

Since $\ell_{A_1}(N') = \delta_0 \ell_{A_2}(N')$ for $N' = A_2/\mathfrak{a}_t(w)$, the limit

$$\mathrm{vol}_{A_2}(w) = \lim_{t\to+\infty} \frac{r!\ell_{A_2}(A_2/\mathfrak{a}_t(w))}{t^r} = \frac{D_0}{\delta_0}\mathrm{vol}_{A_1}(w_1)$$

exists. The lemma follows. □

Construction-Theorem 12.2. Notation and conditions as in Set-up 7.1 and Construction-Theorem 11.1. Assume that there exist an integer $Q \geq 2$ and a real number $b > 0$ with $D = \delta Q^d$ and $c(\varphi^j) \geq bQ^j$ for any integer $j \geq 0$, and let $\lambda := Q^m$. Then there exist an irreducible closed subset $V \subset T$ and an integer $N \geq 1$ with $(g^N)^{-1}(V) = V$ satisfying the following, where η_V is the generic point of V, $B := \mathcal{O}_{Y,\eta_V}$, $\mathfrak{n} := \mathfrak{m}_B$, $r := \dim B$, $h_B := (g^N)^* : B \to B$, and $\Lambda := \lambda^N$.

(1) $\mathrm{Frac}(B) = \mathrm{Frac}(A)$, $r \geq 1$, h_B is a finite injective local $\mathbb{C}$-algebra homomorphism, K_B is $\mathbb{Q}$-Cartier, and $\mathrm{Spec}\, B$ is klt.

(2) The homomorphism h_B satisfies

$$[\mathrm{Frac}(B) : h_B(\mathrm{Frac}(B))] = \Lambda^r [B/\mathfrak{n} : (h_B)_{B/\mathfrak{n}}(B/\mathfrak{n})].$$

(3) There exists a real number $b_V > 0$ with $c(h_B^j) \geq b_V \Lambda^j$ for any integer $j \geq 0$.

Proof. We divide the proof into several steps. We let $\mathcal{K}$, α, β, and μ be given by Construction-Proposition 11.2. The set $\mathcal{K}$ is compact Hausdorff, and for any integer $i \geq 0$ the map $\mathcal{K} \to \mathcal{K}$, $v \to \lambda^{-i}(g^i)_* v$, is continuous and preserves μ (Construction-Proposition 11.2(2)). Since

$$\pi \circ g = \mathrm{Spec}(\varphi^m) \circ \pi$$

and π is birational, $g^* = \varphi^m$ on $\mathrm{Frac}(A)$. By Lemma 2.2, A and any local ring of Y are normal excellent local domains containing $\mathbb{C}$ and are analytically irreducible.

Step 1. In this step we show that the set

$$\mathcal{K}' := \{v \in \mathcal{K} \mid \mathrm{vol}_A(\lambda^{-i}(g^i)_* v) = \mathrm{vol}_A(v) \text{ for any integer } i \geq 1\}$$

satisfies $\mu(\mathcal{K}') = 1$, and that for any $v \in \mathcal{K}'$ and any integer $i \geq 1$,

$$v \text{ is the only real valuation } z \text{ of } \mathrm{Frac}(A) \text{ with } (g^i)_* z = (g^i)_* v. \tag{12.4}$$

Let $(H, \mathfrak{m}_H)$ be an analytically irreducible Noetherian local domain of positive dimension containing $\mathbb{C}$, and let v be a real valuation of $\mathrm{Frac}(H)$ centered at $\mathfrak{m}_H$, with $\mathfrak{a}_t(v)$ taken in H. Since $v(\mathfrak{m}_H) > 0$, we have

$$\mathfrak{m}_H^{\lceil t/v(\mathfrak{m}_H)\rceil} \subset \mathfrak{a}_t(v) \quad \text{for any real number } t > 0,$$

so $\mathfrak{a}_\bullet(v)$ is an $\mathfrak{m}_H$-filtration and $\mathrm{vol}_H(v)$ exists [BLQ24, (2.1)]. We show

$$\mathrm{vol}_H(v) = \inf\left\{\frac{e(I)}{v(I)^{\dim H}} \;\middle|\; I \subset H \text{ is an } \mathfrak{m}_H\text{-primary ideal}\right\}. \tag{12.5}$$

For any $\mathfrak{m}_H$-primary ideal $I \subset H$ and any integer $n \geq 1$, $I^n \subset \mathfrak{a}_{nv(I)}(v)$, so

$$\ell_H(H/\mathfrak{a}_{nv(I)}(v)) \leq \ell_H(H/I^n),$$

and letting $n \to +\infty$ we obtain $\mathrm{vol}_H(v) \leq e(I)/v(I)^{\dim H}$. Conversely, for any integer $n \geq 1$ the ideal $\mathfrak{a}_n(v)$ is $\mathfrak{m}_H$-primary with $v(\mathfrak{a}_n(v)) \geq n$, and, by [BLQ24, (2.1)],

$$\lim_{n\to+\infty} \frac{e(\mathfrak{a}_n(v))}{n^{\dim H}} = \mathrm{vol}_H(v).$$

This proves (12.5). Since $v \to v(I)$ is the minimum of the values at finitely many generators of I, it is continuous for the topology of pointwise convergence, so the right-hand side of (12.5) is upper semicontinuous in v, in particular Borel.

Let $v \in \mathcal{K}$. By Construction-Proposition 11.2(1),

$$\mathfrak{m}_A^{\lceil \beta t\rceil} \subset \mathfrak{a}_t(v) \subset \mathfrak{m}_A^{\lceil \alpha t\rceil} \quad \text{for any real number } t > 0,$$

so

$$e(\mathfrak{m}_A)\alpha^d \le \mathrm{vol}_A(v) \le e(\mathfrak{m}_A)\beta^d,$$

and vol_A is a bounded Borel function on $\mathcal{K}$ by (12.5) with A in place of H.

Let $i \ge 1$ be an integer. Since field degrees and residue degrees multiply under composition and $D = \delta Q^d$,

$$[\mathrm{Frac}(A) : \varphi^{mi}(\mathrm{Frac}(A))] = D^{mi} = (\lambda^i)^d \delta^{mi} = (\lambda^i)^d [k : (\varphi^{mi})_k(k)],$$

and $(\varphi^{mi})_* v = (g^i)_* v$. By Lemma 3.3(1), with A, φ^{mi}, and λ^i in place of S, φ, and λ, the bounded Borel function $\Phi_i(v) := \mathrm{vol}_A(\lambda^{-i}(g^i)_* v) - \mathrm{vol}_A(v)$ on $\mathcal{K}$ is non-negative. Since μ is invariant under $v \to \lambda^{-i}(g^i)_* v$, $\int_{\mathcal{K}} \Phi_i d\mu = 0$. Thus $\Phi_i = 0$ μ-almost everywhere. Taking the intersection over $i \ge 1$, $\mu(\mathcal{K}') = 1$.

Let $v \in \mathcal{K}'$. Since $\lambda^{-i}(g^i)_* v \in \mathcal{K}$, Construction-Proposition 11.2(1) implies

$$\mathfrak{a}_t(\lambda^{-i}(g^i)_* v) \subset \mathfrak{m}_A^{\lceil \alpha t \rceil} \quad \text{for any real number } t > 0.$$

Thus Lemma 3.3(2), with A, φ^{mi}, λ^i, v, and α^{-1} in place of S, φ, λ, u, and β, implies (12.4).

Step 2. In this step we choose V and N. Since π is proper and any $v \in \mathcal{K}$ is centered at $\mathfrak{m}_A$, v has a center $\mathrm{center}_Y v \in T$. Since g^* maps the maximal ideal of $\mathcal{O}_{Y,g(\mathrm{center}_Y v)}$ into that of $\mathcal{O}_{Y,\mathrm{center}_Y v}$, $\mathrm{center}_Y(\lambda^{-1} g_* v) = g(\mathrm{center}_Y v)$. The scheme Y is covered by finitely many affine open subsets $\mathrm{Spec}\, S$ with S a finitely generated A-algebra. For each such S, the set $\{v \in \mathcal{K} \mid \mathrm{center}_Y v \in \mathrm{Spec}\, S\}$ is the closed subset defined by $v \ge 0$ at finitely many generators of S. Let $Z \subset Y$ be a closed subset. On $\{v \in \mathcal{K} \mid \mathrm{center}_Y v \in \mathrm{Spec}\, S\}$, $\mathrm{center}_Y v \in Z$ if and only if $\min_x v(x) > 0$, the minimum running over finitely many generators x of the ideal of $Z \cap \mathrm{Spec}\, S$ in S. Thus the complement $\{v \in \mathcal{K} \mid \mathrm{center}_Y v \notin Z\}$ is the union over the finitely many S of the closed sets $\{v \ge 0 \text{ on } S,\ \min_x v(x) = 0\}$, hence closed, so $\{v \in \mathcal{K} \mid \mathrm{center}_Y v \in Z\}$ is open.

By Lemma 8.1, with $v \to \lambda^{-1} g_* v$ and $\mathrm{center}_Y|_{\mathcal{K}}$ in place of f and ζ, there exists an irreducible closed subset $V \subset Y$ such that the Borel set

$$\mathcal{K}_V := \{v \in \mathcal{K} \mid \mathrm{center}_Y v = \eta_V\}$$

satisfies $\mu(\mathcal{K}_V) > 0$, and there exists an integer $N \ge 1$ with $g^N(\eta_V) = \eta_V$. Moreover, for the least such N, the probability measure $\mu_V := \mu(\mathcal{K}_V)^{-1}\mu|_{\mathcal{K}_V}$ is invariant under $v \to \lambda^{-N}(g^N)_* v$. Since $\eta_V \in T$ and T is closed, $V \subset T$. For $v \in \mathcal{K}_V$ we have $\Lambda^{-1}(g^N)_* v \in \mathcal{K}_V$. Since $\mu(\mathcal{K}') = 1$, $\mu_V(\mathcal{K}_V \cap \mathcal{K}') = 1$.

Step 3. In this step we show $(g^N)^{-1}(V) = V$ and (1). Let $L_1 := (g^N)^* \mathrm{Frac}(A)$. Since L_1 has characteristic 0, there exists a finite Galois extension L' of L_1 containing $\mathrm{Frac}(A)$. Let $\widetilde{B}$ be the integral closure of $(g^N)^*(B)$ in L'. Since Y is normal, $(g^N)^*(B)$ is a normal domain with fraction field $L_1 = (L')^{\mathrm{Gal}(L'/L_1)}$, so $\widetilde{B}^{\mathrm{Gal}(L'/L_1)} = (g^N)^*(B)$. Thus $\mathrm{Gal}(L'/L_1)$ acts transitively on the primes of $\widetilde{B}$ over any prime of $(g^N)^*(B)$ [Sta, Tag 0BRI].

We show that any $y \in Y$ with $g^N(y) = \eta_V$ equals η_V. Let $\mathrm{Spec}\, S \subset Y$ be an affine open neighborhood of η_V. Since g^N is finite, $(g^N)^{-1}(\mathrm{Spec}\, S) = \mathrm{Spec}\, S'$ with S' finite over $(g^N)^*(S)$. Let $\mathfrak{p} \subset S$ and $\mathfrak{P} \subset S'$ be the primes of η_V and y, so $((g^N)^*)^{-1}(\mathfrak{P}) = \mathfrak{p}$, and let

$$B' := \big((g^N)^*(S \backslash \mathfrak{p})\big)^{-1} S' \subset \mathrm{Frac}(A).$$

Then B' is integral over $(g^N)^*(B) = ((g^N)^*(S\backslash\mathfrak{p}))^{-1}(g^N)^*(S)$, so $B' \subset \widetilde{B}$. Moreover, $\mathfrak{P}B'$ is a prime of B' over $(g^N)^*(\mathfrak{n})$ and $B'_{\mathfrak{P}B'} = \mathcal{O}_{Y,y}$. Since $\widetilde{B}$ is integral over B', there exists a prime $\widetilde{\mathfrak{P}}$ of $\widetilde{B}$ with $\widetilde{\mathfrak{P}} \cap B' = \mathfrak{P}B'$. Let $u \in \mathcal{K}_V \cap \mathcal{K}'$, which exists by **Step 2**. Since $\mathrm{center}_Y((g^N)_* u) = g^N(\eta_V) = \eta_V$, $u \ge 0$ on $(g^N)^*(B)$ and $u > 0$ on $(g^N)^*(\mathfrak{n})$. Let u' be a valuation of L' extending u. Since $[L' : \mathrm{Frac}(A)] < +\infty$, u' is a real valuation [Sta, Tag 0ASH]. Then $u' \ge 0$ on $\widetilde{B}$, and $\widetilde{\mathfrak{Q}} := \{x \in \widetilde{B} \mid u'(x) > 0\}$ is a prime of $\widetilde{B}$ over $(g^N)^*(\mathfrak{n})$, as is $\widetilde{\mathfrak{P}}$. Let

$\sigma \in \mathrm{Gal}(L'/L_1)$ with $\sigma(\widetilde{\mathfrak{Q}}) = \widetilde{\mathfrak{P}}$, and let z be the restriction of $u' \circ \sigma^{-1}$ to $\mathrm{Frac}(A)$. Then z is a real valuation of $\mathrm{Frac}(A)$, $z \geq 0$ on B', and

$$\{x \in B' \mid z(x) > 0\} = \widetilde{\mathfrak{P}} \cap B' = \mathfrak{P}B',$$

so $\mathrm{center}_Y z = y$. Since σ is the identity on L_1, $z|_{L_1} = u|_{L_1}$, that is, $(g^N)_* z = (g^N)_* u$. By (12.4) with $i = N$, $z = u$, so $y = \mathrm{center}_Y u = \eta_V$.

Let W be an irreducible component of the closed subset $(g^N)^{-1}(V)$ with generic point η_W. We show $g^N(\eta_W) = \eta_V$. Let $\mathrm{Spec}\, S$ be an affine open neighborhood of $g^N(\eta_W)$. Then $\eta_V \in \mathrm{Spec}\, S$. Let $\mathrm{Spec}\, S' = (g^N)^{-1}(\mathrm{Spec}\, S)$, let $\mathfrak{p} \subset S$ be the prime of η_V, and let $\mathfrak{P}_W \subset S'$ be the prime of η_W. Since $(g^N)^* : S \to S'$ is an injective integral homomorphism of domains and S is normal, going down implies that there exists a prime $\mathfrak{P}' \subset \mathfrak{P}_W$ of S' with $((g^N)^*)^{-1}(\mathfrak{P}') = \mathfrak{p}$. The point of $\mathfrak{P}'$ is in $(g^N)^{-1}(V)$ and specializes to η_W. Since W is an irreducible component, $\mathfrak{P}' = \mathfrak{P}_W$, so $g^N(\eta_W) = \eta_V$. By the previous paragraph, $\eta_W = \eta_V$, so $W = V$. Since g^N is finite and $g^N(\eta_V) = \eta_V$, we have $g^N(V) = V$. Thus

$$(g^N)^{-1}(V) = V.$$

With $y = \eta_V$ in the notation above, the maximal ideals of B' are the primes over $(g^N)^*(\mathfrak{n})$ and correspond to the points y with $g^N(y) = \eta_V$. Thus B' is local and $B' = \mathcal{O}_{Y,\eta_V} = B$. Hence $(g^N)^*(B) \subset B$, B is finite over $(g^N)^*(B)$, and $h_B = (g^N)^* : B \to B$ is a finite injective local $\mathbb{C}$-algebra homomorphism. Since π is birational, $\mathrm{Frac}(B) = \mathrm{Frac}(A)$. Since $\eta_V \in T = \pi^{-1}(\mathfrak{m}_A)$ and $d \geq 1$, η_V is not the generic point of Y, so $r \geq 1$. Since Y is $\mathbb{Q}$-factorial and klt (Construction-Theorem 11.1(1)), K_B is $\mathbb{Q}$-Cartier and $\mathrm{Spec}\, B$ is klt. This proves (1).

Step 4. In this step we show that for any $v \in \mathcal{K}_V$, $\mathrm{vol}_B(v)$ exists with $0 < \mathrm{vol}_B(v) < +\infty$, and that the functions

$$v \to \mathrm{vol}_B(v), \qquad v \to v(\mathfrak{n}), \qquad \text{and} \quad v \to \beta_B(v) := \sup_{x \in \mathfrak{n}\setminus\{0\}} \frac{v(x)}{\mathrm{ord}_{\mathfrak{n}}(x)}$$

are Borel on $\mathcal{K}_V$ with values in $(0, +\infty)$. Here $\mathfrak{a}_t(v)$ is taken in B.

Let $v \in \mathcal{K}_V$, so v is centered at $\mathfrak{n}$. By **Step 1** and (12.5), with B and $\mathfrak{n}$ in place of H and I, $\mathrm{vol}_B(v)$ exists and $\mathrm{vol}_B(v) \leq e(\mathfrak{n})v(\mathfrak{n})^{-r} < +\infty$.

We show $\beta_B(v) < +\infty$. Since B is the local ring of Y at $\mathrm{center}_Y v = \eta_V$, $a(v, \mathrm{Spec}\, B) = a(v, Y)$, which is finite by Construction-Proposition 11.2(3). Since K_B is $\mathbb{Q}$-Cartier (Y is $\mathbb{Q}$-factorial by Construction-Theorem 11.1(1)), Lemma 4.6 gives a real number $\kappa_v > 0$ with

$$\mathfrak{a}_t(v) \subset \mathfrak{n}^{\lceil t/\kappa_v \rceil} \quad \text{for any real number } t > 0.$$

Taking $t = v(x)$ for $x \in \mathfrak{n}\setminus\{0\}$ gives $\beta_B(v) \leq \kappa_v < +\infty$. Hence

$$\mathfrak{a}_t(v) \subset \mathfrak{n}^{\lceil t/\beta_B(v) \rceil} \quad \text{for any real number } t > 0,$$

and $\mathrm{vol}_B(v) \geq e(\mathfrak{n})\beta_B(v)^{-r} > 0$.

Any $x \in B$ is an element of $\mathrm{Frac}(A)$, so $v \to v(x)$ is continuous on $\mathcal{K}_V$. Thus $v \to v(\mathfrak{n})$ is continuous, β_B is lower semicontinuous as a supremum of continuous functions, and vol_B is upper semicontinuous on $\mathcal{K}_V$ by (12.5) with B in place of H. Since $\mathcal{K}_V$ is Borel, the three functions are Borel.

Step 5. In this step we show (2). Let

$$\delta_V := [B/\mathfrak{n} : (h_B)_{B/\mathfrak{n}}(B/\mathfrak{n})].$$

Since $h_B = (g^N)^* = \varphi^{mN}$ on $\mathrm{Frac}(B) = \mathrm{Frac}(A)$,

$$[\mathrm{Frac}(B) : h_B(\mathrm{Frac}(B))] = D^{mN}, \qquad \rho := \frac{D^{mN}}{\delta_V \Lambda^r} > 0.$$

We show that for any $v \in \mathcal{K}_V \cap \mathcal{K}'$,

$$\mathrm{vol}_B(v) = \rho\, \mathrm{vol}_B(\Lambda^{-1}(g^N)_* v). \tag{12.6}$$

Let $v' := \Lambda^{-1}(g^N)_* v \in \mathcal{K}_V$. We use Lemma 12.1 with $h_B(B) \subset B$ and v in place of $A_1 \subset A_2$ and w. Then $L_1 = h_B(\mathrm{Frac}(B))$, $D_0 = D^{mN}$, $\delta_0 = \delta_V$, and $\dim B = r \geq 1$. Any real valuation z of $\mathrm{Frac}(B)$ with $z|_{L_1} = v|_{L_1}$ satisfies $(g^N)_* z = (g^N)_* v$, so $z = v$ by (12.4). The isomorphism $h_B : B \to h_B(B)$ carries $v|_{L_1}$ to $v \circ h_B = (g^N)_* v = \Lambda v'$. Since $\mathfrak{a}_t(\Lambda v') = \mathfrak{a}_{t/\Lambda}(v')$, we have

$$\mathrm{vol}_{h_B(B)}(v|_{L_1}) = \mathrm{vol}_B(\Lambda v') = \Lambda^{-r} \mathrm{vol}_B(v'),$$

which exists and is finite by **Step 4**. Thus Lemma 12.1 implies $\mathrm{vol}_B(v) = (D^{mN}/\delta_V)\Lambda^{-r} \mathrm{vol}_B(v')$, which is (12.6).

Let $\Psi := \log \mathrm{vol}_B$, a real-valued Borel function on $\mathcal{K}_V$ (**Step 4**). Since $\mu_V(\mathcal{K}_V \cap \mathcal{K}') = 1$, (12.6) implies $\Psi(v) = \Psi(\Lambda^{-1}(g^N)_* v) + \log \rho$ for μ_V-almost every v. Since μ_V is invariant under $v \to \Lambda^{-1}(g^N)_* v$, for any $t \in \mathbb{R}$,

$$\mu_V(\{\Psi \leq t\}) = \mu_V(\{v \mid \Psi(\Lambda^{-1}(g^N)_* v) \leq t - \log \rho\}) = \mu_V(\{\Psi \leq t - \log \rho\}).$$

Suppose that $\log \rho > 0$. Then $\mu_V(\{\Psi \leq t\}) = \mu_V(\{\Psi \leq t - n \log \rho\})$ for any integer $n \geq 1$, and the right-hand side tends to 0 as $n \to +\infty$ since Ψ is real-valued. Thus $\mu_V(\{\Psi \leq t\}) = 0$ for any t, contradicting $\mu_V(\mathcal{K}_V) = 1$. Suppose that $\log \rho < 0$. Then

$$\mu_V(\{\Psi \leq t\}) = \mu_V(\{\Psi \leq t + n|\log \rho|\}) \to 1 \quad \text{as } n \to +\infty,$$

so $\mu_V(\{\Psi \leq t\}) = 1$ for any t, contradicting $\bigcap_{n \geq 1} \{\Psi \leq -n\} = \emptyset$. Thus $\rho = 1$, that is,

$$[\mathrm{Frac}(B) : h_B(\mathrm{Frac}(B))] = D^{mN} = \Lambda^r \delta_V.$$

This proves (2).

Step 6. In this step we show (3). By **Step 4**, β_B and $v \to v(\mathfrak{n})$ are Borel functions on $\mathcal{K}_V$ with values in $(0, +\infty)$, so there exist real numbers $M_1 > 0$ and $\epsilon > 0$ such that the Borel sets

$$\mathcal{K}_1 := \{v \in \mathcal{K}_V \mid \beta_B(v) \leq M_1\}, \qquad \mathcal{K}_2 := \{v \in \mathcal{K}_V \mid v(\mathfrak{n}) \geq \epsilon\}$$

satisfy $\mu_V(\mathcal{K}_1) > 2/3$ and $\mu_V(\mathcal{K}_2) > 2/3$. Let $j \geq 0$ be an integer. Since μ_V is invariant under $v \to \Lambda^{-j}(g^{Nj})_* v$, the set

$$\mathcal{K}_2^{(j)} := \{v \in \mathcal{K}_V \mid \Lambda^{-j}(g^{Nj})_* v \in \mathcal{K}_2\}$$

satisfies $\mu_V(\mathcal{K}_2^{(j)}) = \mu_V(\mathcal{K}_2)$, so

$$\mu_V(\mathcal{K}_1 \cap \mathcal{K}_2^{(j)}) \geq \mu_V(\mathcal{K}_1) + \mu_V(\mathcal{K}_2) - 1 > \frac{1}{3},$$

and there exists $v_j \in \mathcal{K}_1 \cap \mathcal{K}_2^{(j)}$. Let $c_j := c(h_B^j)$. Since $h_B^j(\mathfrak{n})B \not\subset \mathfrak{n}^{c_j+1}$, there exists $x \in \mathfrak{n}$ with $\mathrm{ord}_{\mathfrak{n}}(h_B^j(x)) = c_j$. Since $h_B^j = (g^{Nj})^*$ on B and $\Lambda^{-j}(g^{Nj})_* v_j \in \mathcal{K}_2$, and since $\beta_B(v_j) \leq M_1$,

$$\epsilon \Lambda^j \leq \Lambda^j \big(\Lambda^{-j}(g^{Nj})_* v_j\big)(x) = v_j(h_B^j(x)) \leq M_1 c_j.$$

Thus

$$c(h_B^j) \geq \frac{\epsilon}{M_1} \Lambda^j \quad \text{for any integer } j \geq 0,$$

and (3) holds with $b_V := \epsilon/M_1$. □

13. Proof of the main theorem

13.1. **Parts (a) and (b).**

Proof of Theorem 1.4(a) and (b). Since F is finite, R is finite over $F^*(R)$, so $\dim F^*(R) = n = \dim R$, hence $\ker F^* = 0$ as R is a domain, and $F^* : R \to R$ is a finite injective local $\mathbb{C}$-algebra homomorphism. Since F^* is a $\mathbb{C}$-algebra homomorphism and $R/\mathfrak{m} = \mathbb{C}$, we have $[R/\mathfrak{m} : F^*_{R/\mathfrak{m}}(R/\mathfrak{m})] = 1$. Thus Lemma 2.12, with R, F^*, and $\mathfrak{m}$ in place of S, φ, and I, implies

$$d(F) = \ell_R(R/F^*(\mathfrak{m})R) = [\mathrm{Frac}(R) : F^*(\mathrm{Frac}(R))].$$

Since $d(F) = c_\infty(F)^n = q^n$, we have

$$[\mathrm{Frac}(R) : F^*(\mathrm{Frac}(R))] = q^n \quad \text{and} \quad c_\infty(F^*) = q. \tag{13.1}$$

We show that $\inf_{j\geq 0} c(F^j)/q^j > 0$. Assume that $\inf_{j\geq 0} c(F^j)/q^j = 0$. Since $c(F^j) = c((F^*)^j)$, Construction-Theorem 10.6, with F^* in place of φ, implies that there exist an integer $a \geq 1$ and prime ideals $0 \neq C \subsetneq P$ of R, both totally invariant under $(F^a)^*$, satisfying (1)–(4) of Construction-Theorem 10.6, where $Q := q^a$, $A := (R/C)_{P/C}$, $\mathfrak{m}_A := (P/C)A$, $k := A/\mathfrak{m}_A$, and $h : A \to A$ is the local homomorphism induced by $(F^a)^*$. By Construction-Theorem 10.6(2), A is normal. Let A_0 be the normalization of R/C, let $\psi : A_0 \to A_0$ be given by Lemma 10.4 with $(F^a)^*$ in place of φ, and let $\mathfrak{q}$ be the prime ideal of A_0 over P/C. The homomorphism of $(A_0)_\mathfrak{q} = A$ induced by ψ is h, since both extend the endomorphism of R/C induced by $(F^a)^*$. Since A is of klt type (Construction-Theorem 10.6(2)), there exists a $\mathbb{Q}$-divisor $\Delta \geq 0$ on $\operatorname{Spec} A$ such that $K_A + \Delta$ is $\mathbb{Q}$-Cartier and $(\operatorname{Spec} A, \Delta)$ is klt. Since $\dim A \geq 2$ by Construction-Theorem 10.6(1), Set-up 7.1 holds with h in place of φ, and Construction-Theorem 10.6(1) reads $D = \delta Q^{\dim A}$. Let m, Y, $\pi : Y \to \operatorname{Spec} A$, $g : Y \to Y$, and $T = \pi^{-1}(\mathfrak{m}_A)$ be as in Construction-Theorem 11.1, with h in place of φ. Thus $\pi \circ g = \operatorname{Spec}(h^m) \circ \pi$. By (1) and (3) of Construction-Theorem 10.6, the assumptions of Construction-Theorem 12.2 hold with Q and the real number of Construction-Theorem 10.6(3). Let V, N, η_V, $B = \mathcal{O}_{Y,\eta_V}$, $\mathfrak{n} = \mathfrak{m}_B$, $r = \dim B$, $h_B = (g^N)^*$, and $\Lambda = Q^{mN}$ be as in Construction-Theorem 12.2.

By Construction-Theorem 11.1 and (1)–(3) of Construction-Theorem 12.2, Set-up 4.8 holds with η_V and h_B in place of y and h. By Theorem 6.5, there exists a primitive divisorial valuation u of $\mathrm{Frac}(B)$ centered at $\mathfrak{n}$ with $(h_B)_* u = \Lambda u$.

By Theorem 6.5, u is also a primitive divisorial valuation of $\mathrm{Frac}(A_0)$ centered at $\mathfrak{n} \cap A_0 = \mathfrak{q}$, since $\eta_V \in T$. Since $A = (A_0)_\mathfrak{q}$, the base change to $\operatorname{Spec} A$ of a model over $\operatorname{Spec} A_0$ on which u is the order of vanishing along a prime divisor shows that u is a primitive divisorial valuation of $\mathrm{Frac}(A)$ centered at $\mathfrak{m}_A$.

By Construction-Theorem 11.1(1), $g_* v = (h^m)_* v$ for any real valuation v of $\mathrm{Frac}(A)$. Since $(h_B)_* u = u \circ (g^*)^N = (g^N)_* u$, we have

$$(h^{mN})_* u = (g^N)_* u = \Lambda u.$$

Since $\mathfrak{q}$ is totally invariant under ψ, it is totally invariant under ψ^{mN}, so Set-up 7.1 holds with ψ^{mN} in place of ψ. Since field degrees and residue degrees multiply under composition,

$$[\mathrm{Frac}(A) : h^{mN}(\mathrm{Frac}(A))] = D^{mN} = \delta^{mN} Q^{mN \dim A} = \Lambda^{\operatorname{ht} \mathfrak{m}_A} [k : (h^{mN})_k(k)].$$

Since h^{mN} is finite and local, $\mathfrak{m}_A$ is totally invariant under h^{mN}. By Proposition 9.3(2), with ψ^{mN}, h^{mN}, Λ, $\mathfrak{m}_A$, and u in place of ψ, φ, Q, P, and w, there exists an $\mathfrak{m}_A$-primary ideal I of A with

$$h^{mN}(I)A \subset I^\Lambda \subset \overline{I^{Q^{mN}}}.$$

This contradicts Construction-Theorem 10.6(4) with $i = mN$. Thus

$$b := \inf_{j\geq 0} c(F^j)/q^j > 0,$$

and $c(F^j) \geq bq^j$ for any integer $j \geq 0$.

We prove (b). Since R is regular, Set-up 4.8 holds with R, $\mathfrak{m}$, $\operatorname{Spec} R$, the closed point of $\operatorname{Spec} R$, F^*, and q in place of A_0, $\mathfrak{q}$, Y, y, h, and Λ. Moreover, the degree identity holds by (13.1), and $c((F^*)^j) \geq bq^j$ for any integer $j \geq 0$. By Theorem 6.5, there exists a primitive divisorial valuation w of $\mathrm{Frac}(R)$ centered at $\mathfrak{m}$ with $F_* w = qw$.

Since F^* is finite and local, $\mathfrak{m}$ is totally invariant under F^*. Since moreover (13.1) holds, $[R/\mathfrak{m} : F^*_{R/\mathfrak{m}}(R/\mathfrak{m})] = 1$, and $\operatorname{ht} \mathfrak{m} = n$, (1) and (2) of Construction-Proposition 9.4, with F^*, $\mathfrak{m}$, and q in place of φ, J, and Q, give a normal integral scheme W, a projective birational morphism $\pi_W : W \to \operatorname{Spec} R$, and a prime divisor E on W such that w is the order of vanishing along E, π_W is an isomorphism over $\operatorname{Spec} R \backslash \{\mathfrak{m}\}$, and $\pi_W^{-1}(\mathfrak{m}) = \operatorname{Supp} E$.

Since π_W is projective, W is the closed subscheme of $\mathbb{P}^M_R$ defined by a finite set $\mathcal{G}$ of homogeneous polynomials in $R[y_0,\dots,y_M]$ for some integer $M \geq 1$. Let $U \subset \mathbb{C}^n$ be a polydisc centered at 0 on which the coefficients of all $G \in \mathcal{G}$ converge, let $\mathcal{X} \subset \mathbb{P}^M \times U$ be the closed complex subspace defined by $\mathcal{G}$, and let $\pi_{\mathcal{X}} : \mathcal{X} \to U$ be the projection, so $\pi_{\mathcal{X}}$ is projective. Let x be a closed point of W. Then $\pi_W(x) = \mathfrak{m}$ and x has residue field $\mathbb{C}$, so $x \in \pi_{\mathcal{X}}^{-1}(0)$. Possibly reordering indices, we may assume that $x \in \{y_0 \neq 0\}$. Let $\mathbf{s} := (y_1/y_0,\dots,y_M/y_0)$, let $\mathfrak{a} \subset R[\mathbf{s}]$ be the ideal of $W \cap \{y_0 \neq 0\}$, and let $z := (z_1,\dots,z_n)$. Then $\mathcal{O}_{W,x} = (R[\mathbf{s}]/\mathfrak{a})_x$ and

$$\mathcal{O}_{\mathcal{X},x} = \mathbb{C}\{z, \mathbf{s}-\mathbf{s}(x)\}/\mathfrak{a}\mathbb{C}\{z,\mathbf{s}-\mathbf{s}(x)\},$$

where $\mathbf{s}(x) \in \mathbb{C}^M$ is the value of $\mathbf{s}$ at x. The completions of $\mathcal{O}_{W,x}$ and $\mathcal{O}_{\mathcal{X},x}$ both equal

$$\mathbb{C}[[z,\mathbf{s}-\mathbf{s}(x)]]/\mathfrak{a}\mathbb{C}[[z,\mathbf{s}-\mathbf{s}(x)]].$$

Thus $\mathcal{O}_{W,x} \to \mathcal{O}_{\mathcal{X},x}$ induces an isomorphism of completions, so it is faithfully flat. For each $i \in \{1,\dots,n\}$, π_W is an isomorphism over $D(z_i) \subset \operatorname{Spec} R$, and its inverse is $(\sigma_{i0} : \cdots : \sigma_{iM})$ with $\sigma_{i0},\dots,\sigma_{iM} \in R[z_i^{-1}]$. After shrinking U, each σ_{ij}, some $\tau_{i0},\dots,\tau_{iM} \in R[z_i^{-1}]$ with $\sum_{j=0}^M \tau_{ij}\sigma_{ij} = 1$, and the finitely many coefficients in $R[z_i^{-1}]$ through which the identities (13.2) hold in the coordinate rings of W are holomorphic on $U \cap \{z_i \neq 0\}$, so $\sigma_{i0},\dots,\sigma_{iM}$ have no common zero there, and the polynomial identities

$$G(\sigma_{i0},\dots,\sigma_{iM}) = 0 \text{ for } G \in \mathcal{G} \quad \text{and} \quad y_j\sigma_{ij'} = y_{j'}\sigma_{ij} \text{ for } 0 \leq j,j' \leq M, \tag{13.2}$$

which say that $(\sigma_{i0} : \cdots : \sigma_{iM})$ is the inverse of π_W over $D(z_i)$, hold on $\mathcal{X} \cap \{z_i \neq 0\}$: the first hold in $R[z_i^{-1}]$, and the second, after multiplication by a power of any one of $y_0,\dots,y_M$, hold in the ideal generated by $\mathcal{G}$ in $R[z_i^{-1}][y_0,\dots,y_M]$. After multiplication by a power of z_i, all of them become identities of convergent power series on the polydisc U. Since $U\backslash\{0\} = \bigcup_{i=1}^n (U \cap \{z_i \neq 0\})$, $\pi_{\mathcal{X}}$ is an isomorphism over $U\backslash\{0\}$. Since W is normal and excellent, $\widehat{\mathcal{O}_{W,x}}$ is a domain of dimension n for any closed point x of W, so $\mathcal{X}$ is locally irreducible of dimension n at each point of $\pi_{\mathcal{X}}^{-1}(0)$. Since $\pi_{\mathcal{X}}^{-1}(0)$ is the analytic space of $\pi_W^{-1}(\mathfrak{m}) = \operatorname{Supp} E$, of dimension $n-1$, $\pi_{\mathcal{X}}^{-1}(U\backslash\{0\})$ is dense in $\mathcal{X}$, so $\pi_{\mathcal{X}}$ is proper bimeromorphic.

Since E is an integral projective scheme over $R/\mathfrak{m} = \mathbb{C}$ and W is normal, there exists a closed point x of E at which W and E are regular. Let t be a local equation of E at x. Since $\mathcal{O}_{W,x}/(t)$ is regular, $\widehat{\mathcal{O}_{W,x}}/t\widehat{\mathcal{O}_{W,x}}$ is a domain. Since $\widehat{\mathcal{O}_{W,x}} = \widehat{\mathcal{O}_{\mathcal{X},x}}$, the ring $\mathcal{O}_{\mathcal{X},x}/(t)$ embeds in $\widehat{\mathcal{O}_{W,x}}/t\widehat{\mathcal{O}_{W,x}}$, so t generates a prime ideal of the regular local ring $\mathcal{O}_{\mathcal{X},x}$. By faithful flatness,

$$t^j\mathcal{O}_{\mathcal{X},x} \cap \mathcal{O}_{W,x} = t^j\mathcal{O}_{W,x} \quad \text{for any integer } j \geq 0.$$

Thus for any $f \in R\backslash\{0\}$,

$$w(f) = \max\{j \in \mathbb{N} \mid t^j \text{ divides } f \text{ in } \mathcal{O}_{\mathcal{X},x}\},$$

which is the order of vanishing of f along the irreducible component E_1 of $\pi_{\mathcal{X}}^{-1}(0)$ containing the germ at x of $\{t=0\}$. Hence w is the order of vanishing along the prime divisor E_1 of $\mathcal{X}$, and (b) holds.

We prove (a). Since w is divisorial, w is quasi-monomial (by a log resolution of (W,E) [Tem18, Theorem 1.1.6]), and (a) holds with $v := w$. □

13.2. **Part (c).**

Proof of Theorem 1.4(c). The coordinates of f are homogeneous of degree q and have no common zero, since $U_0^q = U_1^q = U_3^q = \cdots = U_n^q = 0$ forces $U_1U_0^{q-1} - U_2^q = -U_2^q$. Thus f is a morphism with

$$f^*\mathcal{O}_{\mathbb{P}^n}(1) \cong \mathcal{O}_{\mathbb{P}^n}(q),$$

hence finite and surjective, and f is a finite q-polarized endomorphism.

The subvariety Z_r is a linear subspace isomorphic to $\mathbb{P}^{n-r}$, so Z_r is irreducible of dimension $n-r$. A closed point $[U_0 : \cdots : U_n]$ is mapped by f into Z_r if and only if

$$U_1^q = 0, \quad U_1 U_0^{q-1} - U_2^q = 0, \quad \text{and} \quad U_3^q = \cdots = U_r^q = 0,$$

that is, if and only if $U_1 = \cdots = U_r = 0$. Thus $f^{-1}(Z_r) = Z_r$ as sets, so Z_r is totally invariant.

Let $H_3 := \{U_3 = 0\}$, and on the chart $\{U_0 \neq 0\}$ let

$$x := U_1/U_0, \quad y := U_2/U_0, \quad u_3 := U_3/U_0, \ \ldots, \quad \text{and} \quad u_n := U_n/U_0.$$

Since the first coordinate of f is U_0^q, we have $f^{-1}(\{U_0 \neq 0\}) = \{U_0 \neq 0\}$. Let $k \geq 1$ be an integer. On the chart $\{U_0 \neq 0\}$, f^k is given by

$$(x, y, u_3, \ldots, u_n) \to \bigl(G^k(x,y), u_3^{q^k}, \ldots, u_n^{q^k}\bigr), \quad \text{where } G(x,y) := (x - y^q, x^q),$$

so the Jacobian determinant of f^k in the coordinates $x, y, u_3, \ldots, u_n$ is

$$\det D(G^k)(x,y) \cdot q^{k(n-2)} (u_3 \cdots u_n)^{q^k - 1}.$$

Since f^k is finite and surjective, the Jacobian determinant of f^k is a non-zero polynomial, so $\det D(G^k)(x,y)$ is a non-zero polynomial in x and y alone. Since f^k is a finite morphism between smooth varieties, $R(f^k) \geq 0$ and $R(f^k)|_{\{U_0 \neq 0\}}$ is the divisor of the Jacobian determinant. Since $H_3 \cap \{U_0 \neq 0\} = \{u_3 = 0\}$ and none of $\det D(G^k)(x,y), u_4, \ldots, u_n$ vanishes identically on $\{u_3 = 0\}$, the coefficient of H_3 in $R(f^k)$ is $q^k - 1$. Thus

$$R(f^k) - (q^k - 1)H_3 \geq 0 \quad \text{for any integer } k \geq 1. \tag{13.3}$$

Let $p \in H_3$ be a closed point. By (13.3), for any integer $k \geq 1$,

$$\operatorname{mult}_p R(f^k) \geq q^k - 1 \geq (1 - q^{-1}) q^k.$$

By the definition of $\mathcal{E}_{\mathrm{Jac}}(f)$ with $\delta = 1 - q^{-1}$, we have $H_3 \subset \mathcal{E}_{\mathrm{Jac}}(f)$. Since $r \geq 3$, $Z_r \subset H_3$, so $Z_r \subset \mathcal{E}_{\mathrm{Jac}}(f)$.

It remains to prove the last part of Theorem 1.4(c). Let η_{Z_r} be the generic point of Z_r. Since $[1 : 0 : \cdots : 0] \in Z_r$, η_{Z_r} is contained in $\{U_0 \neq 0\}$. For any integer $i \geq 0$, let

$$x_i := (f^i)^* x \quad \text{and} \quad y_i := (f^i)^* y.$$

Since $f^{-i}(\{U_0 \neq 0\}) = \{U_0 \neq 0\}$ and $f^{-i}(Z_r) = Z_r$, both x_i and y_i are non-zero regular functions on $\{U_0 \neq 0\}$ vanishing on Z_r, so they are non-zero elements of the maximal ideal of $\mathcal{O}_{\mathbb{P}^n, \eta_{Z_r}}$. Since $f^* x = x - y^q$ and $f^* y = x^q$, we have $x_{i+1} = x_i - y_i^q$ and $y_{i+1} = x_i^q$ for any integer $i \geq 0$. Thus

$$x_{i+1} = x_i - x_{i-1}^{q^2} \quad \text{for any integer } i \geq 1. \tag{13.4}$$

Suppose that there exist an integer $k \geq 1$ and a real valuation v of the function field of $\mathbb{P}^n$ centered at η_{Z_r} such that $(f^k)_* v = q^k v$, that is, $v((f^k)^* g) = q^k v(g)$ for any non-zero rational function g on $\mathbb{P}^n$. Since v is centered at η_{Z_r} and x_i is a non-zero element of the maximal ideal of $\mathcal{O}_{\mathbb{P}^n, \eta_{Z_r}}$, we have $0 < v(x_i) < +\infty$ for any integer $i \geq 0$. Since $x_{i+k} = (f^k)^* x_i$,

$$v(x_{i+k}) = q^k v(x_i) \quad \text{for any integer } i \geq 0. \tag{13.5}$$

Suppose that $v(x_i) \neq q^2 v(x_{i-1})$ for some integer $i \geq 1$. By (13.4),

$$v(x_{i+1}) = \min\{v(x_i), q^2 v(x_{i-1})\} \leq v(x_i) < q^2 v(x_i).$$

Since $v(x_{i+1}) < q^2 v(x_i)$, (13.4) implies $v(x_{i+2}) = v(x_{i+1}) < q^2 v(x_{i+1})$. By induction, $v(x_{i+1+m}) = v(x_{i+1})$ for any integer $m \geq 0$, which contradicts (13.5) with $i+1$ in place of i, since $q^k > 1$ and $v(x_{i+1}) > 0$. Thus $v(x_i) = q^2 v(x_{i-1})$ for any integer $i \geq 1$, so $v(x_k) = q^{2k} v(x_0)$, whereas $v(x_k) = q^k v(x_0)$ by (13.5). Since $v(x_0) > 0$, $q \geq 2$, and $k \geq 1$, this is impossible.

Thus no such k and v exist, and Theorem 1.4(c) follows. □

13.3. **The non-klt locus of $T_{f,0}$.**

Lemma 13.1. *Let $n \geq 3$ and $q \geq 2$ be integers, let $f : \mathbb{P}^n \to \mathbb{P}^n$ be given by*

$$f([U_0 : \cdots : U_n]) = [U_0^q : U_1 U_0^{q-1} - U_2^q : U_1^q : U_3^q : \cdots : U_n^q],$$

let $H_3 := \{U_3 = 0\}$, and let $T_{f,0} := \sum_{j\geq 1}[q^{-nj}(f^j)_* R(f)]$ be the closed positive $(1,1)$-current of [Yos26, Definition 3.1(4) and Theorem 4.5]*, where $[D]$ is the current of integration of an $\mathbb{R}$-divisor D. Then $T_{f,0} - [H_3]$ is a closed positive $(1,1)$-current, and $H_3 \subset \mathrm{Nklt}(\mathbb{P}^n, 0; T_{f,0})$, where $\mathrm{Nklt}(\mathbb{P}^n, 0; T_{f,0})$ is the non-klt locus of* [Yos26, Definition 4.2].

Proof. Since the coordinates of f are homogeneous of degree q and have no common zero, f is a finite morphism with

$$f^* \mathcal{O}_{\mathbb{P}^n}(1) \cong \mathcal{O}_{\mathbb{P}^n}(q).$$

Since $q > 1$, $(\mathbb{P}^n, 0, f)$ is an int-amplified dynamical triple [Yos26, Definition 3.1(1)(2)], so $T_{f,0}$ admits local psh potentials by [Yos26, Theorem 4.5].

By (13.3) in the proof of Theorem 1.4(c), with $k = 1$, $R(f) - (q-1)H_3 \geq 0$. Since $f^* U_3 = U_3^q$, we have $f^* H_3 = q H_3$. Since $\deg f = q^n$,

$$f_* H_3 = q^{-1} f_* f^* H_3 = q^{n-1} H_3,$$

and $(f^j)_* H_3 = q^{(n-1)j} H_3$ for any integer $j \geq 1$. For any integer $j \geq 1$,

$$q^{-nj}(f^j)_* R(f) - (q-1)q^{-j} H_3 = q^{-nj}(f^j)_* \big(R(f) - (q-1)H_3\big) \geq 0.$$

Since $\sum_{j=1}^N (q-1)q^{-j} = 1 - q^{-N}$ for any integer $N \geq 1$, the current

$$\sum_{j=1}^N \big[q^{-nj}(f^j)_* R(f)\big] - (1 - q^{-N})[H_3] = \sum_{j=1}^N \big[q^{-nj}(f^j)_* R(f) - (q-1)q^{-j} H_3\big]$$

is $[D]$ for a $\mathbb{Q}$-divisor $D \geq 0$, hence closed and positive [Yos26, Notation 4.1]. As $N \to +\infty$, the left-hand side converges weakly to $T_{f,0} - [H_3]$, since $\sum_{j\geq 1}(q-1)q^{-j} = 1$. Since a weak limit of closed positive $(1,1)$-currents is a closed positive $(1,1)$-current, $T_{f,0} - [H_3]$ is a closed positive $(1,1)$-current.

Let $p \in H_3$ be a closed point. Let W be an open neighborhood of p in $(\mathbb{P}^n)^{\mathrm{an}}$ and let ϕ be a psh function on W such that $T_{f,0}|_W = dd^c \phi$. Since the boundary is zero and $\mathbb{P}^n$ is smooth, we may take $\mathrm{id}_{\mathbb{P}^n}$ as the log resolution in [Yos26, Definition 4.2]. Thus $(\mathbb{P}^n, 0; T_{f,0})$ is analytically klt at p if and only if $e^{-2\phi}$ is integrable on a neighborhood of p. Possibly shrinking W, let t be a holomorphic function on W which is part of a holomorphic coordinate system at p and satisfies $H_3 \cap W = \{t = 0\}$. Since $dd^c = (\sqrt{-1}/\pi)\partial\overline{\partial}$, the Poincaré–Lelong formula implies $dd^c \log|t| = [H_3]|_W$ [Yos26, Notation 4.1]. Thus

$$dd^c(\phi - \log|t|) = (T_{f,0} - [H_3])|_W \geq 0,$$

so the locally integrable function $\phi - \log|t|$ is equal almost everywhere to a psh function ψ on W [Dem12, Chapter 1, (1.5)(a) and (1.7)]. Possibly shrinking W to a relatively compact neighborhood of p, $\sup_W \psi < +\infty$. Thus, almost everywhere on W,

$$e^{-2\phi} = e^{-2\psi}|t|^{-2} \geq e^{-2\sup_W \psi}|t|^{-2}.$$

Since t is a coordinate at p and, in the variable t,

$$\int_{\{|t|<\epsilon\}} |t|^{-2} = +\infty \quad \text{for any } \epsilon > 0,$$

the function $|t|^{-2}$ is not integrable on any neighborhood of p, so neither is $e^{-2\phi}$. Thus $p \in \mathrm{Nklt}(\mathbb{P}^n, 0; T_{f,0})$. Since p is an arbitrary closed point of H_3, $H_3 \subset \mathrm{Nklt}(\mathbb{P}^n, 0; T_{f,0})$. The lemma follows. □

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